

Microsoft Official Course



AZ-140T00

Configuring and
Operating Microsoft
Azure Virtual Desktop

AZ-140T00

Configuring and Operating Microsoft Azure Virtual Desktop

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Module 0 Welcome

Start here

Welcome to Configuring and Operating Microsoft Azure Virtual Desktop

AZ-140: Configuring and Operating Microsoft Azure Virtual Desktop

This course teaches Azure Virtual Desktop administrators how to plan, deliver, and manage virtual desktop experiences and remote apps, for any device, on Azure.

Students will learn through a mix of demonstrations and hands-on lab experiences deploying virtual desktop experiences and apps on Azure Virtual Desktop and optimizing them to run in virtual environments.

Level: Intermediate

Audience

Students for **AZ-140: Configuring and Operating Microsoft Azure Virtual Desktop** deliver applications on Azure Virtual Desktop that are optimized to run in multi-session virtual environments. As an Azure Virtual Desktop administrator, you will closely work with the Azure Administrators and Architects, along with Microsoft 365 Administrators.

Azure Virtual Desktop administrator responsibilities include planning, deploying, packaging, updating, and maintaining the Azure Virtual Desktop infrastructure. They also create session host images, implement and manage FSLogix, monitor Azure Virtual Desktop performance, and automate Azure Virtual Desktop management tasks.

Prerequisites

Successful Azure Virtual Desktop administrators start this role with experience on operating systems, virtualization, cloud infrastructure, storage structures, and networking.

Additionally, some experience with planning, deploying, packaging, updating, and maintaining a hybrid cloud infrastructure.

This knowledge includes:

- Understanding of on-premises virtualization technologies, including: VMs, virtual networking, and virtual hard disks.
- Understanding of network configuration, including TCP/IP, Domain Name System (DNS), virtual private networks (VPNs), firewalls, and encryption technologies.
- Understanding of Active Directory concepts, including users, groups, role-based access control.
- Understanding of resilience and disaster recovery, including backup and restore operations.

Note:

If you are new to Azure and cloud computing, consider the free online content: **Azure Fundamentals**¹.

Expected learning

After completing this course, students will be able to:

- Select an appropriate licensing model for Azure Virtual Desktop
- Implement networking for Azure Virtual Desktop
- Manage Azure Virtual Desktop session hosts by using Azure Bastion
- Configure storage for FSLogix components
- Create and manage session host images
- Implement Azure roles and role-based access control (RBAC) for Azure Virtual Desktop
- Configure user Azure Virtual Desktop experience settings
- Install and configure apps on a session host
- Implement business continuity and disaster recovery
- Monitor and manage Azure Virtual Desktop performance

Syllabus

The course content includes a mix of content, hands-on labs, reference links, and module review questions.

Module 0: Welcome

- Start here
 - Welcome to Configuring and Operating Microsoft Azure Virtual Desktop
 - Syllabus
 - AZ-140 Certification Exam

¹ <https://docs.microsoft.com/en-us/learn/paths/azure-fundamentals/>

- Microsoft Learn
- Study Resources

Module 1: Plan an Azure Virtual Desktop implementation

- Azure Virtual Desktop Architecture
 - Azure Virtual Desktop for the enterprise
 - Azure Virtual Desktop components
 - Personal and pooled desktops
 - Service updates for AVD desktops
 - Azure limitations for Azure Virtual Desktop
 - VM sizing
 - Azure Virtual Desktop pricing
- Design the Azure Virtual Desktop architecture
 - Assess existing physical and virtual desktop environments
 - Assess network capacity and speed requirements for AVD
 - Azure Virtual Desktop Experience Estimator
 - Recommend an operating system for an AVD implementation
 - Balancing host pools
 - Recommendations for using subscriptions and management groups
 - Configure a location for the AVD metadata
 - Recommend a configuration for performance requirements
 - Calculate and recommend a configuration for Azure VM capacity requirements
- Design for user identities and profiles
 - Select an appropriate licensing model for AVD based on requirements
 - Personal and multi-session desktop scenarios
 - Recommend an appropriate storage solution
 - Plan for a Desktop client deployment
 - Plan for AVD client deployment - RDP
 - Windows Desktop client to multiple devices
 - Hybrid Identity with Azure Active Directory
 - Plan for Azure AD Connect for user identities
- Labs
 - Prepare for deployment of Azure Virtual Desktop (Azure AD DS)
 - Prepare for deployment of Azure Virtual Desktop (AD DS)

- Review questions

- Module review questions

Module 2: Implement an Azure Virtual Desktop infrastructure

- Implement and manage networking for AVD
 - Implement Azure virtual network connectivity
 - Manage connectivity to the internet and on-premises networks
 - Understanding Azure Virtual Desktop network connectivity
 - Implement and manage network security
 - Configure AVD session hosts using Azure Bastion
 - Azure Network Watcher
- Implement and manage storage for AVD
 - Storage for FSLogix components
 - Configure storage for FSLogix components
 - Configure storage accounts
 - Configure disks
 - Create file shares
- Create and configure host pools and session hosts
 - Demonstration - Create and share a desktop with users
 - Configure the personal desktop host pool assignment type
 - Automate creation of AVD host and host pools by using PowerShell
 - Demonstration - Customize the Azure Virtual Desktop workspace
 - Customize RDP properties for a host pool
 - Manage licensing for session hosts that run Windows client or Windows Server
 - Assign users to host pools
 - Apply OS and application updates to a running AVD host
 - Apply security and compliance settings to session hosts
- Create and manage session host image
 - Create a master image
 - Modify a session host image
 - Plan for image update and management
 - Create and use a Shared Image Gallery (SIG) using the portal
 - Install language packs in AVD

- Labs
 - Create and configure host pools and session hosts (Azure AD DS)
 - Deploy host pools and session hosts by using the Azure portal (AD DS)
 - Implement and manage storage for AVD (Azure AD DS)
 - Implement and manage storage for AVD (AD DS)
 - Deploy host pools and hosts by using Azure Resource Manager templates
 - Deploy and manage host pools and hosts by using PowerShell
 - Create and manage session host images (AD DS)
- Review questions
 - Module review questions

Module 3 Manage access and security

- Manage access
 - RBAC for Azure Virtual Desktop
 - Plan and implement Azure roles and RBAC for AVD
 - Using Azure Virtual Desktop with Intune
- Manage security
 - Plan and implement Conditional Access policies for connections to AVD
 - Understand Conditional Access policy components
 - Plan and implement MFA in AVD
 - Manage security by using Azure Security Center
 - Security posture management and threat protection
 - Microsoft Defender Antivirus for session hosts
- Review questions
 - Module review questions

Module 4 Manage user environments and apps

- Implement and manage FSLogix
 - Plan for FSLogix
 - FSLogix profile containers and Azure files
 - Install FSLogix
 - Storage options for FSLogix profile containers
 - Profile Container vs Office Container
 - Configure Cloud Cache

- Configure Profile Containers
- Manage Rule Sets and application masking
- Configure user experience settings
 - Virtual desktop optimization principles
 - Persistent virtual desktop environments
 - Configure user settings through group policies
 - Configure user settings through Endpoint Manager policies
 - Configure session timeout properties
 - Configure device redirections
 - Configure Universal Print
 - Troubleshoot user profile issues
 - Troubleshoot AVD clients
- Install and configure apps on a session host
 - MSIX app attach
 - How MSIX app attach works
 - Set up a file share for MSIX app attach
 - Configure dynamic application delivery by using MSIX App Attach
 - Demonstration - Configure apps for users
 - Using the OneDrive sync app on virtual desktops
 - Using Microsoft Teams on Azure Virtual desktop
 - Publish built-in apps in Azure Virtual Desktop
 - Troubleshoot application issues related to AVD using User Input Delay
- Labs
 - Implement and manage Azure Virtual Desktop profiles (Azure AD DS)
 - Package Windows Azure Desktop applications (AD DS)
- Review Questions
 - Module review questions

Module 5 Monitor and maintain an Azure Virtual Desktop infrastructure

- Plan and implement business continuity and disaster recovery
 - Business continuity and disaster recovery
 - VM replication
 - Virtual networks, user identities, and data

- FSLogix configuration
- Configure Recovery Services vault for FSLogix user profiles, personal desktops, and images
- Configure backup from the Recovery Services vault
- Automate AVD management tasks
 - Scale session hosts using Azure Automation
 - Create or update an Azure Automation account
 - Create an Azure Automation Run As account
 - Create the Azure Logic App and execution schedule
- Monitor and manage performance and health
 - Monitor AVD by using Azure Monitor
 - Log Analytics workspace for Azure Monitor
 - Monitor AVD by using Azure Advisor
 - How to resolve Azure Advisor recommendations
 - Diagnose graphics performance issues
- Lab
 - Implement autoscaling in host pools (AD DS)
- Review questions
 - Module review questions

AZ-140 Certification Exam

Certification exams measure your ability to accomplish certain technical tasks for a job role. The study areas are based on the Job Task Analysis that was conducted for the role in January, 2021.

Each study area has a percentage indicating the relative weight of the area on the exam. The higher the percentage, the more questions you are likely to see in that area.

Study Area	Percentages
Plan an Azure Virtual Desktop architecture	10-15%
Implement an Azure Virtual Desktop infrastructure	25-30%
Manage access and security	10-15%
Manage user environments and apps	20-25%
Monitor and maintain an Azure Virtual Desktop infrastructure	20-25%

Candidates for this exam should have experience in Azure technologies, including virtualization, networking, identity, storage, backups, resilience, and disaster recovery. They should understand on-premises virtual desktop infrastructure technologies as they relate to migrating to Azure Virtual Desktop. These professionals use the Azure portal and Azure Resource Manager (ARM) templates to accomplish many of

their tasks. They might use PowerShell and Azure Command-Line Interface (CLI) for more efficient automation.

For more information, on the skills measured in the exam, please visit the **AZ-140: Configuring and Operating Microsoft Azure Virtual Desktop**² page.

Microsoft Learn

Microsoft Learn provides self paced skills training for Azure Virtual Desktop. Visit the **Deliver remote desktops and apps from Azure with Azure Virtual Desktop**³ learning path for the following modules:

- **Introduction to Azure Virtual Desktop in Microsoft Azure**⁴
- **Prepare for Azure Virtual Desktop in Microsoft Azure**⁵
- **Deploy Azure Virtual Desktop in Microsoft Azure**⁶
- **Optimize Azure Virtual Desktop in Microsoft Azure**⁷
- **Secure an Azure Virtual Desktop deployment**⁸
- **Deploy applications by using MSIX app attach for Azure Virtual Desktop**⁹

Additionally, Microsoft Learn provides self paced skills training on a variety of Azure topics that are relevant to Azure Virtual Desktop. These Learn modules are helpful for shoring up base knowledge of Azure technologies.

Module 01 - Identity

- **Create Azure users and groups in Azure Active Directory**¹⁰
- **Manage users and groups in Azure Active Directory**¹¹
- **Secure your Azure resources with role-based access control**¹²
- **Secure Azure Active Directory users with Multi-Factor Authentication**¹³
- **Allow users to reset their password with Azure Active Directory self-service password reset**¹⁴
- **Secure your application by using OpenID Connect and Azure AD**¹⁵

Module 02 - Governance and Compliance

- **Analyze costs and create budgets with Azure Cost Management**¹⁶
- **Predict costs and optimize spending for Azure**¹⁷

² <https://docs.microsoft.com/en-us/learn/certifications/exams/az-140>

³ <https://docs.microsoft.com/en-us/learn/paths/m365-wvd/>

⁴ <https://docs.microsoft.com/en-us/learn/modules/m365-wvd-intro/>

⁵ <https://docs.microsoft.com/en-us/learn/modules/m365-prepare-for-wvd/>

⁶ <https://docs.microsoft.com/en-us/learn/modules/m365-deploy-wvd/>

⁷ <https://docs.microsoft.com/en-us/learn/modules/m365-optimize-wvd/>

⁸ <https://docs.microsoft.com/en-us/learn/modules/m365-wvd-security/>

⁹ <https://docs.microsoft.com/en-us/learn/modules/m365-wvd-application-management/>

¹⁰ <https://docs.microsoft.com/en-us/learn/modules/create-users-and-groups-in-azure-active-directory/>

¹¹ <https://docs.microsoft.com/en-us/learn/modules/manage-users-and-groups-in-aad/>

¹² <https://docs.microsoft.com/en-us/learn/modules/secure-azure-resources-with-rbac/>

¹³ <https://docs.microsoft.com/en-us/learn/modules/secure-aad-users-with-mfa/>

¹⁴ <https://docs.microsoft.com/en-us/learn/modules/allow-users-reset-their-password/>

¹⁵ <https://docs.microsoft.com/en-us/learn/modules/secure-app-with-oidc-and-azure-ad/>

¹⁶ <https://docs.microsoft.com/en-us/learn/modules/analyze-costs-create-budgets-azure-cost-management/>

¹⁷ <https://docs.microsoft.com/en-us/learn/modules/predict-costs-and-optimize-spending/>

- Control and organize Azure resources with Azure Resource Manager¹⁸
- Apply and monitor infrastructure standards with Azure Policy¹⁹
- Create custom roles for Azure resources with role-based access control²⁰
- Manage access to an Azure subscription by using Azure role-based access control²¹
- Secure your Azure resources with role-based access control²²

Module 03 - Azure Administration

- Core Cloud Services - Manage services with the Azure portal²³
- Control and organize Azure resources with Azure Resource Manager²⁴
- Build Azure Resource Manager templates²⁵
- Automate Azure tasks using scripts with PowerShell²⁶
- Manage virtual machines with the Azure CLI²⁷

Module 04 - Virtual Networking

- Networking Fundamentals - Principals²⁸
- Design an IP addressing schema for your Azure deployment²⁹
- Secure and isolate access to Azure resources by using network security groups and service endpoints³⁰

Module 05 - Intersite Connectivity

- Distribute your services across Azure virtual networks and integrate them by using virtual network peering³¹
- Connect your on-premises network to Azure with VPN Gateway³²
- Connect your on-premises network to the Microsoft global network by using ExpressRoute³³

Module 06 - Network Traffic Management

- Manage and control traffic flow in your Azure deployment with routes³⁴

¹⁸ <https://docs.microsoft.com/en-us/learn/modules/control-and-organize-with-azure-resource-manager/>

¹⁹ <https://docs.microsoft.com/en-us/learn/modules/intro-to-governance/>

²⁰ <https://docs.microsoft.com/en-us/learn/modules/create-custom-azure-roles-with-rbac/>

²¹ <https://docs.microsoft.com/en-us/learn/modules/manage-subscription-access-azure-rbac/>

²² <https://docs.microsoft.com/en-us/learn/modules/secure-azure-resources-with-rbac/>

²³ <https://docs.microsoft.com/en-us/learn/modules/tour-azure-portal/>

²⁴ <https://docs.microsoft.com/en-us/learn/modules/control-and-organize-with-azure-resource-manager/>

²⁵ <https://docs.microsoft.com/en-us/learn/modules/build-azure-vm-templates/>

²⁶ <https://docs.microsoft.com/en-us/learn/modules/automate-azure-tasks-with-powershell/>

²⁷ <https://docs.microsoft.com/en-us/learn/modules/manage-virtual-machines-with-azure-cli/>

²⁸ <https://docs.microsoft.com/en-us/learn/modules/network-fundamentals/>

²⁹ <https://docs.microsoft.com/en-us/learn/modules/design-ip-addressing-for-azure/>

³⁰ <https://docs.microsoft.com/en-us/learn/modules/secure-and-isolate-with-nsg-and-service-endpoints/>

³¹ <https://docs.microsoft.com/en-us/learn/modules/integrate-vnets-with-vnet-peering/>

³² <https://docs.microsoft.com/en-us/learn/modules/connect-on-premises-network-with-vpn-gateway/>

³³ <https://docs.microsoft.com/en-us/learn/modules/connect-on-premises-network-with-expressroute/>

³⁴ <https://docs.microsoft.com/en-us/learn/modules/control-network-traffic-flow-with-routes/>

- Improve application scalability and resiliency by using Azure Load Balancer³⁵
- Load balance your web service traffic with Application Gateway³⁶
- Enhance your service availability and data locality by using Azure Traffic Manager³⁷

Module 07 - Azure Storage

- Create an Azure Storage account³⁸
- Secure your Azure Storage³⁹
- Optimize storage performance and costs using Blob storage tiers⁴⁰
- Make your application storage highly available with read-access geo-redundant storage⁴¹
- Copy and move blobs from one container or storage account to another from the command line and in code⁴²
- Move large amounts of data to the cloud by using Azure Data Box family⁴³
- Monitor, diagnose, and troubleshoot your Azure storage⁴⁴

Module 08 - Azure Virtual Machines

- Build a scalable application with virtual machine scale sets⁴⁵
- Deploy Azure virtual machines from VHD templates⁴⁶
- Choose the right disk storage for your virtual machine workload⁴⁷
- Add and size disks in Azure virtual machines⁴⁸
- Protect your virtual machine settings with Azure Automation State Configuration⁴⁹

Module 09 - Serverless Computing

- Host a web application with Azure App service⁵⁰
- Stage a web app deployment for testing and rollback by using App Service deployment slots⁵¹
- Scale an App Service web app to efficiently meet demand with App Service scale up and scale out⁵²

³⁵ <https://docs.microsoft.com/en-us/learn/modules/improve-app-scalability-resiliency-with-load-balancer/>

³⁶ <https://docs.microsoft.com/en-us/learn/modules/load-balance-web-traffic-with-application-gateway/>

³⁷ <https://docs.microsoft.com/en-us/learn/modules/distribute-load-with-traffic-manager/>

³⁸ <https://docs.microsoft.com/en-us/learn/modules/create-azure-storage-account/>

³⁹ <https://docs.microsoft.com/en-us/learn/modules/secure-azure-storage-account/>

⁴⁰ <https://docs.microsoft.com/en-us/learn/modules/optimize-archive-costs-blob-storage/>

⁴¹ <https://docs.microsoft.com/en-us/learn/modules/ha-application-storage-with-grs/>

⁴² <https://docs.microsoft.com/en-us/learn/modules/copy-blobs-from-command-line-and-code/>

⁴³ <https://docs.microsoft.com/en-us/learn/modules/move-data-with-azure-data-box/>

⁴⁴ <https://docs.microsoft.com/en-us/learn/modules/monitor-diagnose-and-troubleshoot-azure-storage/>

⁴⁵ <https://docs.microsoft.com/en-us/learn/modules/build-app-with-scale-sets/>

⁴⁶ <https://docs.microsoft.com/en-us/learn/modules/deploy-vms-from-vhd-templates/>

⁴⁷ <https://docs.microsoft.com/en-us/learn/modules/choose-the-right-disk-storage-for-vm-workload/>

⁴⁸ <https://docs.microsoft.com/en-us/learn/modules/add-and-size-disks-in-azure-virtual-machines/>

⁴⁹ <https://docs.microsoft.com/en-us/learn/modules/protect-vm-settings-with-dsc/>

⁵⁰ <https://docs.microsoft.com/en-us/learn/modules/host-a-web-app-with-azure-app-service/>

⁵¹ <https://docs.microsoft.com/en-us/learn/modules/stage-deploy-app-service-deployment-slots/>

⁵² <https://docs.microsoft.com/en-us/learn/modules/app-service-scale-up-scale-out/>

- Dynamically meet changing web app performance requirements with autoscale rules⁵³
- Capture and view page load times in your Azure web app with Application Insights⁵⁴
- Run Docker containers with Azure Container Instances⁵⁵
- Introduction to the Azure Kubernetes Service⁵⁶

Module 10 - Data Protection

- Protect your virtual machines by using Azure Backup⁵⁷
- Back up and restore your Azure SQL database⁵⁸
- Protect your Azure infrastructure with Azure Site Recovery⁵⁹
- Protect your on-premises infrastructure from disasters with Azure Site Recovery⁶⁰

Module 11 - Monitoring

- Analyze your Azure infrastructure by using Azure Monitor logs⁶¹
- Improve incident response with alerting on Azure⁶²
- Monitor the health of your Azure virtual machine by collecting and analyzing diagnostic data⁶³
- Monitor, diagnose, and troubleshoot your Azure storage⁶⁴

Study Resources

There are a lot of additional resources to help you learn about Azure. We recommend you bookmark these pages.

- For Azure Virtual Desktop videos from the Microsoft Mechanics series, see: www.aka.ms/wvdplaylist.
- **Azure Virtual Desktop docs**⁶⁵. Deliver a virtual desktop experience and remote apps to any device. Bring together Microsoft 365 and Azure to provide users with the only multi-session Windows 10 experience—with exceptional scale and reduced IT costs.
- **What's new in Azure Virtual Desktop?**⁶⁶. A monthly article revealing recent Azure Virtual Desktop updates. Make sure to check back here often to keep up with new updates.
- **Migrate or deploy Azure Virtual Desktop instances to Azure**⁶⁷. Guidance from the Cloud Adoption Framework for migrating an organization's end-user desktops to the cloud.

⁵³ <https://docs.microsoft.com/en-us/learn/modules/app-service-autoscale-rules/>

⁵⁴ <https://docs.microsoft.com/en-us/learn/modules/capture-page-load-times-application-insights/>

⁵⁵ <https://docs.microsoft.com/en-us/learn/modules/run-docker-with-azure-container-instances/>

⁵⁶ <https://docs.microsoft.com/en-us/learn/modules/intro-to-azure-kubernetes-service/>

⁵⁷ <https://docs.microsoft.com/en-us/learn/modules/protect-virtual-machines-with-azure-backup/>

⁵⁸ <https://docs.microsoft.com/en-us/learn/modules/backup-restore-azure-sql/>

⁵⁹ <https://docs.microsoft.com/en-us/learn/modules/protect-infrastructure-with-site-recovery/>

⁶⁰ <https://docs.microsoft.com/en-us/learn/modules/protect-on-premises-infrastructure-with-azure-site-recovery/>

⁶¹ <https://docs.microsoft.com/en-us/learn/modules/analyze-infrastructure-with-azure-monitor-logs/>

⁶² <https://docs.microsoft.com/en-us/learn/modules/incident-response-with-alerting-on-azure/>

⁶³ <https://docs.microsoft.com/en-us/learn/modules/monitor-azure-vm-using-diagnostic-data/>

⁶⁴ <https://docs.microsoft.com/en-us/learn/modules/monitor-diagnose-and-troubleshoot-azure-storage/>

⁶⁵ <https://docs.microsoft.com/en-us/azure/virtual-desktop/>

⁶⁶ <https://docs.microsoft.com/en-us/azure/virtual-desktop/whats-new>

⁶⁷ <https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/scenarios/wvd/>

- **Azure Migration Program**⁶⁸. Get the guidance and expert help you need at every stage of your Azure Virtual Desktop cloud migration journey. Migrate infrastructure, databases, and apps—and move forward with confidence.
- **Azure forums**⁶⁹. The Azure forums are very active. You can search the threads for a specific area of interest. You can also browse categories like Azure Storage, Pricing and Billing, Azure Virtual Machines, and Azure Migrate.
- **Microsoft Learning Community Blog**⁷⁰. Get the latest information about the certification tests and exam study groups.
- **Channel 9**⁷¹. Channel 9 provides a wealth of informational videos, shows, and events.
- **Azure Fridays**⁷². Join Scott Hanselman as he engages one-on-one with the engineers who build the services that power Microsoft Azure, as they demo capabilities, answer Scott's questions, and share their insights.
- **Microsoft Azure Blog**⁷³. Keep current on what's happening in Azure, including what's now in preview, generally available, news & updates, and more.
- **Azure Documentation**⁷⁴. Stay informed on the latest products, tools, and features. Get information on pricing, partners, support, and solutions.
- **Azure Architecture Center**⁷⁵. The Azure Architecture Center provides best practices for running your workloads on Azure.

⁶⁸ <https://azure.microsoft.com/en-us/migration/migration-program/>

⁶⁹ <https://social.msdn.microsoft.com/Forums/en-US/home?category=windowsazureplatform>

⁷⁰ <https://www.microsoft.com/en-us/learning/community-blog.aspx>

⁷¹ <https://channel9.msdn.com/>

⁷² <https://channel9.msdn.com/Shows/Azure-Friday>

⁷³ <https://azure.microsoft.com/en-us/blog/>

⁷⁴ <https://docs.microsoft.com/en-us/azure/>

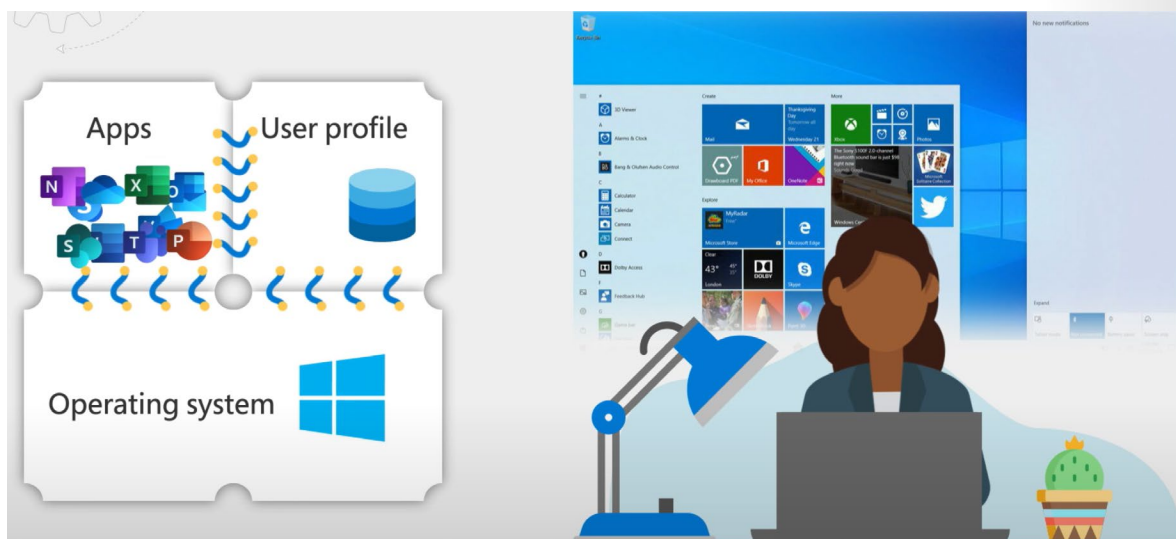
⁷⁵ <https://docs.microsoft.com/en-us/azure/architecture/>

Module 1 Plan an Azure Virtual Desktop implementation

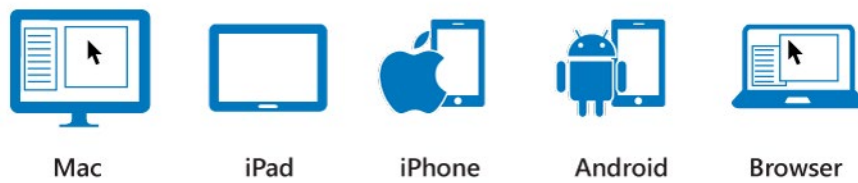
Azure Virtual Desktop Architecture

Azure Virtual Desktop for the enterprise

Azure Virtual Desktop is a desktop and application virtualization service that runs in the Azure cloud. Azure Virtual Desktop works across devices (Windows, Mac, iOS, Android, and Linux) with apps that you can use to access remote desktops and apps.



You can also use most modern browsers to access Azure Virtual Desktop-hosted experiences.

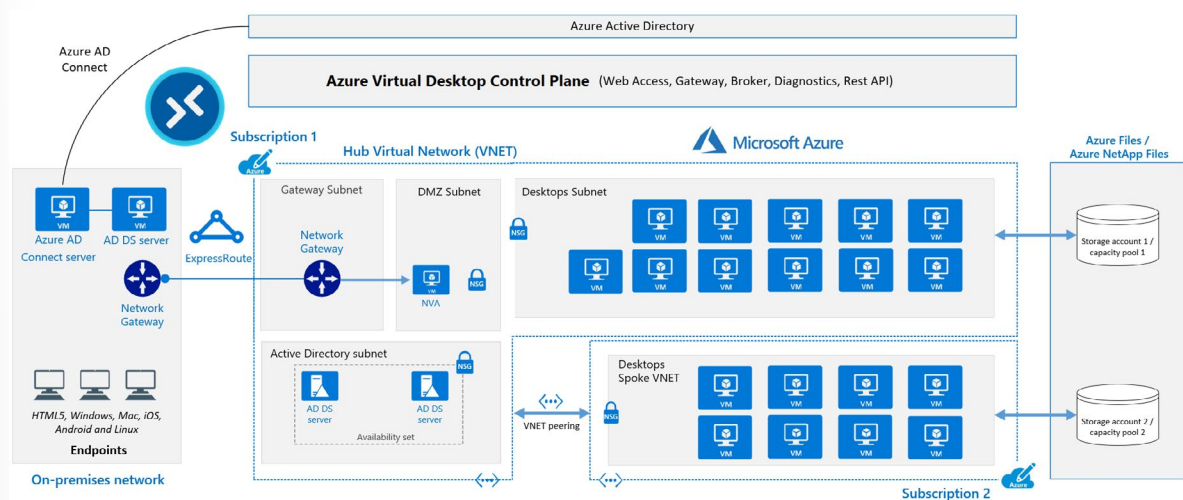


This course helps Desktop Infrastructure Architects, Cloud Architects, Desktop Administrators, or System Administrators explore Azure Virtual Desktop and build virtualized desktop infrastructure (VDI) solutions at enterprise scale. Enterprise-scale solutions generally cover 1,000 virtual desktops and above.

Most demand for enterprise virtual desktop solutions comes from:

- Security and regulation applications like financial services, healthcare, and government.
- Elastic workforce needs like remote work, mergers and acquisition, short-term employees, contractors, and partner access.
- Specific employees like bring your own device (BYOD) and mobile users, call centers, and branch workers.
- Specialized workloads like design and engineering, legacy apps, and software development test.

Architecture



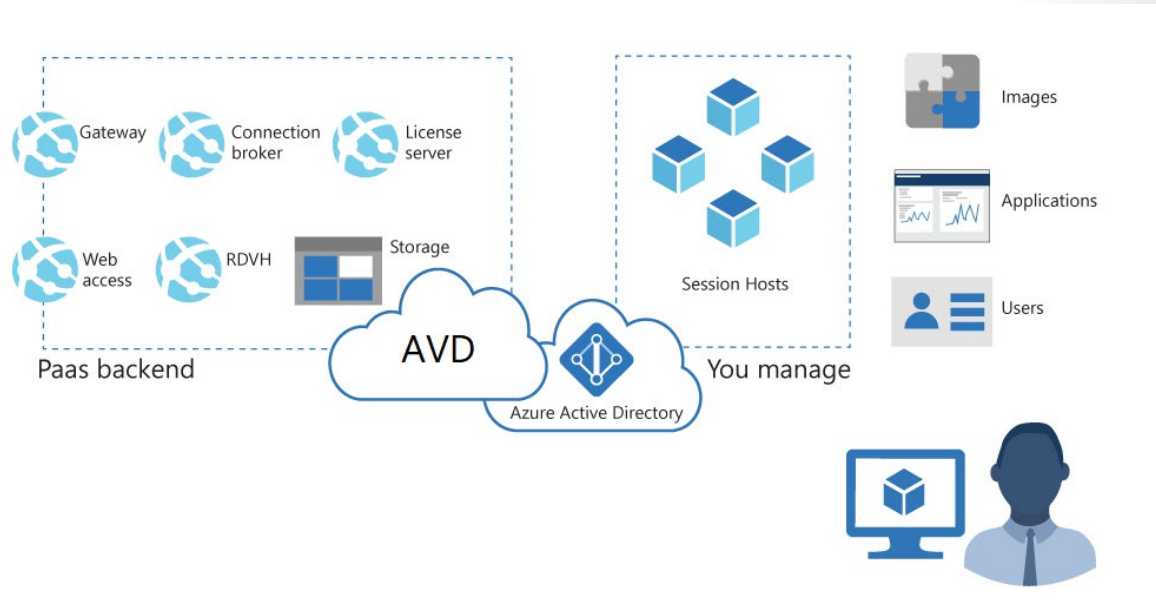
The diagram above shows a typical architectural setup for Azure Virtual Desktop.

- The application endpoints are in the customer's on-premises network. ExpressRoute extends the on-premises network into the Azure cloud, and Azure AD Connect integrates the customer's Active Directory Domain Services (AD DS) with Azure Active Directory (Azure AD).
- The Azure Virtual Desktop control plane handles Web Access, Gateway, Broker, Diagnostics, and extensibility components like REST APIs.
- The customer manages AD DS and Azure AD, Azure subscriptions, virtual networks, Azure Files or Azure NetApp Files, and the Azure Virtual Desktop host pools and workspaces.

- To increase capacity, the customer uses two Azure subscriptions in a hub-spoke architecture, and connects them via virtual network peering.

Azure Virtual Desktop components

Azure Virtual Desktop service architecture is similar to Windows Server Remote Desktop Services. Microsoft manages the infrastructure and brokering components, while enterprise customers manage their own desktop host virtual machines (VMs), data, and clients.



Components Microsoft manages

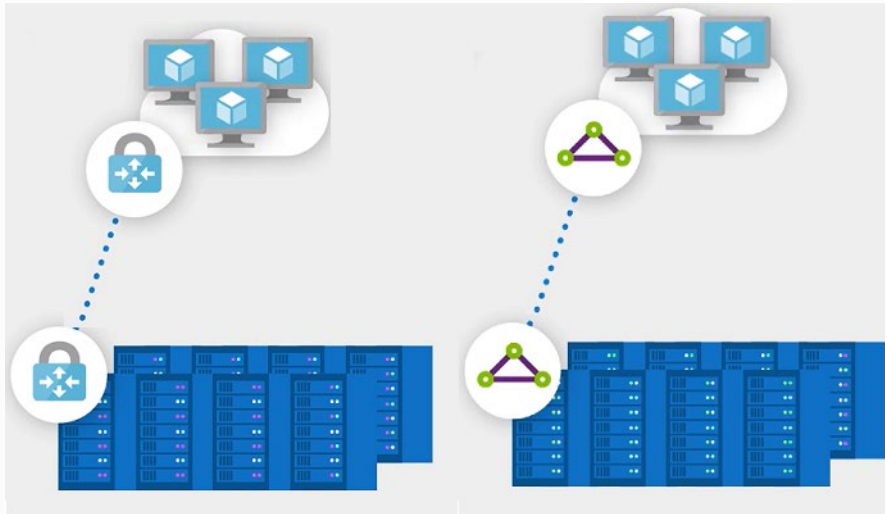
Microsoft manages the following Azure Virtual Desktop services as part of Azure:

- **Web Access:** The Web Access service within Window Virtual Desktop lets users access virtual desktops and remote apps through an HTML5-compatible web browser as they would with a local PC, from anywhere on any device. You can secure Web Access using multifactor authentication in Azure Active Directory.
- **Gateway:** The Remote Connection Gateway service connects remote users to Azure Virtual Desktop apps and desktops from any internet-connected device that can run an Azure Virtual Desktop client. The client connects to a gateway, which then orchestrates a connection from a VM back to the same gateway.
- **Connection Broker:** The Connection Broker service manages user connections to virtual desktops and remote apps. The Connection Broker provides load balancing and reconnection to existing sessions.
- **Diagnostics:** Remote Desktop Diagnostics is an event-based aggregator that marks each user or administrator action on the Azure Virtual Desktop deployment as a success or failure. Administrators can query the event aggregation to identify failing components.
- **Extensibility components:** Azure Virtual Desktop includes several extensibility components. You can manage Azure Virtual Desktop using Windows PowerShell or with the provided REST APIs, which also enable support from third-party tools.

Components you manage

Customers manage these components of Azure Virtual Desktop solutions:

- **Azure Virtual Network:** Azure Virtual Network lets Azure resources like VMs communicate privately with each other and with the internet. By connecting Azure Virtual Desktop host pools to an Active Directory domain, you can define network topology to access virtual desktops and virtual apps from the intranet or internet, based on organizational policy. You can connect an Azure Virtual Desktop to an on-premises network using a virtual private network (VPN), or use Azure ExpressRoute to extend the on-premises network into the Azure cloud over a private connection.



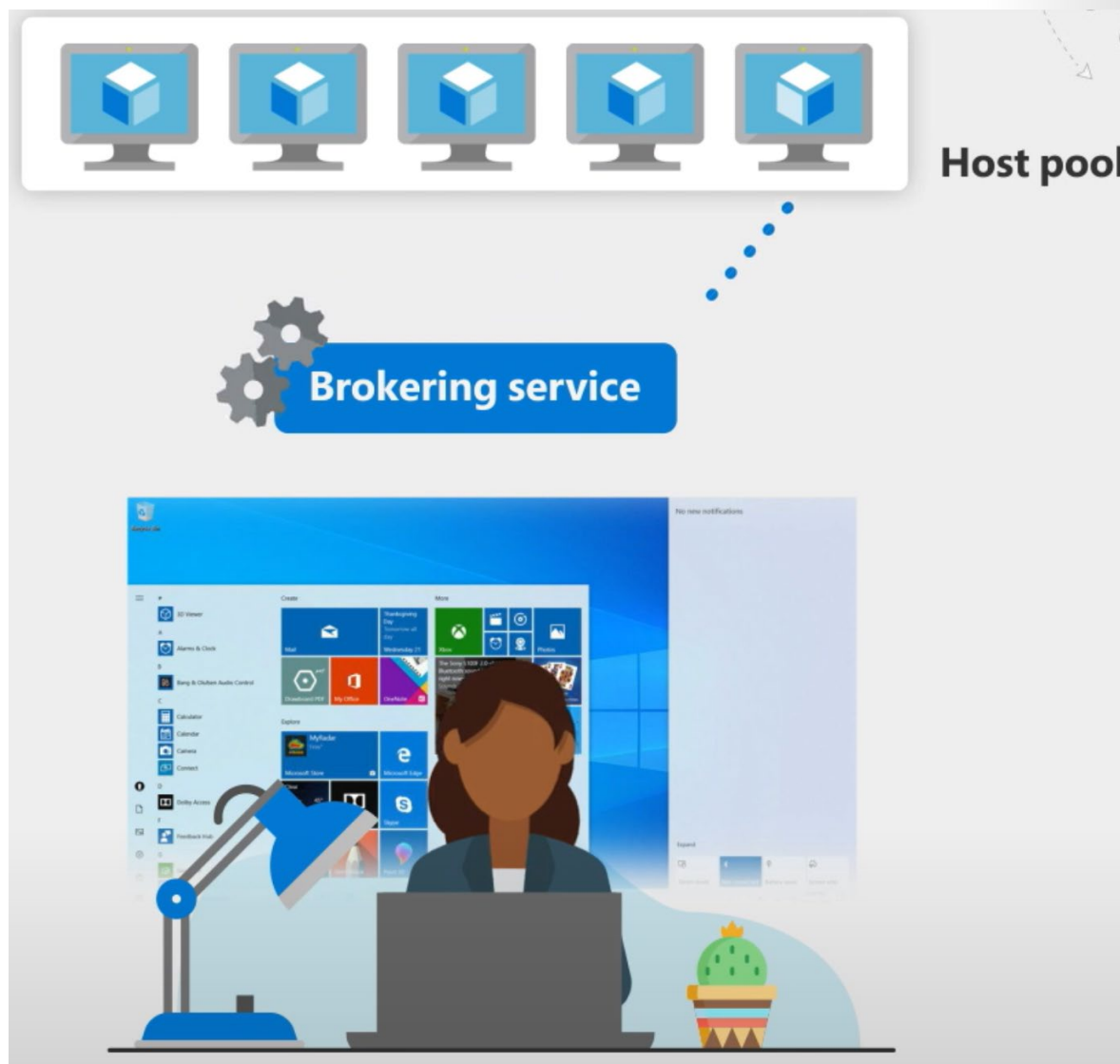
- **Azure AD:** Azure Virtual Desktop uses Azure AD for identity and access management. Azure AD integration applies Azure AD security features like conditional access, multi-factor authentication, and the Intelligent Security Graph, and helps maintain app compatibility in domain-joined VMs.
- **AD DS:** Azure Virtual Desktop VMs must domain-join an AD DS service, and the AD DS must be in sync with Azure AD to associate users between the two services. You can use Azure AD Connect to associate AD DS with Azure AD.
- **Azure Virtual Desktop session hosts:** A host pool can run the following operating systems:
 - Windows 7 Enterprise
 - Windows 10 Enterprise
 - Windows 10 Enterprise Multi-session
 - Windows Server 2012 R2 and above
 - Custom Windows system images with pre-loaded apps, group policies, or other customizations

You can choose VM sizes, including GPU-enabled VMs. Each session host has an Azure Virtual Desktop host agent, which registers the VM as part of the Azure Virtual Desktop workspace or tenant. Each host pool can have one or more app groups, which are collections of remote applications or desktop sessions that users can access.

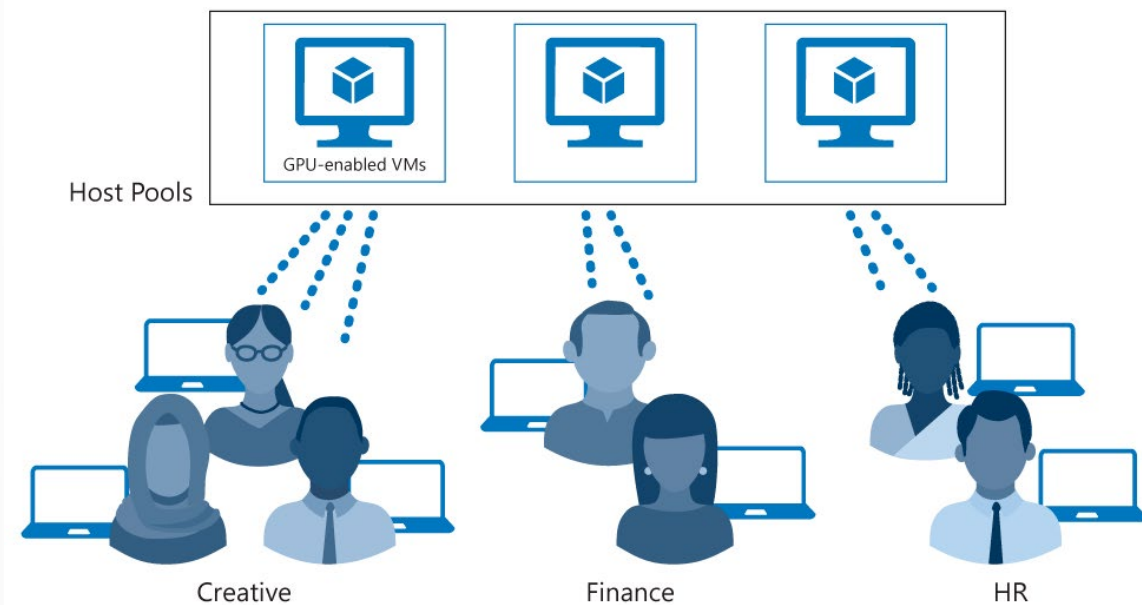
- **Azure Virtual Desktop workspace:** The Azure Virtual Desktop workspace or tenant is a management construct to manage and publish host pool resources.

Personal and pooled desktops

Host pools are a collection of one or more identical virtual machines (VMs) within Azure Virtual Desktop environments.

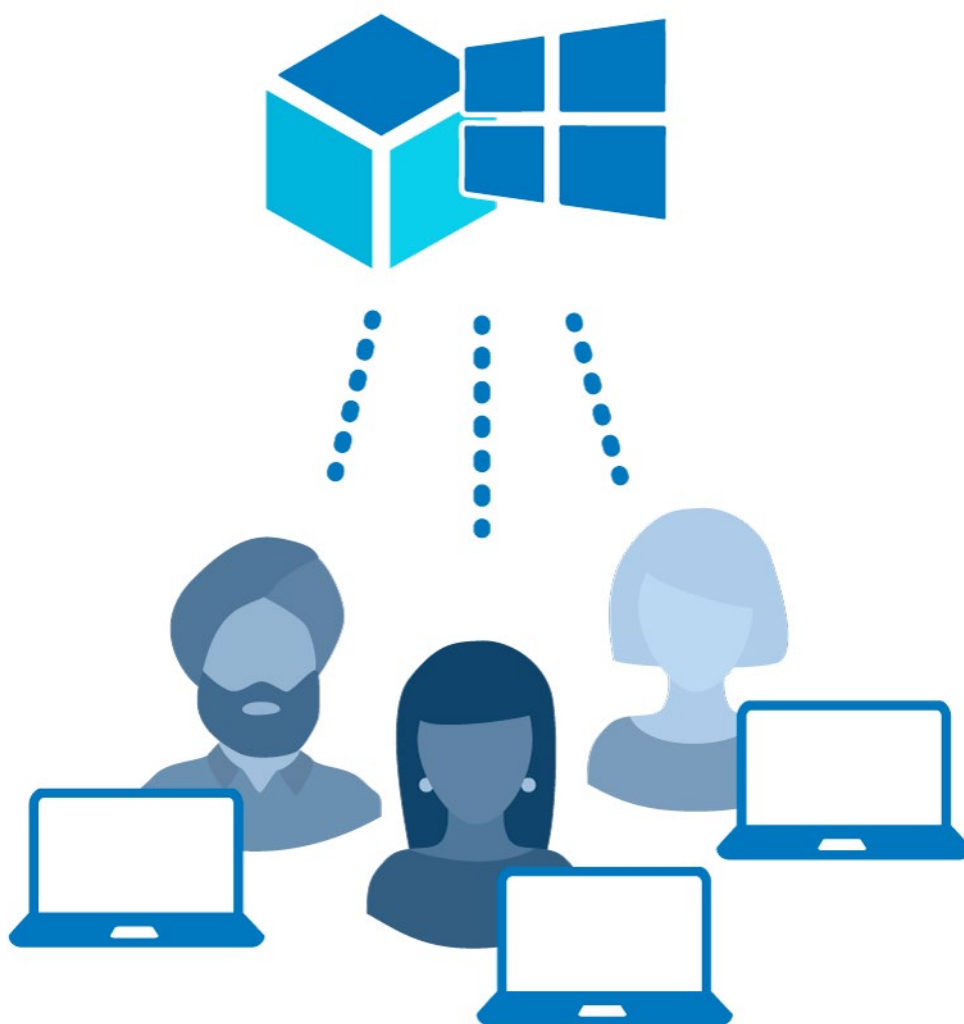


Each host pool can contain an app group that users can interact with as they would on a physical desktop.

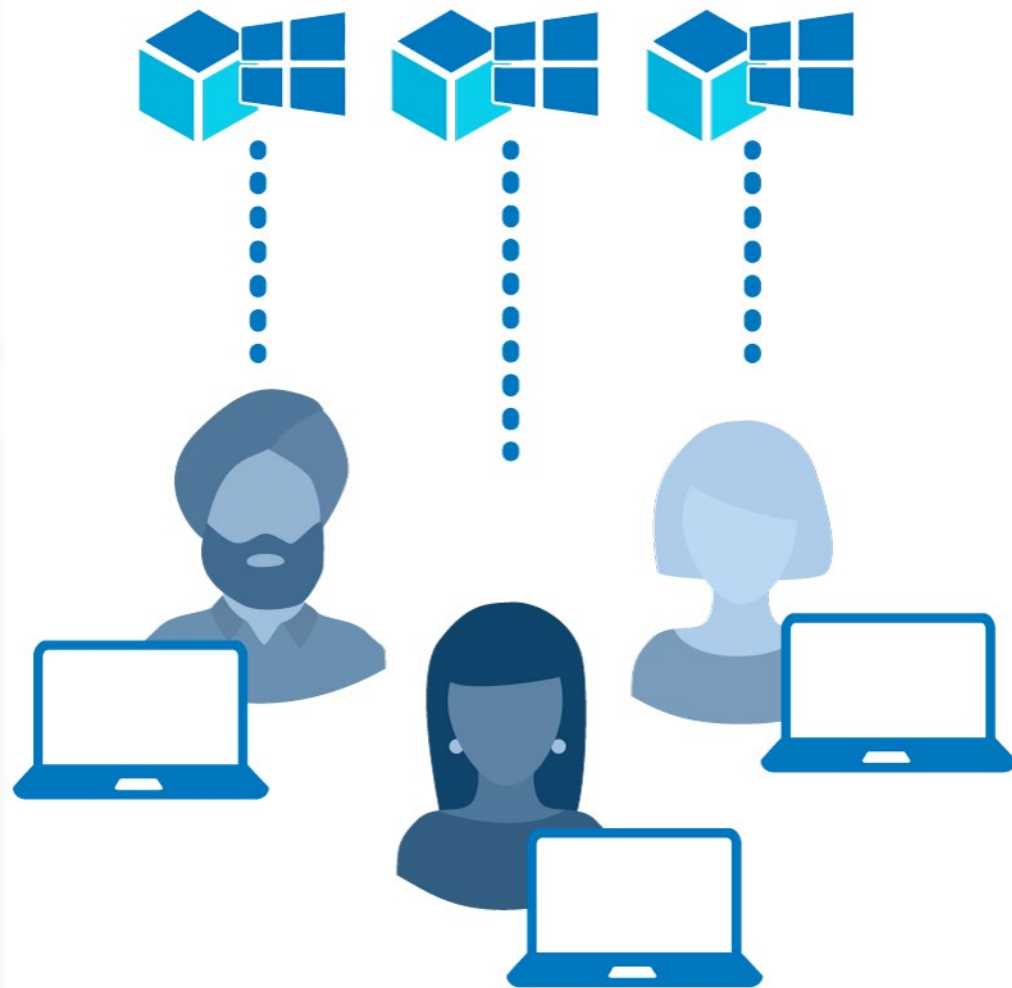


Users obtain access to host pools by being allocated to a host pool using an assigned Application Group:

- **Pooled:** You can configure a pooled host pool where several users sign in and share a VM. Typically, none of those users would be a local administrator on the pooled VM. With pooled, you can use one of the recommended images that includes Windows 10 Enterprise multisession. This operating system is exclusive to Azure Virtual Desktop. You can also use your own custom image.



- **Personal:** A personal host pool is where each user has their own dedicated VM. Those users would typically be local administrators for the VM. This enables the user to install or uninstall apps without impacting other users.



Personal desktop solutions (sometimes called persistent desktops) allow users to always connect to the same specific session host. Users can typically modify their desktop experience to meet personal preferences, and save files in the desktop environment. Personal desktop solutions:

- Let users customize their desktop environment, including user-installed applications and saving files within the desktop environment.
- Allow assigning dedicated resources to a specific user, which can be helpful for some manufacturing or development use cases.

Pooled desktop solutions assign users to whichever session host is currently available, depending on the load-balancing algorithm. Because the users don't always return to the same session host each time they connect, they have limited ability to customize the desktop environment and don't usually have administrator access.

Service updates for AVD desktops

There are several options for updating Azure Virtual Desktop desktops. Deploying an updated image every month guarantees compliance and state.

- **Microsoft Endpoint Configuration Manager (MECM)**¹ updates server and desktop operating systems.
- **Windows Updates for Business**² updates desktop operating systems like Windows 10 multi-session.
- **Azure Update Management**³ updates server operating systems.
- **Azure Log Analytics**⁴ checks compliance.
- Deploy a new (custom) image to session hosts every month for the latest Windows and applications updates. You can use an image from the Azure Marketplace or a **custom Azure managed image**⁵.

Azure limitations for Azure Virtual Desktop

The Azure Virtual Desktop service is scalable to more than 10,000 session hosts per workspace. You can address some Azure platform and Azure Virtual Desktop control plane limitations in the design phase to avoid changes in the scaling phase.

Numbers in the following sections are approximate. The numbers are based on a variety of large custom-er deployments, and they might change over time.

Also, note that:

- You **can't create more than 200 application groups** per single Azure AD tenant.
- We recommend that you **don't publish more than 50 applications** per application group.
- We recommend to deploy **not more than 5,000 VMs per Azure subscription per region**, this recommendation applies to both personal and pooled host pools based on Windows 10 Enterprise single and multi-session. Most customers use Windows 10 Enterprise multi-session, which allows multiple users to log on to each VM. You can increase the resources of individual session host VMs to accommodate more user sessions.
- For automated session host scaling tools, **the limits are around 1,200 VMs per Azure subscription per region**, because VM status interaction consumes more resources.
- To manage enterprise environments with **more than 5,000 VMs per Azure subscription in the same region**, you can create multiple Azure subscriptions in a hub-spoke architecture and connect them via virtual network peering, as in the preceding example architecture. You could also deploy VMs in a different region in the same subscription to increase the number of VMs.
- Azure Resource Manager (ARM) subscription **API throttling limits don't allow more than 600 Azure VM reboots per hour** via the Azure portal. You can reboot all your machines at once via the operating system, which doesn't consume any Azure Resource Manager subscription API calls.
- You **can currently deploy 399 VMs per Azure Virtual Desktop ARM template deployment without Availability Sets**, or 200 VMs per Availability Set. You can increase the number of VMs per deployment by switching off Availability Sets in either the ARM template or the Azure portal host pool enrollment.

¹ <https://docs.microsoft.com/en-us/mem/configmgr/>

² <https://docs.microsoft.com/en-us/windows/deployment/update/waas-manage-updates-wufb>

³ <https://docs.microsoft.com/en-us/azure/automation/update-management/overview>

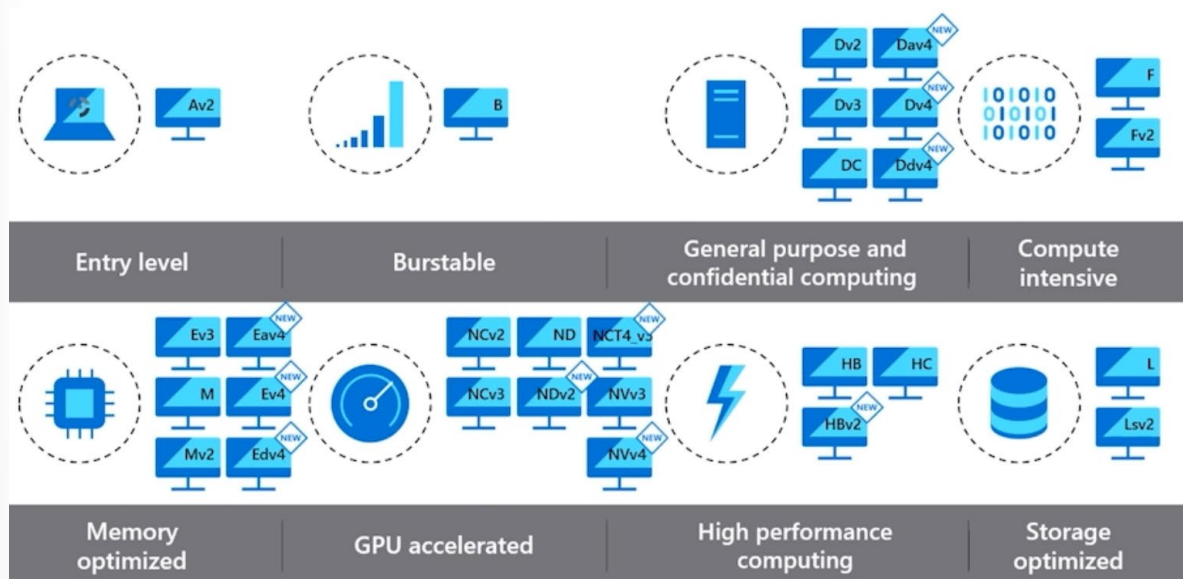
⁴ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/log-analytics-agent>

⁵ <https://docs.microsoft.com/en-us/azure/virtual-machines/windows/capture-image-resource>

- Azure **VM session host name prefixes can't exceed 11 characters**, due to auto-assigning of instance names and the NetBIOS limit of 15 characters per computer account.
- By default, **you can deploy up to 800 instances of most resource types in a resource group**. Azure Compute doesn't have this limit.

VM sizing

Virtual machine sizing guidelines⁶ lists the maximum suggested number of users per virtual central processing unit (vCPU) and minimum VM configurations for different workloads.



This data helps estimate the VMs you need in your host pool.

Use simulation tools to test deployments with both stress tests and real-life usage simulations. Make sure the system is responsive and resilient enough to meet user needs, and remember to vary the load sizes when testing.

Azure Virtual Desktop pricing

Architect your Azure Virtual Desktop solution to realize cost savings. Here are five different options to help manage costs for enterprises:

- **Windows 10 multi-session:** By delivering a multi-session desktop experience for users that have identical compute requirements, you can let more users log onto a single VM at once, resulting in considerable cost savings.
- **Azure Hybrid Benefit:** If you have Software Assurance, you can use **Azure Hybrid Benefit for Windows Server**⁷ to save on the cost of your Azure infrastructure.
- **Azure Reserved Instances:** You can prepay for your VM usage and save money. Combine **Azure Reserved Instances**⁸ with Azure Hybrid Benefit for up to 80 percent savings over list prices.
- **Session host load-balancing:** When setting up session hosts, **Breadth-first** is the standard default mode, which spreads users randomly across session hosts. **Depth-first** mode fills up a session host

⁶ <https://docs.microsoft.com/en-us/windows-server/remote/remote-desktop-services/virtual-machine-recs>

⁷ <https://docs.microsoft.com/en-us/azure/virtual-machines/windows/hybrid-use-benefit-licensing>

⁸ <https://azure.microsoft.com/pricing/reserved-vm-instances/>

server with the maximum number of users before it moves on to the next session host. You can adjust this setting for maximum cost benefits.

Design the Azure Virtual Desktop architecture

Assess existing physical and virtual desktop environments

This topic lists just a few of the approved partner providers and independent software vendors for Azure Virtual Desktop integrations.

These solutions can help you to assess an existing physical and virtual desktop environment using automated tools, such as SysTrack and Stratusphere.

For a list of migration partners, see **Migration Opportunities for Partners for Azure Virtual Desktop**⁹.

For a full list of Azure Virtual Desktop partners, see **Azure Virtual Desktop partner integrations**¹⁰.

Lakeside Software



Lakeside Software is an Azure Virtual Desktop value-added services provider that equips IT teams with software for monitoring performance and assessing Azure migration readiness of user workloads. With this software, IT gains clearer visibility into application usage and resource consumption to streamline the migration process. Lakeside Software collects data at every workspace to create a comprehensive report on user environments, enabling quick troubleshooting and optimization of assets.

Lakeside Software's digital experience monitoring solution, SysTrack, can help provide a great user experience by tracking performance and identifying ideal workloads for migration. SysTrack works to extend the value of Azure Virtual Desktop through right-sizing assessments and continuous monitoring of user environments.

- **Go to the partner website**¹¹.

Liquidware



Liquidware is an Azure Virtual Desktop value-added services provider that delivers software that manages and optimizes Azure Virtual Desktop deployment. The Liquidware Essentials suite provides application delivery through layering, user environment management, and key user experience visibility and diagnostics. With solutions for assessing migration readiness and analyzing usage metrics, Liquidware provides a seamless virtual desktop experience for end users.

Liquidware Essentials extends the value of Azure Virtual Desktop by efficiently harvesting user profiles and gathering key user data to streamline migration of user environments to Azure. Additionally, Liquidware Essentials simplifies image management by unifying user profiles and layering apps based on configurable rights management settings.

Liquidware's Stratusphere UX is the industry-leading user experience monitoring and diagnostics solution. An independent and platform-agnostic architecture, the solution provides user-centric visibility that is persistent across platform delivery. The ability to examine and drill down into user, machine and

⁹ <https://www.microsoft.com/azure/partners/practices/migrate-existing-apps#windows-virtual-desktop>

¹⁰ <https://docs.microsoft.com/en-us/azure/virtual-desktop/partners>

¹¹ <https://www.lakesidesoftware.com/assessments/vwd>

application workloads is critically important to optimize the workspace environment and design an Applications Strategy. Stratusphere UX also provides the most comprehensive array of metrics of any user experience monitoring solution on the market today.

- **Go to the partner website¹².**

Citrix



Citrix is an approved provider that offers enterprises centralized hybrid management of virtual apps and desktops workloads in Azure, side by side with on-premises deployments. Citrix Workspace with the Virtual Apps and Desktops service allows users to access apps and desktops from any device, leveraging the advanced Citrix HDX protocol to deliver a high definition experience from anywhere.

Citrix extends the value of Azure Virtual Desktop with robust enterprise tools to improve user density and performance, provision workloads on demand, and simplify image and application management. IT can optimize costs with intelligent scaling tools, while delivering an incredible user experience that's field-tested against the toughest applications across industries. Additionally, Citrix Managed Desktops is an Azure Virtual Desktop-enabled desktops-as-a-service program that provides a simple, cloud-based management solution for delivering virtual apps and desktops to any device.

- **Go to the partner website¹³.**

VMware



VMware Horizon Cloud on Microsoft Azure is a native cloud service that lets organizations quickly deploy remote desktops and applications from their existing Microsoft Azure subscriptions while leveraging all the features of VMware Horizon. Horizon Cloud uses the Horizon Control Plane to provide a single management interface for all Horizon environments, on-premises or in the cloud. This enables hybrid desktop virtualization and lets customers move their workloads to Azure at their own pace.

As an Azure Virtual Desktop approved provider, VMware can help customers that want to use Azure Virtual Desktop while still enjoying the additional functionality that comes with VMware Horizon, such as integrated and easy-to-use power management, cloud-based monitoring, and the Blast Extreme protocol. These features adapt to changing network conditions on the fly to provide a consistently excellent user experience. VMware Horizon Cloud also comes with VMware App Volumes and Dynamic Environment Manager, which add advanced application and user environment management capabilities that work with MSIX app attach and FSLogix.

- **Go to the partner website¹⁴.**

¹² <https://www.liquidware.com/solutions/solutions-platform/microsoft>

¹³ <https://more.citrix.com/wwd>

¹⁴ <https://www.vmware.com/products/horizon-cloud-virtual-desktops.html>

Assess network capacity and speed requirements for AVD

When using a remote Windows session, your network's available bandwidth greatly impacts the quality of your experience. Different applications and display resolutions require different network configurations, so it's important to make sure your network is configured to meet your needs.

The following recommendations apply to networks with less than 0.1% loss. These recommendations apply regardless of how many sessions you're hosting on your virtual machines (VMs).

Applications

The following table lists the minimum recommended bandwidths for a smooth user experience.

Workload type	Recommended bandwidth
Light	1.5 Mbps
Medium	3 Mbps
Heavy	5 Mbps
Power	15 Mbps

Keep in mind that the stress put on your network depends on both your app workload's output frame rate and your display resolution. If either the frame rate or display resolution increases, the bandwidth requirement will also rise. For example, a light workload with a high-resolution display requires more available bandwidth than a light workload with regular or low resolution.

Other scenarios can have their bandwidth requirements change depending on how you use them, such as:

- Voice or video conferencing
- Real-time communication
- Streaming 4K video

Make sure to load test these scenarios in your deployment using simulation tools like Login VSI. Vary the load size, run stress tests, and test common user scenarios in remote sessions to better understand your network's requirements.

Display resolutions

Different display resolutions require different available bandwidths. The following table lists the bandwidths we recommend for a smooth user experience at typical display resolutions with a frame rate of 30 frames per second (fps). These recommendations apply to single and multiple user scenarios. Keep in mind that scenarios involving a frame rate under 30 fps, such as reading static text, require less available bandwidth.

Typical display resolutions at 30 fps	Recommended bandwidth
About 1024 × 768 px	1.5 Mbps
About 1280 × 720 px	3 Mbps
About 1920 × 1080 px	5 Mbps
About 3840 × 2160 px (4K)	15 Mbps

Azure Virtual Desktop Experience Estimator

Use the **Azure Virtual Desktop Experience Estimator**¹⁵ to determine the connection round trip time (RTT) from your current location, through the Azure Virtual Desktop service, to each Azure region in which you can deploy virtual machines.

The highlighted Azure region is the one with the lowest connection RTT from your current location. The times displayed are estimates intended to help assess end user experience quality for your Azure Virtual Desktop deployment. The actual experience will vary depending on network conditions, end user device, and the configuration of the deployed virtual machines.

Azure Region*	Round Trip Time (ms)
West US 2	30
West US	48
West Central US	52
Central US	66
North Central US	73
South Central US	74
Canada Central	85
East US	94
Canada East	96
East US 2	98
Japan East	127
Japan West	134
Korea Central	151
North Europe	158
Korea South	159
UK South	167
UK West	169
East Asia	171
France Central	171
West Europe	176
Germany West Central	179
Switzerland West	181
Switzerland North	184
Germany North	186
France South	187
Southeast Asia	189
Australia East	206
Brazil South	208
Australia Central	210
Australia Central 2	211
Australia Southeast	217
South India	223
West India	242

¹⁵ <https://azure.microsoft.com/services/virtual-desktop/assessment/>

Azure Region*	Round Trip Time (ms)
Central India	244
UAE Central	269
UAE North	269
South Africa West	307
South Africa North	323

Recommend an operating system for a AVD implementation

You can access Azure Virtual Desktop resources on devices with Windows 10, Windows 10 IoT Enterprise, and Windows 7 using the Windows Desktop client.

The client doesn't support Window 8 or Windows 8.1.

Choose the client that matches your version of Windows:

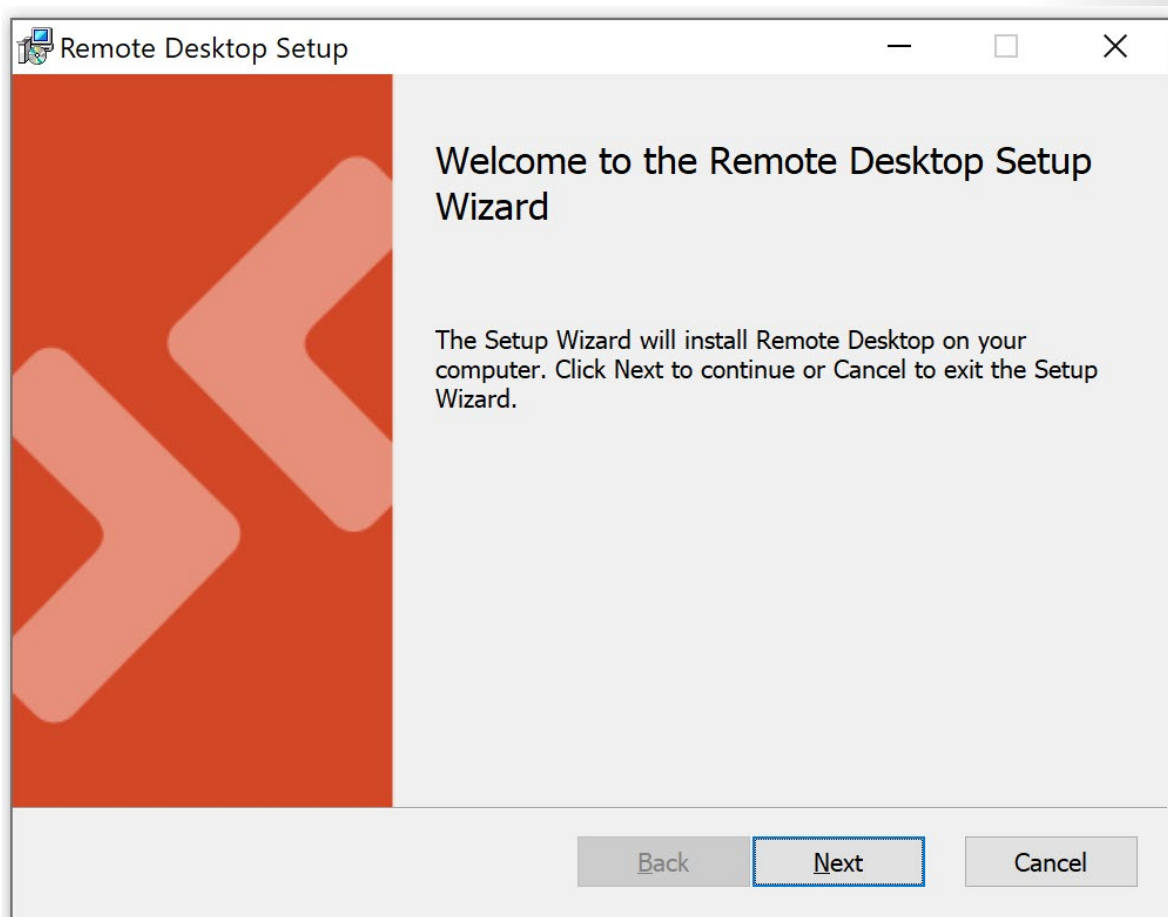
- **Windows 64-bit**¹⁶
- **Windows 32-bit**¹⁷
- **Windows ARM64**¹⁸

You can install the client for the current user, which doesn't require admin rights, or your admin can install and configure the client so that all users on the device can access it.

¹⁶ <https://go.microsoft.com/fwlink/?linkid=2068602>

¹⁷ <https://go.microsoft.com/fwlink/?linkid=2098960>

¹⁸ <https://go.microsoft.com/fwlink/?linkid=2098961>



Once installed, the client can be launched from the Start menu by searching for **Remote Desktop**.

Azure Virtual Desktop doesn't support the RemoteApp and Desktop Connections (RADC) client or the Remote Desktop Connection (MSTSC) client.

Balancing host pools

Azure Virtual Desktop supports two load-balancing methods. Each method determines which session host will host a user's session when they connect to a resource in a host pool.

The following load-balancing methods are available in Azure Virtual Desktop:

- Breadth-first load balancing allows you to evenly distribute user sessions across the session hosts in a host pool.
- Depth-first load balancing allows you to saturate a session host with user sessions in a host pool. Once the first session reaches its session limit threshold, the load balancer directs any new user connections to the next session host in the host pool until it reaches its limit, and so on.

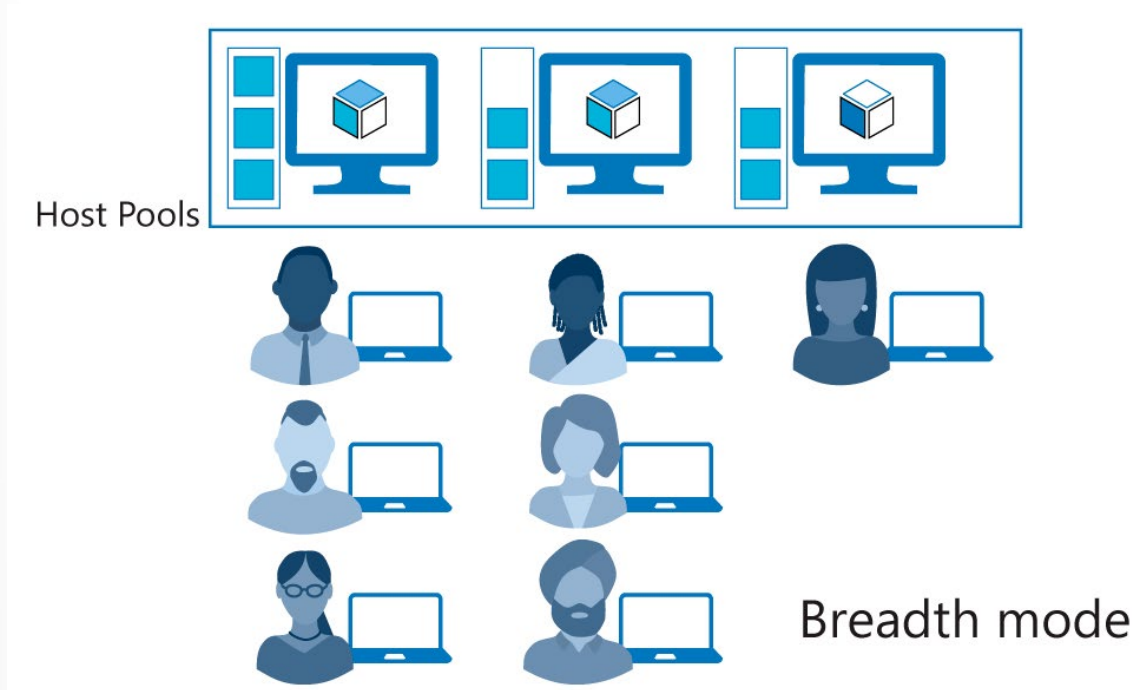
Each host pool can only configure one type of load-balancing specific to it. However, both load-balancing methods share the following behaviors no matter which host pool they're in:

- If a user already has a session in the host pool and is reconnecting to that session, the load balancer will successfully redirect them to the session host with their existing session. This behavior applies even if that session host's `AllowNewConnections` property is set to `False`.

- If a user doesn't already have a session in the host pool, then the load balancer won't consider session hosts whose AllowNewConnections property is set to False during load balancing.

Breadth-first load-balancing method

The breadth-first load-balancing method allows you to distribute user connections to optimize for this scenario. This method is ideal for organizations that want to provide the best experience for users connecting to their pooled virtual desktop environment.



The breadth-first method first queries session hosts that allow new connections. The method then selects a session host randomly from half the set of session hosts with the least number of sessions. For example, if there are nine machines with 11, 12, 13, 14, 15, 16, 17, 18, and 19 sessions, a new session you create won't automatically go to the first machine. Instead, it can go to any of the first five machines with the lowest number of sessions (11, 12, 13, 14, 15).

Depth-first load-balancing method

The depth-first load-balancing method allows you to saturate one session host at a time to optimize for this scenario. This method is ideal for cost-conscious organizations that want more granular control on the number of virtual machines they've allocated for a host pool.



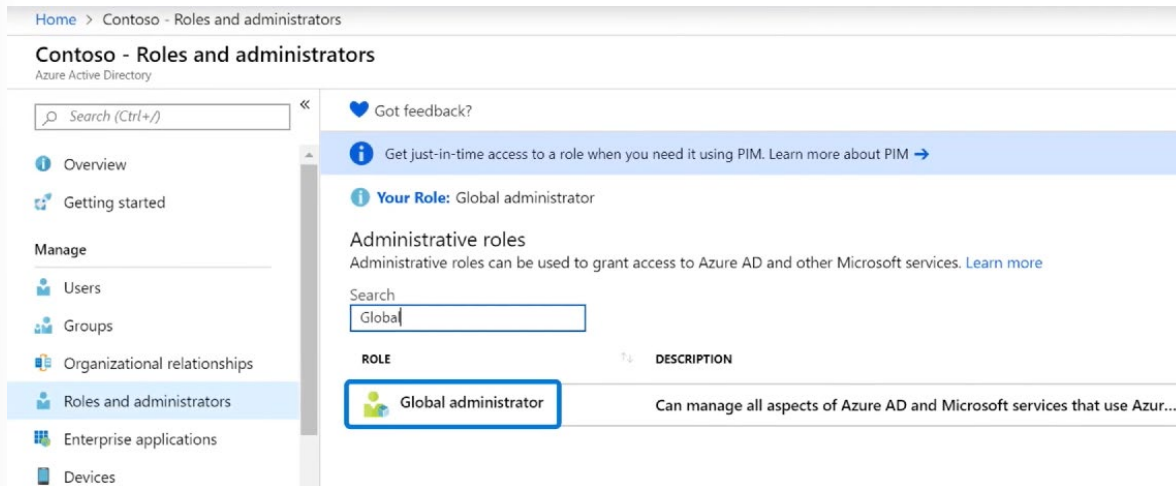
The depth-first method first queries session hosts that allow new connections and haven't gone over their maximum session limit. The method then selects the session host with highest number of sessions. If there's a tie, the method selects the first session host in the query.

The depth-first load balancing algorithm distributes sessions to session hosts based on the maximum session host limit. This parameter is required when you use the depth-first load balancing algorithm. For the best possible user experience, make sure to change the maximum session host limit parameter to a number that best suits your environment.

Recommendations for using subscriptions and management groups

Some Azure Virtual Desktop procedures, such as installing Office on a master VHD image, assume you have elevated access on the VM, whether it's provisioned in Azure or Hyper-V Manager.

As a Global Administrator in Azure Active Directory (Azure AD), you might not have access to all subscriptions and management groups in your directory. This topic describes the ways that you can elevate your access to all subscriptions and management groups.



Why would you need to elevate your access?

If you are a Global Administrator, there might be times when you want to do the following actions:

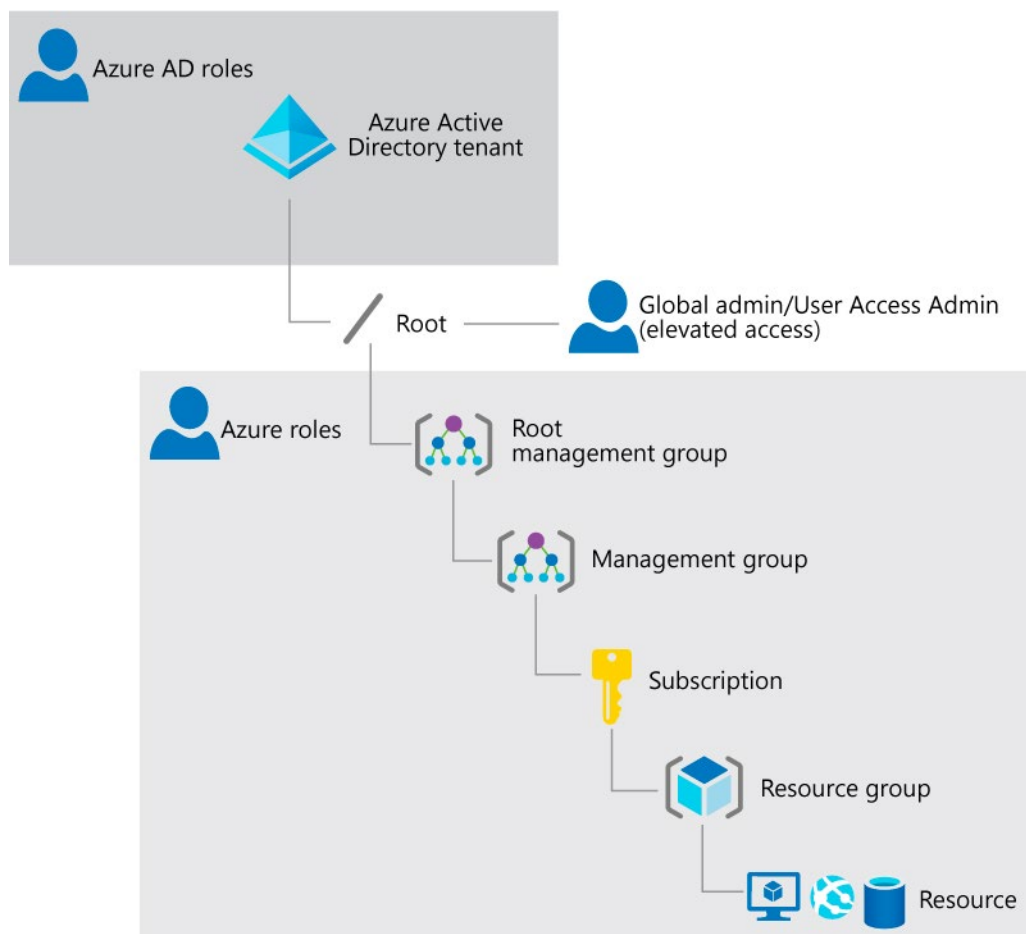
- Regain access to an Azure subscription or management group when a user has lost access
- Grant another user or yourself access to an Azure subscription or management group
- See all Azure subscriptions or management groups in an organization
- Allow an automation app (such as an invoicing or auditing app) to access all Azure subscriptions or management groups

How does elevated access work?

Azure AD and Azure resources are secured independently from one another. That is, Azure AD role assignments do not grant access to Azure resources, and Azure role assignments do not grant access to Azure AD. However, if you are a Global Administrator in Azure AD, you can assign yourself access to all Azure subscriptions and management groups in your directory. Use this capability if you don't have access to Azure subscription resources, such as virtual machines or storage accounts, and you want to use your Global Administrator privilege to gain access to those resources.

When you elevate your access, you will be assigned the User Access Administrator role in Azure at root scope (/). This allows you to view all resources and assign access in any subscription or management group in the directory. User Access Administrator role assignments can be removed using Azure PowerShell, Azure CLI, or the REST API.

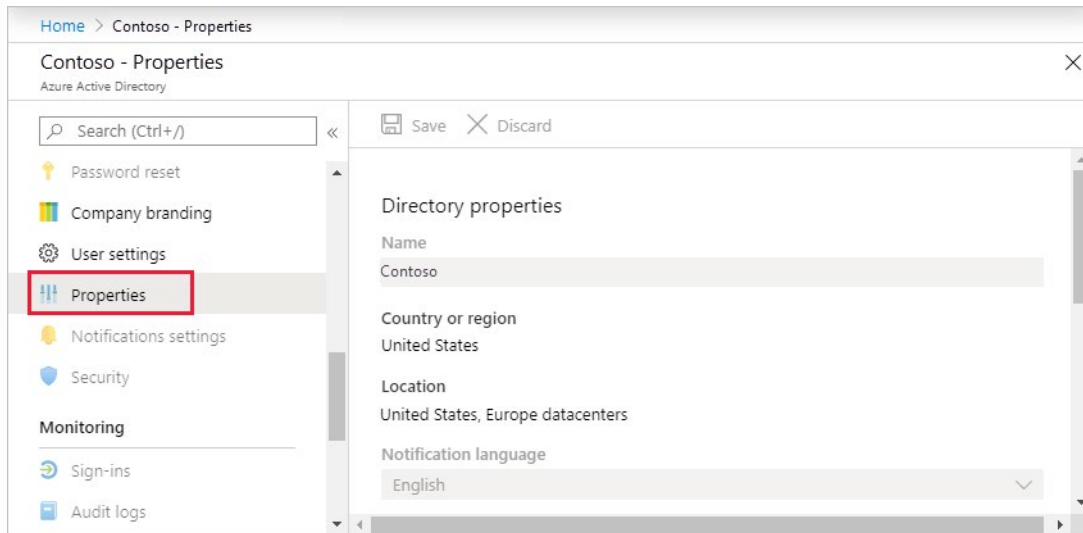
You should remove this elevated access once you have made the changes you need to make at root scope.



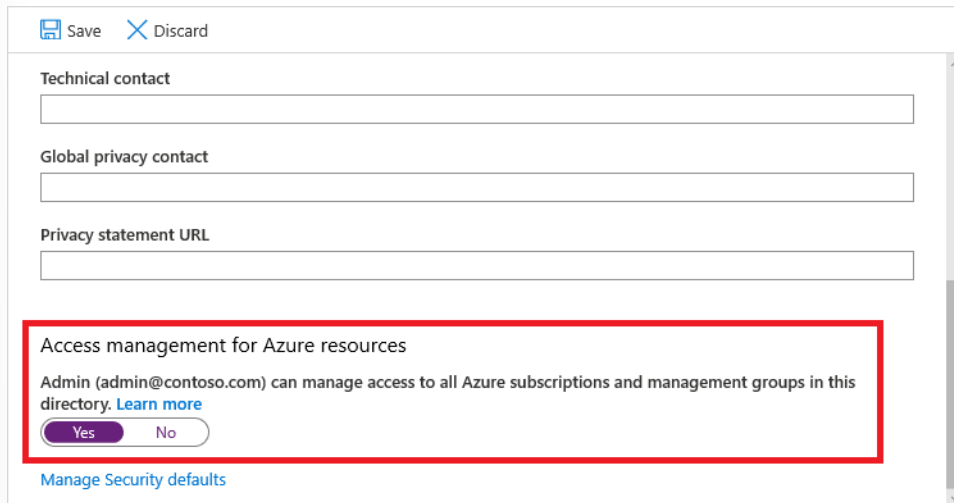
Elevate access for a Global Administrator

Follow these steps to elevate access for a Global Administrator using the Azure portal.

1. Sign in to the Azure portal or the Azure Active Directory admin center as a Global Administrator.
2. Open **Azure Active Directory**.
3. Under **Manage**, select **Properties**.



4. Under **Access management for Azure resources**, set the toggle to **Yes**.



When you set the toggle to **Yes**, you are assigned the User Access Administrator role in Azure RBAC at root scope (/). This grants you permission to assign roles in all Azure subscriptions and management groups associated with this Azure AD directory. This toggle is only available to users who are assigned the Global Administrator role in Azure AD.

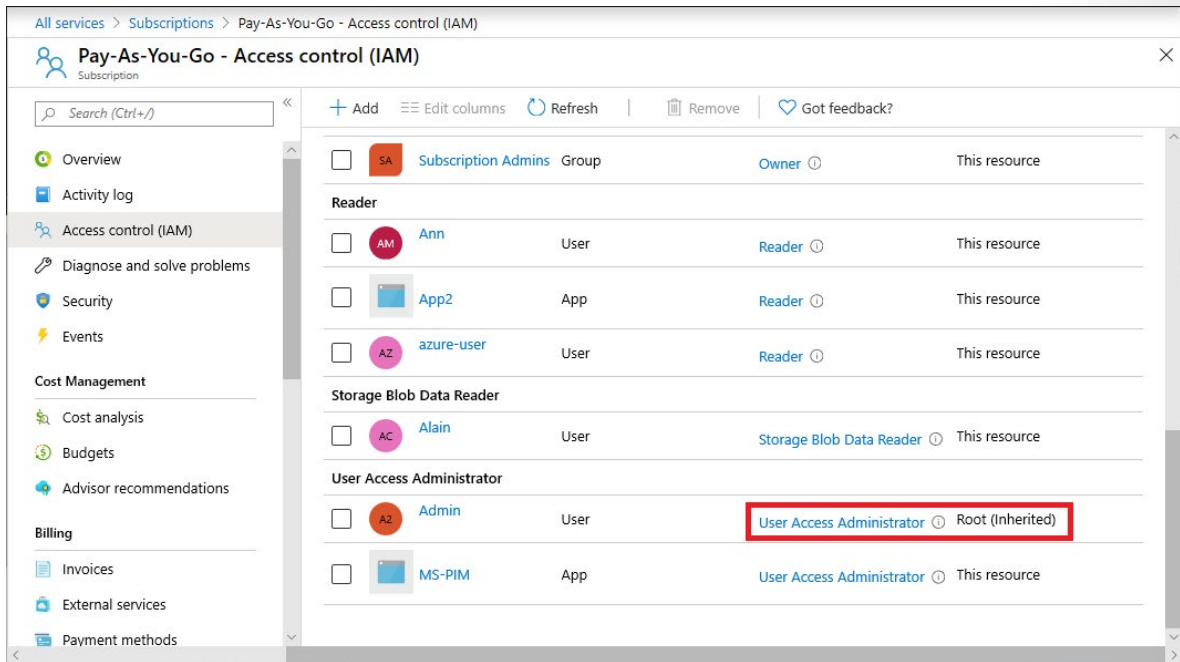
When you set the toggle to **No**, the User Access Administrator role in Azure RBAC is removed from your user account. You can no longer assign roles in all Azure subscriptions and management groups that are associated with this Azure AD directory. You can view and manage only the Azure subscriptions and management groups to which you have been granted access.

5. Click **Save** to save your setting.

This setting is not a global property and applies only to the currently signed in user. You can't elevate access for all members of the Global Administrator role.

6. Sign out and sign back in to refresh your access.

You should now have access to all subscriptions and management groups in your directory. When you view the Access control (IAM) pane, you'll notice that you have been assigned the User Access Administrator role at root scope.



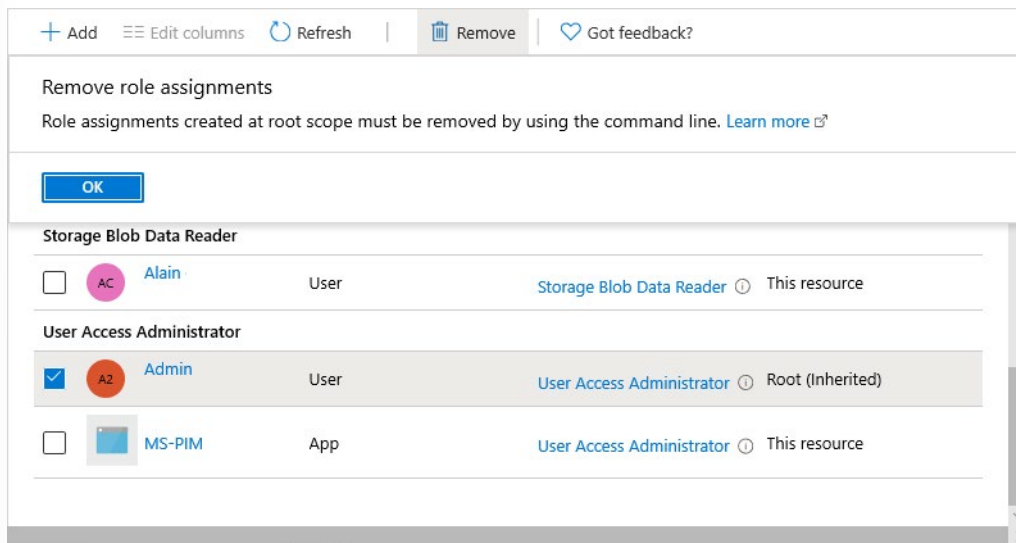
7. Make the changes you need to make at elevated access.

Remove elevated access

To remove the User Access Administrator role assignment at root scope (/), follow these steps.

1. Sign in as the same user that was used to elevate access.
2. In the navigation list, click **Azure Active Directory** and then click **Properties**.
3. Set the **Access management for Azure resources** toggle back to **No**. Since this is a per-user setting, you must be signed in as the same user as was used to elevate access.

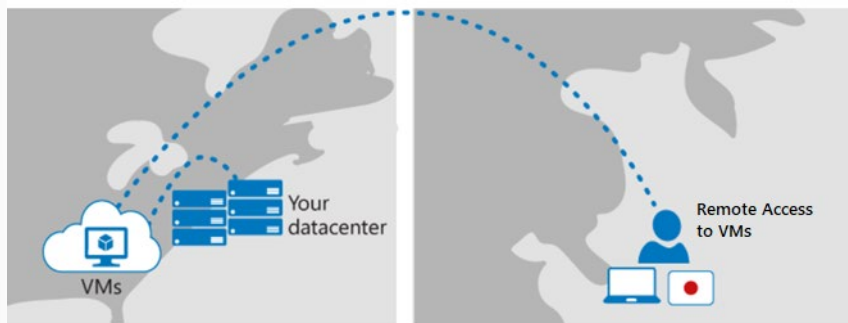
If you try to remove the User Access Administrator role assignment on the Access control (IAM) pane, you'll see the following message. To remove the role assignment, you must set the toggle back to **No** or use Azure PowerShell, Azure CLI, or the REST API.



4. Sign out as Global Administrator.

Configure a location for the AVD metadata

Azure Virtual Desktop is currently available for all geographical locations. Administrators can choose the location to store user data when they create the host pool virtual machines and associated services, such as file servers. Learn more about Azure geographies at the [Azure datacenter map](https://azuredatacentermap.azurewebsites.net/)¹⁹.



Microsoft doesn't control or limit the regions where you or your users can access your user and app-specific data.

Azure Virtual Desktop stores global metadata information like tenant names, host pool names, app group names, and user principal names in a datacenter. Whenever a customer creates a service object, they must enter a location for the service object. The location they enter determines where the metadata for the object will be stored. The customer will choose an Azure region and the metadata will be stored in the related geography.

There is currently support for storing metadata in the following geographies:

- United States (US) (Generally available)
- Europe (EU) (Public preview)

¹⁹ <https://azuredatacentermap.azurewebsites.net/>

When you're selecting a region to create Azure Virtual Desktop service objects in, you'll see regions under both US and EU geographies.

To make sure you understand which region would work best for your deployment, take a look at **our Azure global infrastructure map**²⁰.

The stored metadata is encrypted at rest, and geo-redundant mirrors are maintained within the geography. All customer data, such as app settings and user data, resides in the location the customer chooses and isn't managed by the service. More geographies will become available as the service grows.

Service metadata is replicated within the Azure geography for disaster recovery purposes.

Recommend a configuration for performance requirements

Virtual machines can be monitored for availability and performance with Azure Monitor like any other Azure resource. However, they're unique from other resources since you also need to monitor the guest operating and system and the workloads that run in it.

Differences from other Azure resources

You can collect and act on the same monitoring data from Azure virtual machines as you would monitoring Azure resources with Azure Monitor with the following differences:

- **Platform metrics**²¹ are collected automatically for virtual machines but only for the **virtual machine host**²². You need an agent to collect performance data from the guest operating system.
- Virtual machines don't generate **resource logs**²³ which provide insight into operations that were performed within an Azure resource. You use an agent to collect log data from the guest operating system.
- You can create **diagnostic settings**²⁴ for a virtual machine to send platform metrics to other destinations such as storage and event hubs, but you can't configure these diagnostic settings in the Azure portal.

Monitoring data

Virtual machines in Azure generate **logs**²⁵ and **metrics**²⁶ as shown in the following diagram.

²⁰ <https://azure.microsoft.com/global-infrastructure/geographies/>

²¹ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/data-platform-metrics>

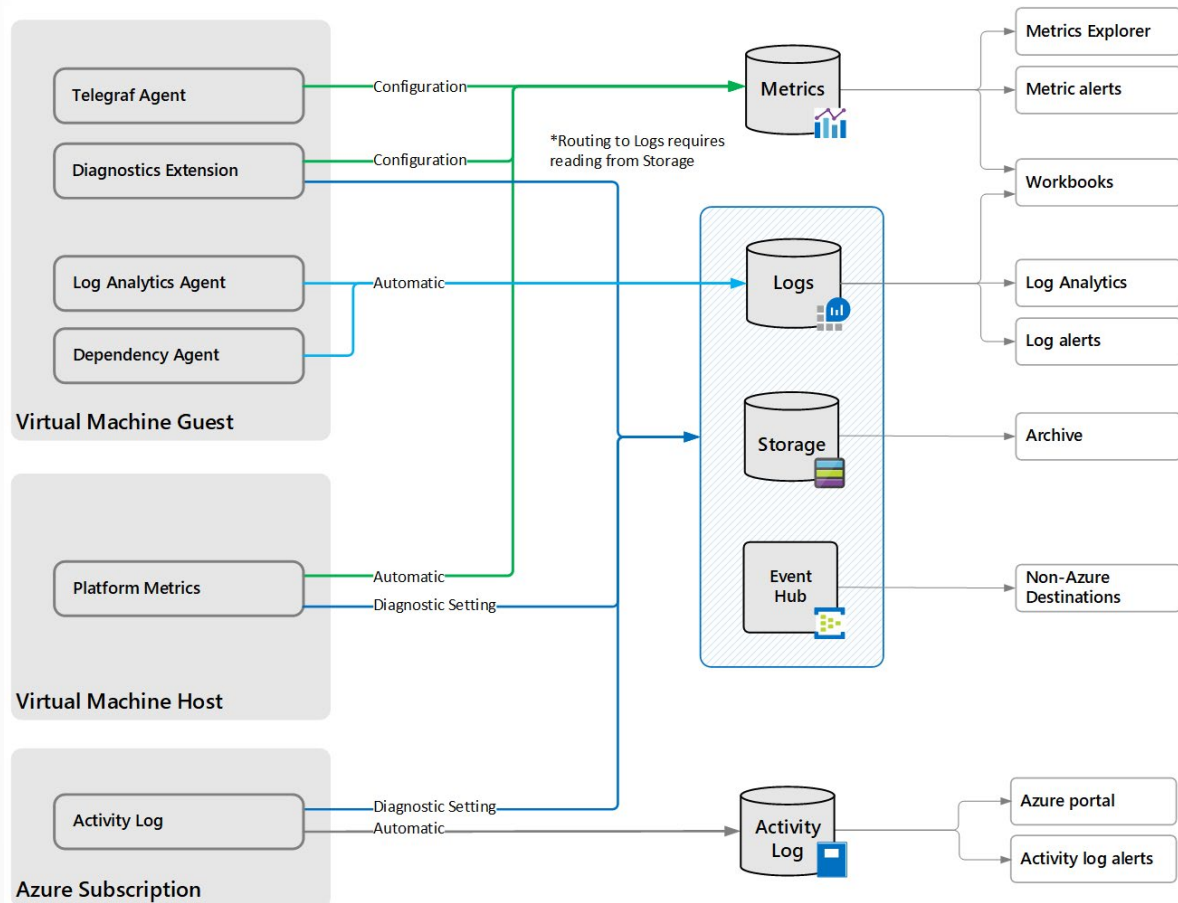
²² <https://docs.microsoft.com/en-us/azure/azure-monitor/vm/monitor-vm-azure>

²³ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/platform-logs-overview>

²⁴ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/diagnostic-settings>

²⁵ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/data-platform-logs>

²⁶ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/data-platform-metrics>



Virtual machine host

Virtual machines in Azure generate the following data for the virtual machine host the same as other Azure resources as described in **Monitoring data**²⁷.

- **Platform metrics**²⁸ - Numerical values that are automatically collected at regular intervals and describe some aspect of a resource at a particular time. Platform metrics are collected for the virtual machine host, but you require the diagnostics extension to collect metrics for the guest operating system.
- **Activity log**²⁹ - Provides insight into the operations on each Azure resource in the subscription from the outside (the management plane). For a virtual machine, this includes such information as when it was started and any configuration changes.

²⁷ <https://docs.microsoft.com/en-us/azure/azure-monitor/insights/monitor-azure-resource>

²⁸ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/data-platform-metrics>

²⁹ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/platform-logs-overview>

Guest operating system

To collect data from the guest operating system of a virtual machine, you require an agent, which runs locally on each virtual machine and sends data to Azure Monitor. Multiple agents are available for Azure Monitor with each collecting different data and writing data to different locations.

- **Log Analytics agent³⁰** - Available for virtual machines in Azure, other cloud environments, and on-premises. Collects data to Azure Monitor Logs. Supports Azure Monitor for VMs and monitoring solutions. This is the same agent used for System Center Operations Manager.
- **Dependency agent³¹** - Collects data about the processes running on the virtual machine and their dependencies. Relies on the Log Analytics agent to transmit data into Azure and supports Azure Monitor for VMs, Service Map, and Wire Data 2.0 solutions.
- **Azure Diagnostic extension³²** - Available for Azure Monitor virtual machines only. Can collect data to multiple locations but primarily used to collect guest performance data into Azure Monitor Metrics for Windows virtual machines.
- **Telegraf agent³³** - Collect performance data from Linux VMs into Azure Monitor Metrics.

Configuration requirements

To enable all features of Azure Monitor for monitoring a virtual machine, you need to collect monitoring data from the virtual machine host and guest operating system to both Azure Monitor Metrics and Azure Monitor Logs. The following table lists the configuration that must be performed to enable this collection.

Configuration step	Actions completed	Features enabled
No configuration	- Host platform metrics collected to Metrics. - Activity log collected.	- Metrics explorer for host. - Metrics alerts for host. - Activity log alerts.
Enable Azure Monitor for VMs	- Log Analytics agent installed. - Dependency agent installed. - Guest performance data collected to Logs. - Process and dependency details collected to Logs.	- Performance charts and workbooks for guest performance data. - Log queries for guest performance data. - Log alerts for guest performance data. - Dependency map.
Install the diagnostics extension and telegraf agent	- Guest performance data collected to Metrics.	- Metrics explorer for guest. - Metrics alerts for guest.
Configure Log Analytics workspace	- Events collected from guest.	- Log queries for guest events. - Log alerts for guest events.
Create diagnostic setting for virtual machine	- Platform metrics collected to Logs. - Activity log collected to Logs.	- Log queries for host metrics. - Log alerts for host metrics. - Log queries for Activity log.

³⁰ <https://docs.microsoft.com/en-us/azure/azure-monitor/agents/agents-overview>

³¹ <https://docs.microsoft.com/en-us/azure/azure-monitor/agents/agents-overview>

³² <https://docs.microsoft.com/en-us/azure/azure-monitor/agents/agents-overview>

³³ <https://docs.microsoft.com/en-us/azure/azure-monitor/platform/collect-custom-metrics-linux-telegraf>

Design for user identities and profiles

Select an appropriate licensing model for AVD based on requirements

You can access Windows 10 Enterprise and Windows 7 Enterprise desktops and apps at no additional cost if you have an eligible Windows or Microsoft 365 license.

Access to desktops powered by Windows Server Remote Desktop Services desktops and apps are available at no additional cost if you are an eligible Microsoft Remote Desktop Services (RDS) Client Access License (CAL) customer.

Reserved Virtual Machine Instances are flexible and can be exchanged or returned.

For most applications, multi-session provides the greatest flexibility and cost savings by sharing the desktop infrastructure, preserving the state of your OS image by operating in a pooled environment, and allowing user data and settings to be saved and accessed using FSLogix.

Personal desktops are typically chosen for the following reasons:

- Users that require administrative rights to modify the operating system and want those changes to be retained if the VM is restarted
- Users which run applications that are not compatible with multi-session.



TYPE	DESCRIPTION	ELIGIBILITY
Virtualize Windows 10 and Windows 7	Access Windows 10 Enterprise and Windows 7 Enterprise desktops and apps at no additional cost if you have an eligible Windows or Microsoft 365 license. Get free Extended Security Updates until January 2023 for your Windows 7 virtual desktop—offering more options to support legacy apps while you transition to Windows 10.	You are eligible to access Windows 10 and Windows 7 with Azure Virtual Desktop if you have one of the following per user licenses*: Microsoft 365 E3/E5 Microsoft 365 A3/A5/Student Use Benefits Microsoft 365 F3 Microsoft 365 Business Premium** Windows 10 Enterprise E3/E5 Windows 10 Education A3/A5 Windows 10 VDA per user

TYPE	DESCRIPTION	ELIGIBILITY
Virtualize Windows Server	Access desktops powered by Windows Server Remote Desktop Services desktops and apps at no additional cost if you are an eligible Microsoft Remote Desktop Services (RDS) Client Access License (CAL) customer.	You are eligible to access Windows Server 2012 R2 and newer desktops and apps if you have a per-user or per-device RDS CAL license with active Software Assurance (SA).

* Customers can access Azure Virtual Desktop from their non-Windows Pro endpoints if they have a Microsoft 365 E3/E5/F3/Business/ A3/A5/Student Use Benefits or Windows 10 VDA per user license.

** Microsoft 365 Business Premium was previously named Microsoft 365 Business prior to April 21, 2020.

Listed below are the Azure components that factor into the price of an Azure Virtual Desktop deployment.

- Virtual machines and operating system (OS) storage
- Data disk (personal desktop only)
- User profile storage
- Networking

Azure Virtual Desktop session host virtual machines (VMs) including Citrix Cloud and VMW Horizon Cloud on Azure deployments are charged at Linux compute rates for Windows 10 single, Windows 10 multi-session and Windows Server.

Personal and multi-session desktop scenarios

Personal Desktop example

Sample use cases for single users accessing a persistent virtual desktop

EXAMPLE WORKLOADS	NUMBER OF USERS IN SCENARIO	TYPE OF USER	VCPUS	RAM	EAST US PRICING	WEST EUROPE PRICING	SOUTH-EAST ASIA PRICING
Graphics Workstation	100	Engineers and graphic designers with 3D modeling, simulations, and CAD workloads. Users spend 5-6 hours a day requiring workstation capability.	12	112 GB	See estimate (https://azure.com/e/)	See estimate (https://)	See estimate (https://)
Microsoft Office	1000	Standard knowledge workers making use of Microsoft Office products. Users work 8-10 hour days.	2	4 GB	See estimate (https://)	See estimate (https://azure)	See estimate (https://)

Multi-session Desktop example

Sample use cases for multiple users sharing a pooled (non-persistent) virtual desktop

EXAMPLE WORK-LOADS	NUMBER OF USERS IN SCENARIO	TYPE OF USER	USER DENSITY	EAST US PRICING	WEST EUROPE PRICING	SOUTHEAST ASIA PRICING
Microsoft Office	1000	Standard knowledge workers making use of Microsoft Office products. 24/7 RI is used to avoid need for management of virtual machines.	2 per vCPU	See estimate (https://	See estimate (https://	See estimate (https:// <i>azure.</i>
Call center/ data entry	1000	Call center users with low intensity workloads, primarily engaged in data entry. Users operate in three 8-hour shifts, making a 24/7 RI instance the most cost effective option.	6 per vCPU	See estimate (https:// <i>azure.</i>	See estimate (https://	See estimate (https:// <i>azure.</i>

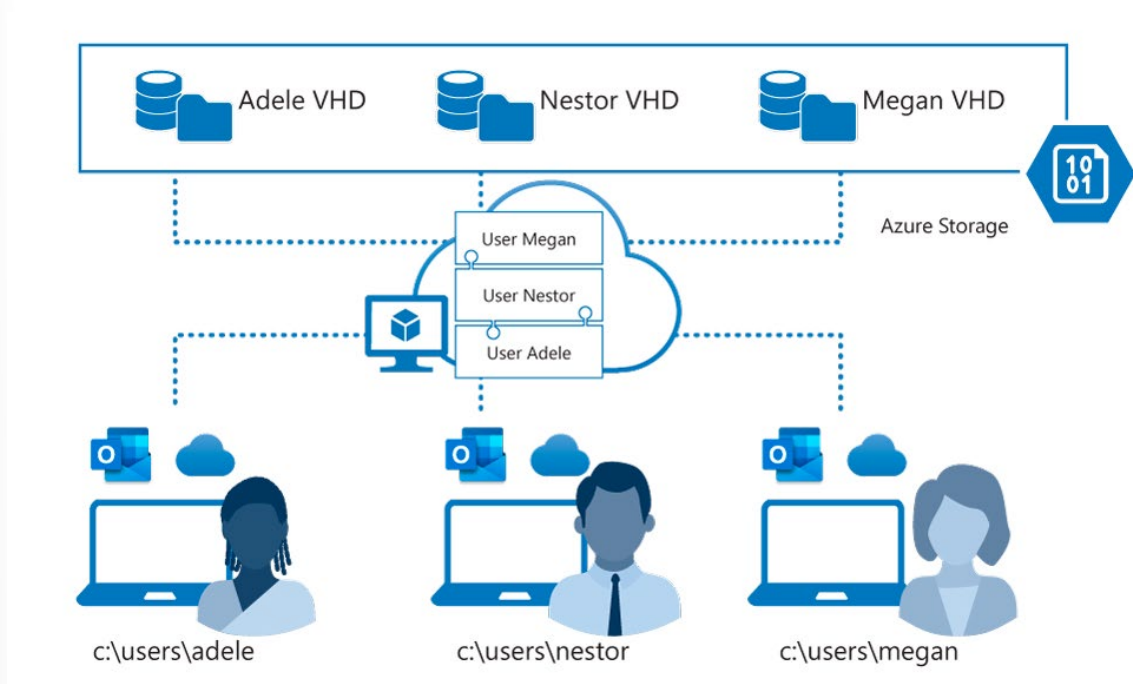
Recommend an appropriate storage solution

Azure offers multiple storage solutions that you can use to store your FSLogix profile container. This topic compares storage solutions that Azure offers for Azure Virtual Desktop FSLogix user profile containers.

We recommend storing FSLogix profile containers on Azure Files for most of our customers.

Azure Virtual Desktop offers FSLogix profile containers as the recommended user profile solution. FSLogix is designed to roam profiles in remote computing environments, such as Azure Virtual Desktop.

At sign-in, this container is dynamically attached to the computing environment using a natively supported Virtual Hard Disk (VHD) and a Hyper-V Virtual Hard Disk (VHDX). The user profile is immediately available and appears in the system exactly like a native user profile.



The following tables compare the storage solutions Azure Storage offers for Azure Virtual Desktop FSLogix profile container user profiles.

Azure platform details

Features	Azure Files	Azure NetApp Files	Storage Spaces Direct
Use case	General purpose	Ultra performance or migration from NetApp on-premises	Cross-platform
Platform service	Yes, Azure-native solution	Yes, Azure-native solution	No, self-managed
Regional availability	All regions	Select regions	All regions
Redundancy	Locally redundant/ zone-redundant/ geo-redundant/ geo-zone-redundant	Locally redundant	Locally redundant/ zone-redundant/ geo-redundant
Tiers and performance	Standard (Transaction optimized) Premium Up to max 100K IOPS per share with 10 GBps per share at about 3 ms latency	Standard Premium Ultra Up to 320k (16K) IOPS with 4.5 GBps per volume at about 1 ms latency	Standard HDD: up to 500 IOPS per-disk limits Standard SSD: up to 4k IOPS per-disk limits Premium SSD: up to 20k IOPS per-disk limits We recommend Premium disks for Storage Spaces Direct
Capacity	100 TiB per share, Upto 5 PiB per general purpose account	100 TiB per volume, up to 12.5 PiB per subscription	Maximum 32 TiB per disk

Features	Azure Files	Azure NetApp Files	Storage Spaces Direct
Required infrastructure	Minimum share size 1 GiB	Minimum capacity pool 4 TiB, min volume size 100 GiB	Two VMs on Azure IaaS (+ Cloud Witness) or at least three VMs without and costs for disks
Protocols	SMB 3.0/2.1, NFSv4.1 (preview), REST	NFSv3, NFSv4.1 (pre-view), SMB 3.x/2.x	NFSv3, NFSv4.1, SMB 3.1

Azure management details

Features	Azure Files	Azure NetApp Files	Storage Spaces Direct
Access	Cloud, on-premises and hybrid (Azure file sync)	Cloud, on-premises (via ExpressRoute)	Cloud, on-premises
Backup	Azure backup snapshot integration	Azure NetApp Files snapshots	Azure backup snapshot integration
Security and compliance	All Azure supported certificates	ISO completed	All Azure supported certificates
Azure Active Directory integration	Native Active Directory and Azure Active Directory Domain Services	Azure Active Directory Domain Services and Native Active Directory	Native Active Directory or Azure Active Directory Domain Services support only

Plan for a Desktop client deployment

You use the Remote Desktop client for Windows Desktop to access Windows apps and desktops remotely from a different Windows device.

NOTE: This topic is **not** intended for the Remote Desktop Connection (MSTSC) client that ships with Windows.

This topic is for the new Remote Desktop (MSRDC) client.

The new Remote Desktop client (MSRDC) supports:

- Windows 10
- Windows 10 IoT Enterprise
- Windows 7 client devices

Install the client

Choose the client that matches the version of Windows.

- **Windows 64-bit**³⁴
- **Windows 32-bit**³⁵
- **Windows ARM64**³⁶

Launch it from the Start menu by searching for **Remote Desktop**.

³⁴ <https://go.microsoft.com/fwlink/?linkid=2068602>

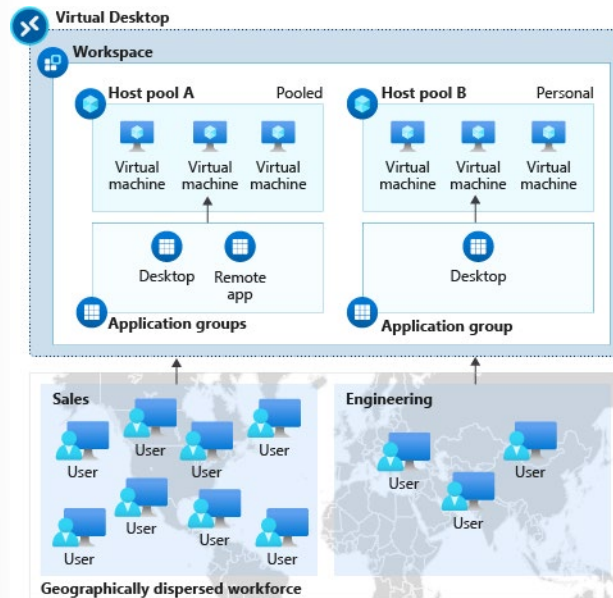
³⁵ <https://go.microsoft.com/fwlink/?linkid=2098960>

³⁶ <https://go.microsoft.com/fwlink/?linkid=2098961>

Workspaces

Get the list of managed resources you can access, such as apps and desktops, by subscribing to the Workspace your admin provided you. When you subscribe, the resources become available on your local PC. The Windows Desktop client currently supports resources published from Azure Virtual Desktop.

The following diagram shows an Azure Virtual Desktop workspace with two host pools.



- **Host pool A** has two application groups: Desktop and RemoteApp. These resources are shared (pooled) across the sales team.
- **Host pool B** has a Desktop application group with personal desktops available to an engineering team.

Subscribe to a Workspace

There are two ways you can subscribe to a Workspace. The client can try to discover the resources available to you from your work or school account or you can directly specify the URL where your resources are for cases where the client is unable to find them. Once you've subscribed to a Workspace, you can launch resources with one of the following methods:

- Go to the Connection Center and double-click a resource to launch it.
- You can also go to the Start menu and look for a folder with the Workspace name or enter the resource name in the search bar.

Subscribe with a user account

1. From the main page of the client, tap **Subscribe**.
2. Sign in with your user account when prompted.
3. The resources will appear in the Connection Center grouped by Workspace.

Subscribe with URL

1. From the main page of the client, tap **Subscribe with URL**.
2. Enter the Workspace URL or your email address:
 - If you use the **Workspace URL**, use the one your admin gave you. If accessing resources from Azure Virtual Desktop, you can use one of the following URL: <https://rdweb.wvd.microsoft.com/api/arm/feeddiscovery>
 - To use **email**, enter your email address. This tells the client to search for a URL associated with your email address if your admin has setup email discovery.
3. Tap **Next**.
4. Sign in with your user account when prompted.
5. The resources will appear in the Connection Center grouped by Workspace.

Plan for AVD client deployment - RDP

The Remote Desktop web client lets you use a compatible web browser to access your organization's remote resources (apps and desktops) published to you by your admin. You'll be able to interact with the remote apps and desktops like you would with a local PC no matter where you are, without having to switch to a different desktop PC.

Once your admin sets up your remote resources, all you need are your domain, user name, password, the URL your admin sent you, and a supported web browser, and you're good to go.

The web client doesn't currently have mobile OS support.

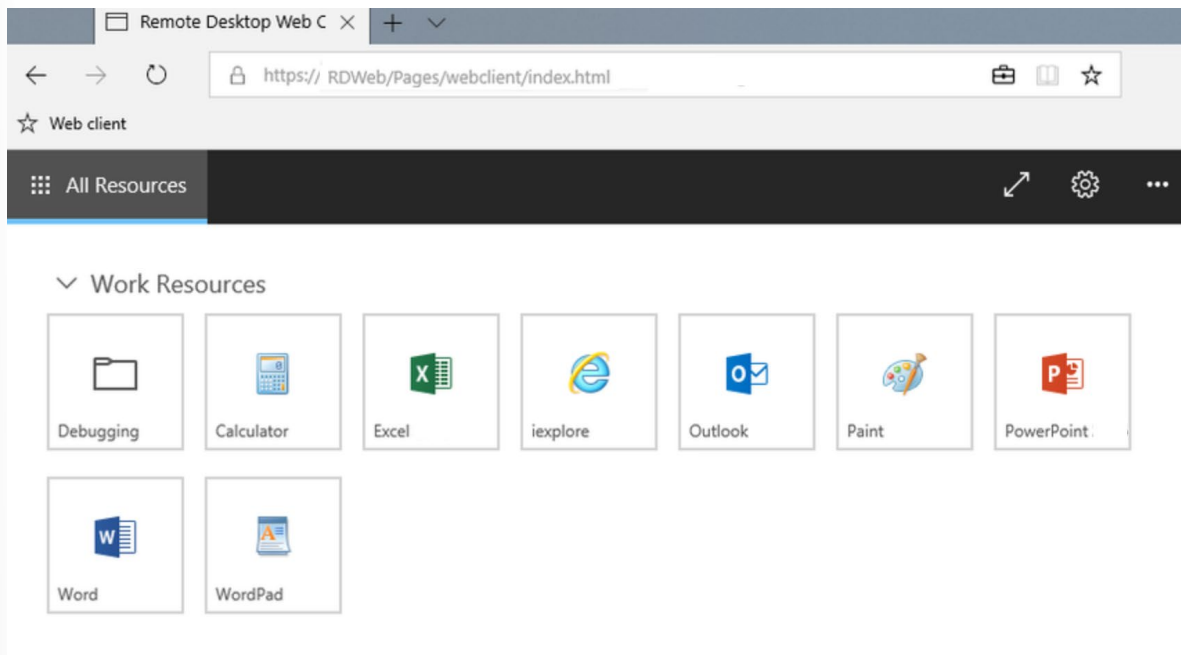
Supported operating systems and browsers

While any HTML5-capable browser should work, we officially support the following operating systems and browsers.

Browser	Supported OS	Notes
Microsoft Edge	Windows	
Internet Explorer	Windows	Version 11 or later
Apple Safari	macOS	
Mozilla Firefox	Windows, macOS, Linux	Version 55 or later
Google Chrome	Windows, macOS, Linux, Chrome OS	

What you'll need to use the web client

- For the web client, you'll need a PC running Windows, macOS, ChromeOS, or Linux. Mobile devices are not supported at this time.
- A modern browser like Microsoft Edge, Internet Explorer 11, Google Chrome, Safari, or Mozilla Firefox (v55.0 and later).
- The URL your admin sent you.



Using the Remote Desktop client

To sign in to the client, go to the URL your admin sent you. At the sign in page, enter your domain and user name in the format `DOMAIN\username`, enter your password, and then select **Sign in**.

After you sign in, the client will take you to the **All Resources** tab, which contains all items published to you under one or more collapsible groups, such as the "Work Resources" group. You'll see several icons representing the apps, desktops, or folders containing more apps or desktops that the admin has made available to the work group. You can come back to this tab at any time to launch additional resources.

To start using an app or desktop, select the item you want to use, enter the same user name and password you used to sign in to the web client if prompted, and then select **Submit**.

You might also be shown a consent dialog to access local resources, like clipboard and printer. You can choose to not redirect either of these, or select **Allow** to use the default settings.

Wait for the web client to establish the connection, and then start using the resource as you would normally.

Access remote resources feed

In a browser, navigate to the Azure Resource Manager-integrated version of the Azure Virtual Desktop web client at <https://rdweb.wvd.microsoft.com/arm/webclient> and sign in with your user account.

Windows Desktop client to multiple devices

Although your users can install the client directly after downloading it, if you're deploying to multiple devices, you may want to also deploy the client to them through other means. Deploying using group policies or the Microsoft Endpoint Configuration Manager lets you run the installer silently using a command line. Run the following commands to deploy the client per-device or per-user.

Per-device installation

```
msiexec.exe /I <path to the MSI> /qn ALLUSERS=1
```

Per-user installation

```
msiexec.exe /i <path to the MSI> /qn ALLUSERS=2 MSIINSTALLPERUSER=1
```

Configuration options

The section below describes the new configuration options for this client.

Configure update notifications

By default, the client notifies you whenever there's an update and automatically updates itself when the client is closed and has no active connections. Even with no active connections, the msrdc.exe process runs in the background to allow you to reconnect quickly when you reopen the client. You can stop **msrdc.exe** by right-clicking on the Azure Virtual Desktop icon in the system tray area and selecting **Disconnect all sessions** in the drop-down menu.

To turn notifications off, set the following registry information:

- **Key:** HKLM\Software\Microsoft\MSRDC\Policies
- **Type:** REG_DWORD
- **Name:** AutomaticUpdates
- **Data:** 0 = Disable notifications and turn off auto-update. 1 = Show notifications and turn off auto-update. 2 = Show notifications and auto-update on close.

Configure user groups

You can configure the client for one of the following types of user groups, which determines when the client receives updates.

Insider group

The Insider group is for early validation, and consists of admins and their selected users. The Insider group serves as a test run to detect any issues in the update that can impact performance before it's released to the Public group.

It's a good idea for each organization to have some users in the Insider group to test updates and catch issues early.

In the Insider group, a new version of the client is released to the users on the second Tuesday of each month for early validation. If the update doesn't have issues, it gets released to the Public group two weeks later. Users in the Insider group will receive update notifications automatically whenever updates are ready.

To configure the client for the Insider group, set the following registry information:

- **Key:** HKLM\Software\Microsoft\MSRDC\Policies

- **Type:** REG_SZ
- **Name:** ReleaseRing
- **Data:** insider

Public group

This group is for all users and is the most stable version. You don't need to do anything to configure this group.

The Public group receives the version of the client that was tested by the Insider group every fourth Tuesday of each month. All users in the Public group will receive an update notification if that setting is enabled.

Hybrid Identity with Azure Active Directory

Organizations are a mixture of on-premises and cloud applications. Users require access to those applications both on-premises and in the cloud.

Microsoft's identity solutions span on-premises and cloud-based capabilities. These solutions create a common user identity for authentication and authorization to all resources, regardless of location. We call this hybrid identity.

With hybrid identity to Azure AD and hybrid identity management these scenarios become possible.

To achieve hybrid identity with Azure AD, one of three authentication methods can be used, depending on your scenarios. The three methods are:

- **Password hash synchronization (PHS)**³⁷
- **Pass-through authentication (PTA)**³⁸
- **Federation (AD FS)**³⁹

These authentication methods also provide single-sign on capabilities. Single-sign on automatically signs your users in when they are on their corporate devices, connected to your corporate network.

Common scenarios and recommendations

Below are common hybrid identity and access management scenarios with recommendations as to which hybrid identity option (or options) might be appropriate for each.

I need to:	PHS and SSO ¹	PTA and SSO ²	AD FS ³
Sync new user, contact, and group accounts created in my on-premises Active Directory to the cloud automatically.	Yes	Yes	Yes
Set up my tenant for Office 365 hybrid scenarios.	Yes	Yes	Yes

³⁷ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/whatis-phs>

³⁸ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/how-to-connect-pta>

³⁹ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/whatis-fed>

I need to:	PHS and SSO ¹	PTA and SSO ²	AD FS ³
Enable my users to sign in and access cloud services using their on-premises password.	Yes	Yes	Yes
Implement single sign-on using corporate credentials.	Yes	Yes	Yes
Ensure no password hashes are stored in the cloud.		Yes	Yes
Enable cloud-based multi-factor authentication solutions.	Yes	Yes	Yes
Enable on-premises multi-factor authentication solutions.			Yes
Support smartcard authentication for my users. ⁴			Yes
Display password expiry notifications in the Office Portal and on the Windows 10 desktop.			Yes

¹ Password hash synchronization with single sign-on.

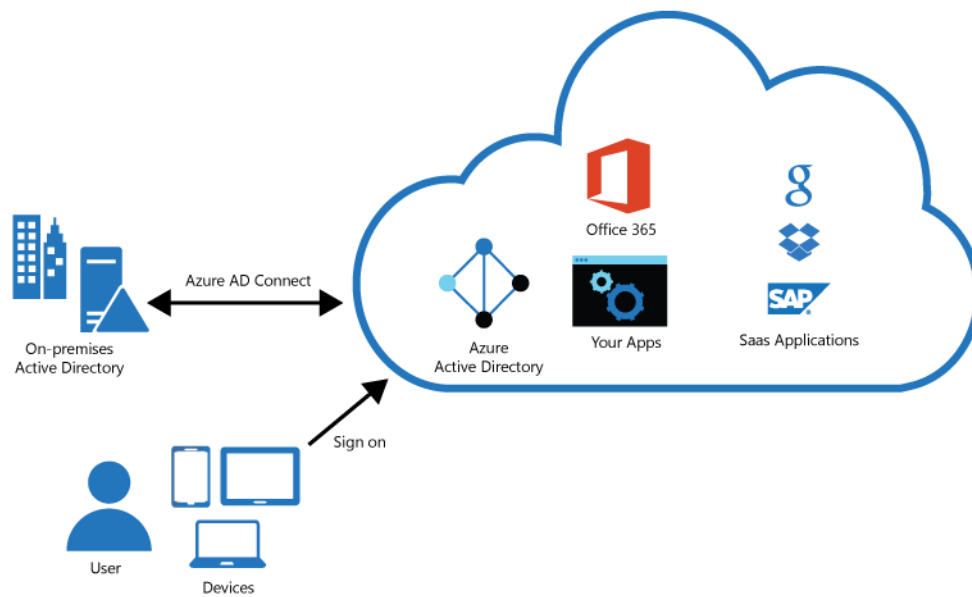
² Pass-through authentication and single sign-on.

³ Federated single sign-on with AD FS.

⁴ AD FS can be integrated with your enterprise PKI to allow sign-in using certificates. These certificates can be soft-certificates deployed via trusted provisioning channels such as MDM or GPO or smartcard certificates (including PIV/CAC cards) or Hello for Business (cert-trust).

Plan for Azure AD Connect for user identities

To keep Windows Server Active Directory in sync with Azure Active Directory, you can configure Azure AD Connect (for hybrid organizations).



Azure AD Connect is the Microsoft tool designed to meet and accomplish your hybrid identity goals. It provides the following features:

- **Password hash synchronization⁴⁰** - A sign-in method that synchronizes a hash of a users on-premises AD password with Azure AD.

⁴⁰ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/whatis-phs>

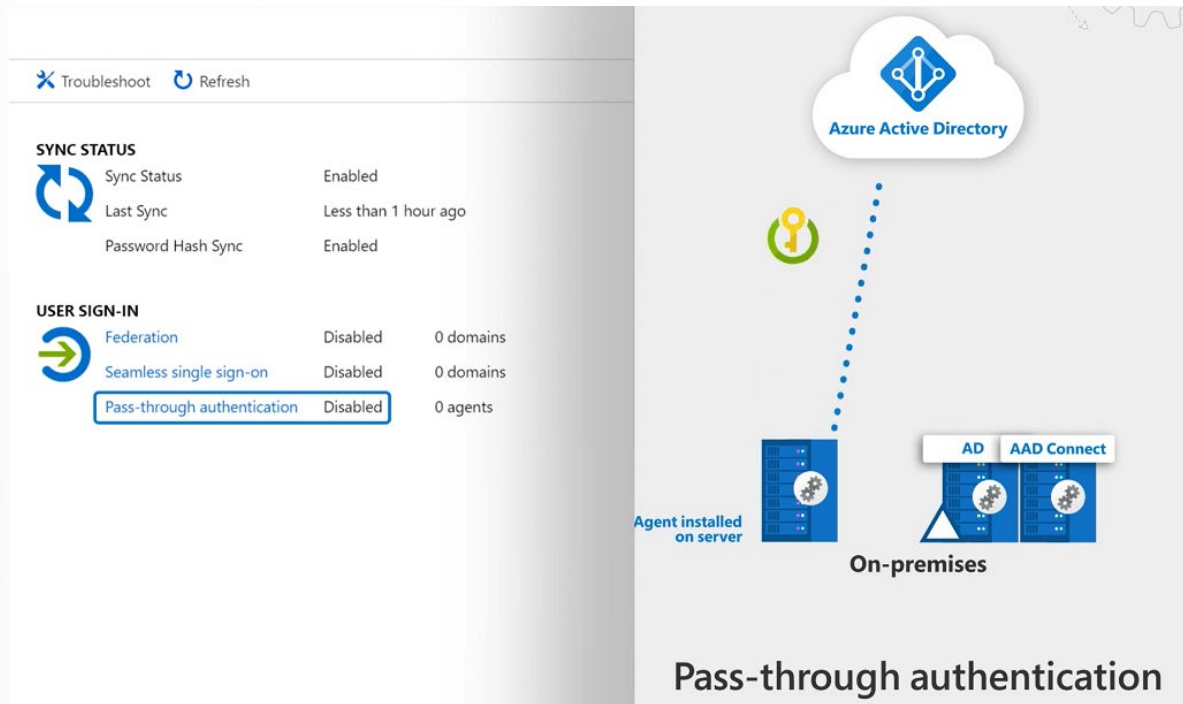
The screenshot displays the Azure Active Directory (AAD) console interface. On the left, the 'SYNC STATUS' section shows 'Sync Status' as 'Enabled' and 'Last Sync' as 'Less than 1 hour ago'. Below this, the 'PASSWORD HASH SYNC' section is highlighted, showing 'Password Hash Sync' as 'Enabled'. The 'USER SIGN-IN' section shows 'Federation', 'Seamless single sign-on', and 'Pass-through authentication' all as 'Disabled' with '0 domains' or '0 agents'.

On the right, a diagram illustrates the 'Password Hash Sync' process. It shows a cloud icon labeled 'Usernames' and 'Password' connected by dotted lines to a server icon labeled 'AD' and 'AAD Connect' under the heading 'On-premises'. Below the server icon, there are two input fields labeled 'Usernames' and 'Password', each containing a series of hash symbols (#####).

Password Hash Sync

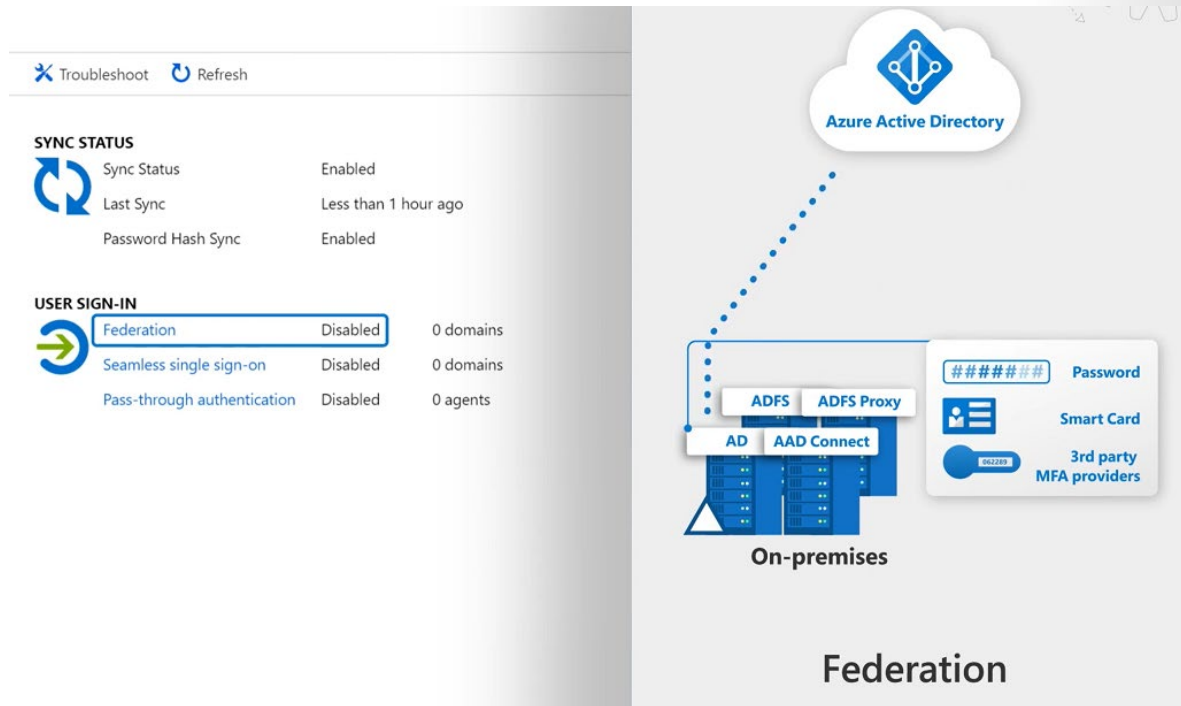
- **Pass-through authentication⁴¹** - A sign-in method that allows users to use the same password on-premises and in the cloud, but doesn't require the additional infrastructure of a federated environment.

⁴¹ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/how-to-connect-pta>



- **Federation integration⁴²** - Federation is an optional part of Azure AD Connect and can be used to configure a hybrid environment using an on-premises AD FS infrastructure. It also provides AD FS management capabilities such as certificate renewal and additional AD FS server deployments.

⁴² <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/how-to-connect-fed-what-is>



- **Synchronization⁴³** - Responsible for creating users, groups, and other objects. As well as, making sure identity information for your on-premises users and groups is matching the cloud. This synchronization also includes password hashes.
- **Health Monitoring⁴⁴** - Azure AD Connect Health can provide robust monitoring and provide a central location in the Azure portal to view this activity.

Azure Virtual Desktop supports hybrid identities through Azure Active Directory (AD), including those federated using Active Directory Federation Services (ADFS).

Since users must be discoverable through Azure AD, Azure Virtual Desktop doesn't support standalone Active Directory deployments with ADFS.

Azure Virtual Desktop currently doesn't support Active Directory Federation Services (ADFS) for SSO.

The only way to avoid being prompted for your credentials for the session host is to save them in the client. We recommend you only do this with secure devices to prevent other users from accessing your resources.

Windows 10 Enterprise multi-session is currently supported to be hybrid Azure AD-joined. After Windows 10 Enterprise multi-session is domain-joined, use the existing Group Policy Object to enable Azure AD registration.

⁴³ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/how-to-connect-sync-what-is>

⁴⁴ <https://docs.microsoft.com/en-us/azure/active-directory/hybrid/whatis-hybrid-identity-health>

Labs

Prepare for deployment of Azure Virtual Desktop (Azure AD DS)

✓ **Important:** To download the most recent version of this lab, please visit the AZ-140 **GitHub repository**⁴⁵.

Direct link to the **Lab - Prepare for deployment of Azure Azure Virtual Desktop (Azure AD DS)**.⁴⁶

Objectives

After completing this lab, you will be able to:

- Implement an Azure AD DS domain
- Configure the Azure AD DS domain environment

Lab prerequisites

- An Azure subscription
- A Microsoft account or an Azure AD account with the Global Administrator role in the Azure AD tenant associated with the Azure subscription and with the Owner role in the Azure subscription

Note: At the time of authoring this course, the MSIX app attach functionality for Azure Virtual Desktop is in public preview. If you intend to run the lab that involves the use of MSIX app attach included in this course, you need to submit a request via an **online form**⁴⁷ to enable MSIX app attach in your subscription. The approval and processing of requests can take up to 24 hours during business days. You'll receive an email confirmation once your request has been accepted and completed.

Estimated time: 150 minutes

Note: Provisioning of an Azure AD DS takes involves about 90-minute wait time.

Lab files

- \\AZ-140\\AllFiles\\Labs\\01\\az140-11_azuredeploycl11a.json
- \\AZ-140\\AllFiles\\Labs\\01\\az140-11_azuredeploycl11a.parameters.json

Exercise 0: Increase the number of vCPU quotas

The main tasks for this exercise are as follows:

1. Identify current vCPU usage
2. Request vCPU quota increase

⁴⁵ https://aka.ms/AZ-140_Labs

⁴⁶ https://aka.ms/AZ-140_01_Lab_01

⁴⁷ <https://aka.ms/enablesixappattach>

Exercise 1: Implement an Azure Active Directory Domain Services (AD DS) domain

The main tasks for this exercise are as follows:

1. Create and configure an Azure AD user account for administration of Azure AD DS domain
2. Deploy an Azure AD DS instance by using the Azure portal
3. Configure the network and identity settings of the Azure AD DS deployment

Exercise 2: Configure the Azure AD DS domain environment

The main tasks for this exercise are as follows:

1. Deploy an Azure VM running Windows 10 by using an Azure Resource Manager QuickStart template
2. Review the default configuration of the Azure AD DS domain
3. Create AD DS users and groups that will be synchronized to Azure AD DS

Prepare for deployment of Azure Virtual Desktop (AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 **GitHub repository**⁴⁸.

Direct link to the **Lab - Prepare for deployment of Azure Virtual Desktop (AD DS)**.⁴⁹

Objectives

After completing this lab, you will be able to:

- Deploy an Active Directory Domain Services (AD DS) single-domain forest by using Azure VMs
- Integrate an AD DS forest with an Azure Active Directory (Azure AD) tenant

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.

Estimated time: 60 minutes

Lab files

- \\AZ-140\\AllFiles\\Labs\\01\\az140-11_azuredeploydc11.parameters.json
- \\AZ-140\\AllFiles\\Labs\\01\\az140-11_azuredeploycl11.json
- \\AZ-140\\AllFiles\\Labs\\01\\az140-11_azuredeploycl11.parameters.json

⁴⁸ https://aka.ms/AZ-140_Labs

⁴⁹ https://aka.ms/AZ-140_01_Lab_02

Exercise 0: Increase the number of vCPU quotas

The main tasks for this exercise are as follows:

1. Identify current vCPU usage
2. Request vCPU quota increase

Exercise 1: Deploy an Active Directory Domain Services (AD DS) domain

The main tasks for this exercise are as follows:

1. Identify an available DNS name for an Azure VM deployment
2. Deploy an Azure VM running an AD DS domain controller by using an Azure Resource Manager QuickStart template
3. Deploy an Azure VM running Windows 10 by using an Azure Resource Manager QuickStart template

Exercise 2: Integrate an AD DS forest with an Azure AD tenant

The main tasks for this exercise are as follows:

1. Create AD DS users and groups that will be synchronized to Azure AD
2. Configure AD DS UPN suffix
3. Create an Azure AD user that will be used to configure synchronization with Azure AD
4. Install Azure AD Connect
5. Configure hybrid Azure AD join

Review questions

Module review questions

Review Question 1

You manage an office where all your users work.

Half of the users in the office work from home, and the remainder work in the office day-to-day.

The employees working from home do not connect to the office using a VPN.

You are designing an AVD instance, and you need to estimate the user workload using the following data.

Light workload:

- 40 users
- 1.5 Mbps

Medium workload:

- 20 users
- 3 Mbps

Heavy workload

- 20 users
- 5 Mbps

How much bandwidth will you need to allocate between the office and Azure to support all three workloads?

- ☐ A. 80 Mbps
- ☐ B. 220 Mbps
- ☐ C. 110 Mbps
- ☐ D. 120 Mbps

Review Question 2

You are planning an AVD instance that includes the following:

West Coast Sales Team:

- AVD single-session desktops
- 10 users
- GPU (not required)

East Coast Sales Team:

- AVD multi-session desktops
- 50 users

- GPU (required)

Southern Sales Team:

- AVD multi-session desktops
- 50 users
- GPU (not required)

Northern Coast Sales Team:

- Remote App
- 10 users
- GPU (not required)

*You are planning on using AVD host pools with autoscaling and load balancing.
You need to determine a design for the host pools with an eye towards minimizing the costs.
What is the minimum number of host pool you should plan for?*

- ☐ A. 2
- ☐ B. 3
- ☐ C. 4
- ☐ D. 5

Review Question 3

You are managing the following computer devices:

Client_Device_A (Win 10 Home)

Client_Device_B (Win 8.1 Pro)

Client_Device_C (Win 10 IoT Ent)

You want your users to access virtualized apps remotely. Of the devices listed above, which ones will support the remote desktop client?

- ☐ A. Client_Device_A, Client_Device_B, and Client_Device_C
- ☐ B. Client_Device_B
- ☐ C. Client_Device_A
- ☐ D. Client_Device_A and Client_Device_C

Review Question 4

You are an administrator for a medium-sized organization where you support 50,000 users.

You are in the middle of planning an AVD deployment.

You need to plan for using FSLogix profile containers. You must find a storage solution for low latency and high Input/output operations per second (IOPS).

What should you use?

- ☐ A. Cache Azure file share on-premises with Azure File Sync
- ☐ B. Azure NetApp files
- ☐ C. A General purpose version 2 (GPv2) account
- ☐ D. Configure an Azure File Sync (Storage Sync Service)

Review Question 5

You are planning an AVD deployment.

You are measuring the latency in the network between where your users are located and where you are planning your deployment.

What could you use to determine the optimal Azure region for deploying a host pool?

- ☐ A. Deploy a WAN optimization network virtual appliance
- ☐ B. Azure Virtual Desktop Experience Estimator
- ☐ C. Diagnose with Network Watcher
- ☐ D. Apply a Network Security Group (NSG) filter

Review Question 6

You manage an existing AVD instance.

You need to provide your external users access to the AVD instance.

Your users have Win 10 Pro and Win 10 Ent running on their computers.

Your users do not have the ability to install apps.

What are you going to recommend your users use to connect to the AVD deployment?

- ☐ A. Microsoft Edge browser
- ☐ B. Modify desktop RDP properties for device redirection
- ☐ C. Launch the Connection Center and click Subscribe
- ☐ D. Implement the RD Connection Broker

Review Question 7

You manage a network that has an on-premises domain that has a universal security group named SecurityUsers.

SecurityUsers syncs with AAD, where there is a hybrid AAD tenant.

You manage an AVD host pool that has three Win 10 Enterprise multi-session hosts.

You want to make sure that only members of SecurityUsers can establish AVD sessions to the host pool.

What needs to be done to meet your goal?

- ☐ A. Create a new role assignment for the host pool
- ☐ B. Modify the RDP properties on the host pool
- ☐ C. Configure role assignment for each of the three VMs
- ☐ D. Assign SecurityUsers to an application group

Answers

Review Question 1

You manage an office where all your users work.

Half of the users in the office work from home, and the remainder work in the office day-to-day.

The employees working from home *do not connect* to the office using a VPN.

You are designing an AVD instance, and you need to estimate the user workload using the following data.

Light workload:

Medium workload:

Heavy workload

How much bandwidth will you need to allocate between the office and Azure to support all three workloads?

- ☐ A. 80 Mbps
- ☐ B. 220 Mbps
- ☒ C. 110 Mbps
- ☐ D. 120 Mbps

Explanation

The answer is C. 110 Mbps.

Because half the users work from home and aren't connecting using a VPN, the other half the users on the office network. Thus, $(401.5 + 203 + 20 \times 5)/2 = 110$ Mbps

See topic: Assess network capacity and speed requirements for AVD.

Review Question 2

You are planning an AVD instance that includes the following:

West Coast Sales Team:

East Coast Sales Team:

Southern Sales Team:

Northern Coast Sales Team:

You are planning on using AVD host pools with autoscaling and load balancing.
 You need to determine a design for the host pools with an eye towards minimizing the costs.
 What is the minimum number of host pool you should plan for?

- ☐ A. 2
- ☒ B. 3
- ☐ C. 4
- ☐ D. 5

Explanation

As seen in the *Azure Virtual Desktop Components* topic, *Azure Virtual Desktop session hosts*: A host pool can run the following operating systems:

Review Question 3

You are managing the following computer devices:

Client_Device_A (Win 10 Home)

Client_Device_B (Win 8.1 Pro)

Client_Device_C (Win 10 IoT Ent)

You want your users to access virtualized apps remotely. Of the devices listed above, which ones will support the remote desktop client?

- ☐ A. Client_Device_A, Client_Device_B, and Client_Device_C
- ☐ B. Client_Device_B
- ☐ C. Client_Device_A
- ☒ D. Client_Device_A and Client_Device_C

Explanation

The answer is D, Client_Device_A and Client_Device_C. As seen in topic *Plan for Windows Desktop client deployment*, the *Remote Desktop client (MSRDC)* supports the following:

Review Question 4

You are an administrator for a medium-sized organization where you support 50,000 users.

You are in the middle of planning an AVD deployment.

You need to plan for using FSLogix profile containers. You must find a storage solution for low latency and high Input/output operations per second (IOPS).

What should you use?

- ☐ A. Cache Azure file share on-premises with Azure File Sync
- ☒ B. Azure NetApp files
- ☐ C. A General purpose version 2 (GPv2) account
- ☐ D. Configure an Azure File Sync (Storage Sync Service)

Explanation

The answer is B, Azure NetApp Files. As seen in the topic *Recommend an appropriate storage solution*, *Azure NetApp Files* provides up to 320k (16K) IOPS with 4.5 GBps per volume at about 1 ms latency.

Review Question 5

You are planning an AVD deployment.

You are measuring the latency in the network between where your users are located and where you are planning your deployment.

What could you use to determine the optimal Azure region for deploying a host pool?

- ☐ A. Deploy a WAN optimization network virtual appliance
- ☒ B. Azure Virtual Desktop Experience Estimator
- ☐ C. Diagnose with Network Watcher
- ☐ D. Apply a Network Security Group (NSG) filter

Explanation

The answer is B, Azure Virtual Desktop Experience Estimator. As seen in the *Azure Virtual Desktop Experience Estimator* topic, use the Azure Virtual Desktop Experience Estimator to determine the connection round trip time (RTT) from your current location, through the Azure Virtual Desktop service, to each Azure region in which you can deploy virtual machines.

Review Question 6

You manage an existing AVD instance.

You need to provide your external users access to the AVD instance.

You users have Win 10 Pro and Win 10 Ent running on their computers.

Your users do not have the ability to install apps.

What are you going to recommend your users use to connect to the AVD deployment?

- ☒ A. Microsoft Edge browser
- ☐ B. Modify desktop RDP properties for device redirection
- ☐ C. Launch the Connection Center and click Subscribe
- ☐ D. Implement the RD Connection Broker

Explanation

The answer is A, Microsoft Edge browser. As seen in the topic *Plan for AVD client deployment - RDP*, to run a web client you need an HTML5-capable browser and a PC running Windows, macOS, ChromeOS, or Linux.

Review Question 7

You manage a network that has an on-premises domain that has a universal security group named SecurityUsers.

SecurityUsers syncs with AAD, where there is a hybrid AAD tenant.

You manage an AVD host pool that has three Win 10 Enterprise multi-session hosts.

You want to make sure that only members of SecurityUsers can establish AVD sessions to the host pool.

What needs to be done to meet your goal?

- ☐ A. Create a new role assignment for the host pool
- ☐ B. Modify the RDP properties on the host pool
- ☐ C. Configure role assignment for each of the three VMs
- ☒ D. Assign SecurityUsers to an application group

Explanation

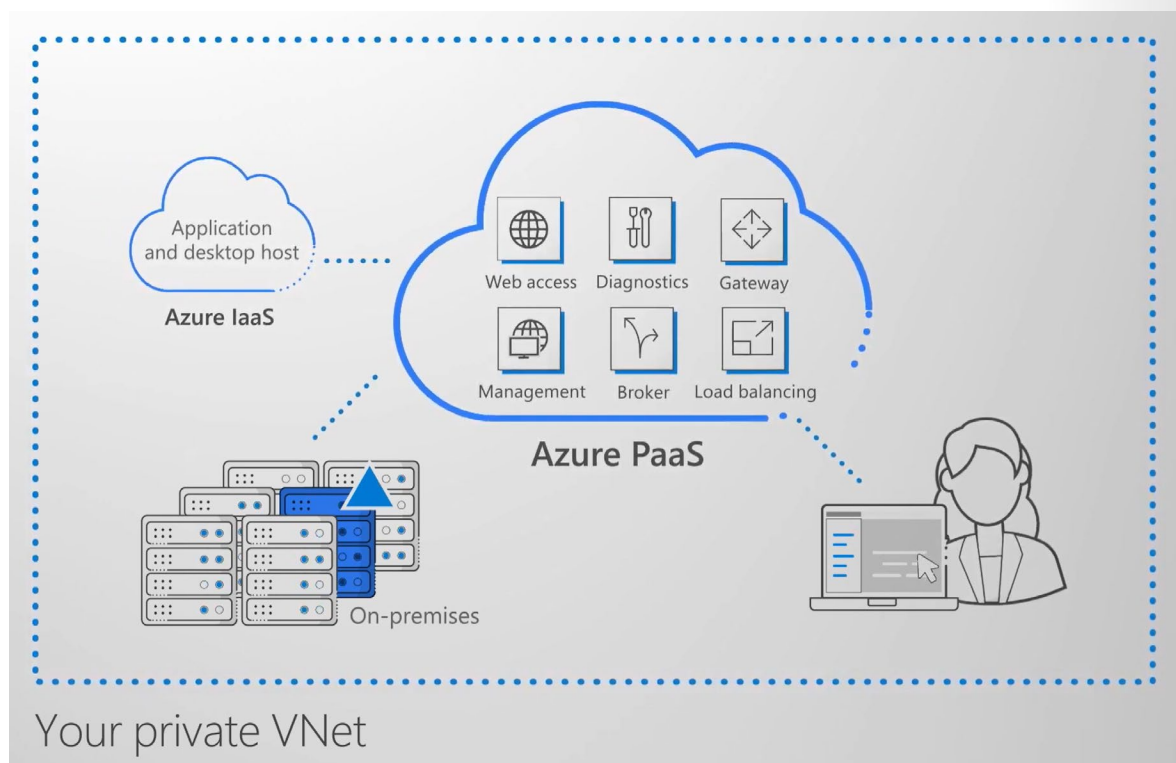
The answer is D, Assign SecurityUsers to an application group. As seen in the topic *Assign SecurityUsers to an application group*, users obtain access to host pools by being allocated to a host pool using an assigned Application Group.

Module 2 Implement an Azure Virtual Desktop infrastructure

Implement and manage networking for AVD

Implement Azure virtual network connectivity

Azure Virtual Network (VNet) is basis for a private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks.



Azure virtual network enables Azure resources to securely communicate with each other, the internet, and on-premises networks.

Key scenarios that you can accomplish using a virtual network include:

- Communication of Azure resources with the internet
- Communication between Azure resources
- Communication with on-premises resources
- Filtering network traffic
- Routing network traffic
- Integration with Azure services

Communicate with the internet

All resources in a VNet can communicate outbound to the internet, by default. You can communicate inbound to a resource by assigning a public IP address or a public Load Balancer. You can also use public IP or public Load Balancer to manage your outbound connections.

When using only an internal Standard Load Balancer, outbound connectivity is not available until you define how you want outbound connections to work with an instance-level public IP or a public Load Balancer.

Communicate between Azure resources

Azure resources communicate securely with each other in one of the following ways:

- **Through a virtual network:** You can deploy VMs, and several other types of Azure resources to a virtual network, such as Azure App Service Environments, the Azure Kubernetes Service (AKS), and Azure Virtual Machine Scale Sets.
- **Through a virtual network service endpoint:** Extend your virtual network private address space and the identity of your virtual network to Azure service resources, such as Azure Storage accounts and Azure SQL Database, over a direct connection. Service endpoints allow you to secure your critical Azure service resources to only a virtual network.
- **Through VNet Peering:** You can connect virtual networks to each other, enabling resources in either virtual network to communicate with each other, using virtual network peering. The virtual networks you connect can be in the same, or different, Azure regions.

Manage connectivity to the internet and on-premises networks

You can connect your on-premises computers and networks to a virtual network using any combination of the following options:

- **Point-to-site virtual private network (VPN):** Established between a virtual network and a single computer in your network.
 - Each computer that wants to establish connectivity with a virtual network must configure its connection.
 - Ideal for just getting started with Azure, or for developers, because it requires little or no changes to your existing network.

- The communication between your computer and a virtual network is sent through an encrypted tunnel over the internet.
- **Site-to-site VPN:** Established between your on-premises VPN device and an Azure VPN Gateway that is deployed in a virtual network.
 - Enables any on-premises resource that you authorize to access a virtual network.
 - The communication between your on-premises VPN device and an Azure VPN gateway is sent through an encrypted tunnel over the internet.
- **Azure ExpressRoute:** Established between your network and Azure, through an ExpressRoute partner.
 - This connection is private. Traffic does not go over the internet.

Filter network traffic

You can filter network traffic between subnets using either or both of the following options:

- **Network security groups (NSGs):** Network security groups and application security groups can contain multiple inbound and outbound security rules that enable you to filter traffic to and from resources by source and destination IP address, port, and protocol.
- **Network virtual appliance (NVA):** A network virtual appliance is a VM that performs a network function, such as a firewall, WAN optimization, or other network function.

Route network traffic

Azure routes traffic between subnets, connected virtual networks, on-premises networks, and the Internet, by default. You can implement either or both of the following options to override the default routes Azure creates:

- **Route tables:** You can create custom route tables with routes that control where traffic is routed to for each subnet.
- **Border gateway protocol (BGP) routes:** If you connect your virtual network to your on-premises network using an Azure VPN Gateway or ExpressRoute connection, you can propagate your on-premises BGP routes to your virtual networks.

Virtual network integration for Azure services

Integrating Azure services to an Azure virtual network enables private access to the service from virtual machines or compute resources in the virtual network. You can integrate Azure services in your virtual network with the following options:

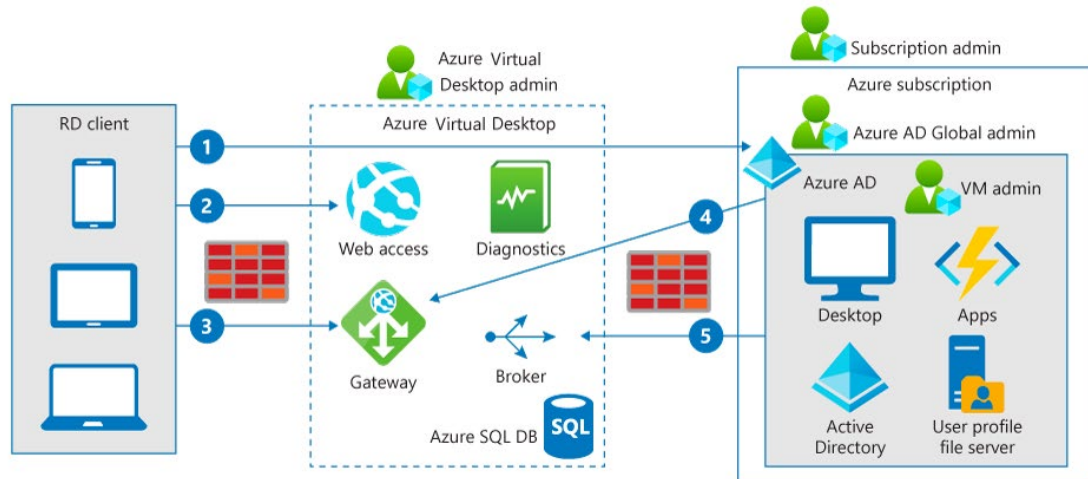
- Deploying dedicated instances of the service into a virtual network. The services can then be privately accessed within the virtual network and from on-premises networks.
- Using Private Link to access privately a specific instance of the service from your virtual network and from on-premises networks.
- You can also access the service using public endpoints by extending a virtual network to the service, through service endpoints. Service endpoints allow service resources to be secured to the virtual network.

Understanding Azure Virtual Desktop network connectivity

Azure Virtual Desktop uses Remote Desktop Protocol (RDP) to provide remote display and input capabilities over network connections.

The connection data flow for Azure Virtual Desktop starts with a DNS lookup for the closest Azure datacenter.

The following image shows the five-step connection process for Azure Virtual Desktop running in Azure.

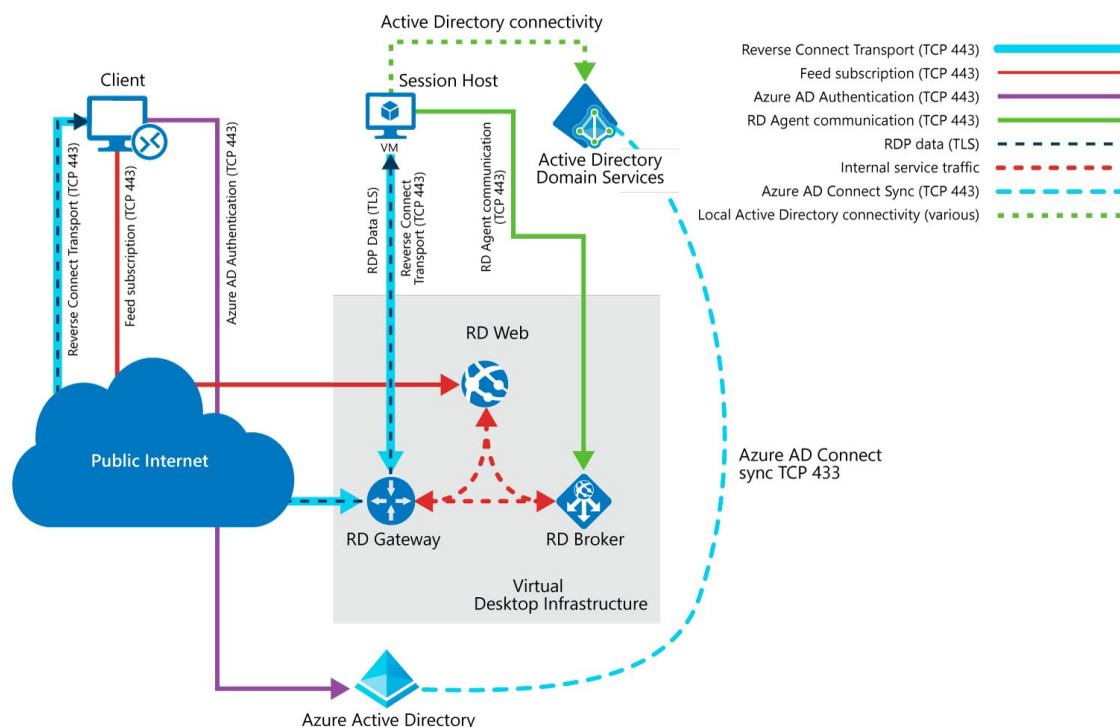


- 1. When authenticated in Azure Active Directory, a token is returned to the Remote Desktop Services client.**
- 2. The gateway checks the token with the connection broker.**
- 3. The broker queries the Azure SQL database for resources assigned to the user.**
- 4. The gateway and the broker select the session host for the connected client.**
- 5. The session host creates a reverse connection to the client by using the Azure Virtual Desktop gateway.**

No inbound ports are opened. In this version, the gateway is acting as an intelligent reverse proxy. The gateway manages all session connectivity, with nothing but pixels reaching the client.

Azure Virtual Desktop hosts client sessions on the session hosts running on Azure. Microsoft manages portions of the services on the customer's behalf and provides secure endpoints for connecting clients and session hosts. The diagram below gives a high-level overview of the network connections used by Azure Virtual Desktop

Azure Virtual Desktop Network Connections



Session connectivity

Azure Virtual Desktop uses Remote Desktop Protocol (RDP) to provide remote display and input capabilities over network connections. RDP has initially released with Windows NT 4.0 Terminal Server Edition and was continuously evolving with every Microsoft Windows and Windows Server release. From the beginning, RDP developed to be independent of its underlying transport stack, and today it supports multiple types of transport.

Reverse connect transport

Azure Virtual Desktop is using reverse connect transport for establishing the remote session and for carrying RDP traffic. Unlike the on-premises Remote Desktop Services deployments, reverse connect transport doesn't use a TCP listener to receive incoming RDP connections. Instead, it is using outbound connectivity to the Azure Virtual Desktop infrastructure over the HTTPS connection.

Session host communication channel

Upon startup of the Azure Virtual Desktop session host, the Remote Desktop Agent Loader service establishes the Azure Virtual Desktop broker's persistent communication channel. This communication channel is layered on top of a secure Transport Layer Security (TLS) connection and serves as a bus for service message exchange between session host and Azure Virtual Desktop infrastructure.

Client connection sequence

Client connection sequence described below:

1. Using supported Azure Virtual Desktop client user subscribes to the Azure Virtual Desktop Workspace
2. Azure Active Directory authenticates the user and returns the token used to enumerate resources available to a user
3. Client passes token to the Azure Virtual Desktop feed subscription service
4. Azure Virtual Desktop feed subscription service validates the token
5. Azure Virtual Desktop feed subscription service passes the list of available desktops and RemoteApps back to the client in the form of digitally signed connection configuration
6. Client stores the connection configuration for each available resource in a set of .rdp files
7. When a user selects the resource to connect, the client uses the associated .rdp file and establishes the secure TLS 1.2 connection to the closest Azure Virtual Desktop gateway instance and passes the connection information
8. Azure Virtual Desktop gateway validates the request and asks the Azure Virtual Desktop broker to orchestrate the connection
9. Azure Virtual Desktop broker identifies the session host and uses the previously established persistent communication channel to initialize the connection
10. Remote Desktop stack initiates the TLS 1.2 connection to the same Azure Virtual Desktop gateway instance as used by the client
11. After both client and session host connected to the gateway, the gateway starts relaying the raw data between both endpoints, this establishes the base reverse connect transport for the RDP
12. After the base transport is set, the client starts the RDP handshake

Connection security

TLS 1.2 is used for all connections initiated from the clients and session hosts to the Azure Virtual Desktop infrastructure components.

For reverse connect transport, both client and session host connect to the Azure Virtual Desktop gateway. After establishing the TCP connection, the client or session host validates the Azure Virtual Desktop gateway's certificate.

After establishing the base transport, RDP establishes a nested TLS connection between client and session host using the session host's certificates.

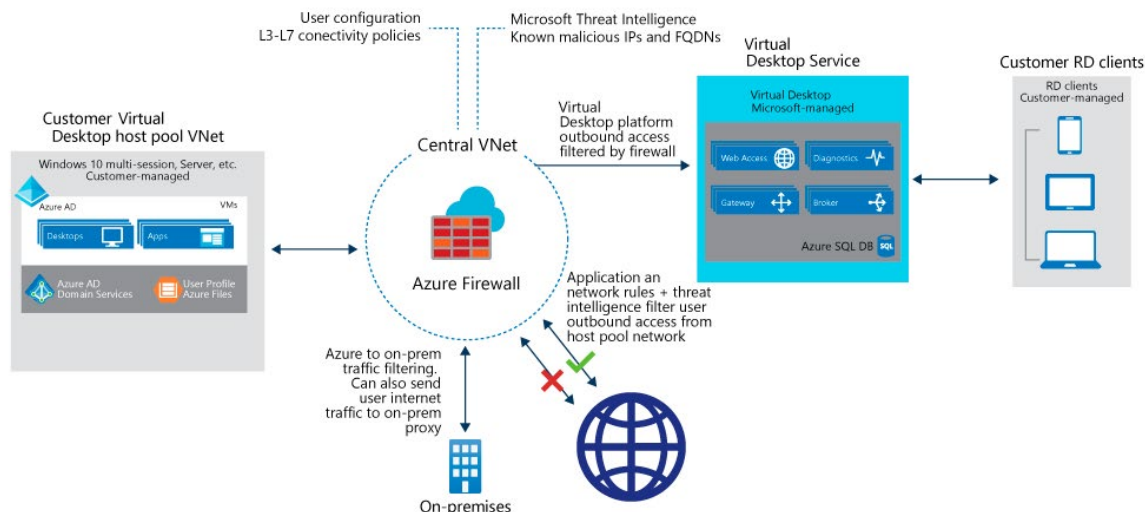
By default, the certificate used for RDP encryption is self-generated by the OS during the deployment

Implement and manage network security

As a refresher from the previous module, when an end user connects to an Azure Virtual Desktop environment, their session is run by a host pool.

- A host pool is a collection of Azure virtual machines that register to Azure Virtual Desktop as session hosts.
- These virtual machines run in your virtual network and are subject to the virtual network security controls.

- They need outbound Internet access to the Azure Virtual Desktop service to operate properly and might also need outbound Internet access for end users.
- Azure Firewall can help you lock down your environment and filter outbound traffic.



The guidelines below provide additional protection for your Azure Virtual Desktop host pool using Azure Firewall.

Host pool outbound access to Azure Virtual Desktop

The Azure virtual machines you create for Azure Virtual Desktop must have access to several Fully Qualified Domain Names (FQDNs) to function properly. Azure Firewall provides an Azure Virtual Desktop FQDN Tag to simplify this configuration. Use the following steps to allow outbound Azure Virtual Desktop platform traffic:

- Deploy Azure Firewall and configure your Azure Virtual Desktop host pool subnet User Defined Route (UDR) to route all traffic via the Azure Firewall. Your default route now points to the firewall.
- Create an application rule collection and add a rule to enable the *WindowsVirtualDesktop* FQDN tag. The source IP address range is the host pool virtual network, the protocol is **https**, and the destination is **WindowsVirtualDesktop**.
- The set of required storage and service bus accounts for your Azure Virtual Desktop host pool is deployment specific, so it isn't yet captured in the *WindowsVirtualDesktop* FQDN tag. You can address this in one of the following ways:
 - Allow https access from your host pool subnet to **xt.blob.core.windows.net*, **eh.servicebus.windows.net* and **xt.table.core.windows.net*. These wildcard FQDNs enable the required access, but are less restrictive.
 - Use the following log analytics query to list the exact required FQDNs, and then allow them explicitly in your firewall application rules:

```
AzureDiagnostics
| where Category == "AzureFirewallApplicationRule"
| search "Deny"
```

```
| search "gsm*eh.servicebus.windows.net" or "gsm*xt.blob.core.windows.net"
or "gsm*xt.table.core.windows.net"
| parse msg_s with Protocol " request from " SourceIP ":" SourcePort:int "
to " FQDN ":" *
| project TimeGenerated,Protocol,FQDN
```

- Create a network rule collection add the following rules:
 - Allow DNS – allow traffic from your ADDS private IP address to * for TCP and UDP ports 53.
 - Allow KMS – allow traffic from your Azure Virtual Desktop virtual machines to Windows Activation Service TCP port 1688.

Some deployments may not need DNS rules, for example Azure Active Directory Domain controllers forward DNS queries to Azure DNS at 168.63.129.16.

Host pool outbound access to the Internet

Depending on your organization needs, you may want to enable secure outbound Internet access for your end users. In cases where the list of allowed destinations is well-defined (for example, Microsoft 365 access) you can use Azure Firewall application and network rules to configure the required access. This routes end-user traffic directly to the Internet for best performance.

If you want to filter outbound user Internet traffic using an existing on-premises secure web gateway, you can configure web browsers or other applications running on the Azure Virtual Desktop host pool with an explicit proxy configuration. These proxy settings only influence your end-user Internet access, allowing the Azure Virtual Desktop platform outbound traffic directly via Azure Firewall.

Configure AVD session hosts using Azure Bastion

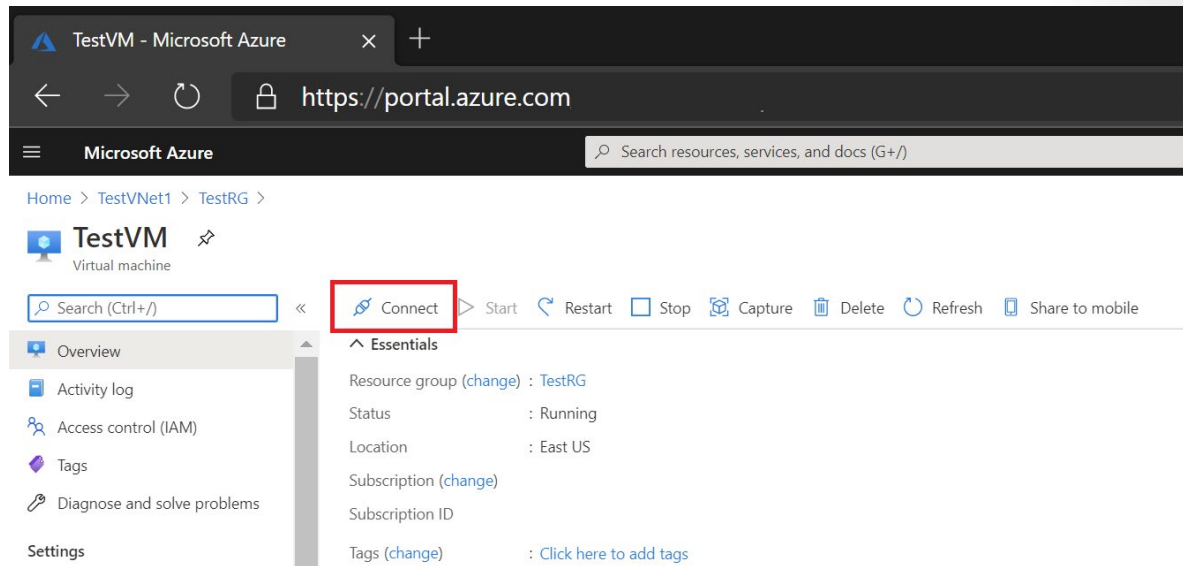
Azure Bastion provides secure connectivity to all VMs in a virtual network in which it is provisioned. Using Azure Bastion protects your virtual machines from exposing RDP/SSH ports to the outside world, while still providing secure access using RDP/SSH.

Before you begin, verify that you have met the following criteria:

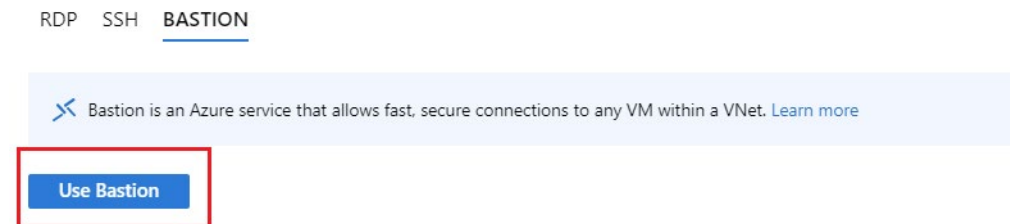
- A VNet with the Bastion host already installed. Make sure that you have set up an Azure Bastion host for the virtual network in which the VM is located. Once the Bastion service is provisioned and deployed in your virtual network, you can use it to connect to any VM in the virtual network.
- A Windows virtual machine in the virtual network.
- The following required roles:
 - Reader role on the virtual machine.
 - Reader role on the NIC with private IP of the virtual machine.
 - Reader role on the Azure Bastion resource.
- To connect to the Windows VM, you must have the following ports open on your Windows VM:
 - Inbound ports: RDP (3389)

Connect

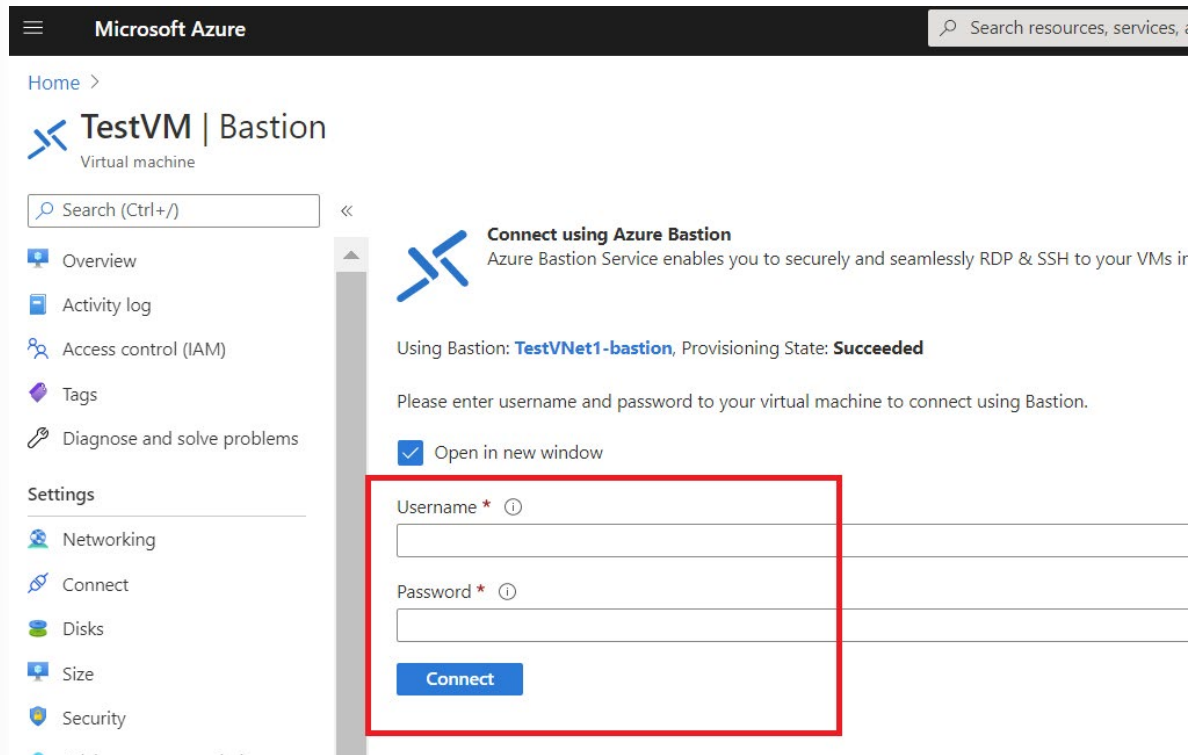
1. Open the Azure portal. Navigate to the virtual machine that you want to connect to, then select **Connect**. Select **Bastion** from the dropdown.



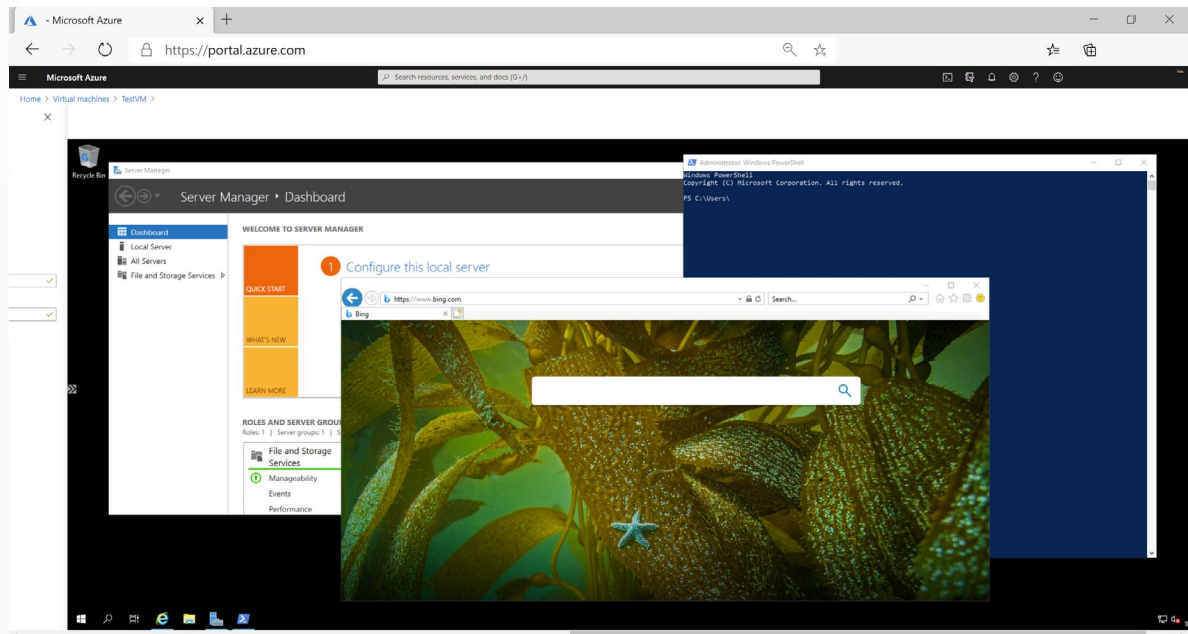
2. After you select Bastion from the dropdown, a side bar appears that has three tabs: RDP, SSH, and Bastion. Because Bastion was provisioned for the virtual network, the Bastion tab is active by default. Select **Use Bastion**.



3. On the **Connect using Azure Bastion** page, enter the username and password for your virtual machine, then select **Connect**.



4. The RDP connection to this virtual machine via Bastion will open directly in the Azure portal (over HTML5) using port 443 and the Bastion service.



Azure Network Watcher

Azure Network Watcher¹ provides tools to monitor, diagnose, view metrics, and enable or disable logs for resources in an Azure virtual network.

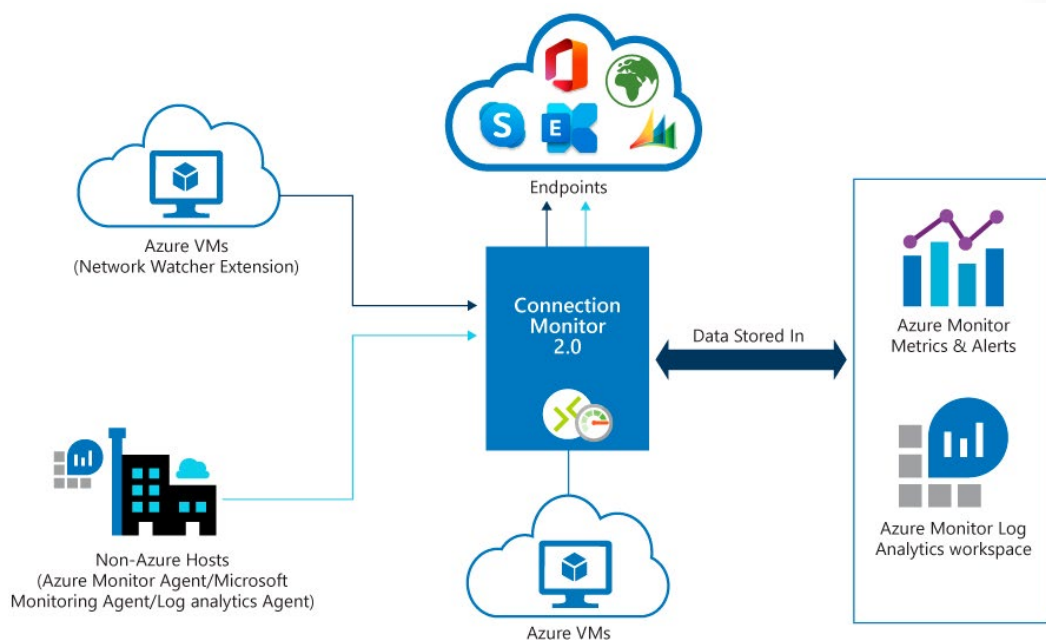
Network Watcher is designed to monitor and repair the network health of IaaS (Infrastructure-as-a-Service) products which includes Virtual Machines, Virtual Networks, Application Gateways, and Load balancers.

Monitor communication between a virtual machine and an endpoint

Endpoints can be another virtual machine (VM), a fully qualified domain name (FQDN), a uniform resource identifier (URI), or IPv4 address.

Connection Monitor 2.0 monitors communication at a regular interval and informs you of reachability, latency, and network topology changes between the VM and the endpoint.

If an endpoint becomes unreachable, connection troubleshoot informs you of the reason. Potential reasons are a DNS name resolution problem, the CPU, memory, or firewall within the operating system of a VM, or the hop type of a custom route, or security rule for the VM or subnet of the outbound connection.

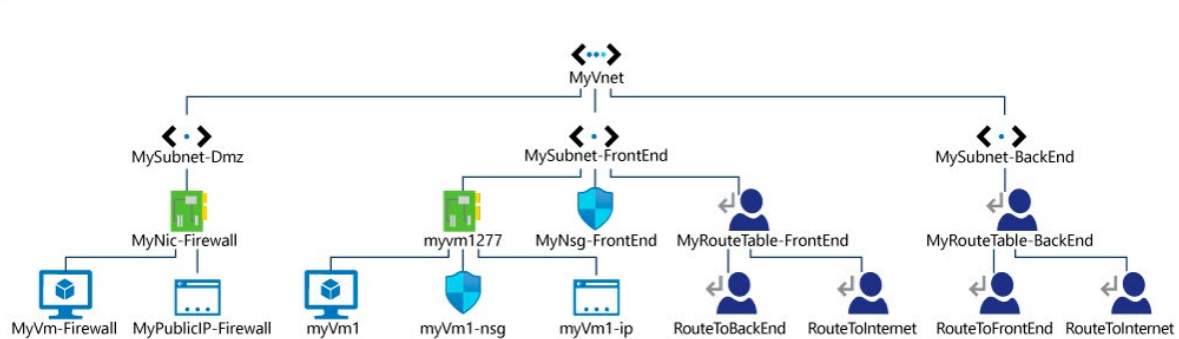


Connection monitor also provides the minimum, average, and maximum latency observed over time. After learning the latency for a connection, you may find that you're able to decrease the latency by moving your Azure resources to different Azure regions.

¹ <https://docs.microsoft.com/en-us/azure/network-watcher/>

View resources in a virtual network and their relationships

As resources are added to a virtual network, it can become difficult to understand what resources are in a virtual network and how they relate to each other. The *topology* capability enables you to generate a visual diagram of the resources in a virtual network, and the relationships between the resources. The following picture shows an example topology diagram for a virtual network that has three subnets, two VMs, network interfaces, public IP addresses, network security groups, route tables, and the relationships between the resources:



Diagnose network traffic filtering problems to or from a VM

When you deploy a VM, Azure applies several default security rules to the VM that allow or deny traffic to or from the VM. You might override Azure's default rules, or create additional rules. At some point, a VM may become unable to communicate with other resources, because of a security rule. The *IP flow verify* capability enables you to specify a source and destination IPv4 address, port, protocol (TCP or UDP), and traffic direction (inbound or outbound). IP flow verify then tests the communication and informs you if the connection succeeds or fails. If the connection fails, IP flow verify tells you which security rule allowed or denied the communication, so that you can resolve the problem.

Network Watcher - IP flow verify
Microsoft

Search (Ctrl+/)

Overview

MONITORING

Topology

NETWORK DIAGNOSTIC TOOLS

IP flow verify

Next hop

Security group view

Packet capture

METRICS

Network subscription limit

LOGS

NSG flow logs

Dagnostic logs

Specify a target virtual machine with associated network security groups, then run an inbound or outbound packet to see if access is allowed or denied.

Subscription*

Microsoft Azure

Resource group*

FabrikamRG

Virtual machine*

fabrikamvm1

Network interface*

fabrikamvm1161

Packet details

Protocol

☒ TCP ☐ UDP

Direction

☒ Inbound ☐ Outbound

Local IP address*

10.1.0.4 ✓

Local port*

443 ✓

Remote IP address*

204.79.197.200 ✓

Remote port*

443 ✓

Check

Result

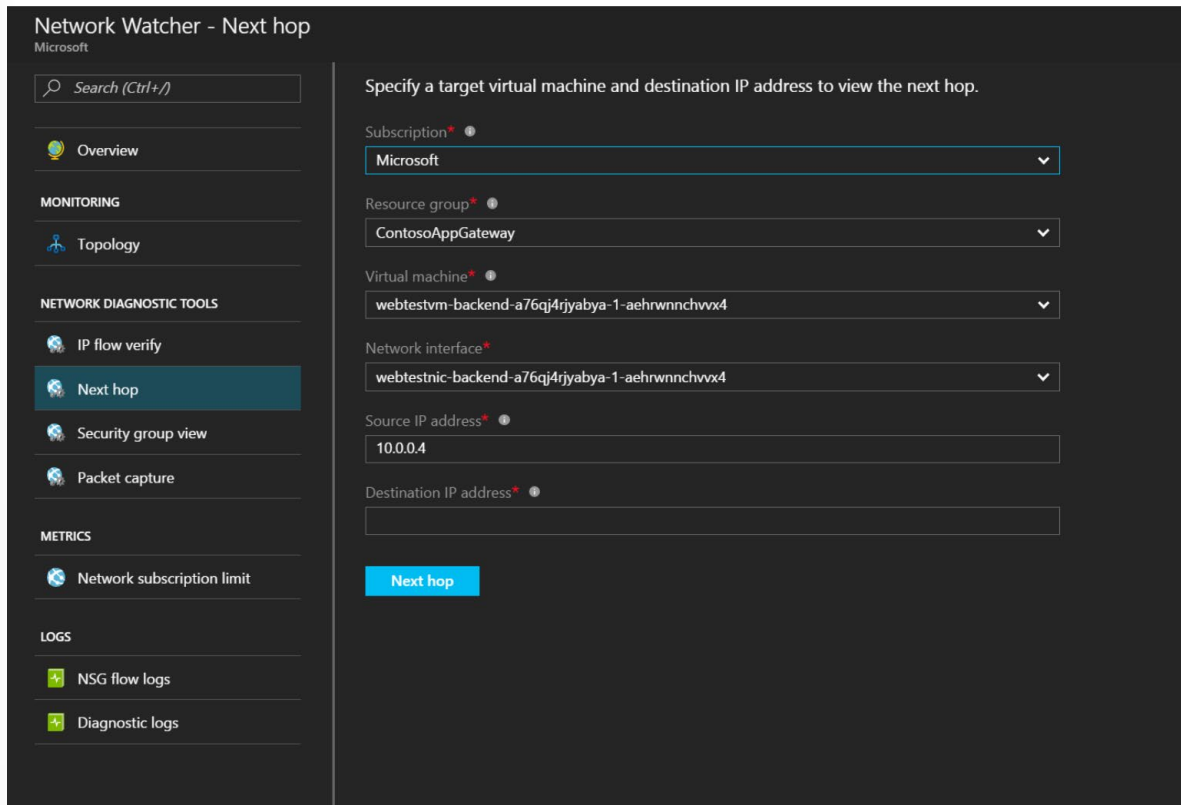
✓ Access allowed

Security rule

AllowInternetOutBound

Diagnose network routing problems from a VM

When you create a virtual network, Azure creates several default outbound routes for network traffic. The outbound traffic from all resources, such as VMs, deployed in a virtual network, are routed based on Azure's default routes. You might override Azure's default routes, or create additional routes. You may find that a VM can no longer communicate with other resources because of a specific route. The *next hop* capability enables you to specify a source and destination IPv4 address. Next hop then tests the communication and informs you what type of next hop is used to route the traffic. You can then remove, change, or add a route, to resolve a routing problem.



Diagnose outbound connections from a VM

The *connection troubleshoot* capability enables you to test a connection between a VM and another VM, an FQDN, a URI, or an IPv4 address. The test returns similar information returned when using the connection monitor capability, but tests the connection at a point in time, rather than monitoring it over time, as connection monitor does.

Capture packets to and from a VM

Advanced filtering options and fine-tuned controls, such as the ability to set time and size limitations, provide versatility. The capture can be stored in Azure Storage, on the VM's disk, or both. You can then analyze the capture file using several standard network capture analysis tools.

Diagnose problems with an Azure Virtual network gateway and connections

Virtual network gateways provide connectivity between on-premises resources and Azure virtual networks. Monitoring gateways and their connections are critical to ensuring communication is not broken. The *VPN diagnostics* capability provides the ability to diagnose gateways and connections. VPN diagnostics diagnoses the health of the gateway, or gateway connection, and informs you whether a gateway and gateway connections, are available. If the gateway or connection is not available, VPN diagnostics tells you why, so you can resolve the problem.

Network Watcher - VPN Diagnostics

Microsoft

Search (Ctrl+/) << ▶ Start troubleshooting

Network Watcher VPN Troubleshoot diagnoses the health of the virtual network gateway or connection. This request is a long running transaction, and the results are returned once the diagnosis is complete. You can select multiple gateways or connections to troubleshoot simultaneously. [Learn more.](#)

Choose a subscription ⓘ Resource group ⓘ Location ⓘ

* Storage account <https://mysavpndiag.blob.cor...>

	NAME ⓘ	TROUBLESHOOTING ST... ⓘ	RESOURCE STATUS ⓘ	RESOURCE GROUP ⓘ	LOCATION ⓘ
☒	▼ VNet1GW	Not started	Succeeded	TestRG1	East US
☐	VNet1toSite1	-	Succeeded	TestRG1	East US

Details

Status Action

Resource VNet1GW

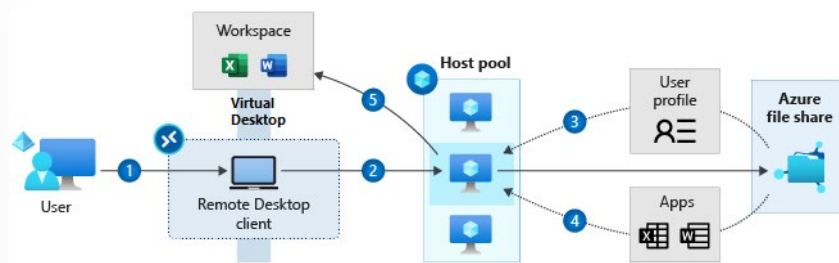
Implement and manage storage for AVD

Storage for FSLogix components

Azure Virtual Desktop service recommends FSLogix profile containers as a user profile solution. FSLogix is designed to roam profiles in remote computing environments, such as Azure Virtual Desktop. It stores a complete user profile in a single container. At sign in, this container is dynamically attached to the computing environment using natively supported Virtual Hard Disk (VHD) and Hyper-V Virtual Hard disk (VHDX).

The VHD or VHDX files are stored to this location and attached to users the next time they sign in.

The following diagram shows the process of getting the user profile after a user signs into the Remote Desktop client.

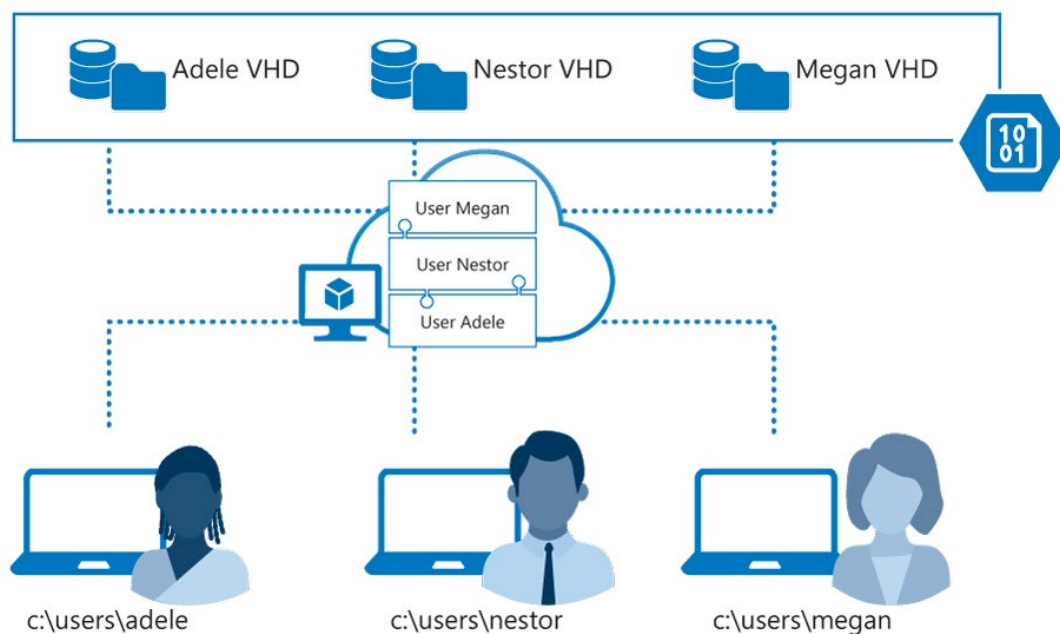


1. User signs into the Remote Desktop client
2. User gets assigned to a session host virtual machine (VM)
3. VM gets the user profile from the Azure file share
4. (Preview) If you have MSIX app attach configured, apps are dynamically delivered to the session host VM. MSIX app attach uses FSLogix storage concepts, but for applications
5. User gets their Azure Virtual Desktop workspace populated with their assigned app(s) or session desktop

The user profile is immediately available and appears in the system exactly like a native user profile.

User profiles

A user profile contains data elements about an individual, including configuration information like desktop settings, persistent network connections, and application settings. By default, Windows creates a local user profile that is tightly integrated with the operating system.



A remote user profile provides a partition between user data and the operating system. It allows the operating system to be replaced or changed without affecting the user data. In Remote Desktop Session Host (RDSH) and Virtual Desktop Infrastructures (VDI), the operating system may be replaced for the following reasons:

- An upgrade of the operating system
- A replacement of an existing Virtual Machine (VM)
- A user being part of a pooled (non-persistent) RDSH or VDI environment

Microsoft products operate with several technologies for remote user profiles, including these technologies:

- Roaming user profiles (RUP)
- User profile disks (UPD)
- Enterprise state roaming (ESR)

UPD and RUP are the most widely used technologies for user profiles in Remote Desktop Session Host (RDSH) and Virtual Hard Disk (VHD) environments.

FSLogix profile containers

FSLogix addresses many profile container challenges. Key among them are:

- **Performance:** The FSLogix profile containers are high performance and resolve performance issues that have historically blocked cached exchange mode.
- **OneDrive:** Without FSLogix profile containers, OneDrive for Business is not supported in non-persistent RDSH or VDI environments.
- **Additional folders:** FSLogix provides the ability to extend user profiles to include additional folders.

Microsoft has started replacing existing user profile solutions, like UPD, with FSLogix profile containers.

Azure Files integration with Azure Active Directory Domain Service

Microsoft Azure Files recently announced the general availability of **Azure Files authentication with Azure Active Directory Domain Service (AD DS)**².

Configure storage for FSLogix components

The Azure Virtual Desktop service offers FSLogix profile containers as the recommended user profile solution. We don't recommend using the User Profile Disk (UPD) solution, which will be deprecated in future versions of Azure Virtual Desktop.

This topic explains how to set up a FSLogix profile container share for a host pool using a virtual machine-based file share.

Note: See Module 4 for more in-depth information on FSLogix installation, configuration, and management.

Create a new virtual machine that will act as a file share

When creating the virtual machine, be sure to place it on either the same virtual network as the host pool virtual machines or on a virtual network that has connectivity to the host pool virtual machines.

After creating the virtual machine, join it to the domain by doing the following things:

1. Connect to the virtual machine with the credentials you provided when creating the virtual machine.
2. On the virtual machine, launch **Control Panel** and select **System**.
3. Select **Computer name**, select **Change settings**, and then select **Change...**
4. Select **Domain** and then enter the Active Directory domain on the virtual network.
5. Authenticate with a domain account that has privileges to domain-join machines.

Prepare the virtual machine to act as a file share for user profiles

The following are general instructions about how to prepare a virtual machine to act as a file share for user profiles:

1. Add the Azure Virtual Desktop Active Directory users to an Active Directory security group. This security group will be used to authenticate the Azure Virtual Desktop users to the file share virtual machine you just created.
2. Connect to the file share virtual machine.
3. On the file share virtual machine, create a folder on the **C drive** that will be used as the profile share.
4. Right-click the new folder, select **Properties**, select **Sharing**, then select **Advanced sharing....**
5. Select **Share this folder**, select **Permissions...**, then select **Add...**
6. Search for the security group to which you added the Azure Virtual Desktop users, then make sure that group has **Full Control**.

² <https://docs.microsoft.com/en-us/azure/storage/files/storage-files-active-directory-overview>

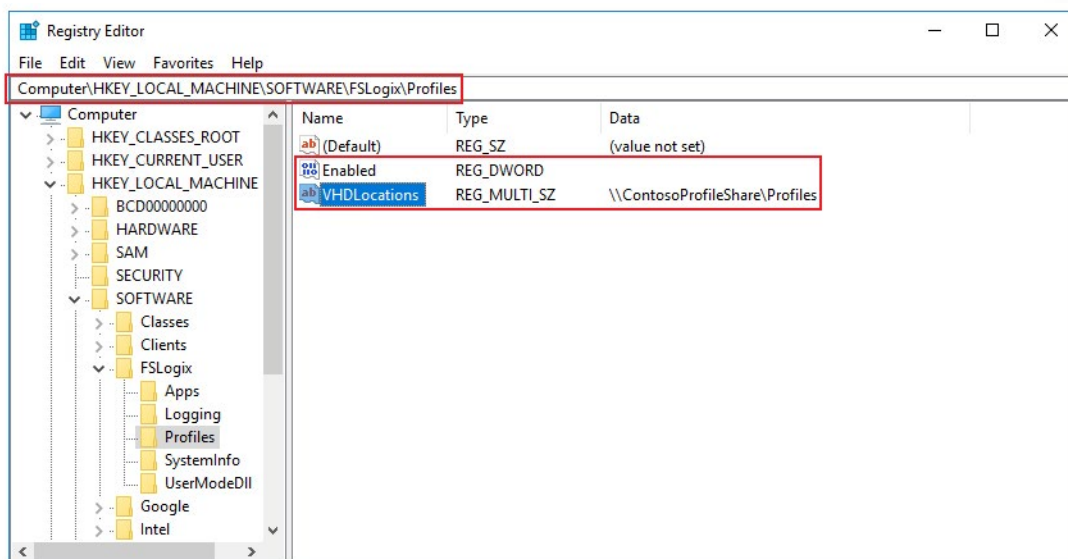
- After adding the security group, right-click the folder, select **Properties**, select **Sharing**, then copy down the **Network Path** to use for later.

Configure the FSLogix profile container

To configure the virtual machines with the FSLogix software, do the following on each machine registered to the host pool:

- Connect to the virtual machine with the credentials you provided when creating the virtual machine.
- Launch an internet browser and navigate to [this link³](#) to download the FSLogix agent.
- Navigate to either `\Win32\Release` or `\X64\Release` in the .zip file and run **FSLogixAppsSetup** to install the FSLogix agent.
- Navigate to **Program Files > FSLogix > Apps** to confirm the agent installed.
- From the start menu, run **RegEdit** as an administrator. Navigate to **Computer\HKEY_LOCAL_MACHINE\software\FSLogix**.
- Create a key named **Profiles**.
- Create the following values for the Profiles key:

Name	Type	Data/Value
Enabled	DWORD	1
VHDLocations	Multi-String Value	"Network path for file share"



Configure storage accounts

Azure supports multiple types of storage accounts for different storage scenarios that AVD customers may have, but there are two main types of storage accounts for Azure Files. Which storage account type you need to create depends on whether you want to create a standard file share or a premium file share:

- General purpose version 2 (GPv2) storage accounts:** GPv2 storage accounts allow you to deploy Azure file shares on standard/hard disk-based (HDD-based) hardware. In addition to storing Azure file

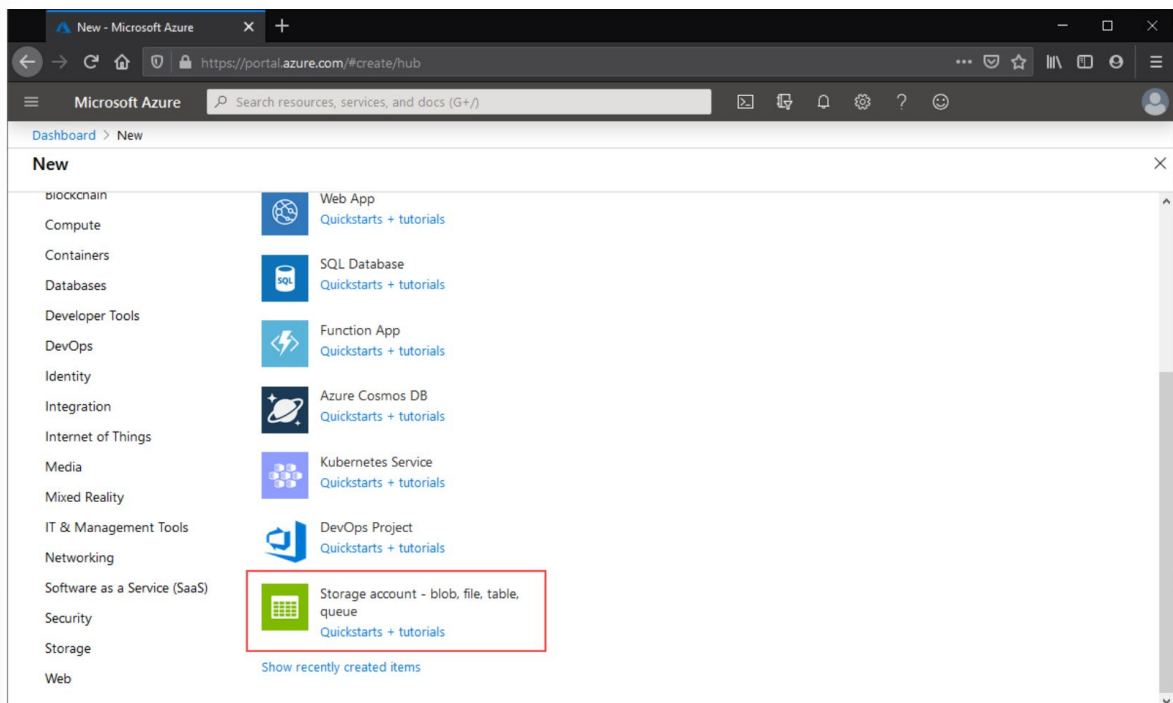
³ <https://go.microsoft.com/fwlink/?linkid=2084562>

shares, GPv2 storage accounts can store other storage resources such as blob containers, queues, or tables. File shares can be deployed into the transaction optimized (default), hot, or cool tiers.

- **FileStorage storage accounts:** FileStorage storage accounts allow you to deploy Azure file shares on premium/solid-state disk-based (SSD-based) hardware. FileStorage accounts can only be used to store Azure file shares; no other storage resources (blob containers, queues, tables, etc.) can be deployed in a FileStorage account.

NOTE: There is an online lab later in this module where you configure Azure Files to store profile containers for Azure Virtual Desktop in Azure AD DS environment (*Lab - Implement and manage storage for AVD (Azure AD DS)*). This topic is meant to provide an overview of the procedure.

To create a storage account via the Azure portal, select **+ Create a resource** from the dashboard. In the resulting Azure Marketplace search window, search for **storage account** and select the resulting search result. This will lead to an overview page for storage accounts; select **Create** to proceed with the storage account creation wizard.



The Basics section

The first section to complete to create a storage account is labeled **Basics**. This contains all the required fields to create a storage account. To create a GPv2 storage account, ensure the **Performance** radio button is set to Standard and the **Account kind** drop-down list is selected to *StorageV2 (general purpose v2)*.

Performance ⓘ

☒ Standard
 ☐ Premium

Account kind ⓘ

StorageV2 (general purpose v2) ▼

To create a FileStorage storage account, ensure the **Performance** radio button is set to Premium and the **Account kind** drop-down list is selected to *FileStorage*.

Performance ⓘ ☐ Standard ☒ Premium

Account kind ⓘ FileStorage ▼

ⓘ This account kind allows you to create Azure file shares with premium performance characteristics that can be tailored to your needs. [Learn more about file storage](#)

The other basics fields are independent from the choice of storage account:

- **Subscription:** The subscription for the storage account to be deployed into.
- **Resource group:** The resource group for the storage account to be deployed into. You may either create a new resource group or use an existing resource group. A resource group is a logical container for grouping your Azure services. When you create a storage account, you have the option to either create a new resource group, or use an existing resource group.
- **Storage account name:** The name of the storage account resource to be created. This name must be globally unique, but otherwise can any name you desire. The storage account name will be used as the server name when you mount an Azure file share via SMB.
- **Location:** The region for the storage account to be deployed into. This can be the region associated with the resource group, or any other available region.
- **Replication:** Although this is labeled replication, this field actually means**redundancy; ** this is the desired redundancy level: locally redundancy (LRS), zone redundancy (ZRS), geo-redundancy (GRS), and geo-zone-redundancy. This drop-down list also contains read-access geo-redundancy (RA-GRS) and read-access geo-zone redundancy (RA-GZRS), which do not apply to Azure file shares; any file share created in a storage account with these selected will be either geo-redundant or geo-zone-redundant, respectively. Depending on your region or selected storage account type, some redundancy options may not be allowed.
- **Blob access tier:** This field does not apply to Azure Files, so you can choose either one of the radio buttons.

Configure disks

This topic shows you how to add and configure a new managed data disk to a Windows virtual machine (VM) by using the Azure portal. The size of the VM determines how many data disks you can attach.

Add a data disk

1. Go to the Azure portal to add a data disk. Search for and select **Virtual machines**.
2. Select a virtual machine from the list.
3. On the **Virtual machine** page, select **Disks**.
4. On the **Disks** page, select **Add data disk**.
5. In the drop-down for the new disk, select **Create disk**.
6. In the **Create managed disk** page, type in a name for the disk and adjust the other settings as necessary. When you're done, select **Create**.

Microsoft Azure

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[Home](#) > [Virtual machines](#) > [Create a virtual machine](#) >

Create a new disk ...

Create a new disk to store applications and data on your VM. Disk pricing varies based on factors including disk size, storage type, and number of transactions. [Learn more](#)

Name *

vm1_DataDisk_0

Source type * ⓘ

None (empty disk) ▼

Size * ⓘ

1024 GiB

Premium SSD LRS

[Change size](#)

Encryption type *

(Default) Encryption at-rest with a platform-managed key ▼

Enable shared disk

☐ Yes
 ☒ No

- In the **Disks** page, select **Save** to save the new disk configuration for the VM.
- After Azure creates the disk and attaches it to the virtual machine, the new disk is listed in the virtual machine's disk settings under **Data disks**.

Initialize a new data disk

- Connect to the VM.
- Select the Windows **Start** menu inside the running VM and enter **diskmgmt.msc** in the search box. The **Disk Management** console opens.
- Disk Management recognizes that you have a new, uninitialized disk and the **Initialize Disk** window appears.
- Verify the new disk is selected and then select **OK** to initialize it.
- The new disk appears as **unallocated**. Right-click anywhere on the disk and select **New simple volume**. The **New Simple Volume Wizard** window opens.
- Proceed through the wizard, keeping all of the defaults, and when you're done select **Finish**.
- Close **Disk Management**.
- A pop-up window appears notifying you that you need to format the new disk before you can use it. Select **Format disk**.
- In the **Format new disk** window, check the settings, and then select **Start**.
- A warning appears notifying you that formatting the disks erases all of the data. Select **OK**.
- When the formatting is complete, select **OK**.

Create file shares

Once you've created a storage account, all that is left is to create your file share. This process is mostly the same regardless of whether you're using a premium file share or a standard file share. You should consider the following differences.

Standard file shares may be deployed into one of the standard tiers: transaction optimized (default), hot, or cool. This is a per file share tier that is not affected by the **blob access tier** of the storage account.

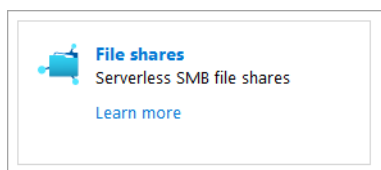
You can change the tier of the share at any time after it has been deployed. Premium file shares cannot be directly converted to standard file shares in any standard tier.

You can move file shares between tiers within GPv2 storage account types (transaction optimized, hot, and cool).

The **quota** property means something slightly different between premium and standard file shares:

- For standard file shares, it's an upper boundary of the Azure file share, beyond which end-users cannot go. The primary purpose for quota for a standard file share is budgetary: "I don't want this file share to grow beyond this point."
 - If a quota is not specified, standard file share can span up to 100 TiB.
- For premium file shares, quota is overloaded to mean **provisioned size**. The provisioned size is the amount that you will be billed for, regardless of actual usage. When you provision a premium file share, you want to consider two factors:
 - The future growth of the share from a space utilization perspective
 - The IOPS required for your workload. Every provisioned GiB entitles you to additional reserved and burst IOPS.

If you just created your storage account, you can navigate to it from the deployment screen by selecting **Go to resource**. If you have previously created the storage account, you can navigate to it via the resource group containing it. Once in the storage account, select the tile labeled **File shares**.



In the file share listing, you should see any file shares you have previously created in this storage account; an empty table if no file shares have been created yet. Select **+ File share** to create a new file share.

The new file share blade should appear on the screen. Complete the fields in the new file share blade to create a file share:

- **Name:** the name of the file share to be created.
- **Quota:** the quota of the file share for standard file shares; the provisioned size of the file share for premium file shares.
- **Tiers:** the selected tier for a file share. This field is only available in a **general purpose (GPv2) storage account**. You can choose transaction optimized, hot, or cool. The share's tier can be changed at any time.

Select **Create** to finishing creating the new share. Note that if your storage account is in a virtual network, you will not be able to successfully create an Azure file share unless your client is also in the virtual

network. You can also work around this point-in-time limitation by using the Azure PowerShell `New-Az-RmStorageShare` cmdlet.

Create and configure host pools and session hosts

Configure host pool assignment type

You can configure the assignment type of your personal desktop host pool to adjust your Windows Virtual Desktop environment to better suit your needs. In this topic, we'll show you how to configure automatic or direct assignment for your users.

The instructions below apply to personal desktop host pools, not pooled host pools, since users in pooled host pools aren't assigned to specific session hosts.

This applies to Windows Virtual Desktop with Azure Resource Manager Windows Virtual Desktop objects.

Configure automatic assignment

Automatic assignment is the default assignment type for new personal desktop host pools created in your Windows Virtual Desktop environment. Automatically assigning users doesn't require a specific session host.

To automatically assign users, first assign them to the personal desktop host pool so that they can see the desktop in their feed. When an assigned user launches the desktop in their feed, they will claim an available session host if they have not already connected to the host pool, which completes the assignment process.

To configure a host pool to automatically assign users to VMs, run the following PowerShell cmdlet:

```
Update-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <host-poolname> -PersonalDesktopAssignmentType Automatic``
```

To assign a user to the personal desktop host pool, run the following PowerShell cmdlet:

```
New-AzRoleAssignment -SignInName <userupn> -RoleDefinitionName "Desktop Virtualization User" -ResourceName <appgroupname> -ResourceGroupName <resourcegroupname> -ResourceType 'Microsoft.DesktopVirtualization/applicationGroups'
```

Configure direct assignment

Unlike automatic assignment, when you use direct assignment, you must assign the user to both the personal desktop host pool and a specific session host before they can connect to their personal desktop. If the user is only assigned to a host pool without a session host assignment, they won't be able to access resources.

To configure a host pool to require direct assignment of users to session hosts, run the following PowerShell cmdlet:

```
Update-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <host-poolname> -PersonalDesktopAssignmentType Direct
```

To assign a user to the personal desktop host pool, run the following PowerShell cmdlet:

```
New-AzRoleAssignment -SignInName <userupn> -RoleDefinitionName "Desktop
Virtualization User" -ResourceName <appgroupname> -ResourceGroupName <re-
sourcegroupname> -ResourceType 'Microsoft.DesktopVirtualization/applica-
tionGroups'
```

To assign a user to a specific session host, run the following PowerShell cmdlet:

powershell

```
Update-AzWvdSessionHost -HostPoolName <hostpoolname> -Name <sessionhost-
name> -ResourceGroupName <resourcegroupname> -AssignedUser <userupn>
```

To directly assign a user to a session host in the Azure portal:

1. Sign in to the Azure portal at <https://portal.azure.com>.
2. Enter **Windows Virtual Desktop** into the search bar.
3. Under **Services**, select **Windows Virtual Desktop**.
4. At the Windows Virtual Desktop page, go the menu on the left side of the window and select **Host pools**.
5. Select the name of the host pool you want to update.
6. Next, go to the menu on the left side of the window and select **Application groups**.
7. Select the name of the desktop app group you want to edit, then select **Assignments** in the menu on the left side of the window.
8. Select **+ Add**, then select the users or user groups you want to publish this desktop app group to.
9. Select **Assign VM** in the Information bar to assign a session host to a user.
10. Select the session host you want to assign to the user, then select **Assign**.
11. Select the user you want to assign the session host to from the list of available users.
12. When you're done, select **Select**.

Automate creation of an AVD host pool using PowerShell

This topic shows you how to use your PowerShell client to create a host pool.

NOTE: There is an online lab later in this module where you create a host pool using PowerShell (*Lab - Deploy and manage host pools and hosts by using PowerShell*). This topic is meant to provide an overview of the procedure.

Run the following cmdlet to sign in to the Windows Virtual Desktop environment:

```
New-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <hostpool-
name> -WorkspaceName <workspace> -HostPoolType <Pooled|Personal> -Load-
BalancerType <BreadthFirst|DepthFirst|Persistent> -Location <region> -Desk-
topAppGroupName <appgroupname>
```

This cmdlet will create the host pool, workspace and desktop app group. Additionally, it will register the desktop app group to the workspace. You can either create a workspace with this cmdlet or use an existing workspace.

Run the next cmdlet to create a registration token to authorize a session host to join the host pool and save it to a new file on your local computer. You can specify how long the registration token is valid by using the `-ExpirationHours` parameter.

The token's expiration date can be no less than an hour and no more than one month. If you set `-ExpirationTime` outside of that limit, the cmdlet won't create the token.

```
New-AzWvdRegistrationInfo -ResourceGroupName <resourcegroupname> -HostPool-
Name <hostpoolname> -ExpirationTime $((get-date).ToUniversalTime().Ad-
dDays(1).ToString('yyyy-MM-ddTHH:mm:ss.fffffffZ'))
```

For example, if you want to create a token that expires in two hours, run this cmdlet:

```
New-AzWvdRegistrationInfo -ResourceGroupName <resourcegroupname> -HostPool-
Name <hostpoolname> -ExpirationTime $((get-date).ToUniversalTime().Ad-
dHours(2).ToString('yyyy-MM-ddTHH:mm:ss.fffffffZ'))
```

After that, run this cmdlet to add Azure Active Directory users to the default desktop app group for the host pool.

```
New-AzRoleAssignment -SignInName <userupn> -RoleDefinitionName "Desktop
Virtualization User" -ResourceName <hostpoolname>"-DAG"> -ResourceGroupName
<resourcegroupname> -ResourceType 'Microsoft.DesktopVirtualization/applica-
tionGroups'
```

Run this next cmdlet to add Azure Active Directory user groups to the default desktop app group for the host pool:

```
New-AzRoleAssignment -ObjectId <usergroupobjectid> -RoleDefinitionName
"Desktop Virtualization User" -ResourceName <hostpoolname>"-DAG"> -Re-
sourceGroupName <resourcegroupname> -ResourceType 'Microsoft.DesktopVirtu-
alization/applicationGroups'
```

Run the following cmdlet to export the registration token to a variable, which will be used later to register the virtual machines to the Windows Virtual Desktop host pool.

```
$token = Get-AzWvdRegistrationInfo -ResourceGroupName <resourcegroupname>
-HostPoolName <hostpoolname>
```

Create virtual machines for the host pool

Now you can create an Azure virtual machine that can be joined to your Windows Virtual Desktop host pool.

You can create a virtual machine in multiple ways:

- Create a virtual machine from an Azure Gallery image
- Create a virtual machine from a managed image
- Create a virtual machine from an unmanaged image

Prepare the virtual machines for Windows Virtual Desktop agent installations

You need to do the following things to prepare your virtual machines before you can install the Windows Virtual Desktop agents and register the virtual machines to your Windows Virtual Desktop host pool:

- You must domain-join the machine. This allows incoming Windows Virtual Desktop users to be mapped from their Azure Active Directory account to their Active Directory account and be successfully allowed access to the virtual machine.
- You must install the Remote Desktop Session Host (RDSH) role if the virtual machine is running a Windows Server OS. The RDSH role allows the Windows Virtual Desktop agents to install properly.

To successfully domain-join, do the following things on each virtual machine:

1. Connect to the virtual machine with the credentials you provided when creating the virtual machine.
2. On the virtual machine, launch **Control Panel** and select **System**.
3. Select **Computer name**, select **Change settings**, and then select **Change...**
4. Select **Domain** and then enter the Active Directory domain on the virtual network.
5. Authenticate with a domain account that has privileges to domain-join machines.

Register the virtual machines to the Windows Virtual Desktop host pool

To register the Windows Virtual Desktop agents, do the following on each virtual machine:

1. Connect to the virtual machine with the credentials you provided when creating the virtual machine.
2. Download and install the Windows Virtual Desktop Agent.
 - Download the **Windows Virtual Desktop Agent**⁴.
 - Run the installer. When the installer asks you for the registration token, enter the value you got from the **Get-AzWvdRegistrationInfo** cmdlet.
3. Download and install the Windows Virtual Desktop Agent Bootloader.
 - Download the **Windows Virtual Desktop Agent Bootloader**⁵.
 - Run the installer.

Customize RDP properties for a host pool

Customizing a host pool's Remote Desktop Protocol (RDP) properties, such as multi-monitor experience and audio redirection, lets you deliver an optimal experience for your users based on their needs. You can customize RDP properties in Azure Virtual Desktop by either using the Azure portal or by using the **-CustomRdpProperty** parameter in the **Update-AzWvdHostPool** cmdlet.

Default RDP file properties

RDP files have the following properties by default:

⁴ <https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RWrmXv>

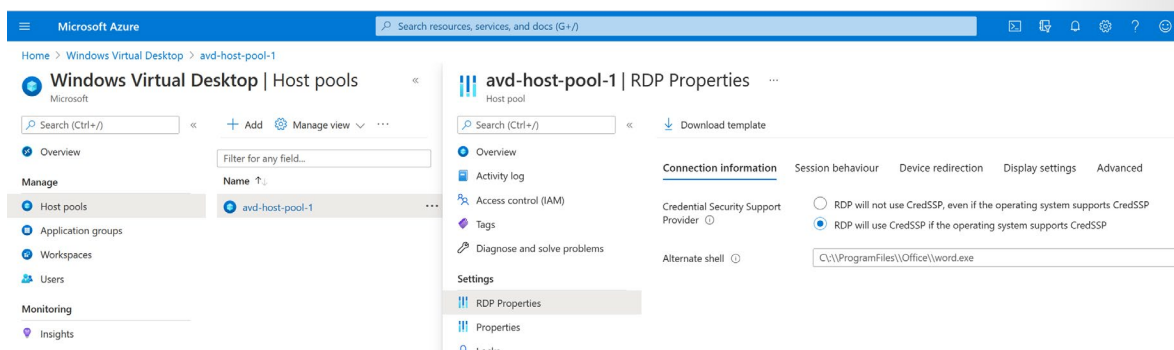
⁵ <https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RWrxrH>

RDP property	On Desktop	As a RemoteApp
Multi-monitor mode	Disabled	Enabled
Drive redirections enabled	Drives, clipboard, printers, COM ports, and smart cards	Drives, clipboard, and printers
Remote audio mode	Play locally	Play locally

Configure RDP properties in the Azure portal

To configure RDP properties in the Azure portal:

1. Sign in to Azure at <https://portal.azure.com>.
2. Enter **azure virtual desktop** into the search bar.
3. Under Services, select **Azure Virtual Desktop**.
4. At the Azure Virtual Desktop page, select **host pools** in the menu on the left side of the screen.
5. Select **the name of the host pool** you want to update.
6. Select **RDP Properties** in the menu on the left side of the screen.



7. Set the property you want.
8. When you're done, select **Save** to save your changes.

The next sections will tell you how to edit custom RDP properties manually in PowerShell.

Add or edit a single custom RDP property

To add or edit a single custom RDP property, run the following PowerShell cmdlet:

PowerShell

```
Update-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <hostpoolname> -CustomRdpProperty <property>
```

To check if the cmdlet you just ran updated the property, run this cmdlet:

PowerShell

```
Get-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <hostpoolname> | format-list Name, CustomRdpProperty
```

```
Name : <hostpoolname>
```

```
CustomRdpProperty : <customRDPpropertystring>
```

For example, if you were checking for the "audiocapturemode" property on a host pool named 0301HP, you'd enter this cmdlet:

PowerShell

```
Get-AzWvdHostPool -ResourceGroupName 0301rg -Name 0301hp | format-list
Name, CustomRdpProperty
```

```
Name                : 0301HP
CustomRdpProperty   : audiocapturemode:i:1;
```

Add or edit multiple custom RDP properties

To add or edit multiple custom RDP properties, run the following PowerShell cmdlets by providing the custom RDP properties as a semicolon-separated string:

PowerShell

```
$properties="<property1>;<property2>;<property3>"
Update-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <host-
poolname> -CustomRdpProperty $properties
```

You can check to make sure the RDP property was added by running the following cmdlet:

PowerShell

```
Get-AzWvdHostPool -ResourceGroupName <resourcegroupname> -Name <hostpool-
name> | format-list Name, CustomRdpProperty

Name                : <hostpoolname>
CustomRdpProperty   : <customRDPpropertystring>
```

Based on our earlier cmdlet example, if you set up multiple RDP properties on the 0301HP host pool, your cmdlet would look like this:

PowerShell

```
Get-AzWvdHostPool -ResourceGroupName 0301rg -Name 0301hp | format-list
Name, CustomRdpProperty

Name                : 0301HP
CustomRdpProperty   : audiocapturemode:i:1;audiomode:i:0;
```

Manage licensing for session hosts that run Windows client

Windows Virtual Desktop licensing allows you to apply a license to any Windows or Windows Server virtual machine that is registered as a session host in a host pool and receives user connections. This

license does not apply to virtual machines that are running as file share servers, domain controllers, and so on.

There are a few ways to use the Windows Virtual Desktop license:

- You can create a host pool and its session host virtual machines using the **Azure Marketplace offering**⁶. Virtual machines created this way automatically have the license applied.
- You can create a host pool and its session host virtual machines using the **GitHub Azure Resource Manager template**⁷. Virtual machines created this way automatically have the license applied.
- You can apply a license to an existing session host virtual machine. To do this, first follow the instructions in **Create a host pool with PowerShell**⁸ to create a host pool and associated VMs.

Apply a Windows license to a session host VM

Run the following PowerShell cmdlet to apply the Windows license:

```
$vm = Get-AzVM -ResourceGroup <resourceGroupName> -Name <vmName>
$vm.LicenseType = "Windows_Client"
Update-AzVM -ResourceGroupName <resourceGroupName> -VM $vm
```

Verify your session host VM is utilizing the licensing benefit

After deploying the VM, run this cmdlet to verify the license type:

```
Get-AzVM -ResourceGroupName <resourceGroupName> -Name <vmName>
```

A session host VM with the applied Windows license will show you something like this:

```
Type                : Microsoft.Compute/virtualMachines
Location            : westus
LicenseType         : Windows_Client
```

VMs without the applied Windows license will show you something like this:

```
Type                : Microsoft.Compute/virtualMachines
Location            : westus
LicenseType         :
```

Run the following cmdlet to see a list of all session host VMs that have the Windows license applied in your Azure subscription:

```
$vms = Get-AzVM
$vms | Where-Object {$_.LicenseType -like "Windows_Client"} | Select-Object
ResourceGroupName, Name, LicenseType
```

⁶ <https://docs.microsoft.com/en-us/azure/virtual-desktop/create-host-pools-azure-marketplace>

⁷ <https://docs.microsoft.com/en-us/azure/virtual-desktop/virtual-desktop-fall-2019/create-host-pools-arm-template>

⁸ <https://docs.microsoft.com/en-us/azure/virtual-desktop/create-host-pools-powershell>

Create and manage session host image

Create a managed VM image

This topic shows you how to prepare a master virtual hard disk (VHD) image for upload to Azure including how to create virtual machines (VMs). These instructions are for an Azure Virtual Desktop-specific configuration that can be used with your organization's existing processes.

We recommend you use an image from the Azure Image Gallery. However, if you do need to use a customized image, make sure you don't already have the Azure Virtual Desktop Agent installed on your device. Using a customized image with the Azure Virtual Desktop Agent can cause problems with the image.

Create a VM

Windows 10 Enterprise multi-session is available in the Azure Image Gallery. There are two options for customizing this image.

- The first option is to provision a virtual machine (VM) in Azure (See: **Create a VM from a managed image**⁹).
- The second option is to create the image locally by downloading the image, provisioning a Hyper-V VM, and customizing it to suit your needs.

Create a VM from a managed image

You can create multiple virtual machines (VMs) from an Azure managed VM image using the Azure portal or PowerShell. A managed VM image contains the information necessary to create a VM, including the OS and data disks. The virtual hard disks (VHDs) that make up the image, including both the OS disks and any data disks, are stored as managed disks.

Before creating a new VM, you'll need to create a managed VM image to use as the source image and grant read access on the image to any user who should have access to the image.

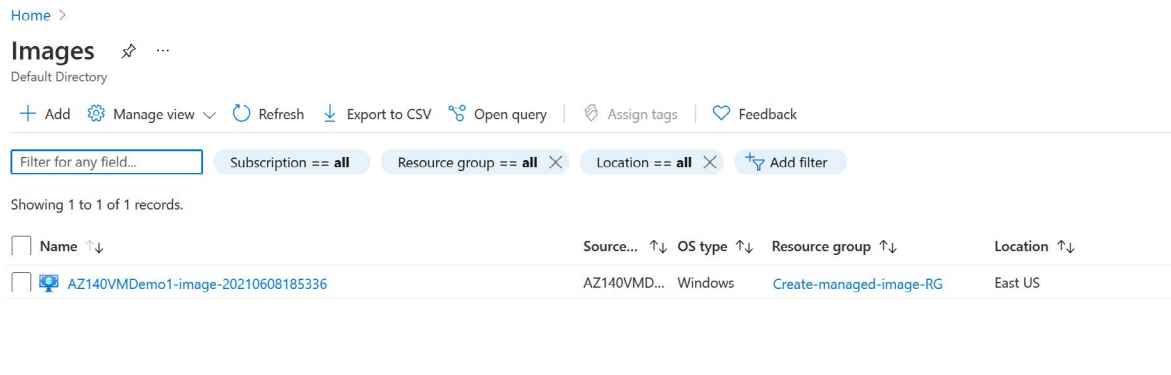
One managed image supports up to 20 simultaneous deployments. Attempting to create more than 20 VMs concurrently, from the same managed image, may result in provisioning timeouts due to the storage performance limitations of a single VHD. To create more than 20 VMs concurrently, use a Shared Image Galleries image configured with 1 replica for every 20 concurrent VM deployments.

Use the portal

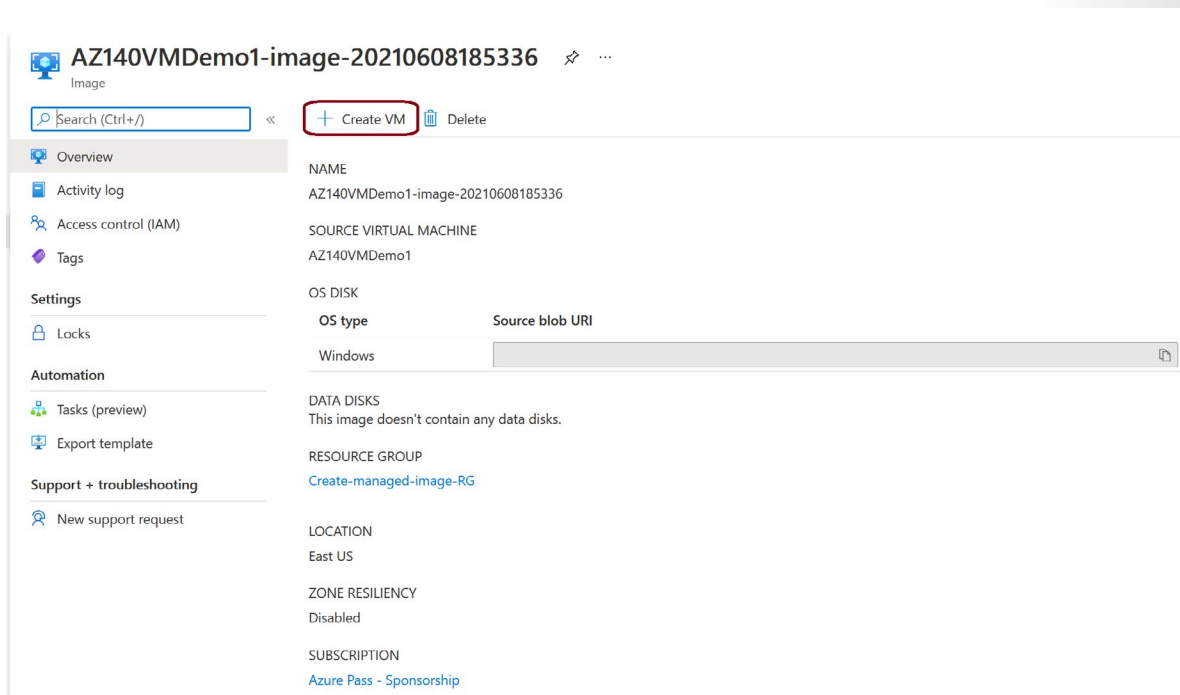
1. Go to the **Azure portal**¹⁰ to find a managed image. Search for and select **Images**.

⁹ <https://docs.microsoft.com/en-us/azure/virtual-machines/windows/create-vm-generalized-managed>

¹⁰ <https://portal.azure.com>



2. Select the image you want to use from the list. The image **Overview** page opens.



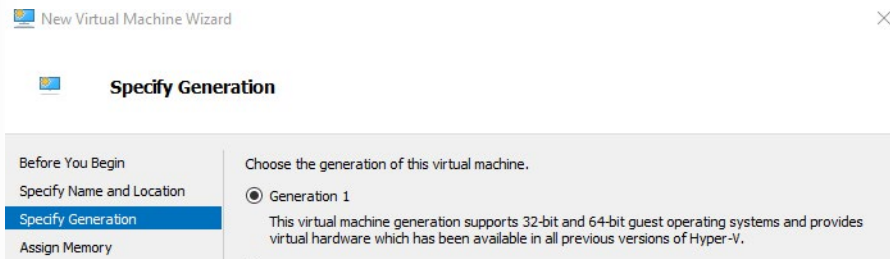
3. Select **Create VM** from the menu.
4. Enter the virtual machine information. The user name and password entered here will be used to log in to the virtual machine. When complete, select **OK**. You can create the new VM in an existing resource group, or choose **Create new** to create a new resource group to store the VM.
5. Select a size for the VM. To see more sizes, select **View all** or change the **Supported disk type** filter.
6. Under **Settings**, make changes as necessary and select **OK**.
7. On the summary page, you should see your image name listed as a **Private image**. Select **Ok** to start the virtual machine deployment.

Local image creation

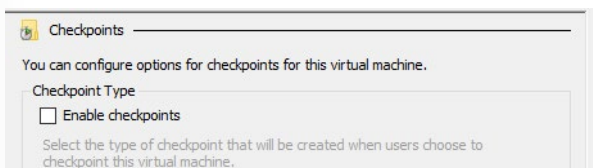
Once you've downloaded the image to a local location, open **Hyper-V Manager** to create a VM with the VHD you copied.

To create a VM with the copied VHD:

1. Open the **New Virtual Machine Wizard**.
2. On the Specify Generation page, select **Generation 1**.



3. Under Checkpoint Type, disable checkpoints by unchecking the check box.



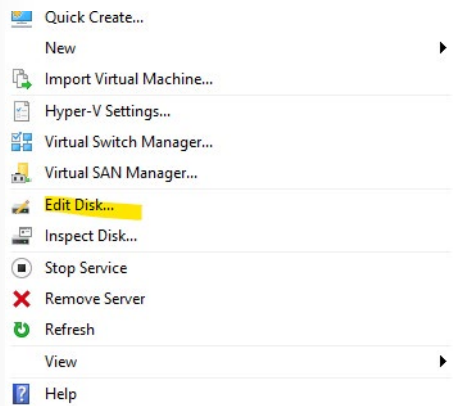
You can also run the following cmdlet in PowerShell to disable checkpoints.

PowerShell

```
Set-VM -Name <VMNAME> -CheckpointType Disabled
```

Fixed disk

If you create a VM from an existing VHD, it creates a dynamic disk by default. It can be changed to a fixed disk by selecting **Edit Disk...** as shown in the following image.



You can also run the following PowerShell cmdlet to change the disk to a fixed disk.

PowerShell

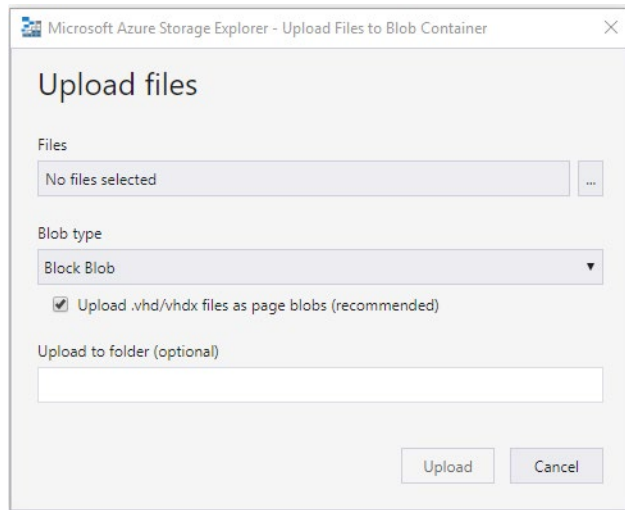
```
Convert-VHD -Path c:\test\MY-VM.vhdx -DestinationPath c:\test\MY-NEW-VM.vhd  
-VHDType Fixed
```

Upload master image to a storage account in Azure

This topic only applies when the master image was created locally.

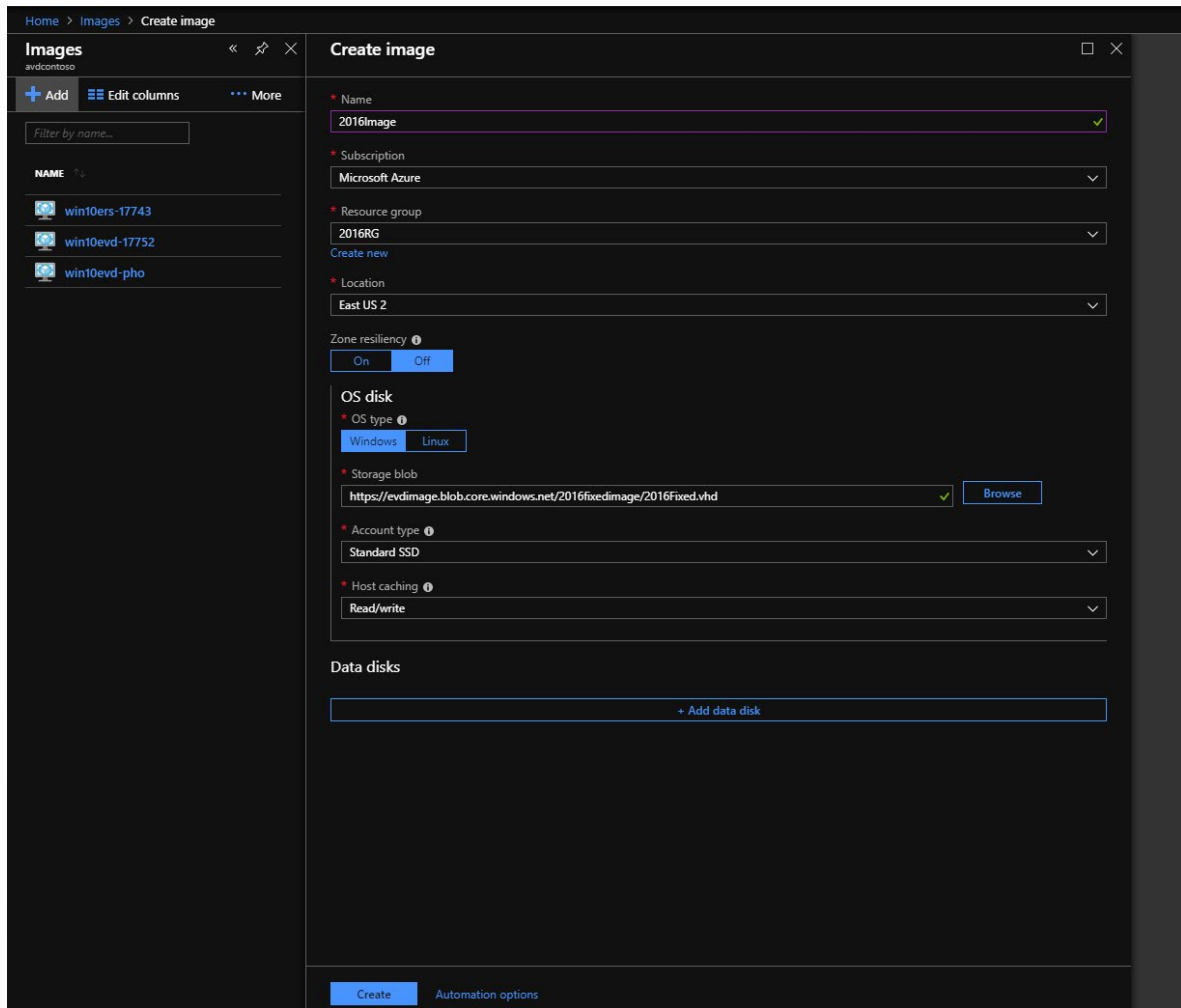
The following instructions apply to a master image was created locally that can be loaded into an Azure storage.

1. Convert the VM image (VHD) to Fixed if you haven't already. If you don't convert the image to Fixed, you can't successfully create the image.
2. Upload the VHD to a blob container in your storage account. You can upload quickly with the Storage

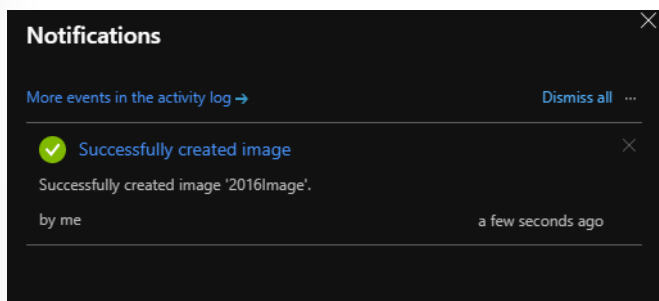


Explorer tool.

3. Next, go to the Azure portal in your browser and search for "Images." Your search should lead you to the **Create image** page, as shown in the following screenshot:



4. Once you've created the image, you should see a notification like the one in the following screenshot:



Modify a session host image

This topic covers how to prepare and modify basic configuration options for apps and your image's registry.

Disable Automatic Updates

To disable Automatic Updates via local Group Policy:

1. Open **Local Group Policy Editor\Administrative Templates\Windows Components\Windows Update**.
2. Right-click **Configure Automatic Update** and set it to **Disabled**.

You can also run the following command on a command prompt to disable Automatic Updates.

```
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU" /v NoAutoUpdate /t REG_DWORD /d 1 /f
```

Specify Start layout for Windows 10 PCs

Run this command to specify a Start layout for Windows 10 PCs.

```
reg add "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer" /v SpecialRoamingOverrideAllowed /t REG_DWORD /d 1 /f
```

Set up time zone redirection

Time zone redirection can be enforced on Group Policy level since all VMs in a host pool are part of the same security group.

To redirect time zones:

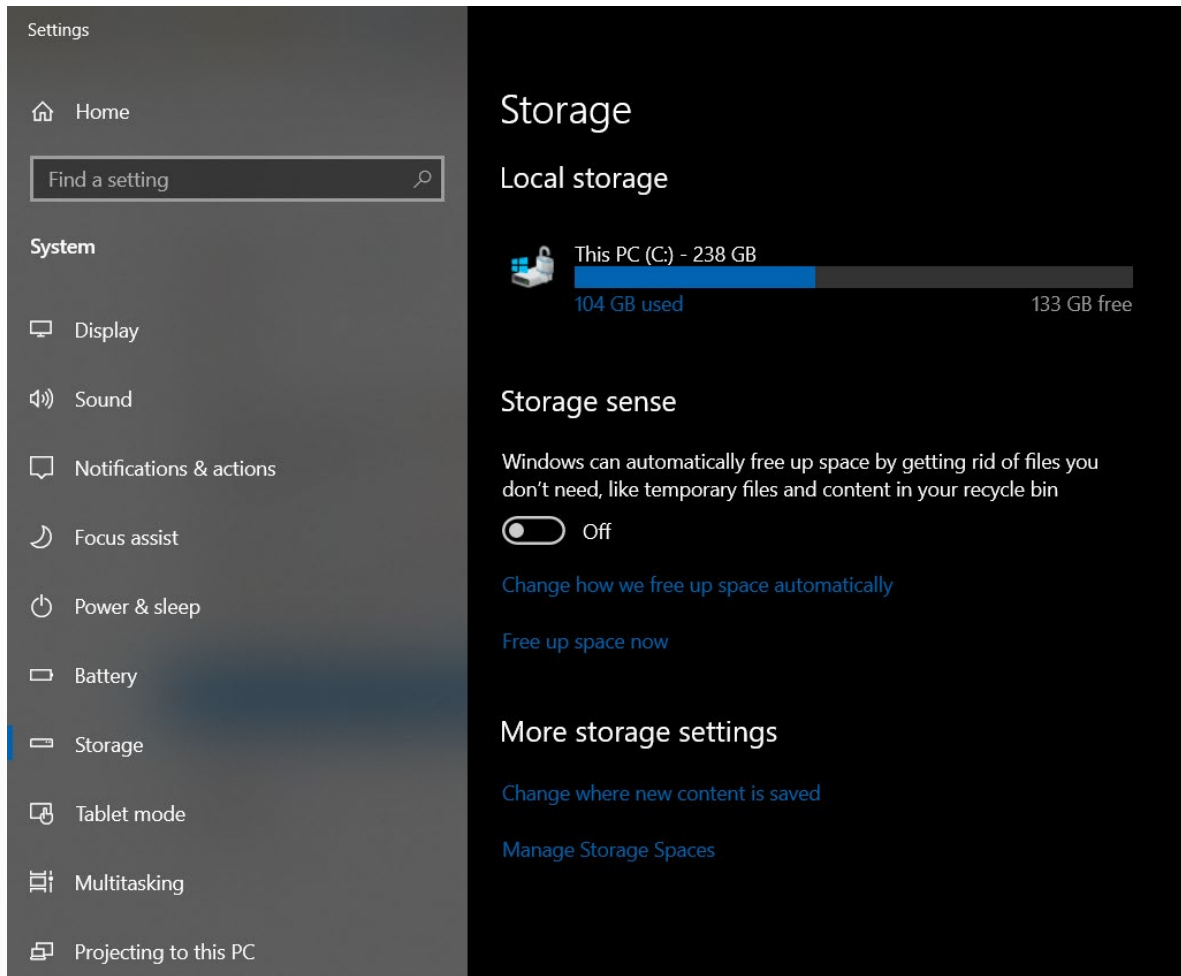
1. On the Active Directory server, open the **Group Policy Management Console**.
2. Expand your domain and Group Policy Objects.
3. Right-click the **Group Policy Object** that you created for the group policy settings and select **Edit**.
4. In the **Group Policy Management Editor**, navigate to **Computer Configuration > Policies > Administrative Templates > Windows Components > Remote Desktop Services > Remote Desktop Session Host > Device and Resource Redirection**.
5. Enable the **Allow time zone redirection** setting.

You can also run this command on the master image to redirect time zones:

```
reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows NT\Terminal Services" /v fEnableTimeZoneRedirection /t REG_DWORD /d 1 /f
```

Disable Storage Sense

For Azure Virtual Desktop session host that use Windows 10 Enterprise or Windows 10 Enterprise multi-session, we recommend disabling Storage Sense. You can disable Storage Sense in the Settings menu under **Storage**, as shown in the following screenshot:



You can also change the setting with the registry by running the following command:

```
reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\StorageSense\Parameters\StoragePolicy" /v 01 /t REG_DWORD /d 0 /f
```

Plan for image update and management

A Shared Image Gallery simplifies custom image sharing across your organization. Custom images are like marketplace images, but you create them yourself. Custom images can be used to bootstrap deployment tasks like preloading applications, application configurations, and other OS configurations.

The Shared Image Gallery lets you share your custom VM images with others in your organization, within or across regions, within an AAD tenant. Choose which images you want to share, which regions you want to make them available in, and who you want to share them with. You can create multiple galleries so that you can logically group shared images.

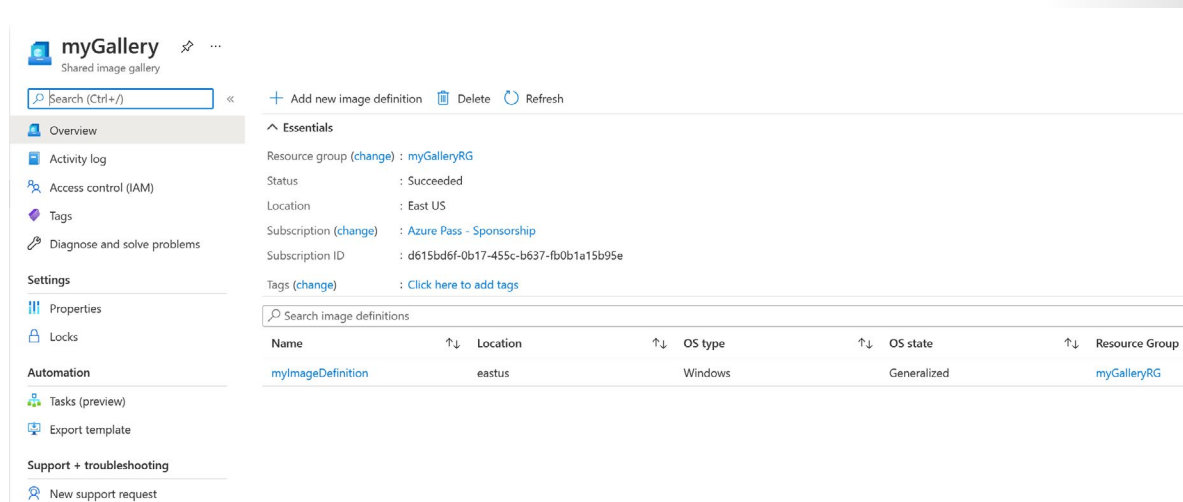
The gallery is a top-level resource that provides full Azure role-based access control (Azure RBAC). Images can be versioned, and you can choose to replicate each image version to a different set of Azure regions. The gallery only works with Managed Images.

Shared Image Gallery is a service that helps you build structure and organization around images. Shared Image Galleries provide:

- Global replication of images.
- Versioning and grouping of images for easier management.
- Highly available images with Zone Redundant Storage (ZRS) accounts in regions that support Availability Zones. ZRS offers better resilience against zonal failures.
- Premium storage support (Premium_LRS).
- Sharing across subscriptions, and even between Active Directory (AD) tenants, using RBAC.
- Scaling your deployments with image replicas in each region.

Using a Shared Image Gallery you can share your images to different users, service principals, or AD groups within your organization. Shared images can be replicated to multiple regions, for quicker scaling of your deployments.

If you have a large number of images that you need to maintain, and would like to make them available throughout your company, you can use a Shared Image Gallery as a repository.



The Shared Image Gallery feature has multiple resource types:

Resource	Description
Image source	This is a resource that can be used to create an image version in an image gallery. An image source can be an existing Azure VM that is either generalized or specialized, a managed image, a snapshot, a VHD or an image version in another image gallery.
Image gallery	Like the Azure Marketplace, an image gallery is a repository for managing and sharing images, but you control who has access.

Resource	Description
Image definition	Image definitions are created within a gallery and carry information about the image and requirements for using it internally. This includes whether the image is Windows or Linux, release notes, and minimum and maximum memory requirements. It is a definition of a type of image.
Image version	An image version is what you use to create a VM when using a gallery. You can have multiple versions of an image as needed for your environment. Like a managed image, when you use an image version to create a VM, the image version is used to create new disks for the VM. Image versions can be used multiple times.

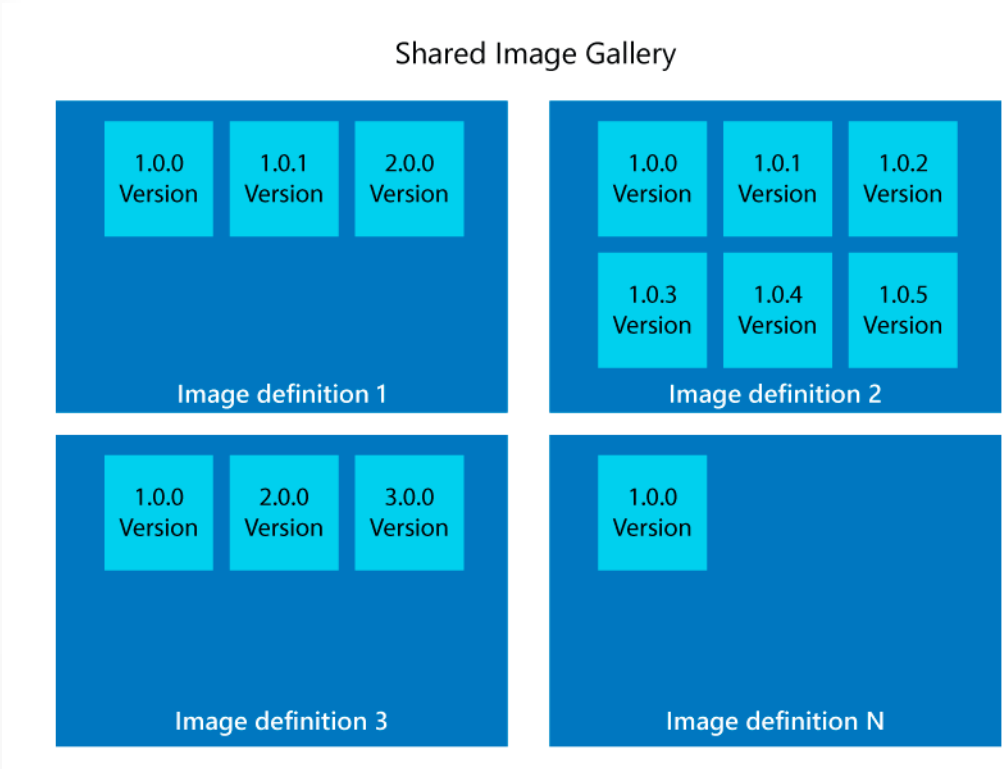


Image definitions

Image definitions are a logical grouping for versions of an image. The image definition holds information about why the image was created, what OS it is for, and other information about using the image. An image definition is like a plan for all of the details around creating a specific image. You don't deploy a VM from an image definition, but from the image versions created from the definition.

There are three parameters for each image definition that are used in combination - Publisher, Offer and SKU. These are used to find a specific image definition. You can have image versions that share one or two, but not all three values. For example, here are three image definitions and their values:

Image Definition	Publisher	Offer	SKU
myImage1	Contoso	Finance	Backend
myImage2	Contoso	Finance	Frontend
myImage3	Testing	Finance	Frontend

All three of these have unique sets of values. The format is similar to how you can currently specify publisher, offer, and SKU for Azure Marketplace images in Azure PowerShell to get the latest version of a Marketplace image. Each image definition needs to have a unique set of these values.

Regional Support

All public regions can be target regions, but certain regions require that customers go through a request process in order to gain access. To request that a subscription is added to the allow list for a region such as Australia Central or Australia Central 2, submit an access request

Limits

There are limits, per subscription, for deploying resources using Shared Image Galleries:

- 100 shared image galleries, per subscription, per region
- 1,000 image definitions, per subscription, per region
- 10,000 image versions, per subscription, per region
- 10 image version replicas, per subscription, per region
- Any disk attached to the image must be less than or equal to 1TB in size

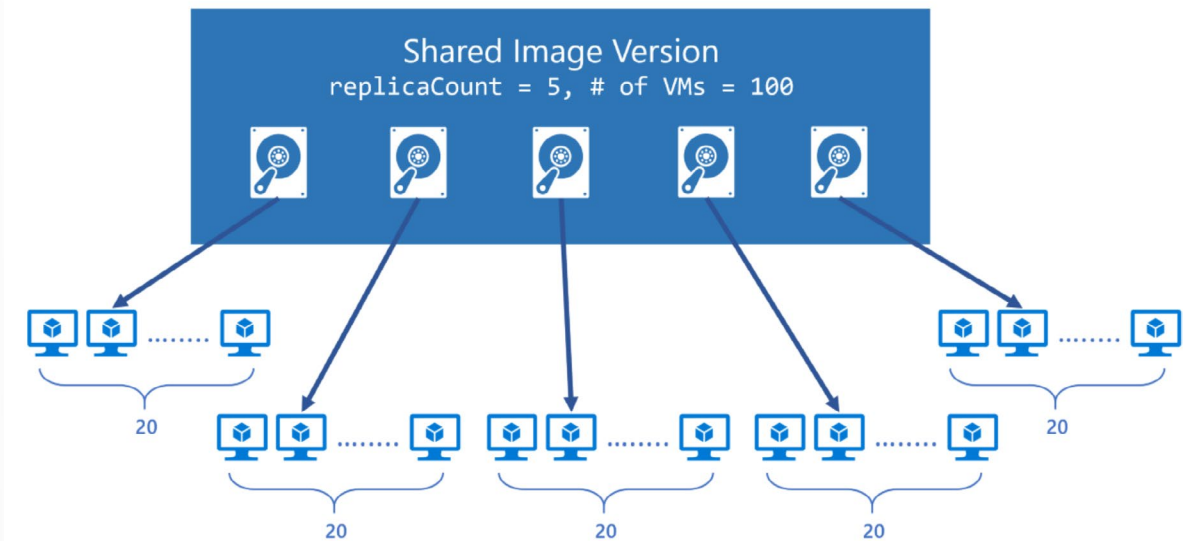
Scaling

Shared Image Gallery allows you to specify the number of replicas you want Azure to keep of the images. This helps in multi-VM deployment scenarios as the VM deployments can be spread to different replicas reducing the chance of instance creation processing being throttled due to overloading of a single replica.

With Shared Image Gallery, you can now deploy up to a 1,000 VM instances in a virtual machine scale set (up from 600 with managed images). Image replicas provide for better deployment performance, reliability and consistency. You can set a different replica count in each target region, based on the scale needs for the region. Since each replica is a deep copy of your image, this helps scale your deployments linearly with each extra replica. While we understand no two images or regions are the same, here's our general guideline on how to use replicas in a region:

- For non-Virtual Machine Scale Set deployments - For every 20 VMs that you create concurrently, we recommend you keep one replica. For example, if you are creating 120 VMs concurrently using the same image in a region, we suggest you keep at least 6 replicas of your image.
- For Virtual Machine Scale Set deployments - For every scale set deployment with up to 600 instances, we recommend you keep at least one replica. For example, if you are creating 5 scale sets concurrently, each with 600 VM instances using the same image in a single region, we suggest you keep at least 5 replicas of your image.

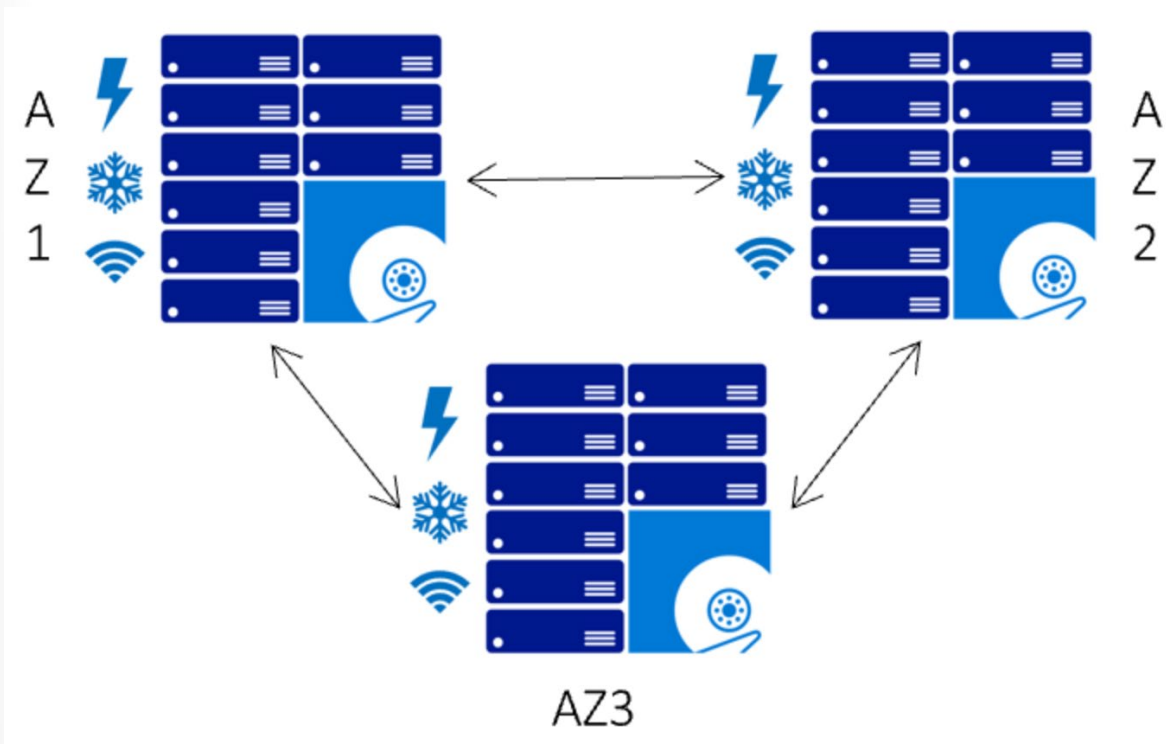
We always recommend you to overprovision the number of replicas due to factors like image size, content and OS type.



Make your images highly available

Azure Zone Redundant Storage (ZRS) provides resilience against an Availability Zone failure in the region. With the general availability of Shared Image Gallery, you can choose to store your images in ZRS accounts in regions with Availability Zones.

You can also choose the account type for each of the target regions. The default storage account type is Standard_LRS, but you can choose Standard_ZRS for regions with Availability Zones. Check the regional availability of ZRS [here](#).



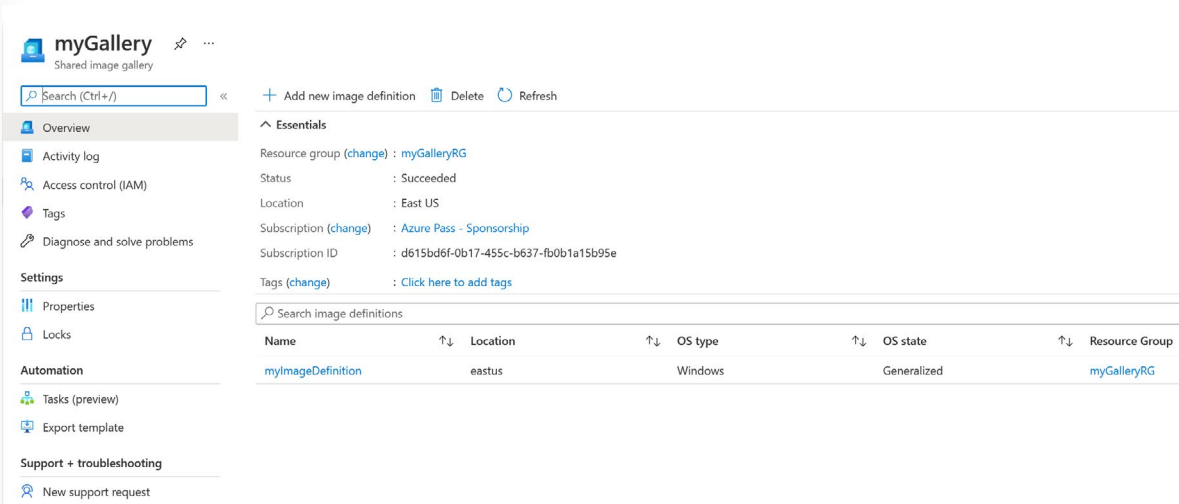
Create and use a Shared Image Gallery (SIG) using the portal

Create an image gallery

An image gallery is the primary resource used for enabling image sharing. Allowed characters for Gallery name are uppercase or lowercase letters, digits, dots, and periods. The gallery name cannot contain dashes. Gallery names must be unique within your subscription.

The following example creates a gallery named *myGallery* in the *myGalleryRG* resource group.

1. Sign in to the Azure portal.
2. Use the type **Shared image gallery** in the search box and select **Shared image gallery** in the results.
3. In the **Shared image gallery** page, click **Add**.
4. On the **Create shared image gallery** page, select the correct subscription.
5. In **Resource group**, select **Create new** and type *myGalleryRG* for the name.
6. In **Name**, type *myGallery* for the name of the gallery.
7. Leave the default for **Region**.
8. You can type a short description of the gallery, like *My image gallery for testing*. and then click **Review + create**.
9. After validation passes, select **Create**.
10. When the deployment is finished, select **Go to resource**.

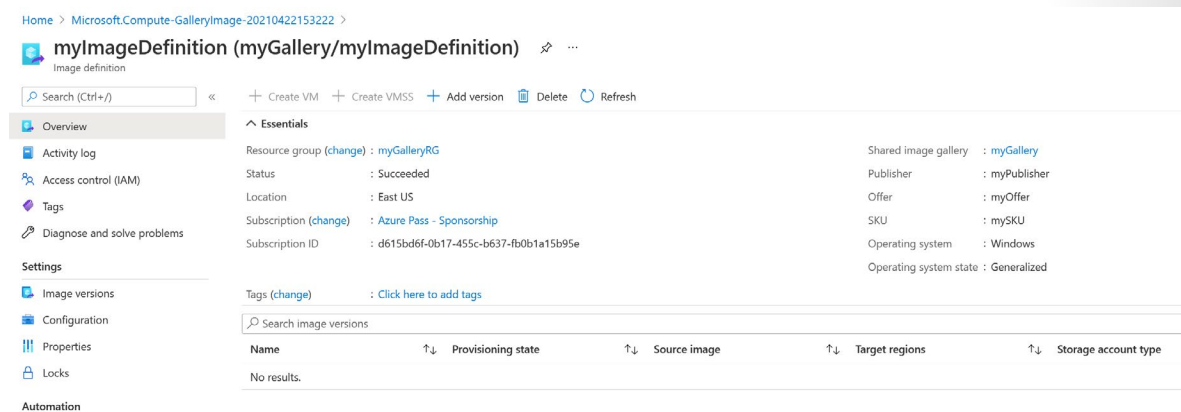


Create an image definition

Image definitions create a logical grouping for images. They are used to manage information about the image versions that are created within them. Image definition names can be made up of uppercase or lowercase letters, digits, dots, dashes and periods.

Create the gallery image definition inside of your gallery. In this example, the gallery image is named *myImageDefinition*.

1. On the page for your new image gallery, select **Add a new image definition** from the top of the page.
2. In the **Add new image definition to shared image gallery**, for **Region**, select *East US*.
3. For **Image definition name**, type *myImageDefinition*.
4. For **Operating system**, select the correct option based on your source VM.
5. For **VM generation**, select the option based on your source VM. In most cases, this will be *Gen 1*.
6. For **Operating system state**, select the option based on your source VM.
7. For **Publisher**, type *myPublisher*.
8. For **Offer**, type *myOffer*.
9. For **SKU**, type *mySKU*.
10. When finished, select **Review + create**.
11. After the image definition passes validation, select **Create**.
12. When the deployment is finished, select **Go to resource**.



Create an image version

Create an image version from a managed image.

When choosing target regions for replication, remember that you also have to include the *source* region as a target for replication.

Allowed characters for image version are numbers and periods. Numbers must be within the range of a 32-bit integer. Format: *MajorVersion.MinorVersion.Patch*.

1. In the page for your image definition, select **Add version** from the top of the page.
2. In **Region**, select the region where your managed image is stored. Image versions need to be created in the same region as the managed image they are created from.
3. For **Name**, type *1.0.0*. The image version name should follow *major.minor.patch* format using integers.
4. In **Source image**, select your source managed image from the drop-down.
5. In **Exclude from latest**, leave the default value of *No*.

6. For **End of life date**, select a date from the calendar that is a couple of months in the future.
7. In **Replication**, leave the **Default replica count** as 1. You need to replicate to the source region, so leave the first replica as the default and then pick a second replica region to be *East US*.
8. When you are done, select **Review + create**. Azure will validate the configuration.
9. When image version passes validation, select **Create**.
10. When the deployment is finished, select **Go to resource**.

It can take a while to replicate the image to all of the target regions.

Share the gallery

We recommend that you share access at the image gallery level. The following walks you through sharing the gallery that you just created.

1. On the page for your new image gallery, in the menu on the left, select **Access control (IAM)**.
2. Under **Add a role assignment**, select **Add**. The **Add a role assignment** pane will open.
3. Under **Role**, select **Reader**.
4. Under **assign access to**, leave the default of **Azure AD user, group, or service principal**.
5. Under **Select**, type in the email address of the person that you would like to invite.
6. If the user is outside of your organization, you will see the message **This user will be sent an email that enables them to collaborate with Microsoft**. Select the user with the email address and then click **Save**.

If the user is outside of your organization, they will get an email invitation to join the organization. The user needs to accept the invitation, then they will be able to see the gallery and all of the image definitions and versions in their list of resources.

Install language packs in AVD

You want your users be able to customize which language their Windows 10 Enterprise multi-session image displays.

There are two ways you can accommodate the language needs of your users:

- Build dedicated host pools with a customized image for each language.
- Have users with different language and localization requirements in the same host pool, but customize their images to ensure they can select whichever language they need.

The latter method is a lot more efficient and cost-effective. However, it's up to you to decide which method best suits your needs. This topic will show you how to customize languages for your images.

You need the following to customize Windows 10 Enterprise multi-session images to add multiple languages:

- An Azure virtual machine (VM) with Windows 10 Enterprise multi-session.
- The Language ISO, Feature on Demand (FOD) Disk 1, and Inbox Apps ISO of the OS version the image uses.
- An Azure Files Share or a file share on a Windows File Server Virtual Machine

The file share (repository) must be accessible from the Azure VM you plan to use to create the custom image.

Create a content repository for language packages and features on demand

To create the content repository for language packages and FODs and a repository for the Inbox Apps packages:

1. On an Azure VM, download the Windows 10 Multi-Language ISO, FODs, and Inbox Apps for Windows 10 Enterprise multi-session, version 1903/1909, and 2004 images.
2. Open and mount the ISO files on the VM.
3. Go to the language pack ISO and copy the content from the **LocalExperiencePacks** and **x64\lang-packs** folders, then paste the content into the file share.
4. **Go to the FOD ISO file**, copy all of its content, then paste it into the file share.
5. Go to the **amd64fre** folder on the Inbox Apps ISO and copy the content in the repository for the inbox apps that you've prepared.

```
Microsoft-Windows-LanguageFeatures-Fonts-Arab-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Beng-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Cans-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Cher-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Deva-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Ethi-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Gujr-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Guru-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Hans-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Hant-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Hebr-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Jpan-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Khmr-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Knda-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Kore-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Lao-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Mlym-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Orya-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-PanEuropeanSupplementalFonts-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Sinh-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Syrc-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Tam-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Tel-Package~31bf3856ad364e35~amd64~~.cab
Microsoft-Windows-LanguageFeatures-Fonts-Thai-Package~31bf3856ad364e35~amd64~~.cab
```

6. Set the permissions on the language content repository share so that you have read access from the VM you'll use to build the custom image.

Create a custom Windows 10 Enterprise multi-session image manually

To create a custom Windows 10 Enterprise multi-session image manually:

1. Deploy an Azure VM, then go to the Azure Gallery and select the current version of Windows 10 Enterprise multi-session you're using.
2. After you've deployed the VM, connect to it using RDP as a local admin.
3. Make sure your VM has all the latest Windows Updates. Download the updates and restart the VM, if necessary.
4. Connect to the language package, FOD, and Inbox Apps file share repository and mount it to a letter drive (for example, drive E).

Finish customizing your image

After you've installed the language packs, you can install any other software you want to add to your customized image.

Once you're finished customizing your image, you'll need to run the system preparation tool (sysprep).

To run sysprep:

1. Open an elevated command prompt and run the following command to generalize the image:

```
C:\Windows\System32\Sysprep\sysprep.exe /oobe /generalize /shutdown
```

2. Stop the VM, then capture it in a managed.
3. You can now use the customized image to deploy an Azure Virtual Desktop host pool.

Enable languages in Windows settings app

Finally, after you deploy the host pool, you'll need to add the language to each user's language list so they can select their preferred language in the Settings menu.

To ensure your users can select the languages you installed, sign in as the user, then run the following PowerShell cmdlet to add the installed language packs to the Languages menu. You can also set up this script as an automated task or logon script that activates when the user signs in to their session.

PowerShell

```
$LanguageList = Get-WinUserLanguageList
$LanguageList.Add("es-es")
$LanguageList.Add("fr-fr")
$LanguageList.Add("zh-cn")
Set-WinUserLanguageList $LanguageList -force
```

After a user changes their language settings, they'll need to sign out of their Azure Virtual Desktop session and sign in again for the changes to take effect.

Labs

Create and configure host pools and session hosts (Azure AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 **GitHub repository**¹¹.

Direct link to the **Lab - Create and configure host pools and session hosts (Azure AD DS)**.¹²

Objectives

After completing this lab, you will be able to:

- Configure an Azure Azure Virtual Desktop environment in an Azure AD DS domain.
- Validate Azure Azure Virtual Desktop environment in an Azure AD DS domain.

Lab prerequisites

- An Azure subscription
- A Microsoft account or an Azure AD account with the Global Administrator role in the Azure AD tenant associated with the Azure subscription and with the Owner or Contributor role in the Azure subscription
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (Azure AD DS)**

Estimated time: 60 minutes

Lab files

None

Exercise 1: Configure an Azure Azure Virtual Desktop environment

The main tasks for this exercise are as follows:

1. Prepare AD DS domain and the Azure subscription for deployment of an Azure Azure Virtual Desktop host pool
2. Deploy an Azure Azure Virtual Desktop host pool
3. Configure Azure Virtual Desktop application groups
4. Configure Azure Virtual Desktop workspaces

¹¹ https://aka.ms/AZ-140_Labs

¹² https://aka.ms/AZ-140_02_Lab_01

Exercise 2: Validate Azure Azure Virtual Desktop environment

The main tasks for this exercise are as follows:

1. Install Microsoft Remote Desktop client (MSRDC) on a Windows 10 computer
2. Subscribe to an Azure Virtual Desktop workspace
3. Test Azure Virtual Desktop apps

Deploy host pools and session hosts by using the Azure portal (AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)¹³.

Direct link to the **Lab - Deploy host pools and session hosts by using the Azure portal (AD DS)**.¹⁴

Objectives

After completing this lab, you will be able to:

- Implement an Azure Azure Virtual Desktop environment in an AD DS domain
- Validate Azure Azure Virtual Desktop environment in an AD DS domain

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (AD DS)**

Estimated time: 60 minutes

Lab files

None

Exercise 1: Implement an Azure Azure Virtual Desktop environment in an AD DS domain

The main tasks for this exercise are as follows:

1. Prepare AD DS domain and the Azure subscription for deployment of an Azure Azure Virtual Desktop host pool
2. Deploy an Azure Azure Virtual Desktop host pool
3. Manage the Azure Azure Virtual Desktop host pool session hosts

¹³ https://aka.ms/AZ-140_Labs

¹⁴ https://aka.ms/AZ-140_02_Lab_02

4. Configure Azure Virtual Desktop application groups
5. Configure Azure Virtual Desktop workspaces

Exercise 2: Validate Azure Azure Virtual Desktop environment

The main tasks for this exercise are as follows:

1. Install Microsoft Remote Desktop client (MSRDC) on a Windows 10 computer
2. Subscribe to an Azure Virtual Desktop workspace
3. Test Azure Virtual Desktop apps

Implement and manage storage for AVD (Azure AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 **GitHub repository**¹⁵.

Direct link to the **Lab - Implement and manage storage for AVD (Azure AD DS)**.¹⁶

Objectives

After completing this lab, you will be able to:

- Configure Azure Files to store profile containers for Azure Virtual Desktop in Azure AD DS environment

Lab prerequisites

- An Azure subscription
- A Microsoft account or an Azure AD account with the Global Administrator role in the Azure AD tenant associated with the Azure subscription and with the Owner or Contributor role in the Azure subscription
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (Azure AD DS)**

Estimated time: 30 minutes

Lab files

- None

Exercise: Configure Azure Files to store profile containers for Azure Virtual Desktop

The main tasks for this exercise are as follows:

1. Create an Azure Storage account
2. Create an Azure Files share

¹⁵ https://aka.ms/AZ-140_Labs

¹⁶ https://aka.ms/AZ-140_02_Lab_03

3. Enable Azure AD DS authentication for the Azure Storage account
4. Configure the Azure Files share permissions
5. Configure the Azure Files directory and file level permissions

Implement and manage storage for AVD (AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 **GitHub repository**¹⁷.

Direct link to the **Lab - Implement and manage storage for AVD (AD DS)**.¹⁸

Objectives

After completing this lab, you will be able to:

- Configure Azure Files to store profile containers for Azure Virtual Desktop

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (AD DS)**

Estimated time: 30 minutes

Lab files

- None

Exercise: Configure Azure Files to store profile containers for Azure Virtual Desktop

The main tasks for this exercise are as follows:

1. Create an Azure Storage account
2. Create an Azure Files share
3. Enable AD DS authentication for the Azure Storage account
4. Configure the Azure Files RBAC-based permissions
5. Configure the Azure Files file system permissions

¹⁷ https://aka.ms/AZ-140_Labs

¹⁸ https://aka.ms/AZ-140_02_Lab_04

Deploy host pools and hosts by using Azure Resource Manager templates

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)¹⁹.

Direct link to the **Lab - Deploy host pools and hosts by using Azure Resource Manager templates**.²⁰.

Objectives

After completing this lab, you will be able to:

- Deploy Azure Azure Virtual Desktop host pools and hosts by using Azure Resource Manager templates

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (AD DS)** or **Prepare for deployment of Azure Azure Virtual Desktop (Azure AD DS)**
- The completed lab **Deploy host pools and session hosts by using the Azure portal (AD DS)** or **Deploy host pools and session hosts by using the Azure portal (Azure AD DS)**

Estimated time: 45 minutes

Lab files

- \\AZ-140\\AllFiles\\Labs\\02\\az140-23_azuredeployhp23.parameters.json
- \\AZ-140\\AllFiles\\Labs\\02\\az140-23_azuremodifyhp23.parameters.json

Exercise 1: Deploy Azure Azure Virtual Desktop host pools and hosts by using Azure Resource Manager templates

The main tasks for this exercise are as follows:

1. Prepare for deployment of an Azure Azure Virtual Desktop host pool by using an Azure Resource Manager template
2. Deploy an Azure Azure Virtual Desktop host pool and hosts by using an Azure Resource Manager template
3. Verify deployment of the Azure Azure Virtual Desktop host pool and hosts
4. Prepare for adding of hosts to the existing Azure Azure Virtual Desktop host pool by using an Azure Resource Manager template
5. Add hosts to the existing Azure Azure Virtual Desktop host pool by using an Azure Resource Manager template

¹⁹ https://aka.ms/AZ-140_Labs

²⁰ https://aka.ms/AZ-140_02_Lab_05

6. Verify changes to the Azure Azure Virtual Desktop host pool
7. Manage personal desktop assignments in the Azure Azure Virtual Desktop host pool

Exercise 2: Stop and deallocate Azure VMs provisioned in the lab

The main tasks for this exercise are as follows:

1. Stop and deallocate Azure VMs provisioned in the lab

Note: In this exercise, you will deallocate the Azure VMs provisioned in this lab to minimize the corresponding compute charges

Deploy and manage host pools and hosts by using PowerShell

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)²¹.

Direct link to the **Lab - Deploy and manage host pools and hosts by using PowerShell**.²².

Objectives

After completing this lab, you will be able to:

- Deploy Azure Azure Virtual Desktop host pools and hosts by using PowerShell
- Add hosts to the Azure Virtual Desktop host pool by using PowerShell

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (AD DS)** or **Prepare for deployment of Azure Azure Virtual Desktop (Azure AD DS)**

Estimated time: 60 minutes

Lab files

- \\AZ-140\\AllFiles\\Labs\\02\\az140-24_azuredeployhp3.json
- \\AZ-140\\AllFiles\\Labs\\02\\az140-24_azuredeployhp3.parameters.json

²¹ https://aka.ms/AZ-140_Labs

²² https://aka.ms/AZ-140_02_Lab_06

Exercise 1: Implement Azure Azure Virtual Desktop host pools and session hosts by using PowerShell

The main tasks for this exercise are as follows:

1. Prepare for deployment of Azure Virtual Desktop host pool by using PowerShell
2. Create an Azure Virtual Desktop host pool by using PowerShell
3. Perform a template-based deployment of an Azure VM running Windows 10 Enterprise by using PowerShell
4. Add an Azure VM running Windows 10 Enterprise as a session host to the Azure Virtual Desktop host pool by using PowerShell
5. Verify the deployment of the Azure Azure Virtual Desktop session host

Exercise 2: Stop and deallocate Azure VMs provisioned in the lab

The main tasks for this exercise are as follows:

1. Stop and deallocate Azure VMs provisioned in the lab

Note: In this exercise, you will deallocate the Azure VMs provisioned in this lab to minimize the corresponding compute charges

Create and manage session host images (AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)²³.

Direct link to the **Lab - Create and manage session host images (AD DS)**.²⁴

Objectives

After completing this lab, you will be able to:

- Deploy Azure Azure Virtual Desktop host pools and session hosts by using PowerShell
- Add session hosts to the Azure Virtual Desktop host pool by using PowerShell

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.
- The completed lab **Prepare for deployment of Azure Azure Virtual Desktop (AD DS)** or **Prepare for deployment of Azure Azure Virtual Desktop (Azure AD DS)**

Estimated time: 60 minutes

²³ https://aka.ms/AZ-140_Labs

²⁴ https://aka.ms/AZ-140_02_Lab_07

Lab files

- \\AZ-140\\AllFiles\\Labs\\02\\az140-25_azuredeployvm25.json
- \\AZ-140\\AllFiles\\Labs\\02\\az140-25_azuredeployvm25.parameters.json

Exercise 1: Create and manage session host images

The main tasks for this exercise are as follows:

1. Prepare for configuration of an Azure Virtual Desktop host image
2. Configure an Azure Virtual Desktop host image
3. Create an Azure Virtual Desktop host image
4. Provision an Azure Virtual Desktop host pool by using the custom image

Exercise 2: Stop and deallocate Azure VMs provisioned in the lab

The main tasks for this exercise are as follows:

1. Stop and deallocate Azure VMs provisioned in the lab

Note: In this exercise, you will deallocate the Azure VMs provisioned in this lab to minimize the corresponding compute charges

Review questions

Module review questions

Review Question 1

You manage an AAD tenant named westwind.com with an virtual network named AVD-Vnet-A.

You deploy an AAD DS to a domain named westwindA.com to AVD-Vnet-A.

You want to deploy an AVD host pool named AVDHostPool.

You want to make sure that you can deploy Win 10 Ent host pools to AVDHostPool.

What do you do first?

- ☐ A. Configure a private endpoint
- ☐ B. Add an additional network adapter
- ☐ C. Modify the DNS settings on your virtual network (AVD-Vnet-A)
- ☐ D. Implement a RD Session Host

Review Question 2

You are planning on deploying multiple AVD session hosts with private IP addresses.

You want to make sure that your admins can initiate an RDP session to all of the session hosts from the Azure admin portal.

What should you implement?

- ☐ A. An RDP or SSH client on all Win 10 computers
- ☐ B. Azure Bastion
- ☐ C. A path-based redirection on the Application Gateway
- ☐ D. A subnet level NSG on the AzureFirewallSubnet

Review Question 3

You are planning an AVD deployment that will use FSLogix profile containers.

The following is being planned for the Azure Storage account with will be hosting the FSLogix profile containers.

- Account type: StorageV2 (GP v2)
- Performance: Premium
- Name: AVDStorage2

What is needed to make sure the storage account supports the AVD deployment?

- ☐ A. Set block size to 4 MB
- ☐ B. Create a capacity pool
- ☐ C. Set Authentication type to Account key
- ☐ D. Set the Account kind to FileStorage

Review Question 4

You manage a network that has an on-premises domain that has a universal security group named SecurityUsers.

SecurityUsers syncs with AAD, where there is a hybrid AAD tenant.

You manage an AVD host pool that has three Win 10 Enterprise multi-session hosts.

You want to make sure that only members of SecurityUsers can establish AVD sessions to the host pool.

What needs to be done to meet your goal?

- ☐ A. Create a new role assignment for the host pool
- ☐ B. Modify the RDP properties on the host pool
- ☐ C. Configure role assignment for each of the three VMs
- ☐ D. Assign SecurityUsers to an application group

Review Question 5

You manage an AAD tenant named WestwindEast.com

You use an account named Administrator-East to deploy an Azure AD DS managed domain named

A-AD-DS-WestwindEast.com to a virtual network called VNET-East.

You want to deploy the AVD host pool named East-Pool-1 to VNET-East.

You want to make sure that you can use the Administrator-East account to deploy Win 10 Enterprise session hosts to East-Pool-1.

What is the first thing you should do?

- ☐ A. Change the password for the Administrator-East account
- ☐ B. Configure a role assignment for the East-Pool-1 host pool
- ☐ C. Configure a role assignment for each VM in the host pool
- ☐ D. Configure a policy preference in a GPO

Answers

Review Question 1

You manage an AAD tenant named westwind.com with an virtual network named AVD-Vnet-A. You deploy an AAD DS to a domain named westwindA.com to AVD-Vnet-A. You want to deploy an AVD host pool named AVDHostPool. You want to make sure that you can deploy Win 10 Ent host pools to AVDHostPool. What do you do first?

- ☐ A. Configure a private endpoint
- ☐ B. Add an additional network adapter
- ☒ C. Modify the DNS settings on your virtual network (AVD-Vnet-A)
- ☐ D. Implement a RD Session Host

Explanation

The answer is C, Modify the DNS settings on your virtual network (AVD-Vnet-A).

As seen in topic *Create a host pool by using the Azure portal*, a possible solution to deploy host pools with appropriate DNS settings to have the right credentials, or that the DNS configuration is set to the defaults.

Review Question 2

You are planning on deploying multiple AVD session hosts with private IP addresses. You want to make sure that your admins can initiate an RDP session to all of the session hosts from the Azure admin portal. What should you implement?

- ☐ A. An RDP or SSH client on all Win 10 computers
- ☒ B. Azure Bastion
- ☐ C. A path-based redirection on the Application Gateway
- ☐ D. A subnet level NSG on the AzureFirewallSubnet

Explanation

The answer is B, Azure Bastion.

As seen in topic *Configure AVD session hosts using Azure Bastion*, Azure Bastion is a service you deploy that lets you connect to a virtual machine using your browser and the Azure portal. The Azure Bastion service is a fully platform-managed PaaS service that you provision inside your virtual network.

Review Question 3

You are planning an AVD deployment that will use FSLogix profile containers. The following is being planned for the Azure Storage account with will be hosting the FSLogix profile containers.

What is needed to make sure the storage account supports the AVD deployment?

- ☐ A. Set block size to 4 MB
- ☐ B. Create a capacity pool
- ☐ C. Set Authentication type to Account key
- ☒ D. Set the Account kind to FileStorage

Explanation

The answer is D, Set the Account kind to FileStorage.

As seen in topic Set the Account kind to FileStorage when creating a FileStorage storage account, ensure the Performance radio button is set to Premium and the Account kind drop-down list is selected to FileStorage.

Review Question 4

You manage a network that has an on-premises domain that has a universal security group named SecurityUsers.

SecurityUsers syncs with AAD, where there is a hybrid AAD tenant.

You manage an AVD host pool that has three Win 10 Enterprise multi-session hosts.

You want to make sure that only members of SecurityUsers can establish AVD sessions to the host pool.

What needs to be done to meet your goal?

- ☐ A. Create a new role assignment for the host pool
- ☐ B. Modify the RDP properties on the host pool
- ☐ C. Configure role assignment for each of the three VMs
- ☒ D. Assign SecurityUsers to an application group

Explanation

The answer is D, Assign SecurityUsers to an application group.

As seen in the topic Deploy and manage host pools and hosts by using PowerShell, users obtain access to host pools by being allocated to a host pool using an assigned Application Group.

Review Question 5

You manage an AAD tenant named `WestwindEast.com`

You use an account named `Administrator-East` to deploy an Azure AD DS managed domain named `A-AD-DS-WestwindEast.com` to a virtual network called `VNET-East`.

You want to deploy the AVD host pool named `East-Pool-1` to `VNET-East`.

You want to make sure that you can use the `Administrator-East` account to deploy Win 10 Enterprise session hosts to `East-Pool-1`.

What is the first thing you should do?

- ☒ A. Change the password for the `Administrator-East` account
- ☐ B. Configure a role assignment for the `East-Pool-1` host pool
- ☐ C. Configure a role assignment for each VM in the host pool
- ☐ D. Configure a policy preference in a GPO

Explanation

The answer is A, Change the password for the `Administrator-East` account.

As seen in the topic *Create a host pool by using the Azure portal*, the `Administrator` account can't have multi-factored authentication (MFA) enabled. And, when joining to an Azure Active Directory Domain Services (Azure AD DS) domain, the account must be part of the Azure AD DC Administrators group and the account password must work in Azure AD DS.



Module 3 Manage access and security

Manage access

RBAC for Azure Virtual Desktop

Azure Virtual Desktop uses Azure role-based access controls (RBAC) to assign roles to users and admins. These roles give admins permission to carry out certain tasks.

The standard built-in roles for Azure are:

- **Owner**
- **Contributor**
- **Reader**

However, Azure Virtual Desktop has additional roles that let you separate management roles for host pools, app groups, and workspaces.

These roles are named in compliance with Azure's standard roles and least-privilege methodology.

Azure Virtual Desktop doesn't have a specific Owner role. However, you can use a standard Owner role for the service objects.

Azure Virtual Desktop Roles

Desktop Virtualization Contributor

The Desktop Virtualization Contributor role lets you manage all aspects of the deployment. However, it doesn't grant you access to compute resources. You'll also need the User Access Administrator role to publish app groups to users or user groups.

Desktop Virtualization Reader

The Desktop Virtualization Reader role lets you view everything in the deployment but doesn't let you make any changes.

Host Pool Contributor

The Host Pool Contributor role lets you manage all aspects of host pools, including access to resources. You'll need an extra contributor role, Virtual Machine Contributor, to create virtual machines. You will need AppGroup and Workspace contributor roles to create host pool using the portal or you can use Desktop Virtualization Contributor role.

Host Pool Reader

The Host Pool Reader role lets you view everything in the host pool, but won't allow you to make any changes.

Application Group Contributor

The Application Group Contributor role lets you manage all aspects of app groups. If you want to publish app groups to users or user groups, you'll need the User Access Administrator role.

Application Group Reader

The Application Group Reader role lets you view everything in the app group and will not allow you to make any changes.

Workspace Contributor

The Workspace Contributor role lets you manage all aspects of workspaces. To get information on applications added to the app groups, you'll also need to be assigned the Application Group Reader role.

Workspace Reader

The Workspace Reader role lets you view everything in the workspace, but won't allow you to make any changes.

User Session Operator

The User Session Operator role lets you send messages, disconnect sessions, and use the "logoff" function to sign sessions out of the session host. However, this role doesn't let you perform session host management like removing session host, changing drain mode, and so on. This role can see assignments but can't modify admins. We recommend you assign this role to specific host pools. If you give this permission at a resource group level, the admin will have read permission on all host pools under a resource group.

Session Host Operator

The Session Host Contributor role lets you view and remove session hosts, as well as change drain mode. They can't add session hosts using the Azure portal because they don't have write permission for host pool objects. If the registration token is valid (generated and not expired), you can use this role to add session hosts to the host pool outside of Azure portal if the admin has compute permissions through the Virtual Machine Contributor role.

Plan and implement Azure roles and RBAC for WVD

Azure Virtual Desktop has a delegated access model that lets you define the amount of access a particular user is allowed to have by assigning them a role.

A role assignment has three components: security principal, role definition, and scope.

The Azure Virtual Desktop delegated access model is based on the Azure RBAC model.

Azure Virtual Desktop delegated access supports the following values for each element of the role assignment:

Security principal

- Users
- User groups
- Service principals

Role definition

- Built-in roles
- Custom roles

Scope

- Host pools
- App groups
- Workspaces

PowerShell cmdlets for role assignments

Azure Virtual Desktop uses Azure role-based access control (Azure RBAC) while publishing app groups to users or user groups. The Desktop Virtualization User role is assigned to the user or user group and the scope is the app group. This role gives the user special data access on the app group.

Run the following cmdlet to add Azure Active Directory users to an app group:

PowerShell

```
New-AzRoleAssignment -SignInName <userupn> -RoleDefinitionName "Desktop  
Virtualization User" -ResourceName <appgroupname> -ResourceGroupName <re-  
sourcegroupname> -ResourceType 'Microsoft.DesktopVirtualization/applica-  
tionGroups'
```

Run the following cmdlet to add Azure Active Directory user group to an app group:

PowerShell

```
New-AzRoleAssignment -ObjectId <usergroupobjectid> -RoleDefinitionName  
"Desktop Virtualization User" -ResourceName <appgroupname> -ResourceGroupN-  
ame <resourcegroupname> -ResourceType 'Microsoft.DesktopVirtualization/  
applicationGroups'
```

Using Azure Virtual Desktop with Intune

With Microsoft Intune, you can secure and manage your Azure Virtual Desktop VMs with policy and apps at scale, after they're enrolled.

Prerequisites

Currently, Intune supports Azure Virtual Desktop VMs that are:

- Running Windows 10 Enterprise, version 1809 or later.
- Hybrid Azure AD-joined
- Set up as personal remote desktops in Azure.
- Enrolled in Intune in one of the following methods:
 - Configure Active Directory group policy to automatically enroll devices that are hybrid Azure AD joined
 - Configuration Manager co-management
 - User self-enrollment via Azure AD Join

Intune treats Azure Virtual Desktop personal VMs the same as Windows 10 Enterprise physical desktops.

This lets you use some of your existing configurations and secure the VMs with compliance policy and conditional access. Intune management doesn't depend on or interfere with Azure Virtual Desktop management of the same virtual machine.

Limitations

There are some limitations to keep in mind when managing Windows 10 Enterprise remote desktops:

Configuration

All VM limitations listed in **Using Windows 10 virtual machines**¹ also apply to Azure Virtual Desktop VMs.

Also, the following profiles aren't currently supported:

- Domain Join
- Wi-Fi

Remote actions

The following Windows 10 desktop device remote actions aren't supported/recommended for Azure Virtual Desktop VMs:

- Autopilot reset
- BitLocker key rotation
- Fresh Start
- Remote lock

¹ <https://docs.microsoft.com/en-us/mem/intune/fundamentals/windows-10-virtual-machines>

- Reset password
- Wipe

Retirement

Deleting VMs from Azure leaves orphaned device records in Intune. They'll be automatically cleaned up according to the cleanup rules configured for the tenant.

Windows 10 Enterprise multi-session

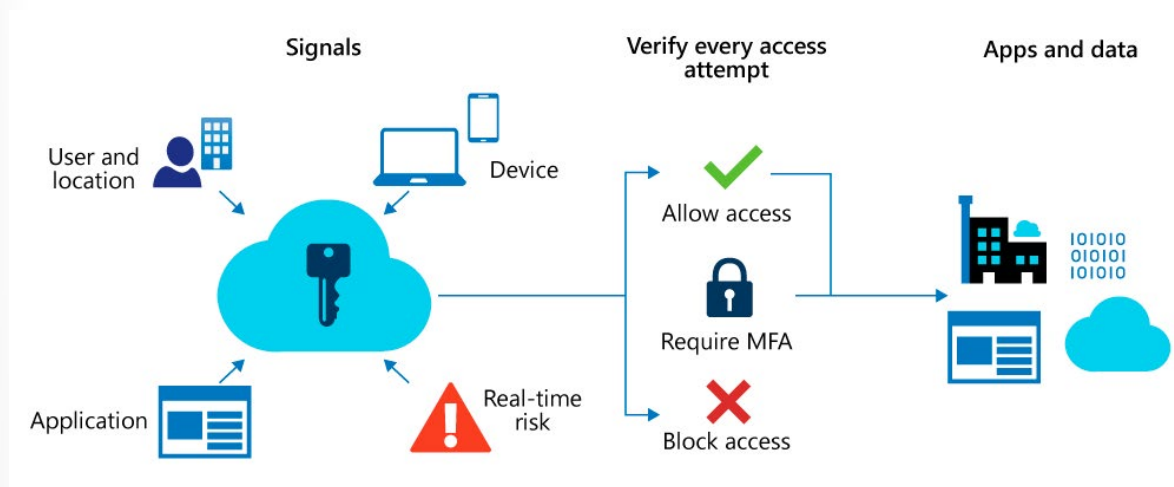
Intune doesn't currently support management of Windows 10 Enterprise multi-session.

Manage security

Plan and implement Conditional Access policies for connections to AVD

Planning your Conditional Access deployment is critical to achieving your organization's access strategy for apps and resources.

Azure Active Directory (Azure AD) Conditional Access analyses signals such as user, device, and location to automate decisions and enforce organizational access policies for resource. You can use Conditional Access policies to apply access controls like Multi-Factor Authentication (MFA). Conditional Access policies allow you to prompt users for MFA when needed for security, and stay out of users' way when not needed.



Microsoft provides standard conditional policies called security defaults that ensure a basic level of security. However, your organization may need more flexibility than security defaults offer. You can use Conditional Access to customize security defaults with more granularity and to configure new policies that meet your requirements.

Benefits

The benefits of deploying Conditional Access are:

- **Increase productivity.** Only interrupt users with a sign-in condition like MFA when one or more signals warrants it. Conditional Access policies allow you to control when users are prompted for MFA, when access is blocked, and when they must use a trusted device.
- **Manage risk.** Automating risk assessment with policy conditions means risky sign-ins are at once identified and remediated or blocked. Coupling Conditional Access with Identity Protection, which detects anomalies and suspicious events, allows you to target when access to resources is blocked or gated.
- **Address compliance and governance.** Conditional Access enables you to audit access to applications, present terms of use for consent, and restrict access based on compliance policies.
- **Manage cost.** Moving access policies to Azure AD reduces the reliance on custom or on-premises solutions for Conditional Access, and their infrastructure costs.

Prerequisites

- A working Azure AD tenant with Azure AD Premium or trial license enabled.
- An account with Conditional Access administrator privileges.
- A non-administrator user with a password you know, such as testuser.
- A group that the non-administrator user is a member of.

Understand Conditional Access policy components

Conditional Access policies are if-then statements: If an assignment is met, then apply these access controls.

When configuring Conditional Access policies, conditions are called *assignments*. Conditional Access policies allow you to enforce access controls on your organization's apps based on certain assignments.

New

Info

Want to switch back to the previous configuration experience? Click to leave the preview. →

Name *

Example: 'Device compliance app policy'

Assignments

Users and groups ⓘ
0 users and groups selected

Cloud apps or actions ⓘ
No cloud apps or actions selected

Conditions ⓘ
0 conditions selected

Access controls

Grant ⓘ
0 controls selected

Session ⓘ
0 controls selected

Device platforms ⓘ
Not configured

Locations ⓘ
Not configured

Client apps (Preview) ⓘ
Not configured

Device state (Preview) ⓘ
Not configured

Assignments define the

- Users and groups to be affected by the policy
- Cloud apps or actions to which the policy will apply
- Conditions under which the policy will apply.

Access controls settings determine how to enforce a policy:

- Grant or Block access to cloud apps.
- Session controls enable limited experiences within specific cloud apps.

Ask the right questions to build your policies

Policies answer questions about who should access your resources, what resources they should access, and under what conditions. Policies can be designed to grant access, or to block access. Be sure to ask the right questions about what your policy is trying to achieve.

Document the answers to questions for each policy before building it out.

Users and Groups

- Which users and groups will be included in or excluded from the policy?
- Does this policy include all users, specific group of users, directory roles, or external users?

Cloud apps or actions

- What application(s) will the policy apply to?
- What user actions will be subject to this policy?

Conditions

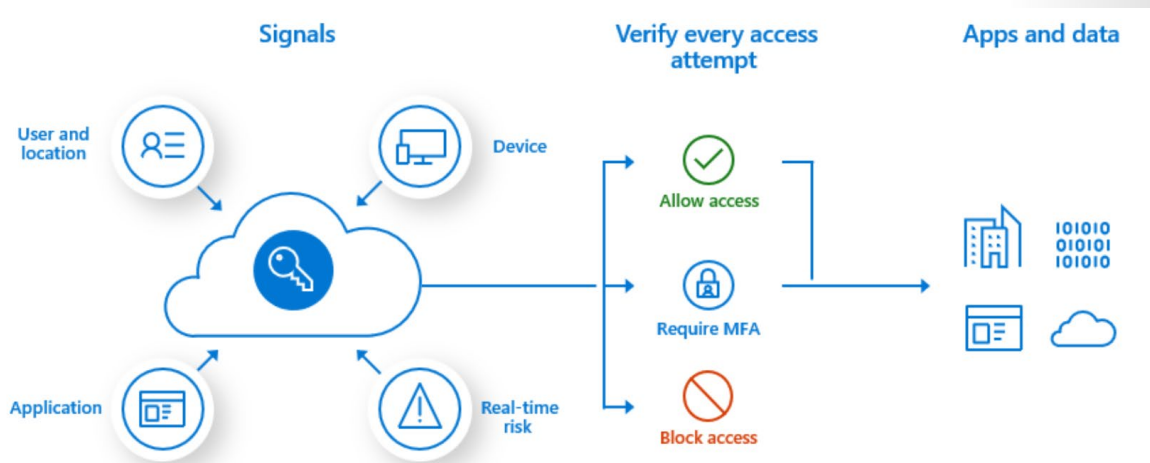
- Which device platforms will be included in or excluded from the policy?
- What are the organization's trusted locations?
- What locations will be included in or excluded from the policy?
- What client app types (browser, mobile, desktop clients, apps with legacy authentication methods) will be included in or excluded from the policy?
- Do you have policies that would drive excluding Azure AD Joined devices or Hybrid Azure AD joined devices from policies?
- If using Identity Protection, do you want to incorporate sign-in risk protection?

Plan and implement MFA in AVD

The Windows client for Azure Virtual Desktop is meant to integrate Azure Virtual Desktop with your local machine. However, when you configure your Azure Virtual Desktop account into the Windows Client, there are certain measures you'll need to take to keep yourself and your users safe.

NOTE: There is an online lab later in this module where you implement Azure AD-based Conditional Access for Azure Virtual Desktop (*Lab - Configure Conditional Access policies for connections to AVD (AD DS)*). This topic is meant to provide an overview of the procedure.

When you first sign in, the client asks for your username, password, and Azure multifactor authentication. The next time you sign in, the client will remember your token from your Azure Active Directory (AD) Enterprise Application. When you select **Remember me** on the prompt for credentials for the session host, your users can sign in after restarting the client without needing to reenter their credentials.



While remembering credentials is convenient, it can also make deployments on Enterprise scenarios or personal devices less secure. To protect your users, you can make sure the client keeps asking for Azure multifactor authentication credentials more frequently.

This topic will show you how to configure the Conditional Access policy for Azure Virtual Desktop to enable this setting.

Here's what you'll need:

- Assign users a license that includes Azure Active Directory Premium P1 or P2.
- An Azure Active Directory group with your users assigned as group members.
- Enable Azure multifactor authentication for all your users.

Create a Conditional Access policy

Here's how to create a Conditional Access policy that requires multifactor authentication when connecting to Azure Virtual Desktop:

1. Sign in to the **Azure portal** as a global administrator, security administrator, or Conditional Access administrator.
2. Browse to **Azure Active Directory > Security > Conditional Access**.
3. Select **New policy**.
4. Give your policy a name. We recommend that organizations create a meaningful standard for the names of their policies.
5. Under **Assignments**, select **Users and groups**.
6. Under **Include**, select **Select users and groups > Users and groups > Choose the group you created**.
7. Select **Done**.
8. Under **Cloud apps or actions > Include**, select **Select apps**.
9. Select one of the following apps based on which version of Azure Virtual Desktop you're using. Choose **Azure Virtual Desktop** (App ID 9cdead84-a844-4324-93f2-b2e6bb768d07)
10. Go to **Conditions > Client apps**, then select where you want to apply the policy to:
 - Select **Browser** if you want the policy to apply to the web client.

- Select **Mobile apps and desktop clients** if you want to apply the policy to other clients.
- Select both check boxes if you want to apply the policy to all clients.

WVD-2FA-Accountants

Conditional access policy

 Delete

Control user access based on conditional access policy to bring signals together, to make decisions, and enforce organizational policies. [Learn more](#)

Name *

WVD-2FA-Accountants

Assignments

Users and groups ⓘ

Specific users included >

Cloud apps or actions ⓘ

1 app included >

Conditions ⓘ

1 condition selected >

Access controls

Grant ⓘ

1 control selected >

Session ⓘ

0 controls selected >

Control user access based on signals from conditions like risk, device platform, location, client apps, or device state. [Learn more](#)

Device platforms ⓘ >

Not configured

Locations ⓘ >

Not configured

Client apps ⓘ >

1 included

Device state (Preview) ⓘ >

Not configured

Client apps

×

Control user access to target specific client applications not using modern authentication. [Learn more](#)

Select the client apps this policy will apply to

Modern authentication clients

☐ Browser

☒ Mobile apps and desktop clients

Legacy authentication clients

☐ Exchange ActiveSync clients

☐ Other clients ⓘ

 Since this policy was created, the default client apps configuration has been updated.

11. Once you've selected your app, choose **Select**, and then select **Done**.

New [X]

Cloud apps or actions [X]

Info

Try out the new configuration experience. Click to enable the preview. →

Name *

wvdrsdemopolcy ✓

Assignments

Users and groups ⓘ >

Specific users included

Cloud apps or actions ⓘ >

No cloud apps or actions sele...

Conditions ⓘ >

0 conditions selected

Access controls

Grant ⓘ >

0 controls selected

Session ⓘ >

0 controls selected

Enable policy

Report-only On Off

Create

Select what this policy applies to

Cloud apps User actions

Include Exclude

☐ None

☐ All cloud apps

☒ Select apps

Select

Windows Virtual Desktop >

WV Windows Virtual Desktop
9cdead84-a844-4324-93f2-b2e6bb768...

Done

12. Under **Access controls > Grant**, select **Grant access, Require multi-factor authentication**, and then **Select**.
13. Under **Access controls > Session**, select **Sign-in frequency**, set the value to the time you want between prompts, and then select **Select**. For example, setting the value to **1** and the unit to **Hours**, will require multifactor authentication if a connection is launched an hour after the last one.
14. Confirm your settings and set **Enable policy** to **On**.
15. Select **Create** to enable your policy.

Manage security by using Azure Security Center

One thing that makes cloud services different from traditional on-premises virtual desktop infrastructures (VDIs) is how they handle security responsibilities.

For example, in a traditional on-premises VDI, the customer would be responsible for all aspects of security. However, in most cloud services, these responsibilities are shared between the customer and the company.

When you use Azure Virtual Desktop, it's important to understand that while some components come already secured for your environment, you'll need to configure other areas yourself to fit your organization's security needs.

Here are the security needs you're responsible for in your Azure Virtual Desktop deployment:

Security need	Is the customer responsible for this?
Identity	Yes
User devices (mobile and PC)	Yes
App security	Yes
Session host OS	Yes
Deployment configuration	Yes
Network controls	Yes
Virtualization control plane	No
Physical hosts	No
Physical network	No
Physical datacenter	No

The security needs the customer isn't responsible for are handled by Microsoft.

Microsoft recommends enabling Azure Security Center Standard for subscriptions, virtual machines, key vaults, and storage accounts.

With Azure Security Center Standard, you can:

- Manage vulnerabilities.
- Assess compliance with common frameworks like PCI.
- Strengthen the overall security of your environment.

Security posture management and threat protection

Azure Security Center provides security posture management and threat protection capabilities for Azure Virtual Desktop VMs in the following ways:

- Secure configuration assessment and Secure Score.
- Industry-tested vulnerability assessment.
- Host level detections.
- Agentless cloud network micro-segmentation & detection.
- File integrity monitoring.
- Just-in-time VM access.
- Adaptive Application Controls.

Using the Azure Security Center portal you can view Azure Virtual Desktop host pool VMs under **Inventory**:

Microsoft Azure

Search resources, services, and docs (G+)

Home > Security Center

Security Center | Inventory

Showing subscription 'Azure Pass - Sponsorship'

Search (Ctrl+/) Refresh Add non-Azure servers Open query Assign tags Download CSV report Trigger logic app

General

Filter by name Subscriptions == All Resource Groups == All Resource types == All Add filter

Overview
Getting started
Recommendations
Security alerts
Inventory
Workbooks
Community

Cloud Security
Secure Score
Regulatory compliance
Azure Defender
Firewall Manager

Management
Pricing & settings
Security policy
Security solutions
Workflow automation
Coverage
Cloud connectors

Total Resources 9 Unhealthy Resources 7 Unmonitored Resources 5 Unregistered subscriptions 0

Resource name ↑↓	Resource type ↑↓	Subscription ↑↓
Azure Pass - Sponsorship	Subscription	Azure Pass - Sponsorship
avd-vm-pool-3	Virtual machines	Azure Pass - Sponsorship
avd-vm-pool-2	Virtual machines	Azure Pass - Sponsorship
avd-vm-pool-1	Virtual machines	Azure Pass - Sponsorship
avd-vm-pool-0	Virtual machines	Azure Pass - Sponsorship
avd-demo-vm1	Virtual machines	Azure Pass - Sponsorship
cs4100320012e8bc6f1	Storage accounts	Azure Pass - Sponsorship
avd-dm-rg1-vnet	Virtual networks	Azure Pass - Sponsorship
default	Subnets	Azure Pass - Sponsorship

Choose a specific VM to view recommendations and Severity:

Home > Security Center >

Resource health

Is this new resource health page helpful to you? Yes No

avd-vm-pool-3
Virtual machine

Unmonitored Monitoring Active recommendations 9 Active alerts 0

Resource information

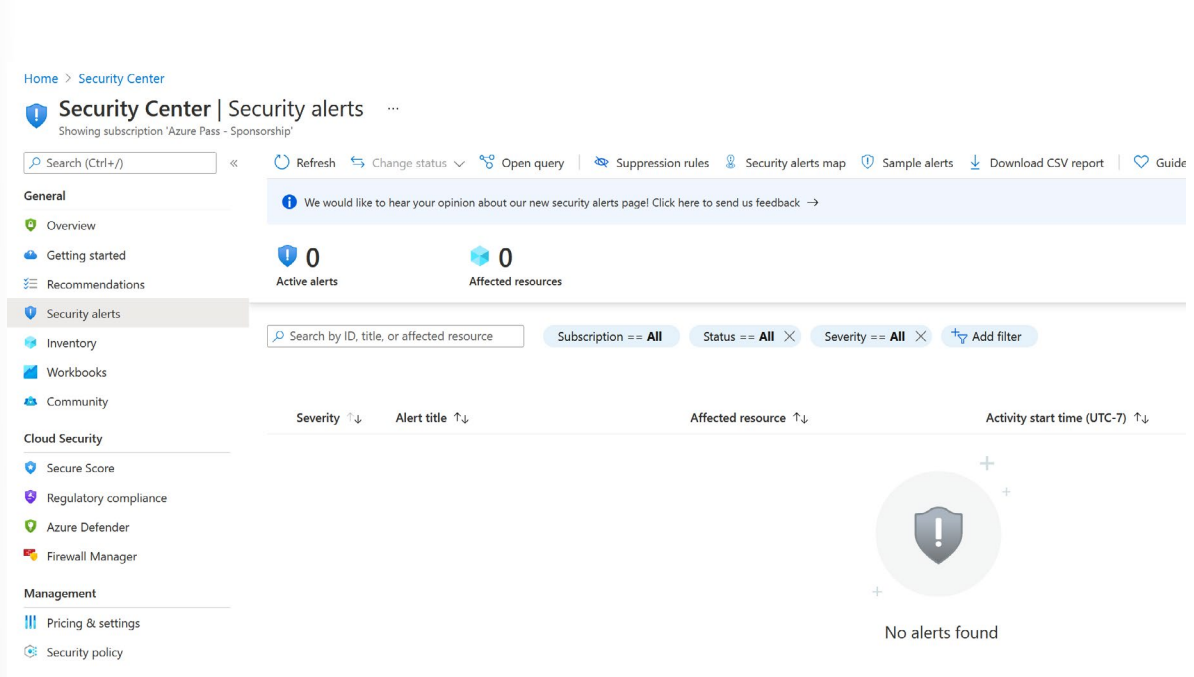
Subscription Azure Pass - Sponsorship Resource Group avd-dm-rg1
Environment Azure Location eastus
Operating System Windows Status VM running

Recommendations Alerts

Search Status == All Severity == All

Severity	Description	Status
High	Disk encryption should be applied on virtual machines	Unhealthy
High	Log Analytics agent should be installed on your virtual machine	Unhealthy
High	Install endpoint protection solution on virtual machines	Unhealthy
High	Windows web servers should be configured to use secure communication protocols Preview	Unhealthy
Medium	A vulnerability assessment solution should be enabled on your virtual machines	Unhealthy
Medium	Network traffic data collection agent should be installed on Windows virtual machines Preview	Unhealthy
Medium	Guest Configuration extension should be installed on your machines Preview	Unhealthy
Low	Non-internet-facing virtual machines should be protected with network security groups	Unhealthy
Low	Azure Backup should be enabled for virtual machines Preview	Unhealthy
High	Virtual machines should be migrated to new Azure Resource Manager resources	Healthy
Medium	Log Analytics agent health issues should be resolved on your machines	Healthy
Medium	IP forwarding on your virtual machine should be disabled	Healthy

Security alerts can be viewed under **General** and **Security Alerts**:



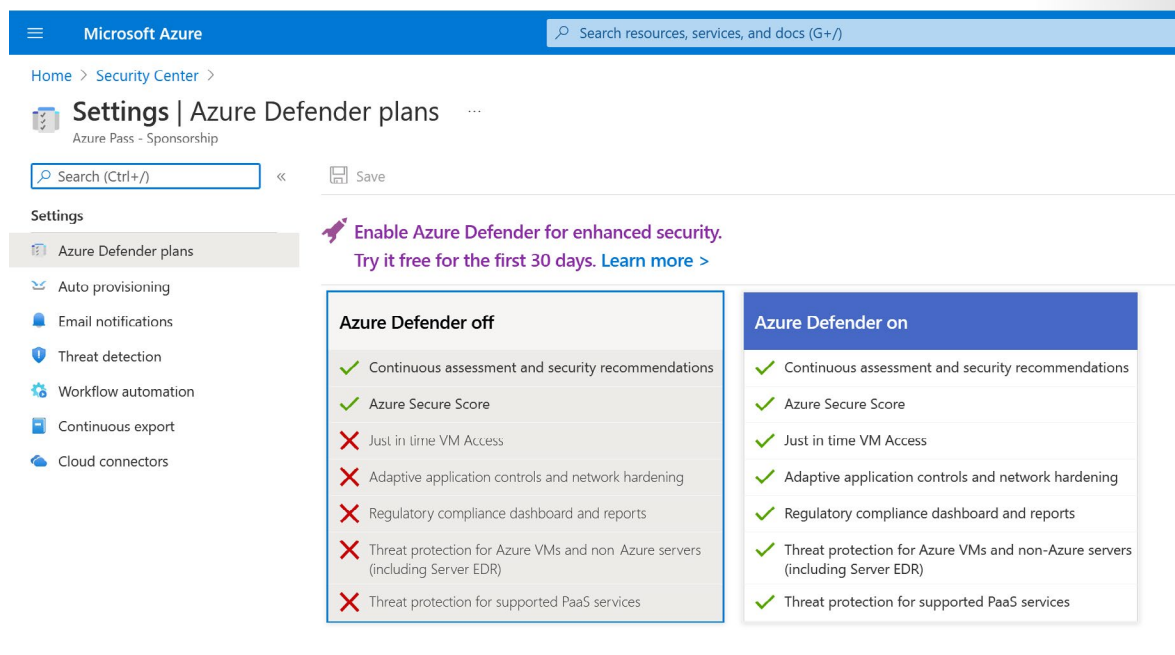
Security alerts and recommendations can be consumed and managed from the Security Center portal or exported to analysis tools for remediation.

Enabling Azure Security Center for Azure Virtual Desktop environment

Azure Security Center Free tier provides security recommendations and Secure Score for Azure Virtual Desktop deployments.

To enable all protection capabilities, do the following:

1. Make sure you have Azure Security Center Standard tier (as shown below).
2. Enable threat protection for Virtual Machines.



Microsoft Defender Antivirus for session hosts

Microsoft Defender for Endpoint is Microsoft's enterprise endpoint security platform which is created to help businesses to prevent, investigate, detect, and respond to threats. This serves to increase the level of security of your whole endpoint configuration. This applies to both virtual desktops running on the Cloud as part of Azure Virtual Desktop or on your physical Windows 10 Endpoints.

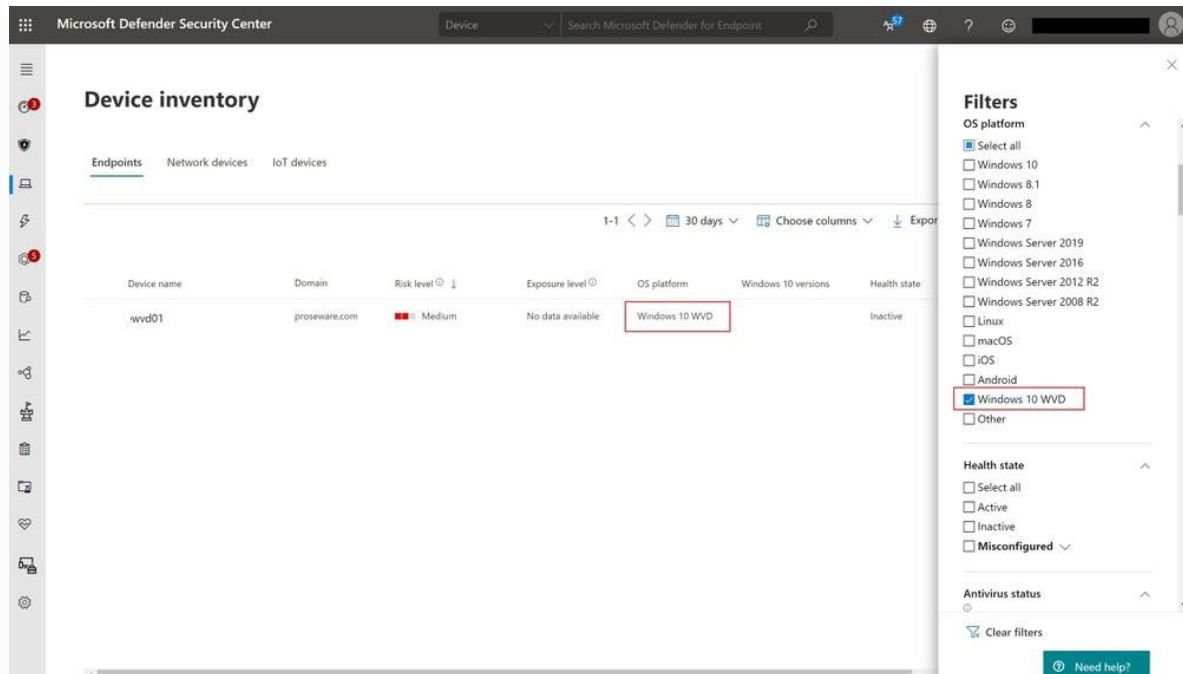
Defender for Endpoint supports Azure Virtual Desktop with up to 50 concurrent user connections for Windows 10 Enterprise multi-session.

Single session scenarios on Windows 10 Enterprise is fully supported for onboarding your Azure Virtual Desktop machines into Defender for Endpoint.

There are several new items in the Microsoft Defender Security Center that support Azure Virtual Desktop detailed in this topic.

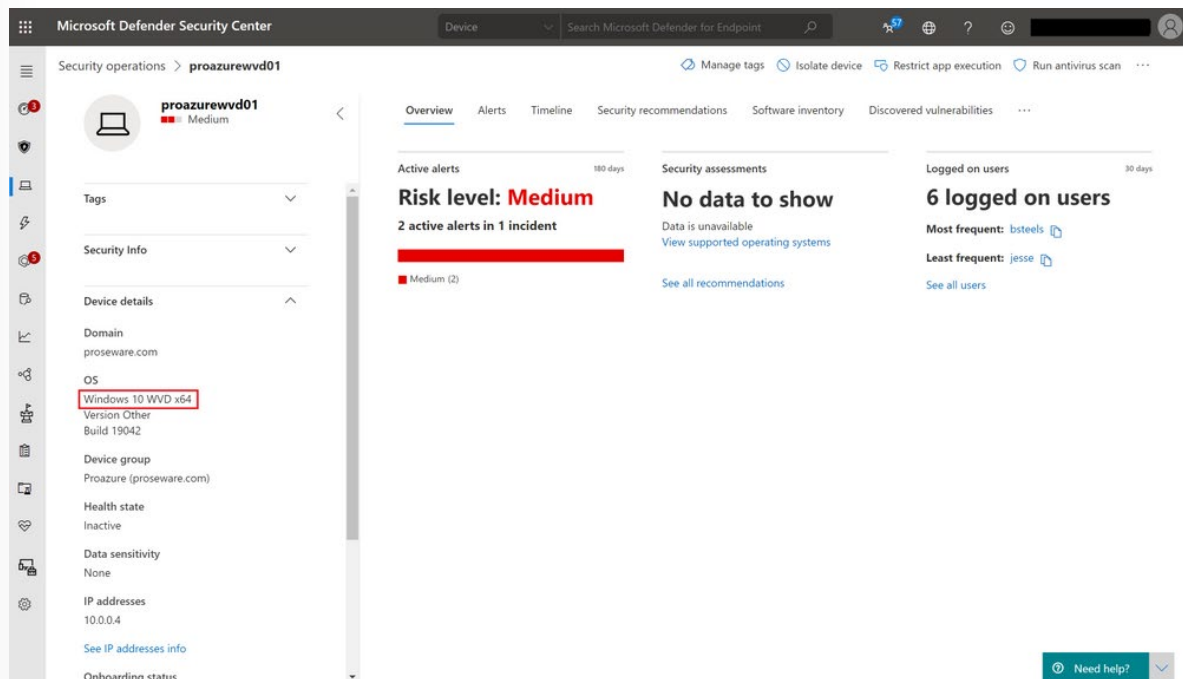
Device Inventory Page

On the device inventory page, select **filters** to see the **Windows 10 AVD** filter for viewing only Azure Virtual Desktop machines. You can identify Azure Virtual Desktop machines by looking for **Windows 10 AVD** in the OS platform column.



Device Page

On the device page, **Azure Virtual Desktop** is seen under the device details section. Under **OS** you'll see **Windows 10 AVD x64** indicating an Azure Virtual Desktop machine.



The device page also shows the number of logged on users in the past 30 days on the overview tab.

Selecting the **See all users** link allows you to view the complete list of users. You'll see a number of columns that include: **Logon Type**, **logon type 10**, and **RemoteInteractive**.

Close ×


Logged-on users (30 days)
proazurewvd01

Choose columns ▼

First seen	Last seen	User type	Logon type
1/12/21, 10:04 PM	1/12/21, 10:04 PM	Local admin	Network, RemoteInteract
1/12/21, 11:23 PM	1/12/21, 11:56 PM	Standard user	RemoteInteractive, Netw
1/12/21, 11:43 PM	1/12/21, 11:55 PM	Standard user	Network, RemoteInteract
1/12/21, 11:57 PM	1/12/21, 11:57 PM	Standard user	Network
1/12/21, 11:57 PM	1/12/21, 11:57 PM	Local admin	Network, RemoteInteract
1/12/21, 11:46 PM	1/12/21, 11:59 PM	Standard user	Network, RemoteInteract

Machine Timeline

The machine timeline is populated with telemetry information for all active user sessions on an Azure Virtual Desktop machine. This allows you to see all events happening on the machine and the ability to investigate timeline events that are specific to a particular user session. In the example below, there are events in the machine timeline for five users who are logged on concurrently to an Azure Virtual Desktop machine:

proazurewvd01
Manage tags Isolate device Restrict app execution Run antivirus scan Collect investigation p...

Overview Alerts Timeline Security recommendations Software inventory Discovered vulnerabilities Missing KBs

Aug 2020 Sep 2020 Oct 2020 Nov 2020 Dec 2020 Jan 2021

Export Search
1 week Choose columns ▼

Event time	Event	Additional information	User	Entities
Jan 13, 2021, 12:25:16.946 AM	msedge.exe remotely changed a region's memory protection for its child pro...	T1055.012: Process Hollowing	btarks	userinit.exe > explorer.exe > msedge.exe > msedge.exe
Jan 13, 2021, 12:25:16.946 AM	msedge.exe remotely changed a region's memory protection for its child pro...	T1055.012: Process Hollowing	btarks	explorer.exe > setup.exe > msedge.exe > msedge.exe
Jan 13, 2021, 12:25:16.946 AM	msedge.exe remotely changed a region's memory protection for its child pro...	T1055.012: Process Hollowing	jesse	svchost.exe > explorer.exe > msedge.exe > msedge.exe
Jan 13, 2021, 12:25:16.946 AM	msedge.exe remotely changed a region's memory protection for its child pro...	T1055.012: Process Hollowing	ungrers	userinit.exe > explorer.exe > msedge.exe > msedge.exe
Jan 13, 2021, 12:25:16.946 AM	msedge.exe remotely changed a region's memory protection for its child pro...	T1055.012: Process Hollowing	btels	userinit.exe > explorer.exe > msedge.exe > msedge.exe
Jan 13, 2021, 12:25:16.946 AM	msedge.exe remotely changed a region's memory protection for its child pro...	T1055.012: Process Hollowing	btels	userinit.exe > explorer.exe > msedge.exe > msedge.exe
Jan 13, 2021, 12:22:06.227 AM	msedge.exe dropped 6 files from the local system into a staging directory: "C:\...	T1074.001: Local Data Stealing T1005: Data from Local S...	btarks	userinit.exe > explorer.exe > msedge.exe > edge_checkout_page_validator.js
Jan 13, 2021, 12:13:20.507 AM	setup.exe opened the LNK file Microsoft EdgeLink	T1204.002: Malicious File	jesse	explorer.exe > msedge.exe > setup.exe > Microsoft EdgeLink
Jan 13, 2021, 12:13:07.014 AM	Windows run specified command on each user login on value name: ETASCC3...	T1112: Modify Registry	btels	explorer.exe > msedge.exe > msedge.exe > 5-1-5-21-265357268-211988757-3346622801-21031.SOF...
Jan 13, 2021, 12:13:06.486 AM	setup.exe opened the LNK file Microsoft EdgeLink	T1204.002: Malicious File	blanner	explorer.exe > msedge.exe > setup.exe > Microsoft EdgeLink
Jan 13, 2021, 12:10:53.988 AM	SDDHelper.exe has initiated a connection to https://newsrules.officeapps.liv...	T1071.001: Web Protocol T1573: Encrypted Channel	blanner	services.exe > svchost.exe > SDDHelper.exe > 52.109.8.19443
Jan 13, 2021, 12:09:50.592 AM	backgroundTaskHost.exe loaded fake ContentDeliveryManager.Background.d...	T1574.001: DLL Search G... T1036.005: Match Legit...	blanner	services.exe > svchost.exe > backgroundTaskHost.exe > ContentDeliveryManager.Background.d...
Jan 13, 2021, 12:05:16.974 AM	RuntimeBroker.exe deleted 169 files	T1070.004: File Deletion	jesse	services.exe > svchost.exe > RuntimeBroker.exe > Microsoft_S49861C3F910_Bvekyb3dbbwerApp

To see all activity related to a specific user, search for the username to display all telemetry:

Microsoft Defender Security Center

Security operations > proazurewvd01

Overview Alerts **Timeline** Security recommendations Software inventory Discovered vulnerabilities Missing KBs

Export **bbanner** 1/13/2021-1/13/2021 Choose columns Filters

Event time	Event	Additional information	User	Entities
Jan 13, 2021, 1:57:16.986 PM	backgroundTaskHost.exe successfully established connection with 40.126.7.1024...		bbanner	svchost.exe > back...
Jan 13, 2021, 12:57:08.670 PM	backgroundTaskHost.exe successfully established connection with 40.126.7.9944...		bbanner	svchost.exe > back...
Jan 13, 2021, 12:33:53.420 PM	backgroundTaskHost.exe successfully established connection with 131.253.33.20...		bbanner	svchost.exe > back...
Jan 13, 2021, 12:27:04.636 PM	backgroundTaskHost.exe successfully established connection with 40.126.7.9744...		bbanner	svchost.exe > back...
Jan 13, 2021, 12:12:51.919 PM	msedge.exe successfully established connection with 13.107.43.23443 (config.ed...		bbanner	msedge.exe > mse...
Jan 13, 2021, 10:55:51.158 AM	svchost.exe created process msioa.exe		system	services.exe > svch...
Jan 13, 2021, 10:55:51.157 AM	msioa.exe successfully established connection with 52.109.88.35443 (nexusrules...		bbanner	svchost.exe > msio...
Jan 13, 2021, 10:26:48.344 AM	backgroundTaskHost.exe successfully established connection with 20.190.151.68...		bbanner	svchost.exe > back...

Load newer results

Need help?

Incidents and Alerts

Below is a sample alert triggered for a user on an Azure Virtual Desktop machine:

Microsoft Defender Security Center

Search results > proazurewvd01 > Attempt to stop Microsoft Defender for Endpoin...

Attempt to stop Microsoft Defender for Endpoint sensor

proazurewvd01.proseware.com Risk level Medium ... proseware/srogers

ALERT STORY

- [3828] explorer.exe
- [16824] powershell.exe
- [15492] cmd.exe
- [6788] net.exe net stop sense
- net.exe is attempting to stop Unknown
- Attempt to stop Microsoft Defender for Endpoint ...** Medium New Detected
- [17024] net1.exe net1 stop sense
- net1.exe is attempting to stop Unknown
- Attempt to stop Microsoft Defender for End...** Medium New Detected

Collapse all

Details

Attempt to stop Microsoft Defender for Endpoint sensor

Medium New

See in timeline Link to another incident

Manage alert

Classify this alert True alert False alert

Status New

Classification Select classification...

Alert details

Incident Attempt to stop Microsoft Defender for Endpoint sensor on one endpoint (open in Microsoft 365 Defender)

Detection source EDR

Detection technology Behavioral

Detection status Detected

Need help?

Review questions

Module review questions

Review Question 1

You have an AVD session host with VMs.

You want to allow a group of user access to the VMs.

What should you do?

- ☐ A. Modify the RDP Properties for the host pool
- ☐ B. Create a role assignment
- ☐ C. Configure conditional access policies in Azure AD
- ☐ D. Configure and NSG

Review Question 2

You manage and on-premises network. You have a subscription that has:

- A virtual network
- an AVD host pool
- An Azure Firewall

The virtual network connects the on-premises network using site-to-site VPN.

You want to make sure that only users in the on-premises network connect to the AVD resources in the host pool.

What should you do?

- ☐ A. Configure a conditional access policy
- ☐ B. Run New-AzRoleAssignment cmdlet
- ☐ C. Run mstsc.exe
- ☐ D. Create and AppLocker policy

Answers

Review Question 1

You have an AVD session host with VMs.
You want to allow a group of user access to the VMs.
What should you do?

- ☐ A. Modify the RDP Properties for the host pool
- ☒ B. Create a role assignment
- ☐ C. Configure conditional access policies in Azure AD
- ☐ D. Configure and NSG

Explanation

The answer is B, Create a role assignment.

As seen in *Plan and implement Azure roles and RBAC for AVD*, Azure Virtual Desktop has a delegated access model that lets you define the amount of access a particular user can have by assigning them a role. A role assignment has three components: security principal, role definition, and scope. The Azure Virtual Desktop delegated access model is based on the Azure RBAC model. Azure Virtual Desktop delegated access supports user group access.

Review Question 2

You manage an on-premises network. You have a subscription that has:

- ☒ A. Configure a conditional access policy
- ☐ B. Run New-AzRoleAssignment cmdlet
- ☐ C. Run mstsc.exe
- ☐ D. Create and AppLocker policy

Explanation

The answer is A, Configure a conditional access policy.

As seen in topic *Understand Conditional Access policy components*, Conditional Access policy assignments define the conditions under which the policy will apply that provides session controls that enable limited experiences.

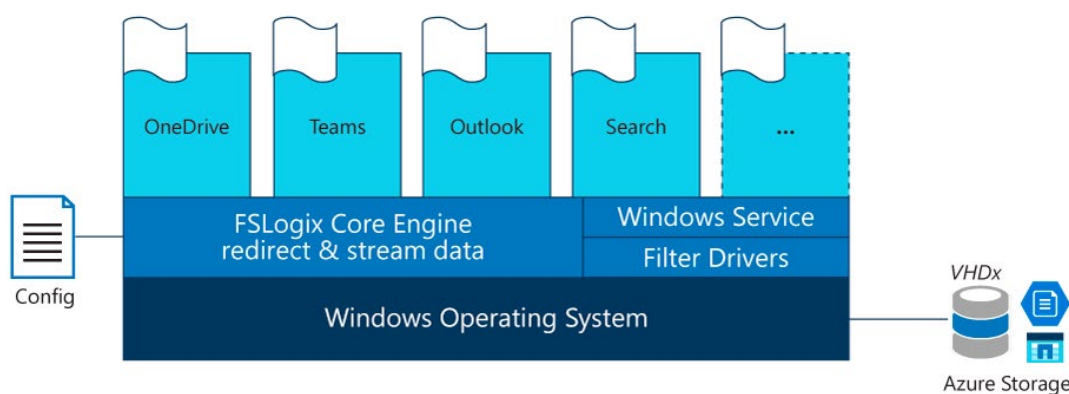
Module 4 Manage user environments and apps

Implement and manage FSLogix

Plan for FSLogix

The Azure Virtual Desktop service recommends FSLogix profile containers as a user profile solution. FSLogix is designed to roam profiles in remote computing environments, such as Azure Virtual Desktop. It stores a complete user profile in a single container.

At sign in, this container is dynamically attached to the computing environment using natively supported Virtual Hard Disk (VHD) and Hyper-V Virtual Hard disk (VHDX). The user profile is immediately available and appears in the system exactly like a native user profile.



User profiles

A user profile contains data elements about an individual, including configuration information like desktop settings, persistent network connections, and application settings. By default, Windows creates a local user profile that is tightly integrated with the operating system.

A remote user profile provides a partition between user data and the operating system. It allows the operating system to be replaced or changed without affecting the user data. In Remote Desktop Session Host (RDSH) and Virtual Desktop Infrastructures (VDI), the operating system may be replaced for the following reasons:

- An upgrade of the operating system
- A replacement of an existing Virtual Machine (VM)
- A user being part of a pooled (non-persistent) RDSH or VDI environment

FSLogix profile containers and Azure files

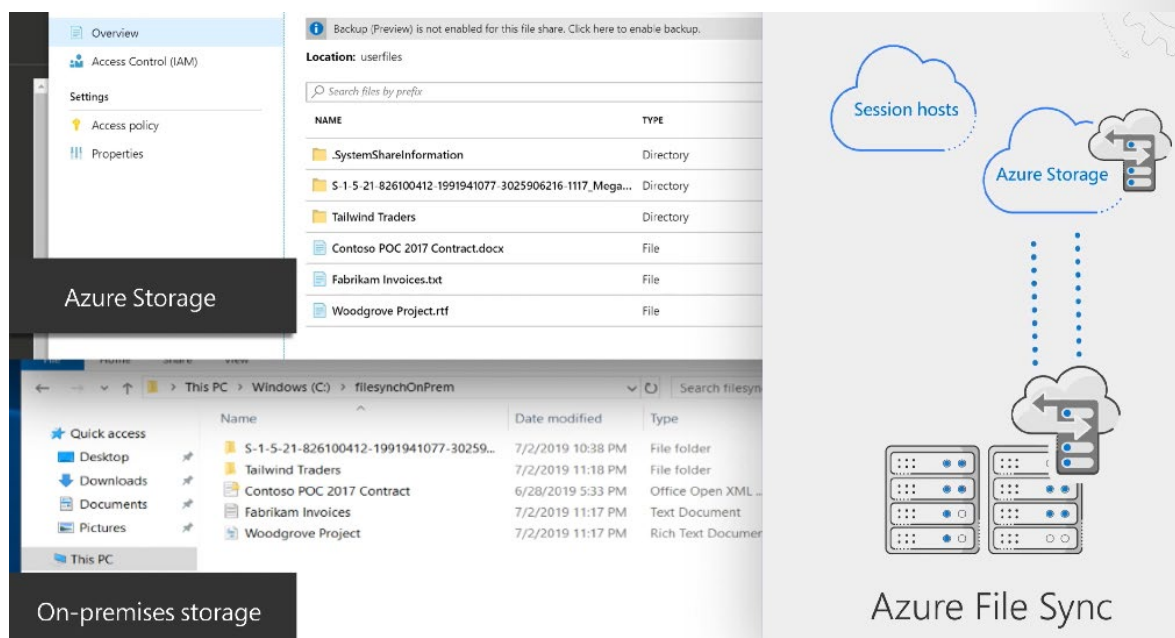
FSLogix addresses many profile container challenges. Key among them are:

- **Performance:** The FSLogix profile containers are high performance and resolve performance issues that have historically blocked cached exchange mode.
- **OneDrive:** Without FSLogix profile containers, OneDrive for Business is not supported in non-persistent RDSH or VDI environments.
- **Additional folders:** FSLogix provides the ability to extend user profiles to include additional folders.

NOTE: There is an online lab later in this module where you configure Azure Files to store profile containers for Azure Virtual Desktop in Azure AD DS environment, and, implement FSLogix based profiles for Azure Virtual Desktop in Azure AD DS environment (*Lab - Implement and manage Azure Virtual Desktop profiles (Azure AD DS)*). This topic is meant to provide an overview of the procedure.

Azure Files integration with Azure Active Directory Domain Service

FSLogix profile containers' performance and features take advantage of the cloud using Azure Files authentication with Azure Active Directory Domain Service (AD DS). By addressing both cost and administrative overhead, Azure Files with Azure AD DS Authentication is a premium solution for user profiles in the Azure Virtual Desktop service.



Best practices for Azure Virtual Desktop

Azure Virtual Desktop offers full control over size, type, and count of VMs that are being used by customers.

To ensure your Azure Virtual Desktop environment follows best practices:

- Azure Files storage account must be in the same region as the session host VMs.
- Azure Files permissions should match permissions described in **Requirements - Profile Containers¹**.
- Each host pool VM must be built of the same type and size VM based on the same master image.
- Each host pool VM must be in the same resource group to aid management, scaling and updating.
- For optimal performance, the storage solution and the FSLogix profile container should be in the same data center location.
- The storage account containing the master image must be in the same region and subscription where the VMs are being provisioned.

Install FSLogix

This topic describes how to download and install FSLogix tools.

The FSLogix software no longer requires license keys. It is recommended that the latest version of FSLogix is downloaded and installed. If legacy FSLogix customers must continue to use an older version, the following key may be used **MSFT0-YXKIX-NVQI4-I6WIA-O4TXE**. All users must be appropriately entitled and agree to license terms before using FSLogix.

¹ <https://docs.microsoft.com/en-us/fslogix/fslogix-storage-config-ht>

Download FSLogix

FSLogix is available for download [here](#)²

Install Microsoft FSLogix components

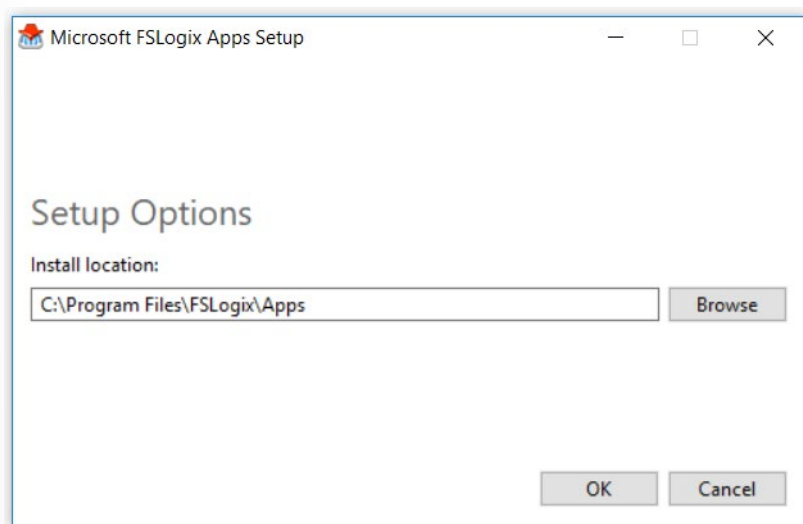
The download for FSLogix includes three installers that are used to install the specific component(s) necessary for your use.

Microsoft FSLogix Apps Installation

Microsoft FSLogix Apps installs the core drivers and components for all FSLogix solutions. Any environment using FSLogix must install FSLogix Apps. After installation configure Profile Container before using for profile redirection.

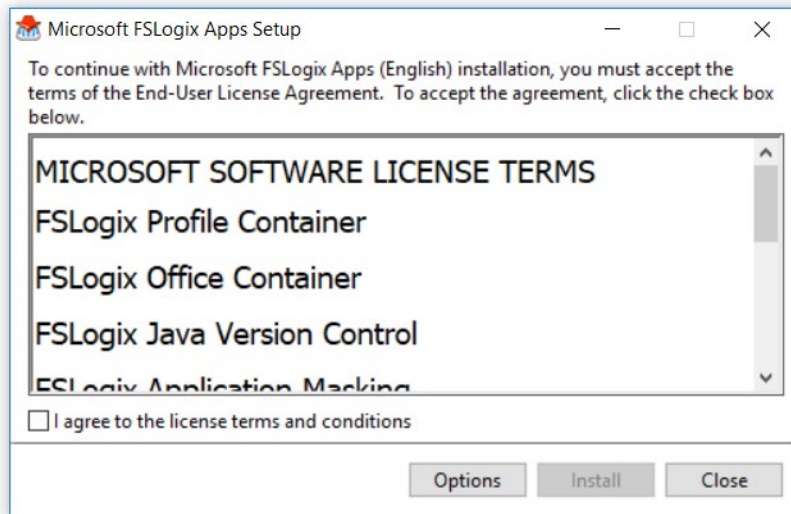
To install FSLogix Applications:

- From the FSLogix download file, select 32 bit or 64 bit depending on your environment
- Run `FSLogixAppSetup.exe`
- Click **Options** to specify an installation folder

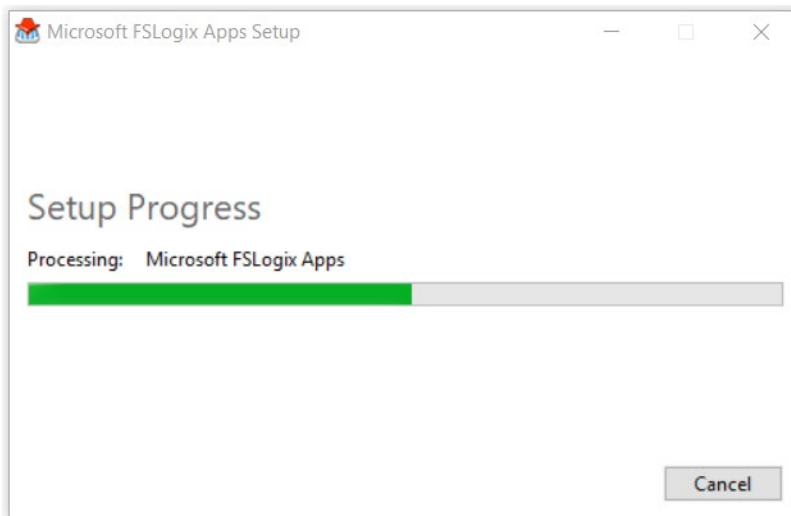


- Accept the license agreement and click **Install**

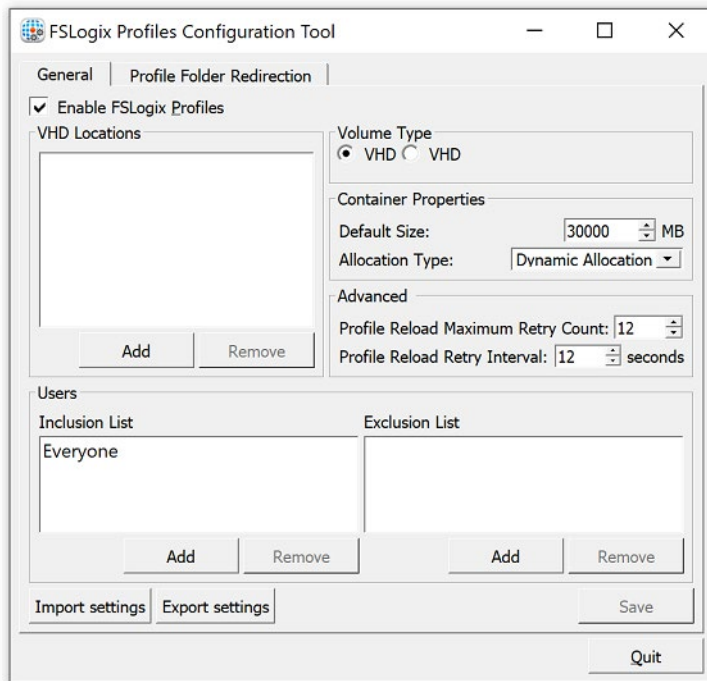
² https://aka.ms/fslogix_download



- Microsoft FSLogix Apps will install



To view the FSLogix Configuration Tool, check `\Program Files\FSLogix\Apps\ConfigurationTool.exe`.



Application Masking Rule Editor Installation

The Application Masking Rule Editor is used to define rules used by **Application Masking**³.

- From the FSLogix Download file, select 32 bit or 64 bit depending on your environment
- Run `FSLogixAppsRuleEditorSetup.exe`
- Use **Options** to specify installation folder (see screenshot for Microsoft FSLogix Apps above)
- Accept the license agreement and click **Install**

Storage options for FSLogix profile containers

This section compares storage solutions that Azure offers for Azure Virtual Desktop FSLogix user profile containers.

As mentioned in Module 2, it's best to store FSLogix profile containers on Azure Files.

FSLogix is designed to roam profiles in remote computing environments, such as Azure Virtual Desktop.

- At sign-in, this container is dynamically attached to the computing environment using a natively supported Virtual Hard Disk (VHD) and a Hyper-V Virtual Hard Disk (VHDX).
- The user profile is immediately available and appears in the system exactly like a native user profile.

The following tables compare the storage solutions Azure Storage offers for Azure Virtual Desktop FSLogix profile container user profiles.

³ <https://docs.microsoft.com/en-us/fslogix/implement-application-masking-tutorial>

Azure platform details

Features	Azure Files	Azure NetApp Files	Storage Spaces Direct
Use case	General purpose	Ultra performance or migration from NetApp on-premises	Cross-platform
Platform service	Yes, Azure-native solution	Yes, Azure-native solution	No, self-managed
Regional availability	All regions	Select regions	All regions
Redundancy	Locally redundant/ zone-redundant/ geo-redundant	Locally redundant	Locally redundant/ zone-redundant/ geo-redundant
Tiers and performance	Standard Premium Up to max 100k IOPS per share with 5 GBps per share at about 3 ms latency	Standard Premium Ultra Up to 320k (16K) IOPS with 4.5 GBps per volume at about 1 ms latency	Standard HDD: up to 500 IOPS per-disk limits Standard SSD: up to 4k IOPS per-disk limits Premium SSD: up to 20k IOPS per-disk limits We recommend Premium disks for Storage Spaces Direct
Capacity	100 TiB per share	100 TiB per volume, up to 12.5 PiB per subscription	Maximum 32 TiB per disk
Required infrastructure	Minimum share size 1 GiB	Minimum capacity pool 4 TiB, min volume size 100 GiB	Two VMs on Azure IaaS (+ Cloud Witness) or at least three VMs without and costs for disks
Protocols	SMB 2.1/3. and REST	NFSv3, NFSv4.1 (pre-view), SMB 3.x/2.x	NFSv3, NFSv4.1, SMB 3.1

Profile Container vs Office Container

It's important to understand the differences between Profile Container and Office Container for proper use and maximum benefit.

Office Container is a subset of Profile Container. Although all of the benefits of Office Container are also delivered from Profile Container, there are times when it may be beneficial to use them together.

Profile Container and Office Container are configured differently. It's important to completely understand the configuration process, especially when using them together.

Profile Container

Profile Container is used to redirect the full user profile. Profile Container is used in non-persistent, virtual environments, such as Virtual Desktops. When using Profile Container the entire user profile, except for data that is excluded using the redirections.xml, is included in the profile container.

For users familiar with managing profiles in non-persistent environments, the function of Profile Container may be compared to Microsoft User Profile Disk, Microsoft Roaming Profiles, or Citrix UPM.

Office Container

Office Container is generally implemented with another profile solution, and is designed to improve the performance of Microsoft Office in non-persistent environments. As opposed to Profile Container, Office Container redirects only the local user files for Microsoft Office. When configuring Office Container, each Office component is independently included based on the selected settings to include data for specific office components.

When Office Container is used with other profile solutions, it's that those solutions are configured to exclude certain data.

The data contained in the Office Container can be re-created from various server locations. As an example, the .OST file is generated from the email server(S), if the file is lost or damaged it may be recovered.

Using Profile Container and Office Container together

There are several reasons why Profile Container and Office Container may be used together. The most common reasons are:

- Discretion is wanted in the storage location for Office Data vs. other profile data
- If the Office Container or Profile Container is damaged, the remaining data remains intact. Storage discretion is useful if there is a problem with Office Data, which can be recovered from the server as the Office Container can be deleted without impacting the rest of the user configuration.
- Office Container may be used with Profile Container as a mechanism to specify which Office components will have their data included in the container

Configure Cloud Cache

Cloud Cache is an optional add-on to Profile Container and Office Container.

In this topic, you learn how to:

- Configure Cloud Cache for SMB
- Configure Cloud Cache for Azure Page Blobs

Prior to the configuration, make sure you've done the following.

- Installed FSLogix
- Verified that users have appropriate access to network file storage
- If using Azure Page Blobs for Cloud Cache verify that an appropriate Azure Storage service is being used.

Configure Cloud Cache for SMB

To configure Cloud Cache for Profile Container, refer to the following.

All settings are applied to **HKLM\SOFTWARE\FSLogix\Profiles**.

Add or verify:

Configuring Cloud Cache for Profile Container		
Registry Value	Type	Value

Configuring Cloud Cache for Profile Container		
CCDLocations	REG_SZ / MULTI_SZ	type=smb,connectionString=<\Location1\Folder1>;type=smb,-connectionString=<\Location2\folder2>
Enabled	DWORD	1

Parameters include:

- <Location for Cloud Cache Provider>
- Each Provider is separated by ;
- This sample is for two SMB Providers

Configuring Cloud Cache for Office Container

All settings are applied to **HKLM\SOFTWARE\Policies\FSTLogix\ODFC**.

Remove any setting for VHDLocations.

Add or verify the following:

Registry Value	Type	Value
CCDLocations	REG_SZ / MULTI_SZ	type=smb,connectionString=<\Location1\Folder1>;type=smb,-connectionString=<\Location2\folder2>
Enabled	DWORD	1

- <Location for Cloud Cache Provider>
- Each provider is separated by ;

Configure Cloud Cache for Azure Page Blobs

The instructions in the section below provide an understanding of the conceptual use of Cloud Cache with Page Blobs. The instructions in this section would expose sensitive Azure credentials to any user with access to the host registry, if implemented in production. When creating a production implementation, care must be used to protect these credentials.

Configuring Cloud Cache for Profile Container

All settings are applied to **HKLM\SOFTWARE\FSTLogix\Profiles**.

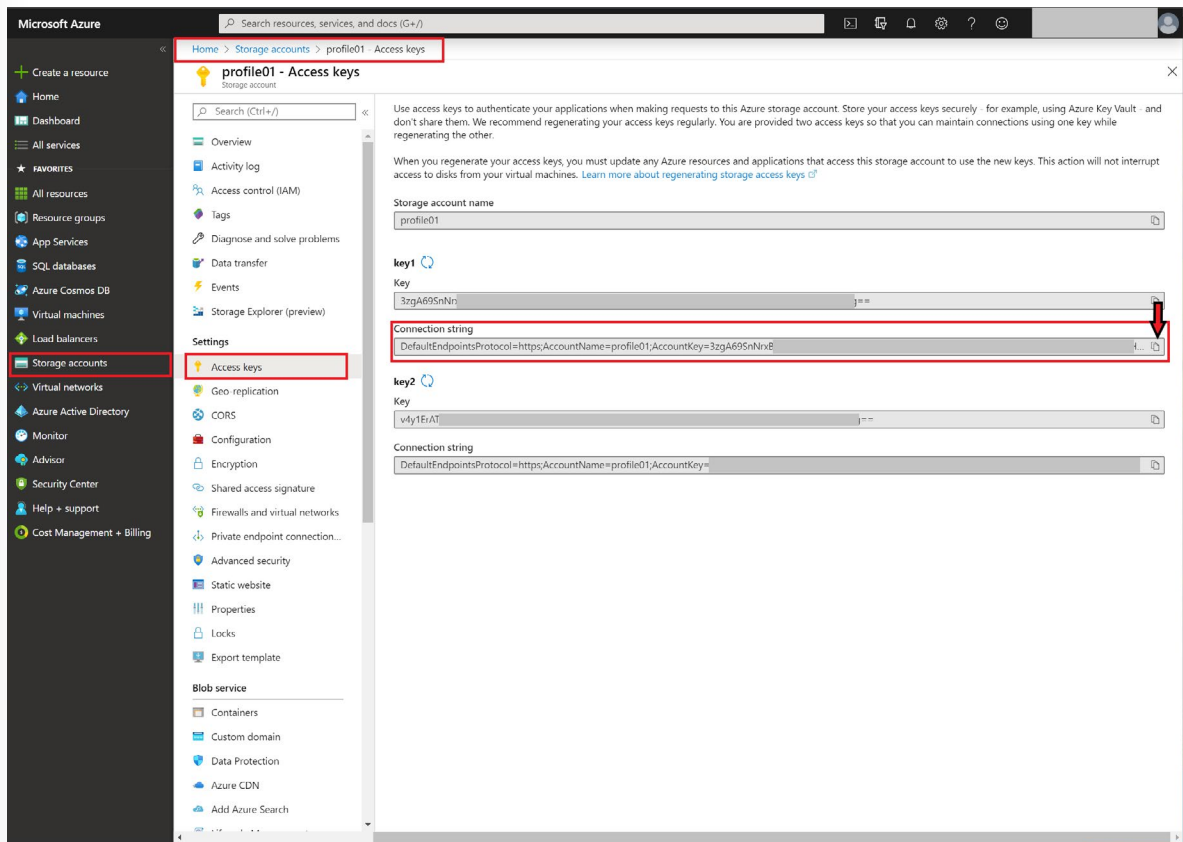
Remove any settings for VHDLocations

Add or verify the following:

Registry Value	Type	Value
CCDLocations	REG_SZ / MULTI_SZ	type=azure,connectionString="-DefaultEndpointsProto-col=https;AccountName=;AccountKey=;EndpointSuffix="

Registry Value	Type	Value
Enabled	DWORD	1

- <Location for Cloud Cache Provider>
- Each provider is separated by ;
- This sample is for one Azure Page Blob provider
- Page Blob connection string should be enclosed in ""
- These settings are used to create the Azure Connection String:
- DefaultEndpointsProtocol=[http or https]
- AccountName=myAccountName
- AccountKey=myAccountKey
- EndpointSuffix=mySuffix
- The following screenshot shows where to find the payload for the connection string in the Azure portal:



Azure Account Keys are sensitive and may be protected using Credential Manager.

Configure Profile Containers

Profile Container is a full remote profile solution for non-persistent environments. Profile Container redirects the entire user profile to a remote location. Profile Container configuration defines how and where the profile is redirected.

Profile Container is inclusive of the benefits found in Office Container.

When using Profile Container, both applications and users see the profile as if it's located on the local drive.

In this topic, learn how to:

- Configure Profile Container Registry Settings
- Set up Include and Exclude User Groups

Before configuring Profile Container:

- **Download and install**⁴ FSLogix Software
- Consider the storage and network requirements for your users' profiles
- Verify that your users have appropriate storage permissions where profiles will be placed
- Profile Container is installed and configured after stopping use of other solutions used to manage remote profiles
- Exclude the VHD(X) files for Profile Containers from Anti Virus (AV) scanning

Configure Profile Container Registry settings

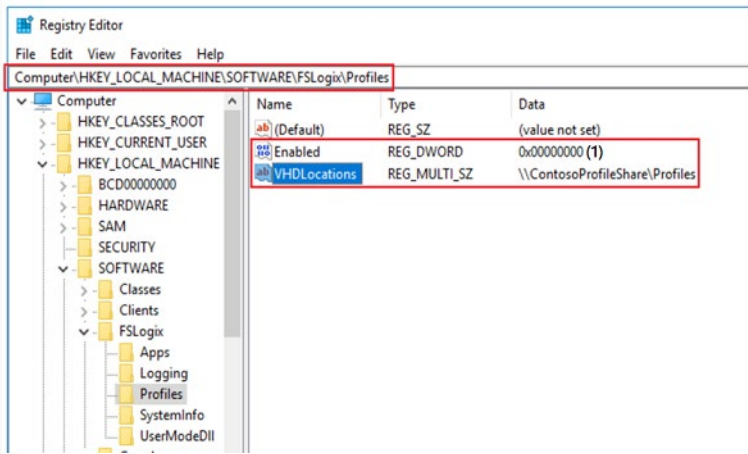
The configuration of Profile Container is accomplished through registry settings and user groups. Registry settings may be managed manually, with GPOs, or using alternate preferred methods. Configuration settings for Profile Container are set in **HKLM\SOFTWARE\FSLogix\Profiles**.

Below are settings are **required** to enable Profile Container and to specify the location for the profile VHD to be stored. The minimum required settings to enable Profile Containers are:

Value	Type	Configured Value	Description
Enabled (required setting)	DWORD	1	0: Profile Containers disabled. 1: Profile Containers enabled

⁴ <https://docs.microsoft.com/en-us/fslogix/install-ht>

Value	Type	Configured Value	Description
VHDLocations (required setting)	MULTI_SZ or REG_SZ		A list of file system locations to search for the user's profile VHD(X) file. If one isn't found, one will be created in the first listed location. If the VHD path doesn't exist, it will be created before it checks if a VHD(X) exists in the path. These values can contain variables that will be resolved. Supported variables are %user-name%, %userdo-main%, %sid%, %osma-jor%, %osminor%, %osbuild%, %osservice-pack%, %profilever-sion%, and any environ-ment variable available at time of use.



VHDLocations may be replaced by CCDLocations when using Cloud Cache.

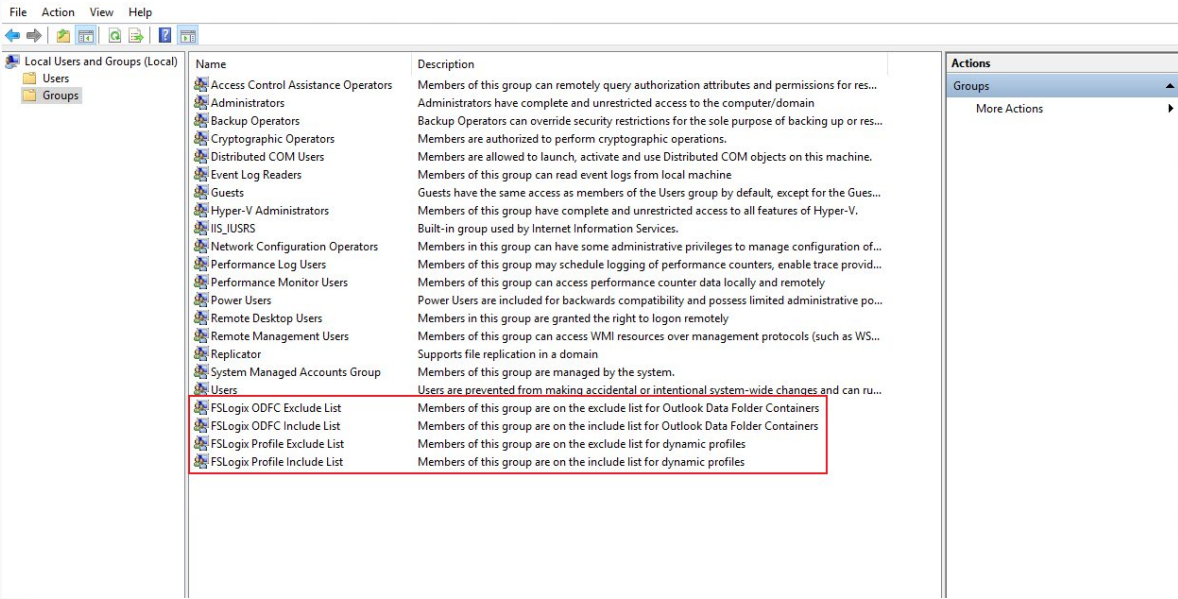
These settings are often helpful when configuring Profile Container but are not required.

Value	Type	Configured Value	Description
DeleteLocalProfile-WhenVHDShouldApply	DWORD	0	0: no deletion. 1: delete local profile if exists and matches the profile being loaded from VHD. NOTE: Use caution with this setting. When the FSLogix Profiles system determines a user should have a FSLogix profile, but a local profile exists, Profile Container permanently deletes the local profile. The user will then be signed in with an FSLogix profile.
FlipFlopProfileDirectoryName	DWORD	0	When set to '1' the SID folder is created as "%username%%sid%" instead of the default "%sid%%username%". This setting has the same effect as setting SIDDirNamePattern = "%username%%sid%" and SIDDirNameMatch = "%username%%sid%".
PreventLoginWithFailure	DWORD	0	If set to 1 Profile Container will load FRXShell if there's a failure attaching to, or using an existing profile VHD(X). The user will receive the FRXShell prompt - default prompt to call support, and the users only option will be to sign out.

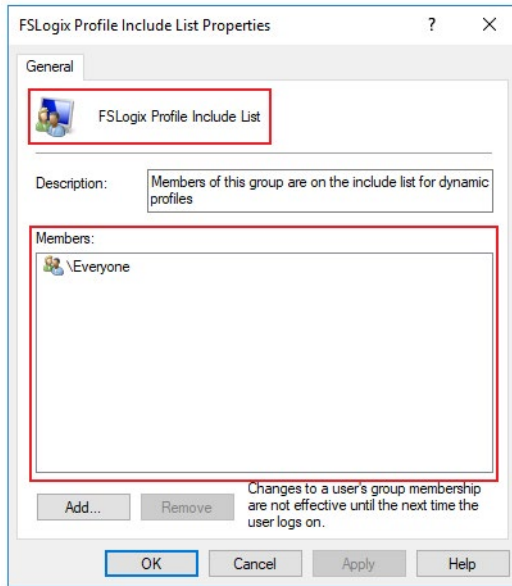
Value	Type	Configured Value	Description
PreventLoginWithTemp-Profile	DWORD	0	If set to 1 Profile Container will load FRXShell if it's determined a temp profile has been created. The user will receive the FRXShell prompt - default prompt to call support, and the users only option will be to sign out.

Set up Include and Exclude User Groups

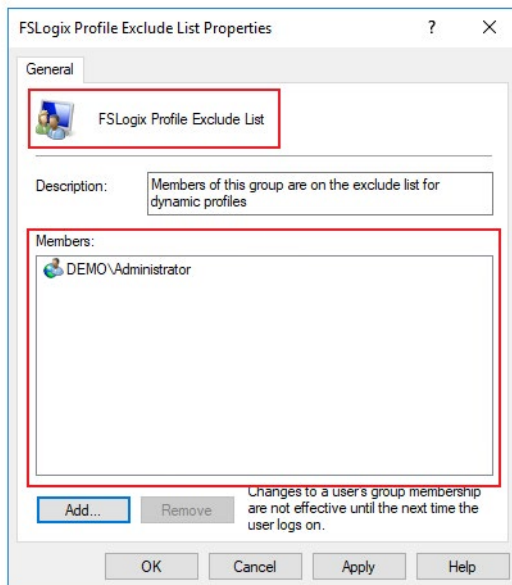
There are often users, such as local administrators, that have profiles that should remain local. During installation, four user groups are created to manage users who's profiles are included and excluded from Profile Container and Office Container redirection.



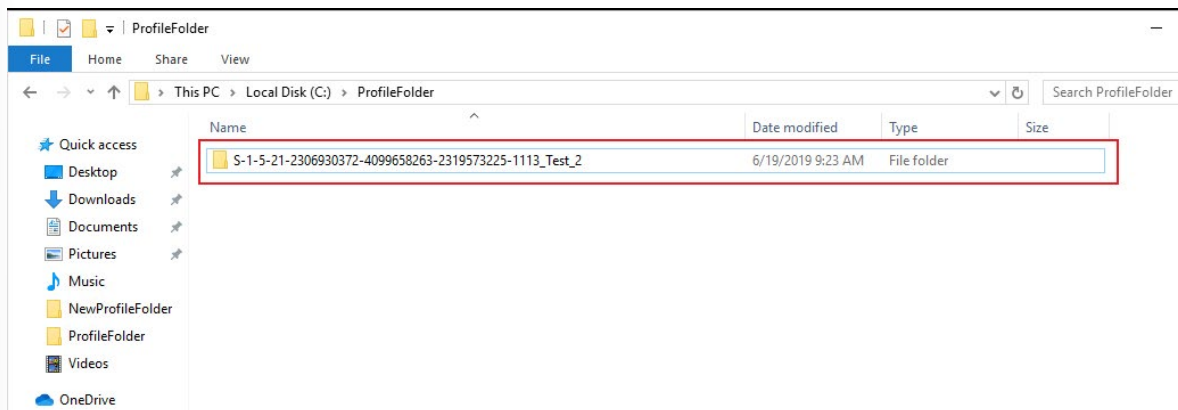
By default Everyone is added to the FSLogix Profile Include List group.



Adding a user to the FSLogix Profile Exclude List group means that the FSLogix agent will not attach a FSLogix profile container for the user. In the case where a user is a member of both the exclude and include groups, exclude takes priority.



Profile Containers is now configured and ready to be used. In order to verify that Profile Container is working, sign in as a user in the Included List group. Using File Manager, navigate to the location specified in VHDLocations. Verify that a folder, with the user name and SID has been created.



Manage Rule Sets and application masking

Application Masking manages access to Applications, Fonts, and other items based on criteria. The Application Rules Editor is used to Describe the item, such as application, to be managed. The Editor is also used to define criteria rules are managed by.

Things you can do with the Apps Rules Editor:

- Create new Rule Sets
- Edit existing Rule Sets
- Manage the user and group assignments for Rule Sets
- Temporarily test rule-sets

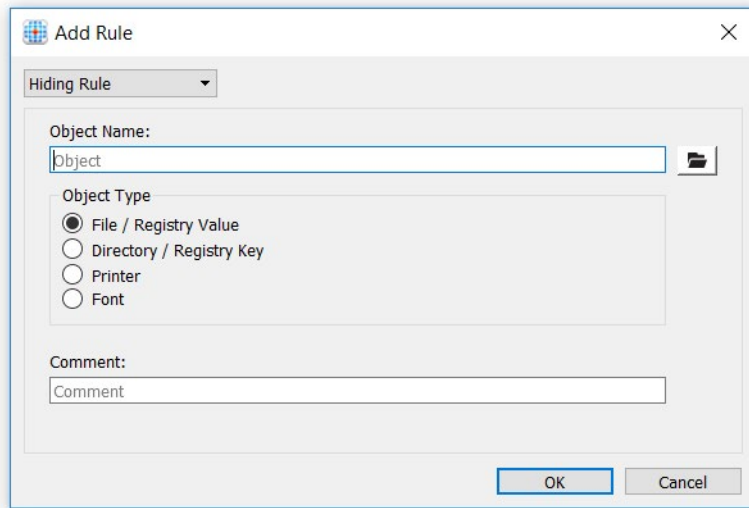
Before using the Application Rules Editor, it must be **installed**⁵

Rule Types

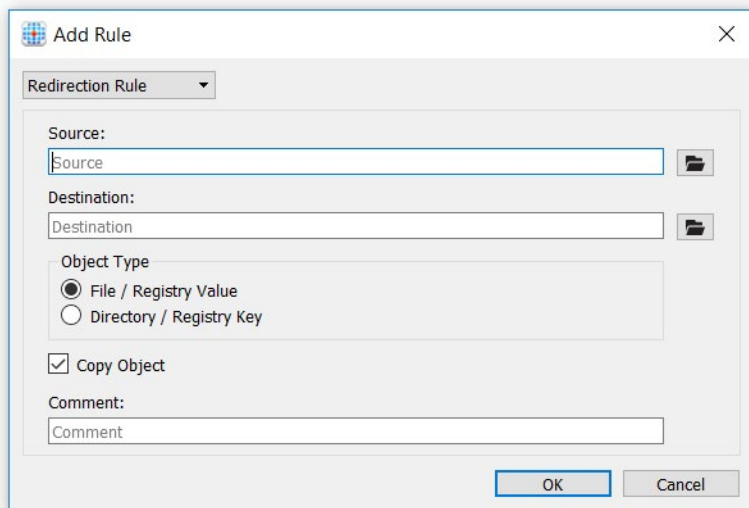
FSlogix supports four rule types:

Hiding Rule - hides the specified items using specified criteria

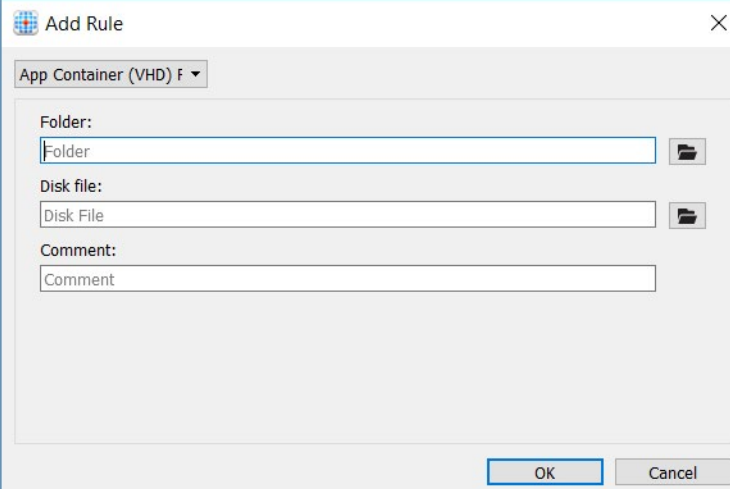
⁵ <https://docs.microsoft.com/en-us/fslogix/install-ht>



Redirect Rule - causes the specified item to be redirected as defined

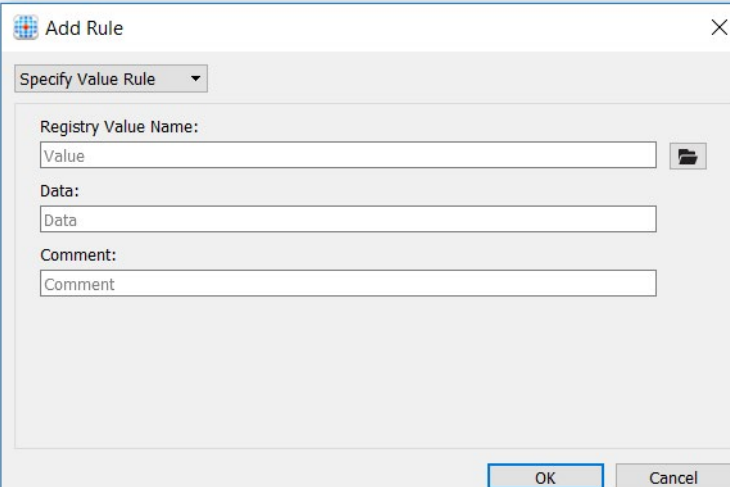


App Container Rule - redirects the specified content into a VHD



The 'Add Rule' dialog box is shown with the 'App Container (VHD)' dropdown menu selected. It contains three input fields: 'Folder' with the text 'Folder', 'Disk file' with the text 'Disk File', and 'Comment' with the text 'Comment'. Each input field has a folder icon to its right. At the bottom right are 'OK' and 'Cancel' buttons.

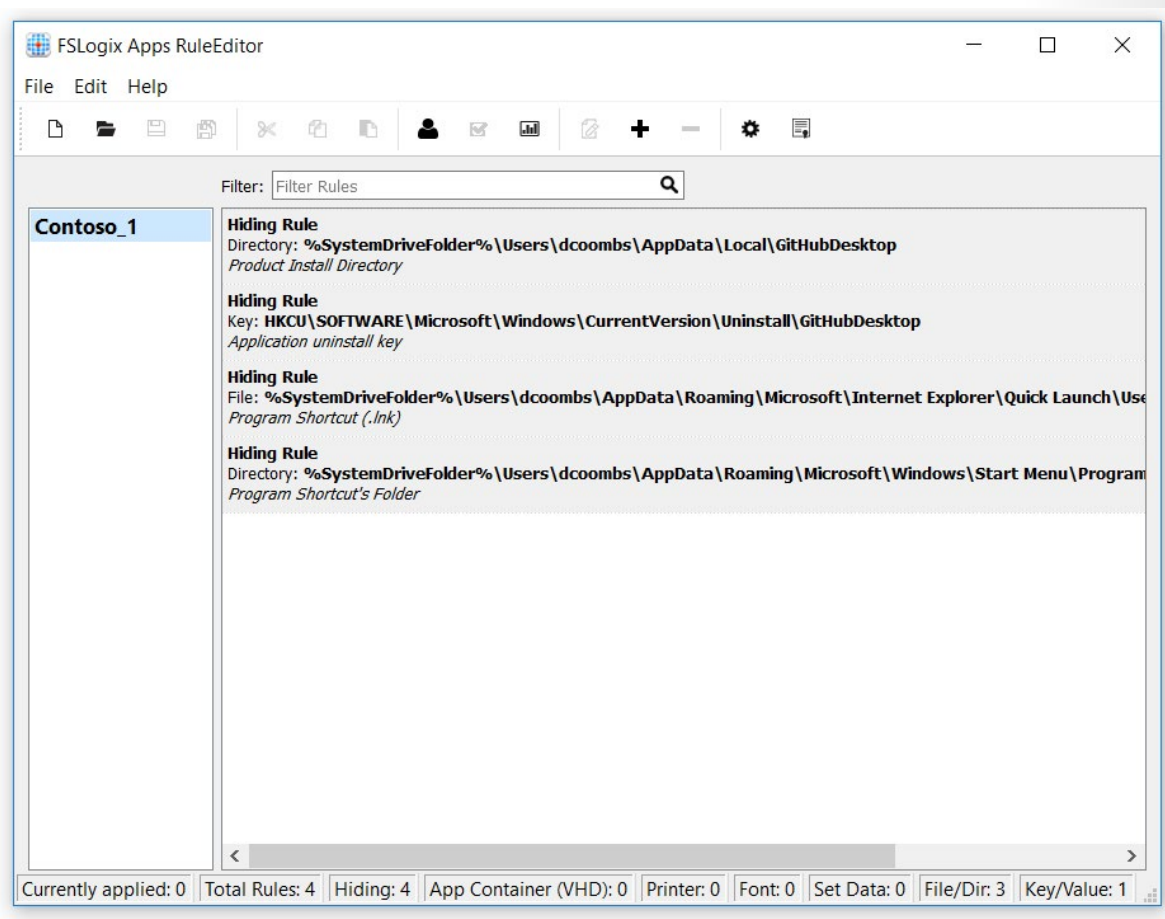
Specify Value Rule - assigns a value for the specified item



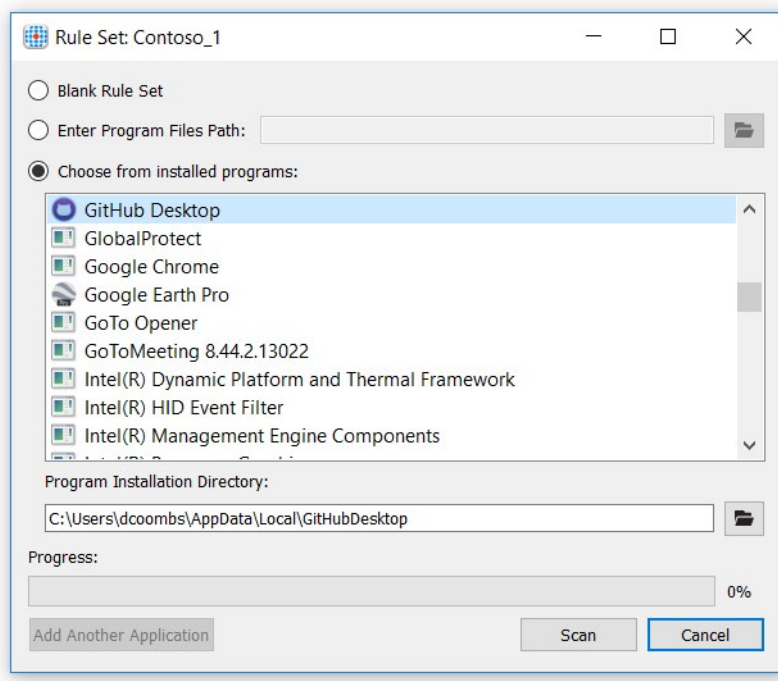
The 'Add Rule' dialog box is shown with the 'Specify Value Rule' dropdown menu selected. It contains three input fields: 'Registry Value Name' with the text 'Value', 'Data' with the text 'Data', and 'Comment' with the text 'Comment'. Each input field has a folder icon to its right. At the bottom right are 'OK' and 'Cancel' buttons.

Create a new Rule Set

1. Open the Apps Rule Editor. The first time you enter the Apps Rules Editor there won't be any rule sets in the left panel. In this example, one rule set has already been created named Contoso_1 with GitHub Desktop added.



2. Click **File** then **New** to create a new Rule Set
3. Provide a name for the Rule Set and click **Enter Filename**
4. After a filename is entered, a selection is made for the type and content of the rule
5. After specifying the parameters wanted, click **Scan** to create a rule (In this example, GitHub Desktop is selected)



Create a new rule

- Select an existing Rule Set from the left panel
- Select **Edit** then **New Rule**
- Specify the **type of rule**⁶
- Enter the required parameters
- Click **OK**

Delete a rule

- Select an existing Rule Set from the left panel
- Select one or more Rules from the right panel
- Select **Edit** then Delete Rule

Edit a rule

- Select an existing Rule Set from the left panel
- Select an existing Rule from the right panel
- Select **Edit** then Edit Rule

⁶ <https://docs.microsoft.com/en-us/fslogix/application-masking-rules-ht>

Redirecting to a network

Files and directories can be redirected to resources located on a network. The user must have appropriate rights to the network resource. To redirect to a network location, enter the path (in UNC format) into the Destination field.

Deploying Rule Sets

Application Masking and Java Version Control rely on Rules and Rule Sets.

By default, Rules and Rule Sets are accessed from **C:\Program Files\FSLogix\Apps\Rules**.

The location where Rules and Rule Sets are accessed differ if the FSLogix installation location is changed.

To deploy a rule set, use any method to copy rule files (.fxr) and assignment files (.fxa) to the rules directory.

Configure user experience settings

Virtual desktop optimization principles

The guiding principles to optimize performance of Windows 10 in a virtual desktop environment are to minimize graphic redraws and effects, background activities that have no major benefit to the virtual desktop environment, and generally reduce running processes to the bare minimum.

A secondary goal is to reduce disk space usage in the base image to the bare minimum.

With virtual desktop implementations, the smallest possible base, or “gold” image size, can slightly reduce memory utilization on the host system, as well as a small reduction in overall network operations required to deliver the desktop environment to the consumer.

Optimization principles

Some implementations of virtual desktop environments use a “base” operating system image, which then becomes the basis for the desktops subsequently presented to the users for work. There are persistent and non-persistent base images.

The persistent type preserves changes to the virtual desktop operating system from one session to the next. The non-persistent type does not preserve changes to the virtual desktop operating system from one session to the next. To the user this desktop is little different than other virtual or physical device, other than it is accessed over a network.

The optimization settings could take place on a reference machine. A virtual machine (VM) would be an ideal place to build the VM, because state can be saved, checkpoints can be made, backups can be made, and so on. A default OS installation is performed to the base VM. That base VM is then optimized by removing unneeded apps, installing Windows updates, installing other updates, deleting temporary files, applying settings, and so on.

Virtual Desktop Security

For virtual desktops, security is not handled much differently than physical devices. Enterprise customers may choose to utilize the built-in to Windows services of Windows Security, which comprises a suite of services that work well connected or not connected to the Internet.

For those virtual desktop environments not connected to the Internet, security signatures can be downloaded proactively several times per day, because Microsoft may release more than one signature update per day. Those signatures can then be provided to the virtual desktop devices and scheduled to be installed during production, regardless of persistent or non-persistent.

Updates

Windows 10 utilizes a monthly update rhythm. In some cases virtual desktop administrators control the process of updating through a process of shutting down VMs based on a “master” or “gold” image, unseal that image which is read-only, patch the image, then reseal it and bring it back into production. Therefore, there is no need to have virtual desktop devices checking Windows Update.

There is a set of scripts available at GitHub.com, that will do all the work items documented in this section of the course.

The Internet URL for the optimization scripts can be found at <https://github.com/The-Virtual-Desktop-Team/Virtual-Desktop-Optimization-Tool>.

Persistent virtual desktop environments

Persistent virtual desktop is at the basic level, a device that saves operating system state in between reboots. Other software layers of the virtual desktop solution provide the users easy and seamless access to their assigned VMs, often with a single sign-on solution.

There are several different implementations of persistent virtual desktop.

- **Traditional VMs**, where the VM has its own virtual disk file, starts up normally, and saves changes from one session to the next. The difference is how the user accesses this VM. There may be a web portal the user signs in to that automatically directs the user to one or more virtual desktop devices (VMs) assigned to them.
- **Image-based persistent VMs**, optionally with personal virtual disks (PVD). In this type of implementation there is a base/gold image on one or more host servers. A VM is created, and one or more virtual disks are created and assigned to this disk for persistent storage.
 - When the VM is started, a copy of the base image is read into the memory space of that VM. At the same time, a persistent virtual disk assigned to that VM, with any previous OS deltas is merged through a complex process.
 - Changes such as event log writes, log writes, and so on. are redirected to the read/write virtual disk assigned to that VM.
 - In this circumstance, OS and app servicing may operate normally, using traditional servicing software such as Windows Server Update Services, or other management technologies.
- **Master/gold image**. The difference between a persistent virtual desktop device and a "normal" virtual desktop device is the relationship to the master/gold image. At some point updates must be applied to the master. It is at this point where organizations decide how the user persistent changes are handled. In some cases, the disk with the user changes is discarded or reset. It may also be that the changes the user makes to the machine are kept through monthly Quality Updates, and the base is reset following a Feature Update.

Non-persistent virtual desktop environments

When a non-persistent virtual desktop implementation is based on a base or "gold" image, the optimizations are mostly performed in the base image, and then through local settings and local policies.

With image-based non-persistent (NP) virtual desktop environments, the base image is read-only. When an NP virtual desktop device (VM) is started, a copy of the base image is streamed to the VM. Activity that occurs during startup and thereafter until the next reboot is redirected to a temporary location. Usually the users are provided network locations to store their data. In some cases, the user's profile is merged with the standard VM to provide the user their settings.

One important aspect of NP virtual desktop that is based on a single image, is servicing. Updates to the operating system (OS) and components of the OS are delivered usually once per month. With image based virtual desktop environment, there is a set of processes that must be performed to get updates to the image:

- On a given host, all the VMs on that host, based from the base image must be shut down or turned off. This means the users are redirected to other VMs.
- In some implementations, this is referred to as "draining." The virtual machine or session host, when set to draining mode, stops accepting new requests, but continues servicing users currently connected to the device.

- In draining mode, when the last user logs off the device, that device is then ready for servicing operations.
- The base image is then opened and started up. All maintenance activities are then performed, such as OS updates, .NET updates, app updates, and so on.
- Any new settings that need to be applied are applied at this time.
- Any other maintenance is performed at this time.
- The base image is then shut down.
- The base image is sealed and set to go back into production.
- Users are allowed to log back on.

One of the challenges with non-persistent virtual desktop is that when a user logs off, nearly all the OS activity is discarded. The user's profile and/or state may be saved to a centralized location, but the virtual machine itself discards nearly all changes that were made since last boot. Therefore, optimizations intended for a Windows computer that saves state from one session to the next are not applicable.

Configure user settings through group policies

Microsoft Desktop Virtualization automatically detects device configurations and network conditions to get users up and running sooner by enabling the instant setup of corporate applications and desktops, and it equips IT to provide access to legacy applications during migration to Windows 10.

There are opportunities for you to refine the Windows 10 operating system specifically for the corporate Microsoft Virtual Desktop Infrastructure (VDI) environment. In the VDI environment, many background services and tasks are disabled from the beginning.

This topic is not a blueprint, but rather a guide or starting point. Some recommendations might disable functionality that you would prefer to use, so you should consider the cost versus the benefit of adjusting any setting in your scenario.

Creating the Windows 10 image

The first step is to install a reference image of Windows 10 on either a physical or virtual machine. Installing to a virtual machine is easy and allows you to save versions of the virtual hard-disk (VHD) file, in case you want to roll back to an earlier version.

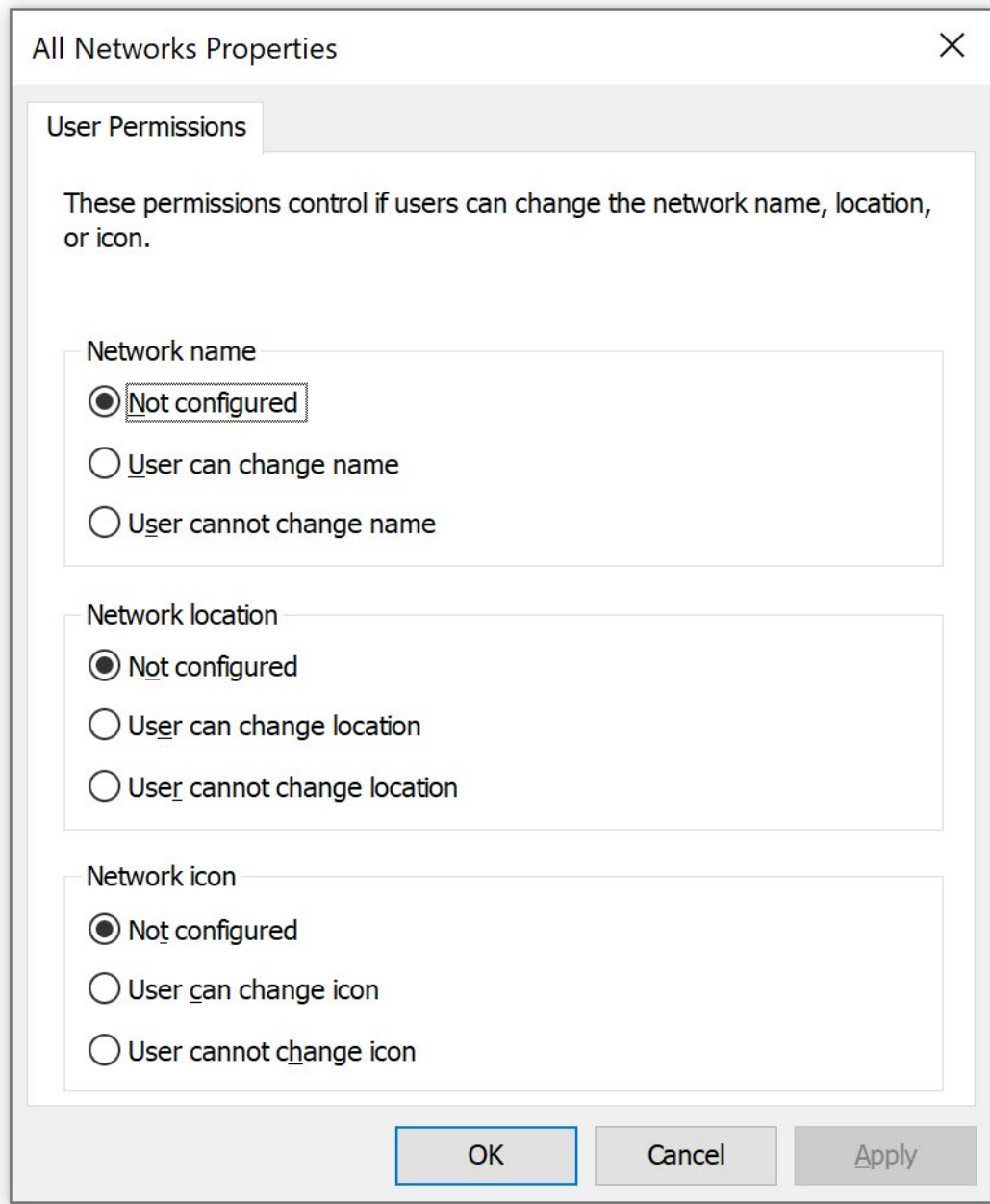
During installation, you can choose either **Express Settings** or **Customize**. The settings offered during the **Customize** option are adjustable by using Group Policy, so the method of installing the base OS is not that important.

Group Policy settings

To edit Group Policy settings, press the Windows button and type *group policy* or *gpedit.msc*. In the results that return, click **Edit group policy** to open Local Group Policy Editor.

Below is a example of configuring Group Policy settings for the Network.

Under **Computer Configuration**, expand **Windows Settings**, and then expand **Security Settings**. Click **Network List Manager Policies**, and then double-click **All Networks**. In the dialog that opens, in the **Network location** area, select the radio button for **User cannot change location**. Click the **OK** button to save.



Collapse **Windows Settings**, and then expand **Administrative Templates**. Click or expand **Network**, and then adjust each setting as follows by double-clicking it, then selecting the radio button for the indicated value and clicking the **OK** button:

Setting area	Setting	Recommended value for VDI use
Background Intelligent Transfer Service (BITS)		
	Do not allow the BITS client to use Windows Branch Cache	Enabled
	Do not allow the computer to act as a BITS Peercaching client	Enabled

Setting area	Setting	Recommended value for VDI use
	Do not allow the computer to act as a BITS Peercaching server	Enabled
	Allow BITS Peercaching	Disabled
BranchCache		
	Turn on BranchCache	Disabled
Hotspot Authentication		
	Enable Hotspot Authentication	Disabled
Microsoft Peer-to-Peer Networking Services		
	Turn off Microsoft Peer-to-Peer Networking Services	Enabled
Offline Files		
	Allow or Disallow use of the Offline Files feature	Disabled

Configure user settings through Endpoint Manager policies

Microsoft Endpoint Manager customers can leverage integration with Azure Virtual Desktop to manage and operate their deployments efficiently and establish a secure remote work solution.

You can now enroll Azure Virtual Desktop Virtual Machines that are hybrid Azure AD joined (joined to your on-premises Active Directory and registered with your Azure Active Directory) with Microsoft Intune and manage them in the Microsoft Endpoint Manager admin center the same way as physical devices.

This simplifies management, provides a centralized view across both physical devices and virtual desktops and opens up new areas of collaboration. The Endpoint Manager integration is generally available for Windows 10 Enterprise desktops. Windows 10 Enterprise multi-session supports policies at device specific scope.

Device name	Managed by	Ownership	Compliance	OS	OS version	Last check-in
WVD-dedic-1	Intune	Corporate	Compliant	Windows	10.0.18363.1016	9/2/2020, 3:53:54 P
WVD-person-0	Intune	Corporate	Compliant	Windows	10.0.18363.1016	9/2/2020, 3:54:34 P
WVD-person-2	Intune	Corporate	Compliant	Windows	10.0.18363.1016	9/2/2020, 3:51:44 P
WVDdedic-0	Intune	Corporate	Compliant	Windows	10.0.18363.1016	9/2/2020, 3:57:21 P
WVDshared-0	Intune	Corporate	Compliant	Windows	10.0.18363.1016	9/2/2020, 3:51:41 P
WVDshared-2	Intune	Corporate	Compliant	Windows	10.0.18363.1016	9/2/2020, 3:55:11 P
ab-2526552	Intune	Corporate	Not Compliant	Windows	10.0.17763.737	10/23/2019, 8:32:31
alg-10-15265	Co-managed	Corporate	See ConfigMgr	Windows	10.0.18362.356	10/23/2019, 8:32:31
alg-10-16523	Co-managed	Corporate	See ConfigMgr	Windows	10.0.18362.356	10/23/2019, 8:32:31
alg-10-97452	Co-managed	Corporate	See ConfigMgr	Windows	10.0.18362.356	10/23/2019, 8:32:31
ameliest_Android_10/...	Intune	Corporate	Not Compliant	Android (work profile)	10.0	10/23/2019, 8:32:41
antbroo_Android_5/13...	Intune	Corporate	Not Compliant	Android (device admin...)	8.0.0	10/23/2019, 8:32:41
matthewin10pro2	Intune	Corporate	Not Compliant	Windows	10.0.17763.864	12/1/2019, 3:39:22
matthewin10pro3	Intune	Corporate	Not Compliant	Windows	10.0.17763.864	12/1/2019, 3:44:19
mrogers_Windows_8/...	Intune	Unknown	Not Compliant	Windows	0.0.0.0	

Configure session timeout properties

Signing users out when they're inactive preserves resources and prevents access by unauthorized users. We recommend that timeouts balance user productivity as well as resource usage. For users that interact with stateless applications, consider more aggressive policies that turn off machines and preserve resources.

The timeout options for RDP is set on the servers in the Local Group Policy.

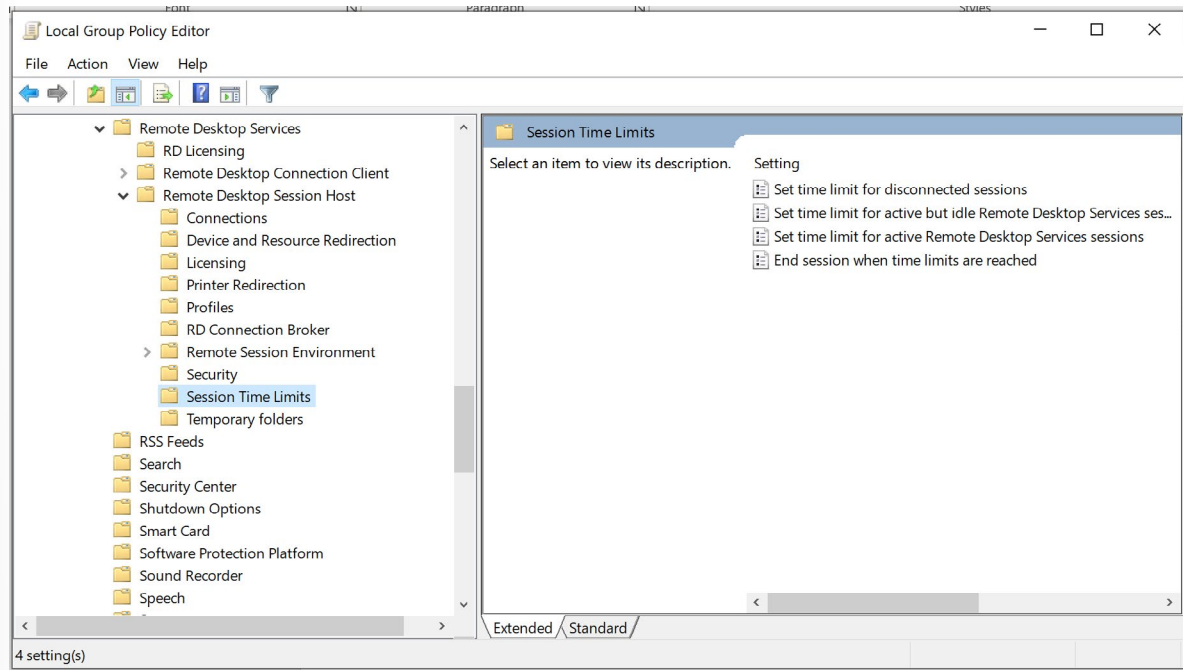
↑

↑ To configure session time limits, do the following:

↑

1. To edit Group Policy settings, press the Windows button and type *group policy* or *gpedit.msc*. In the results that return, click **Edit group policy** to open Local Group Policy Editor.

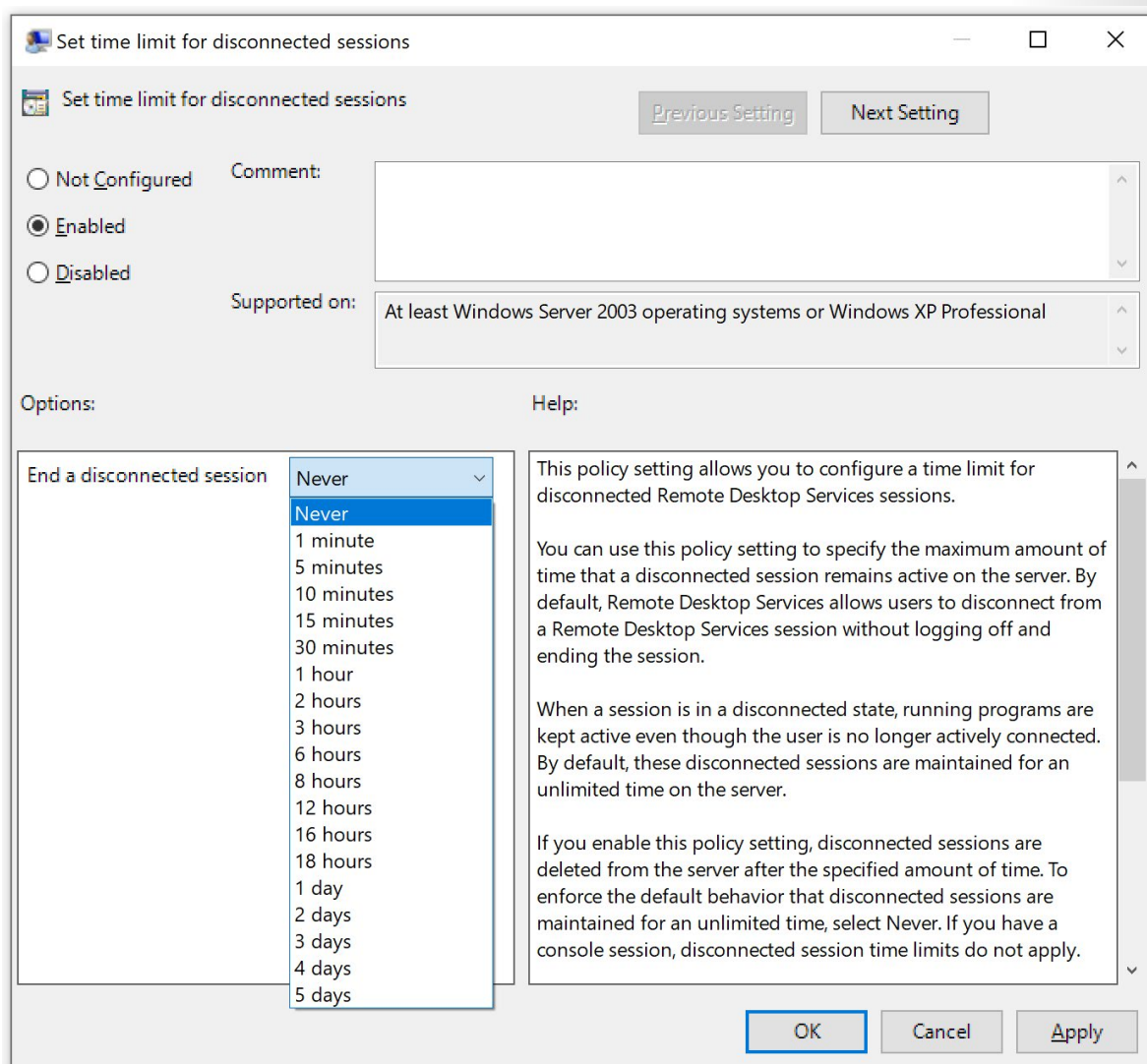
2. Navigate to **Computer Configuration > Administrative Templates > Windows Components > Remote Desktop Services > Remote Desktop Session Host > Session Time Limits**.



3. In the right pane of the Local Group Policy Editor, double-click to configure the following:

- Set time limit for disconnected sessions
- Set time limit for active but idle Remote Desktop Services sessions
- Set time limit for active Remote Desktop Services sessions
- End Session when time limits are reached

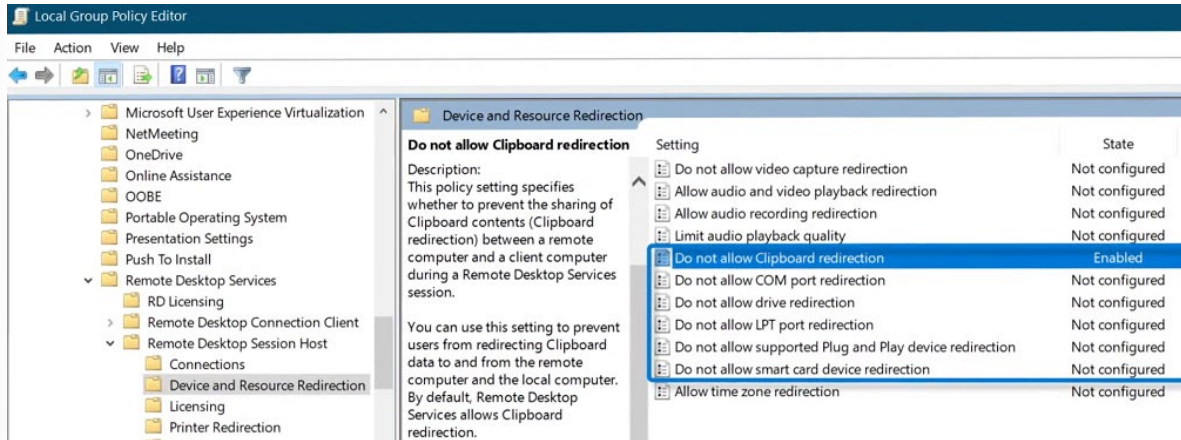
For example the, **Set time limit for logoff of RemoteApp sessions** is seen in the graphic below.



4. Click **Enabled**.
5. Select the desired time for logoff delay, and then click **OK**.
6. At a command prompt, type `gpupdate` and press **ENTER** to force the policy to refresh immediately.

Configure device redirections

Configuring device redirections for your Azure Virtual Desktop environment allows you to use printers, USB devices, microphones and other peripheral devices in the remote session.



Some device redirections require changes to both Remote Desktop Protocol (RDP) properties and Group Policy settings.

Each client supports different device redirections.

Setup device redirections

You can use the following RDP properties and Group Policy settings to configure device redirections.

Audio input (microphone) redirection

Set the following RDP property to configure audio input redirection:

- `audiocapturemode:i:1` enables audio input redirection.
- `audiocapturemode:i:0` disables audio input redirection.

Audio output (speaker) redirection

Set the following RDP property to configure audio output redirection:

- `audiomode:i:0` enables audio output redirection.
- `audiomode:i:1` or `audiomode:i:2` disable audio output redirection.

Camera redirection

Set the following RDP property to configure camera redirection:

- `camerastoredirect:s:*` redirects all cameras.
- `camerastoredirect:s:` disables camera redirection.

Note: Even if the `camerastoredirect:s:` property is disabled, local cameras may be redirected through the `devicestoredirect:s:` property. To fully disable camera redirection set `camerastoredirect:s:` and either set `devicestoredirect:s:` or define some subset of plug and play devices that does not include any camera.

You can also redirect specific cameras using a semicolon-delimited list of KSCATEGORY_VIDEO_CAMERA interfaces, such as `camerastoredirect:s:\?usb#vid_0bda&pid_58b0&mi.`

Clipboard redirection

Set the following RDP property to configure clipboard redirection:

- `redirectclipboard:i:1` enables clipboard redirection.
- `redirectclipboard:i:0` disables clipboard redirection.

COM port redirections

Set the following RDP property to configure COM port redirection:

- `redirectcomports:i:1` enables COM port redirection.
- `redirectcomports:i:0` disables COM port redirection.

USB redirection

First, set the following RDP property to enable USB device redirection:

- `usbdevicestoredirect:s:*` enables USB device redirection.
- `usbdevicestoredirect:s:` disables USB device redirection.

Second, set the following Group Policy on the user's local device:

- Navigate to **Computer Configuration > Policies > Administrative Templates > Windows Components > Remote Desktop Services > Remote Desktop Connection Client > RemoteFX USB Device Redirection**.
- Select **Allows RDP redirection of other supported RemoteFX USB devices from this computer**.
- Select the **Enabled** option, and then select the **Administrators and Users in RemoteFX USB Redirection Access Rights** box.
- Select **OK**.

Plug and play device redirection

Set the following RDP property to configure plug and play device redirection:

- `devicestoredirect:s:*` enables redirection of all plug and play devices.
- `devicestoredirect:s:` disables redirection of plug and play devices.

You can also select specific plug and play devices using a semicolon-delimited list, such as `devicestoredirect:s:root\*PNP0F08`.

Local drive redirection

Set the following RDP property to configure local drive redirection:

- `drivestoredirect:s:*` enables redirection of all disk drives.
- `Drivestoredirect:s:` disables local drive redirection.

You can also select specific drives using a semicolon-delimited list, such as `drivestoredirect:s:C;;E;;`.

Printer redirection

Set the following RDP property to configure printer redirection:

- `redirectprinters:i:1` enables printer redirection.
- `redirectprinters:i:0` disables printer redirection.

Configure Universal Print

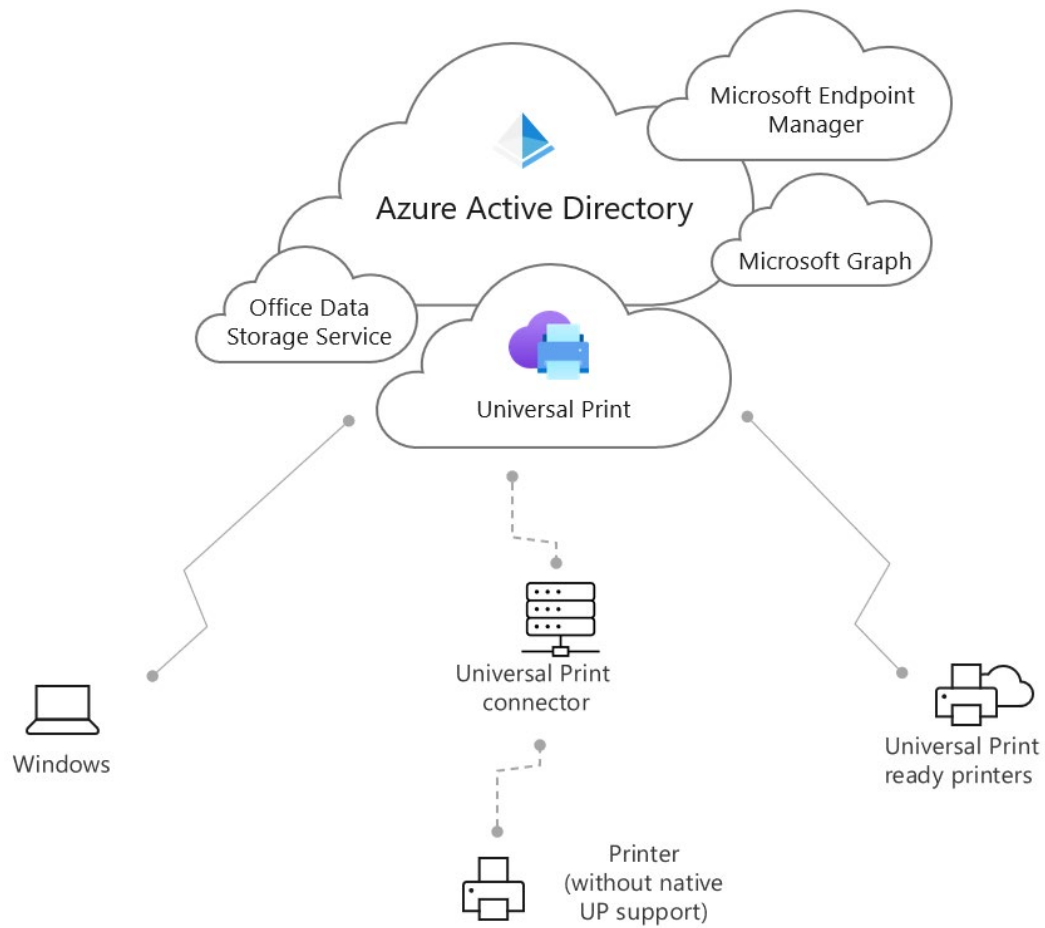
What is Universal Print

Universal Print is a modern print solution that organizations can use to manage their print infrastructure through cloud services from Microsoft.

Universal Print runs entirely on Microsoft Azure. When it's deployed with Universal Print-compatible printers, it doesn't require any on-premises infrastructure.

Universal Print is a Microsoft 365 subscription-based service that organizations use to centralize print management through the Universal Print portal. It's fully integrated with Azure Active Directory and supports single sign-on scenarios.

Universal Print can be deployed with non-compatible printers by using Universal Print connector software.



Component	Description
Universal Print	Cloud print service
Azure Active Directory	User and device identity and authorization service
Office Data Storage Service	Print queue data storage service
Microsoft Endpoint Manager	Client device printer provisioning policy service
Microsoft Graph	Printer management API
Universal Print connector	A component that handles communication between printers and the Universal Print service
Universal Print ready printer	A printer that has built-in support for communicating with Universal Print
Printer (without native UP support)	A printer that needs to be registered using the Universal Print connector to communicate with Universal Print

Enable Universal Print

Universal Print is currently available, see the [Universal Print Public documentation](#)⁷ to learn how to deploy it in your organization.

Assign licenses

Once Universal Print is enabled for a tenant, a Universal Print license needs to be assigned to every user who will be using it, including administrators who manage it.

- A Universal Print license is assigned to the M365 tenant by Global Administrator.
- To configure and manage Universal Print, the corresponding administrator must have a Universal Print license assigned.
- To configure and manage Universal Print, an administrator must be assigned either of the following two Azure Active Directory (Azure AD) roles: Printer Administrator or Global Administrator.
- A client device (to install and print from Universal Print) running Windows client OS.
- An Internet connection.

Step 1: Set up the Universal Print connector

Current in-market printers cannot communicate with Universal Print directly, and require a proxy Universal Print connector. For more information, see the following:

What is Universal Print connector?⁸

Set up Universal Print connector(s)⁹

Step 2: Register printers using Universal Print connector

Printers installed on Universal Print connector(s) need to be registered with Universal Print. For more information, see the following:

Register printers with Universal Print¹⁰

Step 3: Assign permissions and share printer

Registered printers need to have permissions assigned to Azure AD users and security groups. Once the permissions are configured, the printer needs to be shared for the users to be able to add it to their device for printing.

Assign printer permissions and share a printer¹¹

Once the printer is shared, it is ready for users to start printing to it.

⁷ <https://docs.microsoft.com/en-us/universal-print/fundamentals/>

⁸ <https://docs.microsoft.com/en-us/universal-print/fundamentals/universal-print-connector-overview>

⁹ <https://docs.microsoft.com/en-us/universal-print/fundamentals/universal-print-connector-installation>

¹⁰ <https://docs.microsoft.com/en-us/universal-print/fundamentals/universal-print-connector-printer-registration>

¹¹ <https://docs.microsoft.com/en-us/universal-print/portal/share-printers>

Step 4: Add a Universal Print printer to a Windows device

Before you try to add a Universal Print printer to a user's device, ensure that:

- The user's device is connected to internet.
- The user's device is either:
 - Azure AD joined,
 - Azure AD registered, or
 - Hybrid Azure AD joined
- The Universal Print printer has been shared.
- The user has been added to the permissions of Universal Print printer that is to be added on the device.
- The user has been assigned the license to use Universal Print.

To add a Universal Print printer to a Windows client device:

1. Select **Settings > Devices > Printers & scanners**.
2. Select **Add a printer or scanner**.
3. Select the desired printer, and then select **Add device**.
4. Once the printer is added, it will show up in the list of **Printers & scanners** in Settings.

For Universal Print printers, the driver selected on user's Windows device is **Universal Print Class Driver**. Do not change the driver for Universal Print printers.

To test the printer by printing a test page:

1. On the **Printers & scanners** settings page, select the Universal Print printer.
2. Select **Manage**.
3. Select **Open print queue** to monitor the test print job.
4. Select **Print a test page**. You will see the job in print queue window. If the job is accepted by Universal Print, the status will change to `Sent to printer`.

Troubleshoot user profile issues

This topic provides an overview of the issues you may encounter when setting up an Azure Virtual Desktop environment and provides ways to resolve the issues.

Report issues

To report issues or suggest features for Azure Virtual Desktop with Azure Resource Manager integration, visit the **Azure Virtual Desktop Tech Community**¹².

¹² <https://techcommunity.microsoft.com/t5/Windows-Virtual-Desktop/bd-p/WindowsVirtualDesktop>

You can use the Tech Community to discuss best practices or suggest and vote for new features.

When you make a post asking for help or propose a new feature, make sure you describe your topic in as much detail as possible. Detailed information can help other users answer your question or understand the feature you're proposing a vote for.

Escalation tracks

Before doing anything else, make sure to check the **Azure status page**¹³ and **Azure Service Health**¹⁴ to make sure your Azure service is running properly.

Use the following table to identify and resolve issues you may encounter when setting up an environment using Remote Desktop client. Once your environment's set up, you can use our new **Diagnostics service**¹⁵ to identify issues for common scenarios.

Issue	Suggested Solution
Session host pool Azure Virtual Network (VNET) and Express Route settings	Open an Azure support request (https://azure.microsoft.com/support/create-ticket/), then select the appropriate service (under the Networking category).

¹³ <https://status.azure.com/status>

¹⁴ <https://azure.microsoft.com/features/service-health/>

¹⁵ <https://docs.microsoft.com/en-us/azure/virtual-desktop/diagnostics-role-service>

Issue	Suggested Solution
Session host pool Virtual Machine (VM) creation when Azure Resource Manager templates provided with Azure Virtual Desktop aren't being used	Open an Azure support request (https://azure.microsoft.com/support/create-ticket/), then select Azure Virtual Desktop for the service. For issues with the Azure Resource Manager templates that are provided with Azure Virtual Desktop, see Azure Resource Manager template errors section of Host pool creation.
Managing Azure Virtual Desktop session host environment from the Azure portal	Open an Azure support request (https://azure.microsoft.com/support/create-ticket/). For management issues when using Remote Desktop Services/Azure Virtual Desktop PowerShell, see Azure Virtual Desktop PowerShell or open an Azure support request (https://azure.microsoft.com/support/create-ticket/), select Azure Virtual Desktop for the service, select Configuration and management for the problem type, then select Issues configuring environment using PowerShell for the problem subtype.
Managing Azure Virtual Desktop configuration tied to host pools and application groups (app groups)	See Azure Virtual Desktop PowerShell, or open an Azure support request (https://azure.microsoft.com/support/create-ticket/), select Azure Virtual Desktop for the service, then select the appropriate problem type.
Deploying and manage FSLogix Profile Containers	See Troubleshooting guide for FSLogix products and if that doesn't resolve the issue, Open an Azure support request (https://azure.microsoft.com/support/create-ticket/), select Azure Virtual Desktop for the service, select FSLogix for the problem type, then select the appropriate problem subtype.
Remote desktop clients malfunction on start	See Troubleshoot the Remote Desktop client and if that doesn't resolve the issue, Open an Azure support request (https://azure.microsoft.com/support/create-ticket/), select Azure Virtual Desktop for the service, then select Remote Desktop clients for the problem type. If it's a network issue, your users need to contact their network administrator.

Issue	Suggested Solution
Connected but no feed	Troubleshoot using the User connects but nothing is displayed (no feed) section of Azure Virtual Desktop service connections. If your users have been assigned to an app group, open an Azure support request (https://azure.microsoft.com/support/create-ticket/), select Azure Virtual Desktop for the service, then select Remote Desktop Clients for the problem type.
Feed discovery problems due to the network	Your users need to contact their network administrator.
Connecting clients	See Azure Virtual Desktop service connections and if that doesn't solve your issue, see Session host virtual machine configuration.
Responsiveness of remote applications or desktop	If issues are tied to a specific application or product, contact the team responsible for that product.
Licensing messages or errors	If issues are tied to a specific application or product, contact the team responsible for that product.
Issues with third-party authentication methods or tools	Verify that your third-party provider supports Azure Virtual Desktop scenarios and approach them regarding any known issues.
Issues using Log Analytics for Azure Virtual Desktop	For issues with the diagnostics schema, open an Azure support request (https://azure.microsoft.com/support/create-ticket/). For queries, visualization, or other issues in Log Analytics, select the appropriate problem type under Log Analytics.
Issues using M365 apps	Contact the M365 admin center with one of the M365 admin center help options.

Troubleshoot AVD clients

This topic describes common issues with the Remote Desktop client and how to fix them.

Remote Desktop client for Windows 10 stops responding or cannot be opened

You can reset the user data from the About page or using a command.

Use the following command to remove your user data, restore default settings and unsubscribe from all Workspaces.

Cmd

```
msrdcw.exe /reset [/f]
```

Web client won't open

First, test your internet connection by opening another website in your browser; for example, `<a href="http://www.bing.com/" title="" target="_blank" data-generated=''>www.bing.com</a>`.

Use **nslookup** to confirm DNS can resolve the FQDN:

Cmd

```
nslookup rdweb.wvd.microsoft.com
```

Try connecting with another client, like Remote Desktop client for Windows 10, and check to see if you can open the web client.

Can't open other websites while connected to the web client

If you can't open other websites while you're connected to the web client, there might be network connection problems or a network outage. We recommend you contact network support.

Nslookup can't resolve the name

If nslookup can't resolve the name, then there might be network connection problems or a network outage. We recommend you contact network support.

Your client can't connect but other clients on your network can connect

If your browser starts acting up or stops working while you're using the web client, follow these instructions to troubleshoot it:

1. Restart the browser.
2. Clear browser cookies.
3. Clear browser cache.
4. Open browser in Private mode.

Client doesn't show my resources

First, check the Azure Active Directory account you're using. If you've already signed in with a different Azure Active Directory account than the one you want to use for Azure Virtual Desktop, you should either sign out or use a private browser window.

If that doesn't work, make sure your app group is associated with a workspace.

Web client stops responding or disconnects

Try connecting using another browser or client.

Other browsers and clients also malfunction or fail to open

If issues continue even after you've switched browsers, the problem may not be with your browser, but with your network.

Web client keeps prompting for credentials

If the Web client keeps prompting for credentials, follow these instructions:

1. Confirm the web client URL is correct.
2. Confirm that the credentials you're using are for the Azure Virtual Desktop environment tied to the URL.
3. Clear browser cookies.
4. Clear browser cache.
5. Open your browser in Private mode.

Install and configure apps on a session host

MSIX app attach

MSIX is a Windows app package format that provides a modern packaging experience to all Windows apps. The MSIX package format preserves the functionality of existing app packages and/or install files in addition to enabling new, modern packaging and deployment features to Win32, WPF, and Windows Forms apps.

MSIX app attach is a way to deliver MSIX applications to both physical and virtual machines. However, MSIX app attach is different from regular MSIX because it's made especially for Azure Virtual Desktop. This topic will describe what MSIX app attach is and what it can do for you.

Application delivery options in Azure Virtual Desktop

You can deliver apps in Azure Virtual Desktop through one of the following methods:

- Put apps in a master image
- Use tools like SCCM or Intune for central management
- Dynamic app provisioning (AppV, VMWare AppVolumes, or Citrix AppLayering)
- Create custom tools or scripts using Microsoft and a third-party tool

What does MSIX app attach do?

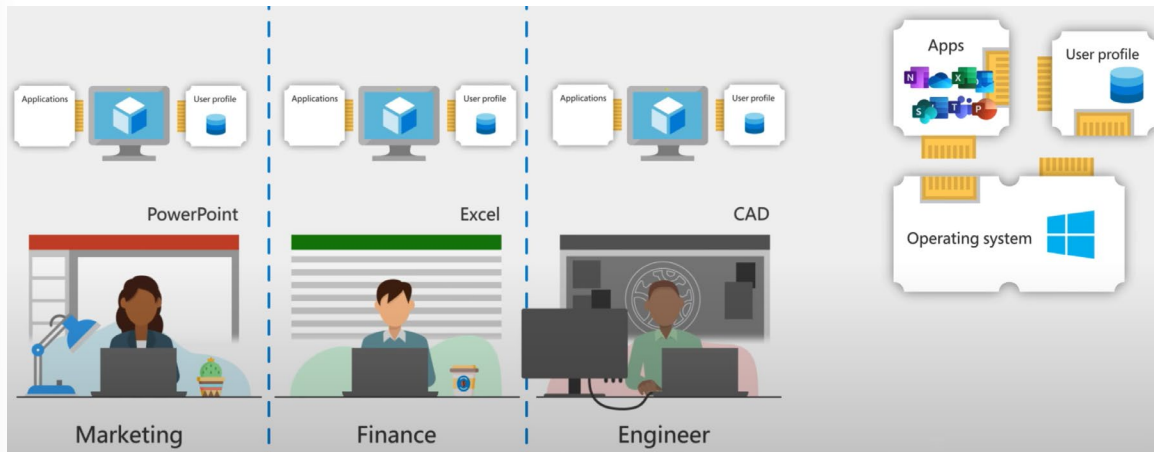
In an Azure Virtual Desktop deployment, MSIX app attach can:

- Create separation between user data, the OS, and apps by using MSIX containers.
- Remove the need for repackaging when delivering applications dynamically.
- Reduce the time it takes for a user to sign in.
- Reduce infrastructure requirements and cost.

How MSIX app attach works

MSIX app attach stores application files in a separate virtual hard disk from the operating system. It registers the regular MSIX package on a device instead of on a physical download and installation. The registration uses existing Windows APIs and has minimal impact on user sign-in times, which enhances the user experience.

When you open MSIX app attach, the application files are accessed from a VHD. You're not even aware that the application isn't locally installed.

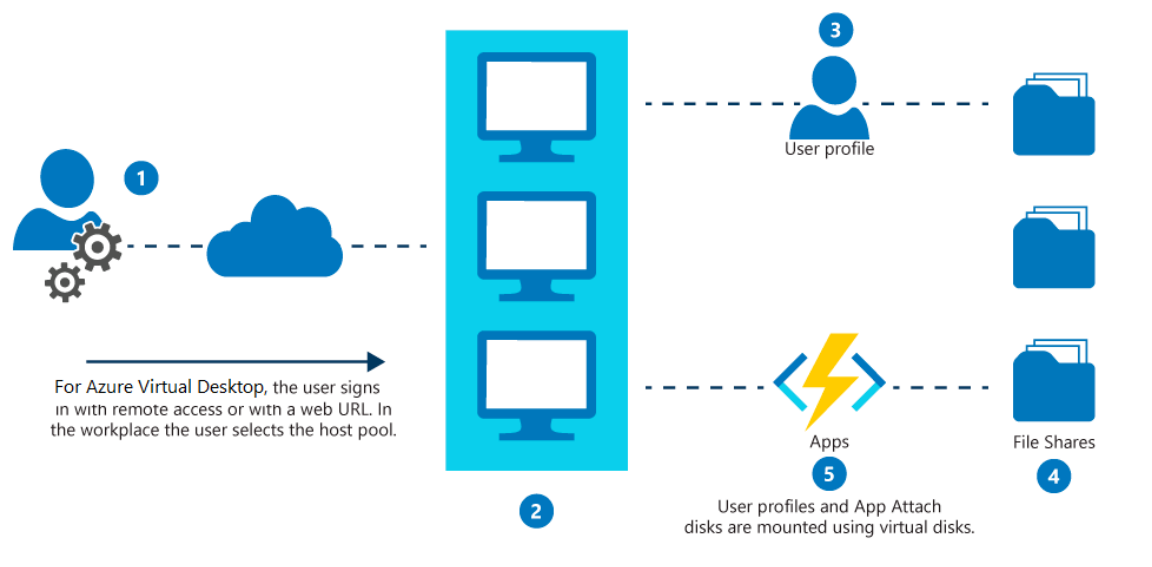


MSIX app attach follows several steps or actions:

Term	Definition
Stage	MSIX app attach notifies the operating system that an application is available, and that the virtual disk that contains the MSIX package (also known as the MSIX image) is mounted.
Registration	MSIX app attach uses a per-user process to make the application available to you.
Delayed registration	Complete registration of the application is delayed until you decide to run the application.
Deregistration	The application is no longer available to you after you sign out.
Destage	The application is no longer available from the virtual machine after shutdown or restart of the machine.

After you open MSIX app attach, you experience the following process:

1. From the Azure Virtual Desktop client, you sign in and select the host pool for which you have access. The process is similar to opening published RemoteApp programs from the Azure Virtual Desktop environment.
2. You're assigned a virtual machine within the host pool, on which a RemoteApp or Remote Desktop session is created. The Azure Virtual Desktop client interacts with that session.
3. If the user profile is configured, the FSLogix agent on the session host provides the user profile from the file share. The file share can be Azure Files, Azure NetApp Files, or an infrastructure as a service (IaaS) file server.
4. Applications that are assigned to you are read from Azure Virtual Desktop.
5. MSIX app attach applications are registered to the virtual machine for you, from the attached MSIX virtual disk. That virtual disk might be on an IaaS file share, Azure Files, or Azure NetApp Files.



Use the following key terms for MSIX app attach as a review and reference.

Feature	Traditional app layering	MSIX app attach
Format	Different app layering technologies require different proprietary formats.	Works with the native MSIX packaging format.
Repackaging overhead	Proprietary formats require sequencing and repackaging per update.	Apps published as MSIX don't require repackaging. However, if the MSIX package isn't available, repackaging overhead still applies.
Ecosystem	N/A (for example, vendors don't ship App-V)	MSIX is Microsoft's mainstream technology that key ISV partners and in-house apps like Office are adopting. You can use MSIX on both virtual desktops and physical Windows computers.
Infrastructure	Additional infrastructure required (servers, clients, and so on)	Storage only
Administration	Requires maintenance and update	Simplifies app updates
User experience	Impacts user sign-in time. Boundary exists between OS state, app state, and user data.	Delivered apps are indistinguishable from locally installed applications.

Set up a file share for MSIX app attach

All MSIX images must be stored on a network share that can be accessed by users in a host pool with read-only permissions.

NOTE: There is an online lab later in this module where you create MSIX app packages and implement a MSIX app attach container for Azure Virtual Desktop in AD DS environment (*Lab - Package Azure Virtual Desktop applications (AD DS)*). This topic is meant to provide an overview of the procedure.

MSIX app attach doesn't have any dependencies on the type of storage fabric the file share uses. The considerations for the MSIX app attach share are same as those for an FSLogix share.

Performance requirements

MSIX app attach image size limits for your system depend on the storage type you're using to store the VHD or VHDX files, as well as the size limitations of the VHD, VHSD or CIM files and the file system.

The following table gives an example of how many resources a single 1 GB MSIX image with one MSIX app inside of it requires for each VM:

Resource	Requirements
Steady state IOPs	1 IOPs
Machine boot sign in	10 IOPs
Latency	400 ms

Requirements can vary widely depending how many MSIX-packaged applications are stored in the MSIX image. For larger MSIX images, you'll need to allocate more bandwidth.

Storage recommendations

Azure offers multiple storage options that can be used for MISX app attach. We recommend using Azure Files or Azure NetApp Files as those options offer the best value between cost and management overhead.

Optimize MSIX app attach performance

Note: The considerations for the MSIX app attach share seems below are same as those for an FSLogix share.

Here are some other things we recommend you do to optimize MSIX app attach performance:

- The storage solution you use for MSIX app attach should be in the same datacenter location as the session hosts.
- To avoid performance bottlenecks, exclude the following VHD, VHDX, and CIM files from antivirus scans:
 - <MSIXAppAttachFileShare>*.VHD
 - <MSIXAppAttachFileShare>*.VHDX
 - \storageaccount.file.core.windows.net\share**.VHD
 - \storageaccount.file.core.windows.net\share**.VHDX
 - .CIM
 - \storageaccount.file.core.windows.net\share**.CIM
- Separate the storage fabric for MSIX app attach from FSLogix profile containers.
- All VM system accounts and user accounts must have read-only permissions to access the file share.

- Any disaster recovery plans for Azure Virtual Desktop must include replicating the MSIX app attach file share in your secondary failover location.

How to set up the file share

The setup process for MSIX app attach file share is largely the same as the setup process for FSLogix profile file shares. However, you'll need to assign users different permissions. MSIX app attach requires read-only permissions to access the file share.

If you're storing your MSIX applications in Azure Files, then for your session hosts, you'll need to assign all session host VMs both storage account role-based access control (RBAC) and file share New Technology File System (NTFS) permissions on the share.

Azure object	Required role	Role function
Session host (VM computer objects)	Storage File Data SMB Share Contributor	Read and Execute, Read, List folder contents
Admins on File Share	Storage File Data SMB Share Elevated Contributor	Full control
Users on File Share	Storage File Data SMB Share Contributor	Read and Execute, Read, List folder contents

To assign session host VMs permissions for the storage account and file share:

- Create an Active Directory Domain Services (AD DS) security group.
- Add the computer accounts for all session host VMs as members of the group.
- Sync the AD DS group to Azure Active Directory (Azure AD).
- Create a storage account.
- Create a file share under the storage account by following the instructions in **Create an Azure file share**¹⁶.
- Join the storage account to AD DS by following the instructions in **enable AD DS authentication for your Azure file shares**¹⁷.
- Assign the synced AD DS group to Azure AD, and assign the storage account the Storage File Data SMB Share Contributor role.
- Mount the file share to any session host by following the instructions in **assign share-level permissions to an identity**¹⁸.
- Grant NTFS permissions on the file share to the AD DS group.
- Set up NTFS permissions for the user accounts. You'll need an operating unit (OU) sourced from the AD DS that the accounts in the VM belong to.

Demonstration - Configure apps for users

From the previous demonstrations in Module 2, there is now a host pool and workspace that can be used to deploy a remote app.

In this demonstration you see how to create a RemoteApp application group to share an application to a different user in the organization.

¹⁶ <https://github.com/MicrosoftDocs/azure-docs/blob/master/articles/storage/files/storage-how-to-create-file-share.md>

¹⁷ <https://github.com/MicrosoftDocs/azure-docs/blob/master/articles/storage/files/storage-files-identity-ad-ds-enable.md>

¹⁸ <https://github.com/MicrosoftDocs/azure-docs/blob/master/articles/storage/files/storage-files-identity-ad-ds-assign-permissions.md>

To complete the exercise, you'll need the credentials for a different non-administrative user account that's in Active Directory.

Step 1: Basics

1. Sign in to the Azure portal.
2. Use the search box to find **Azure Virtual Desktop**.
3. Select **Application groups** > **Add**.
4. Select the subscription, resource group, host pool, and application type.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Virtual Desktop | Application groups >

Create an application group

Basics

Assignments

Applications

Workspace

Tags

Review + create

Subscription * ⓘ

Visual Studio Enterprise

Resource group * ⓘ

wvd-dm-rg1

Create new

Host pool * ⓘ

wvd-host-pool-1

Location ⓘ

Central US

Metadata stored in same location as host pool

Application group type

RemoteApp application groups are where you can add applications. A Desktop application group will grant full desktop access.

Application group type * ⓘ

☒ RemoteApp
 ☐ Desktop

Application group name *

RemoteApp1

Review + create

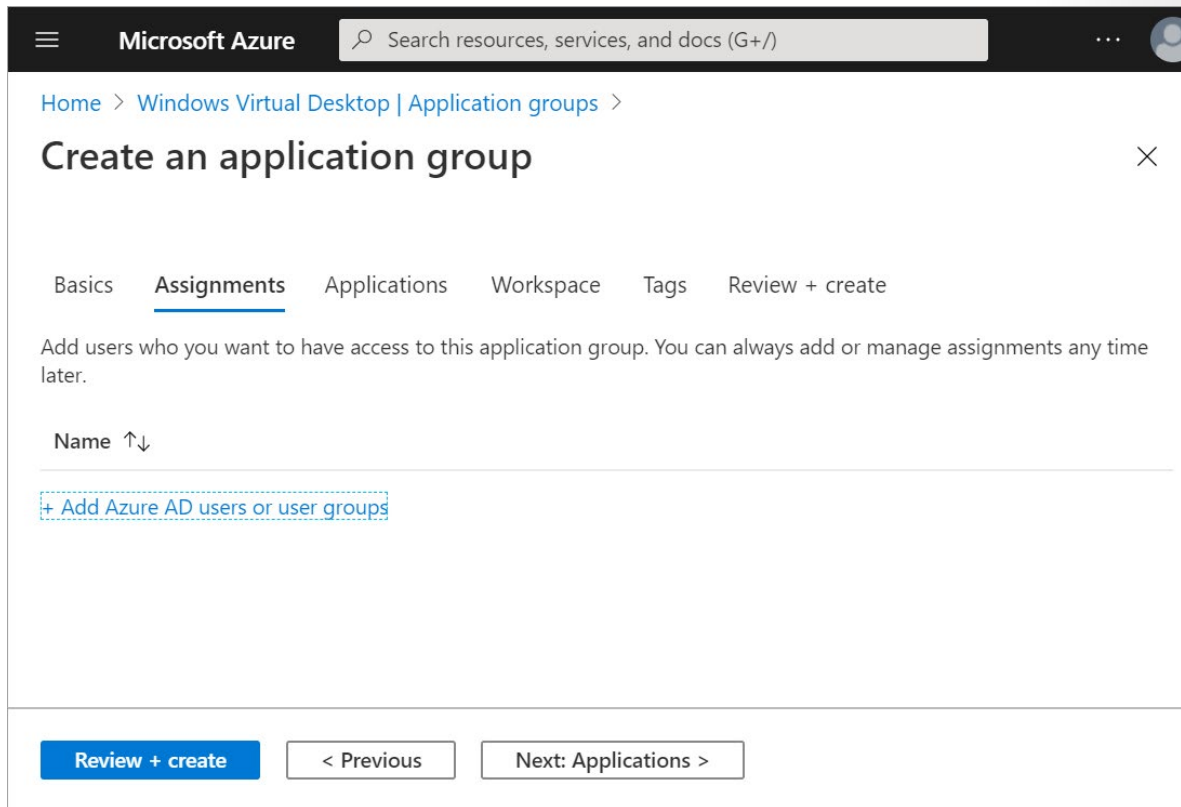
Next: Assignments >

Field	Description
Subscription	Subscription where you want the app group to run
Resource group	Resource group you've created for Azure Virtual Desktop resources
Host pool	avd-host-pool-1
Application group type	RemoteApp
Application group name	RemoteApp1

5. Select **Next: Assignments**.

Step 2: Assignments

1. Select **Add Azure AD users or user groups**.



The screenshot shows the Microsoft Azure portal interface for creating an application group. The top navigation bar includes the Microsoft Azure logo and a search bar. The breadcrumb trail is 'Home > Windows Virtual Desktop | Application groups >'. The main heading is 'Create an application group'. Below this, there are tabs for 'Basics', 'Assignments' (which is selected), 'Applications', 'Workspace', 'Tags', and 'Review + create'. A text block states: 'Add users who you want to have access to this application group. You can always add or manage assignments any time later.' Below this, there is a 'Name' field with an up/down arrow icon. A dashed blue box highlights a link that says '+ Add Azure AD users or user groups'. At the bottom, there are three buttons: 'Review + create' (in blue), '< Previous', and 'Next: Applications >'.

2. Select single or multiple users or you can select user groups.
3. Select **Next: Applications**.

Step 3: Applications

1. Select **Add applications**.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Windows Virtual Desktop | Application groups >

Create an application group

Basics

Assignments

Applications

Workspace

Tags

Review + create

Add applications to this application group. You can always add or manage applications any time later.

Name ↑↓

File path ↑↓

+ Add applications

Review + create

< Previous

Next: Workspace >

2. Use the information in the following table to help you add an application. Accept the rest of the default values where not listed in the table.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Windows Virtual Desktop |

Create an application

Basics

Assignments

Applicat

Add applications to this application group

Name ↑↓

+ Add applications

Add application

Select an application from your start menu or add from a file path.

Application source *

Start menu

Application *

Wordpad

Display name

Wordpad

Description

Application path

C:\Program Files\Windows NT\Accessories\wordpad.exe

Icon path

C:\Program Files\Windows NT\Accessories\wordpad.exe

Icon index

0

Require command line

☒ No ☐ Yes

Review + create

< Previous

Save

Cancel

Field	Value
Application source	Start menu
Application	WordPad
Display name	WordPad

3. Select **Next: Workspace**.

Step 3: Workspace

1. For **Register application group**, select **Yes**.

Microsoft Azure Search resources, services, and docs (G+)

Home > Windows Virtual Desktop | Application groups >

Create an application group

Basics Assignments Applications **Workspace** Tags Review + create

To save some time, you can register the default desktop application group from this host pool, with a new or pre-existing workspace.

Register application group ☐ No ☒ Yes

Register application group ⓘ Contoso virtual workspace

i Another application group in wvd-host-pool-1 has already been registered, so this app group will also be registered to that same workspace.

Review + create < Previous Next: Tags >

2. Select **Review + create**.
3. Review what you've entered and select **Create**.

Verify access to application

1. Go to the **Azure Virtual Desktop web client**¹⁹.
2. Sign in by using the user credentials for the user you assigned to the RemoteApp application group.
3. You should see the application in the workspace.

Using the OneDrive sync app on virtual desktops

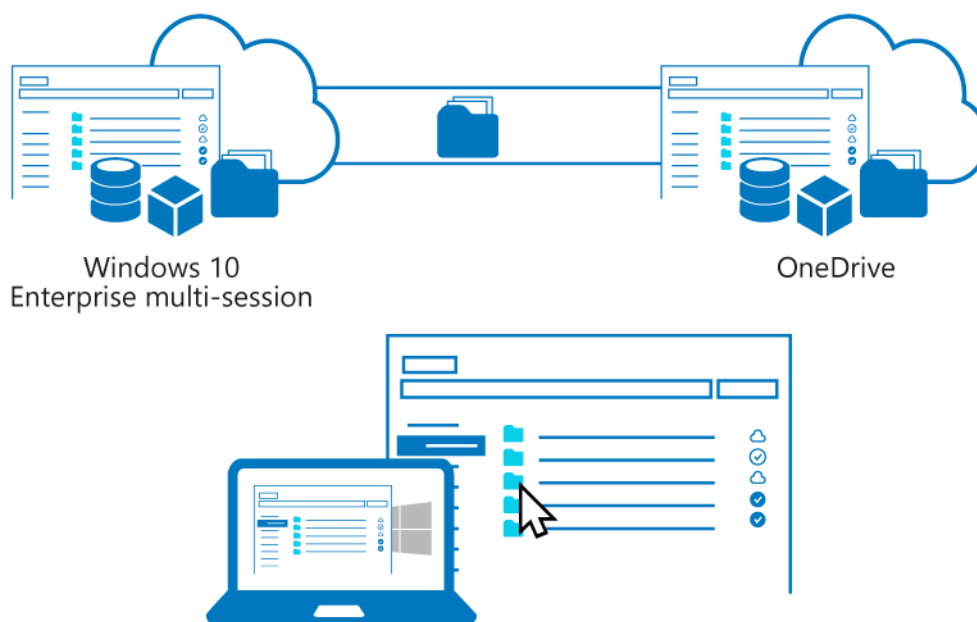
For all supported operating systems, the OneDrive sync app supports:

- Virtual desktops that persist between sessions.
- Non-persistent virtual desktops that use **Azure Virtual Desktop**²⁰.

¹⁹ <https://rdweb.wvd.microsoft.com/arm/webclient/index.html>

²⁰ <https://azure.microsoft.com/services/virtual-desktop/>

- Non-persistent virtual desktops that have **FSLogix Apps**²¹ or **FSLogix Office Container**²², and a Microsoft 365 subscription for all of the following operating systems:
 - Windows 10, 32 or 64-bit (supports VHDX files)
 - Windows 7, 32 or 64-bit (supports VHD files)
 - Windows Server 2019 (supports VHDX)
 - Windows Server 2016 (supports VHDX)
 - Windows Server 2012 R2 (supports VHDX)
 - Windows Server 2008 R2 (supports VHD)



Without FSLogix profile containers, OneDrive for Business is not supported in non-persistent RDSH or VDI environments.

Install the OneDrive sync app per machine

By default, the OneDrive sync app installs per user, meaning OneDrive.exe needs to be installed for each user account on the PC under the %localappdata% folder. With the new per-machine installation option, you can install OneDrive under the **Program Files (x86)** or **Program Files** directory, meaning all profiles on the computer will use the same **OneDrive.exe** binary. Other than where the sync app is installed, the behavior is the same.

The new per-machine sync app provides:

- Automatic transitioning from the previous OneDrive for Business sync app (Groove.exe)
- Automatic conversion from per-user to per-machine
- Automatic updates when a new version is available

²¹ <https://docs.microsoft.com/en-us/fslogix/configure-profile-container-tutorial>

²² <https://docs.microsoft.com/en-us/fslogix/configure-office-container-tutorial>

The per-machine sync app supports syncing OneDrive files in Microsoft 365 and in SharePoint Server 2019.

Deployment instructions

1. Download **OneDriveSetup.exe**.
2. Run `OneDriveSetup.exe /allusers` from a command prompt window or by using Microsoft Endpoint Configuration Manager. This will install the sync app under the `Program Files (x86)\Microsoft OneDrive` directory. When setup completes, OneDrive will start. If accounts were added on the computer, they'll be migrated automatically.

Using Microsoft Teams on Azure Virtual desktop

Media optimization for Microsoft Teams is only available for the Windows Desktop client on Windows 10 machines.

Microsoft Teams on Azure Virtual Desktop supports chat and collaboration. With media optimizations, it also supports calling and meeting functionality.

With media optimization for Microsoft Teams, the Windows Desktop client handles audio and video locally for Teams calls and meetings.

You can still use Microsoft Teams on Azure Virtual Desktop with other clients without optimized calling and meetings.

Teams chat and collaboration features are supported on all platforms. To redirect local devices in your remote session, check out **Customize Remote Desktop Protocol properties for a host pool**²³.

Before you can use Microsoft Teams on Azure Virtual Desktop:

- Install the Windows Desktop client on a Windows 10 or Windows 10 IoT Enterprise.
- Connect to a Windows 10 Multi-session or Windows 10 Enterprise virtual machine (VM).

Install the Teams desktop app

This section will show you how to install the Teams desktop app on your Windows 10 Multi-session or Windows 10 Enterprise VM image.

Prepare your image for Teams

To enable media optimization for Teams, set the following registry key on the host:

1. From the start menu, run **RegEdit** as an administrator. Navigate to **HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Teams**. Create the Teams key if it doesn't already exist.
2. Create the following value for the Teams key:

Name	Type	Data/Value
IsAVDEnvironment	DWORD	1

²³ <https://docs.microsoft.com/en-us/azure/virtual-desktop/teams-on-wvd>

Install the Teams WebSocket Service

Install the latest **Remote Desktop WebRTC Redirector Service**²⁴ on your VM image.

Install Microsoft Teams

You can deploy the Teams desktop app using a per-machine or per-user installation. To install Microsoft Teams in your Azure Virtual Desktop environment:

1. Download the **Teams MSI package**²⁵ that matches your environment. We recommend using the 64-bit installer on a 64-bit operating system.
2. Run one of the following commands to install the MSI to the host VM:

Per-user installation

```
msiexec /i <path_to_msi> /l*v <install_logfile_name>
```

This process is the default installation, which installs Teams to the **%AppData%** user folder. Teams won't work properly with per-user installation on a non-persistent setup.

Per-machine installation

```
msiexec /i <path_to_msi> /l*v <install_logfile_name> ALLUSER=1
```

This installs Teams to the Program Files (x86) folder on a 32-bit operating system and to the Program Files folder on a 64-bit operating system. At this point, the golden image setup is complete. Installing Teams per-machine is required for non-persistent setups.

To uninstall the MSI from the host VM, run this command:

```
msiexec /passive /x <msi_name> /l*v <uninstall_logfile_name>
```

3. This uninstalls Teams from the Program Files (x86) folder or Program Files folder, depending on the operating system environment.

Verify media optimizations loaded

After installing the WebSocket Service and the Teams desktop app, follow these steps to verify that Teams media optimizations loaded:

1. Quit and restart the Teams application.
2. Select your user profile image, then select **About**.
3. Select **Version**.

If media optimizations loaded, the banner will show you **Azure Virtual Desktop Media optimized**.

4. Select your user profile image, then select **Settings**.

²⁴ <https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RE4AQBt>

²⁵ <https://docs.microsoft.com/en-us/microsoftteams/teams-for-vdi>

Publish built-in apps in Azure Virtual Desktop

This topic will tell you how to publish apps, such as Microsoft Edge browser, in your Azure Virtual Desktop environment.

Publish built-in apps

To publish a built-in app:

1. Connect to one of the virtual machines in your host pool.
2. Get the **PackageFamilyName** of the app you want to publish by following the instructions in [this article](#)²⁶.
3. Finally, run the following cmdlet with `<PackageFamilyName>` replaced by the **PackageFamilyName** you found in the previous step:

PowerShell

```
New-AzWvdApplication -Name <applicationname> -ResourceGroupName <resource-groupname> -ApplicationGroupName <appgroupname> -FilePath "shell:appsFolder\<PackageFamilyName>!App" -CommandLineSetting <Allow|Require|DoNotAllow> -IconIndex 0 -IconPath <iconpath> -ShowInPortal:$true
```

Azure Virtual Desktop only supports publishing apps with install locations that begin with `C:\Program Files\WindowsApps`.

Update app icons

After you publish an app, it will have the default Windows app icon instead of its regular icon picture. To change the icon to its regular icon, put the image of the icon you want on a network share. Supported image formats are PNG, BMP, GIF, JPG, JPEG, and ICO.

Publish Microsoft Edge

The process you use to publish Microsoft Edge is a little different from the publishing process for other apps. To publish Microsoft Edge with the default homepage, run this cmdlet:

PowerShell

```
New-AzWvdApplication -Name -ResourceGroupName -ApplicationGroupName -FilePath "shell:AppsFolder\Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge" -CommandLineSetting <Allow|Require|DoNotAllow> -iconPath "C:\Windows\SystemApps\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\microsoftedge.exe" -iconIndex 0 -ShowInPortal:$true
```

Troubleshoot application issues related to AVD using User Input Delay

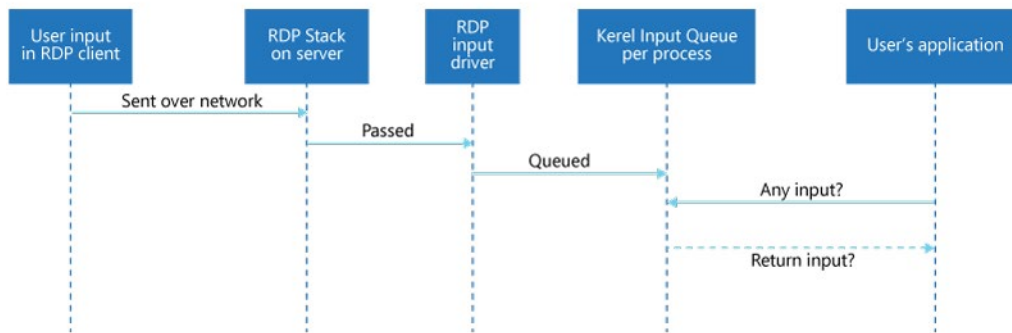
One of the most difficult problems to diagnose is poor application performance—the applications are running slow or don't respond.

²⁶ <https://docs.microsoft.com/en-us/powershell/module/appx/get-appxpackage?view=win10-ps/>

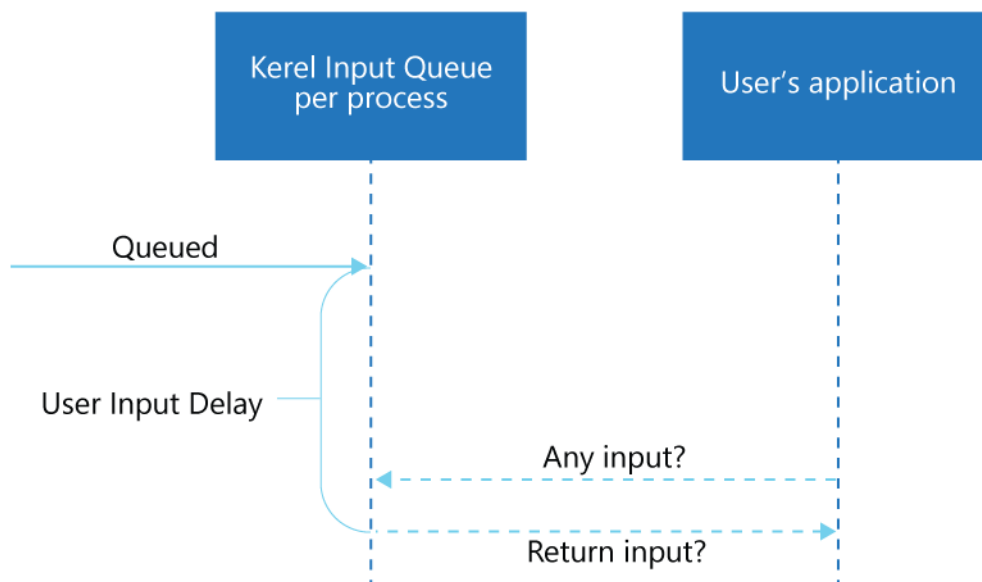
Traditionally, you start your diagnosis by collecting CPU, memory, disk input/output, and other metrics and then use tools like Windows Performance Analyzer to try to figure out what's causing the problem. Unfortunately, in most situations this data doesn't help you identify the root cause because resource consumption counters have frequent and large variations.

The User Input Delay counter can help you quickly identify the root cause for bad end user RDP experiences. This counter measures how long any user input (such as mouse or keyboard usage) stays in the queue before it is picked up by a process, and the counter works in both local and remote sessions.

The following image shows a representation of user input flow from client to application.



The User Input Delay counter measures the max delta (within an interval of time) between the input being queued and when it's picked up by the app in a traditional message loop, as shown in the following flow chart:



One important detail of this counter is that it reports the maximum user input delay within a configurable interval. This is the longest time it takes for an input to reach the application, which can impact the speed of important and visible actions like typing.

For example, in the following table, the user input delay would be reported as 1,000 ms within this interval. The counter reports the slowest user input delay in the interval because the user's perception of "slow" is determined by the slowest input time (the maximum) they experience, not the average speed of all total inputs.

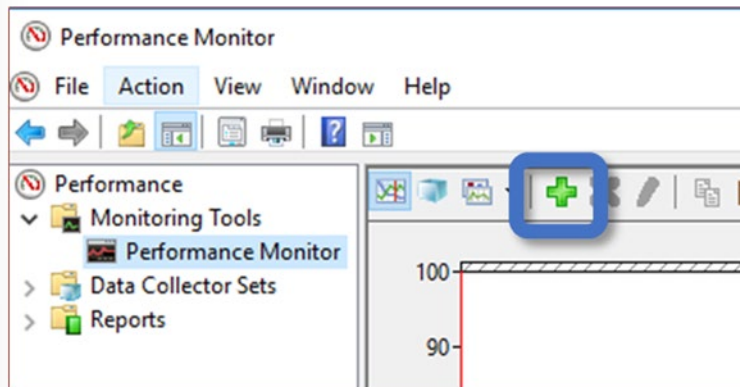
Number	0	1	2
Delay	16 ms	20 ms	1,000 ms

Enable and use the new performance counters

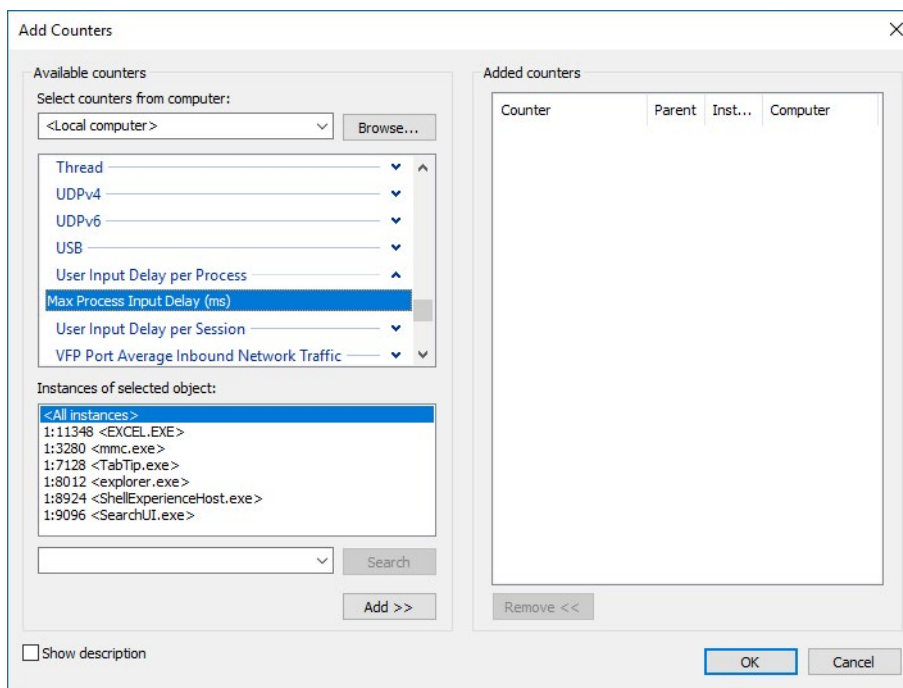
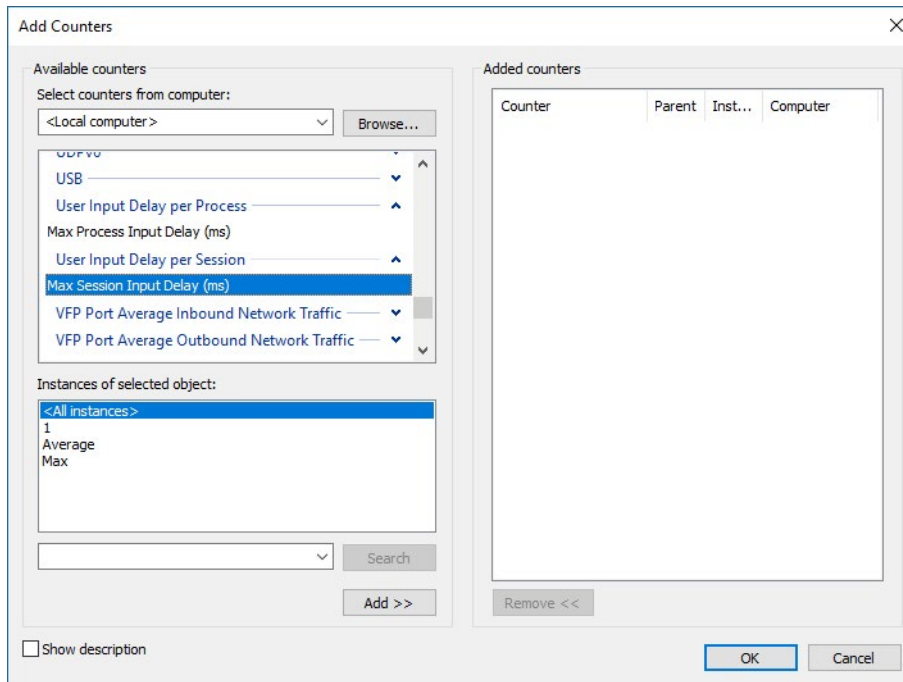
To use these new performance counters, you must first enable a registry key by running this command:

```
reg add "HKLM\System\CurrentControlSet\Control\Terminal Server" /v "Enable-LagCounter" /t REG_DWORD /d 0x1 /f
```

Next, restart the server. Then, open the Performance Monitor, and select the plus sign (+), as shown in the following screen shot.



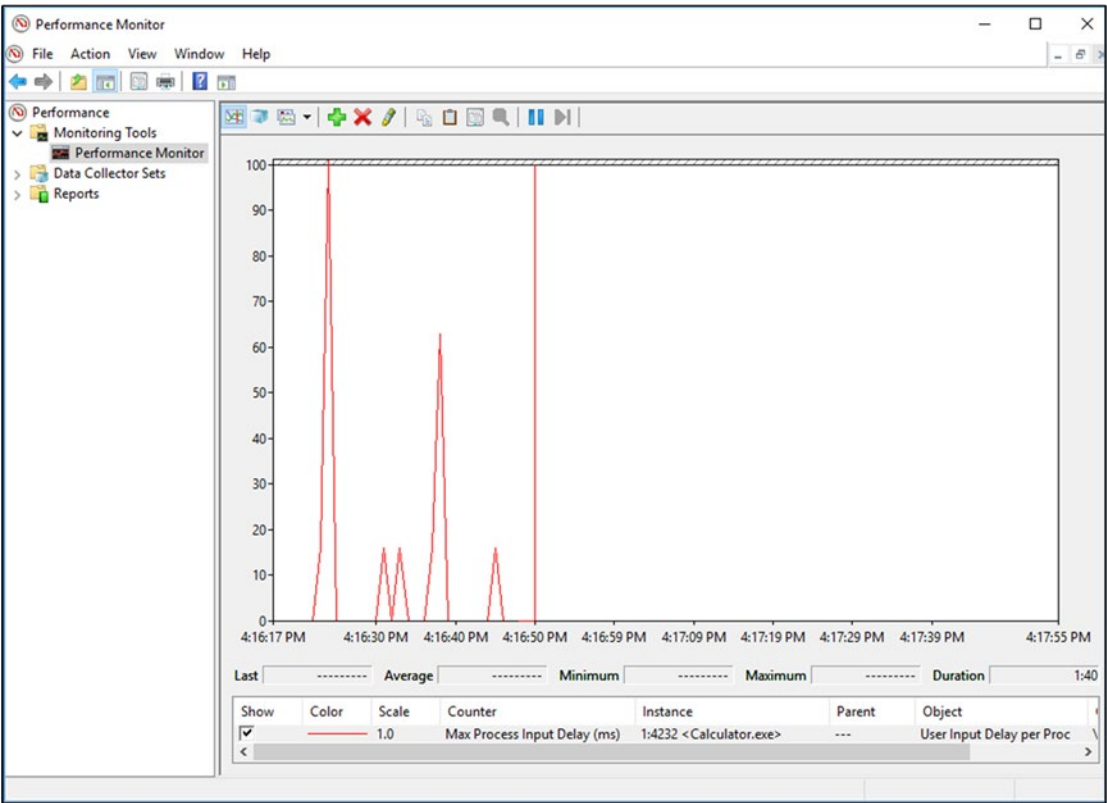
After doing that, you should see the Add Counters dialog, where you can select **User Input Delay per Process** or **User Input Delay per Session**.



If you select **User Input Delay per Process**, you'll see the **Instances of the selected object** (in other words, the processes) in SessionID:ProcessID <Process Image> format.

For example, if the Calculator app is running in a Session ID 1, you'll see 1:4232 <Calculator.exe>.

The counter starts reporting user input delay as soon as you add it. Note that the maximum scale is set to 100 (ms) by default.



User Input Delay per Session

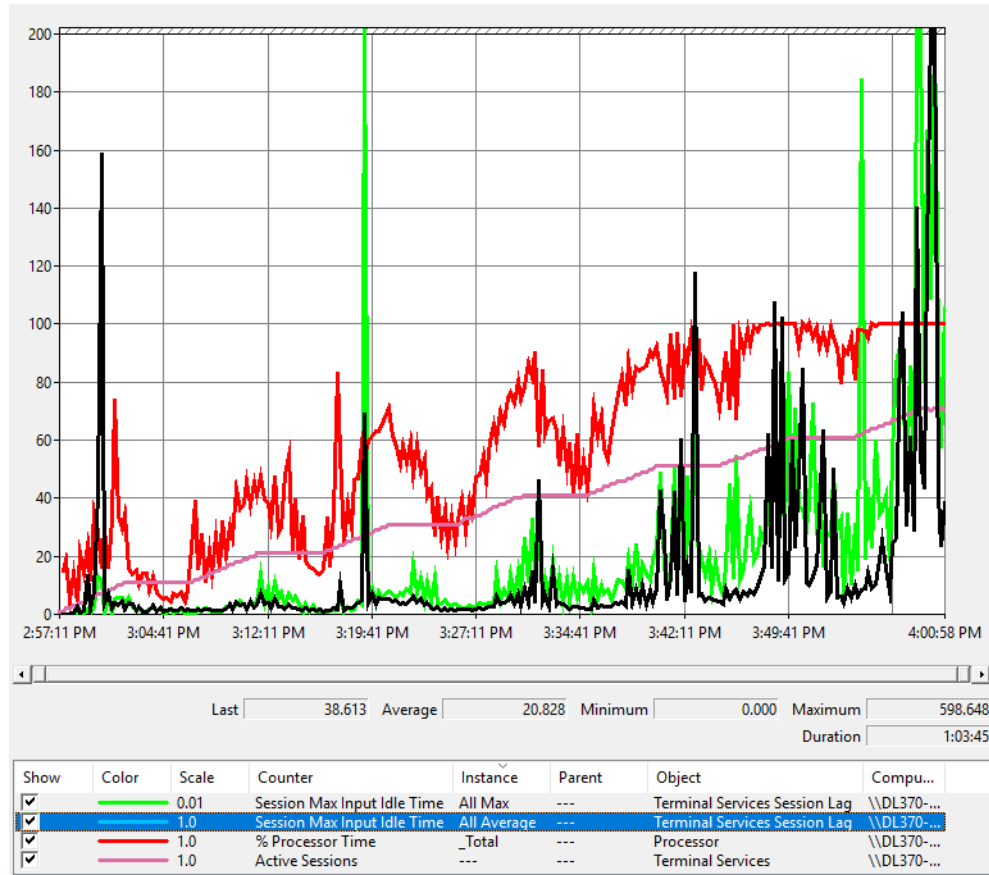
There are instances for each session ID, and their counters show the user input delay of any process within the specified session. In addition, there are two instances called "Max" (the maximum user input delay across all sessions) and "Average" (the average across all sessions).

This table shows a visual example of these instances. (You can get the same information in Perfmon by switching to the Report graph type.)

Type of counter	Instance name	Reported delay (ms)	
User Input Delay per process	1:4232 <Calculator.exe>	200	
User Input Delay per process	2:1000 <Calculator.exe>	16	
User Input Delay per process	1:2000 <Calculator.exe>	32	
User Input Delay per session	1	200	
User Input Delay per session	2	16	
User Input Delay per session	Average	108	
User Input Delay per session	Max	200	

Counters used in an overloaded system

Now let's look at what you'll see in the report if performance for an app is degraded. The following graph shows readings for users working remotely in Microsoft Word. In this case, the RDSH server performance degrades over time as more users log in.



Here's how to read the graph's lines:

- The pink line shows the number of sessions signed in on the server.
- The red line is the CPU usage.
- The green line is the maximum user input delay across all sessions.
- The blue line (displayed as black in this graph) represents average user input delay across all sessions.

Notice there's a correlation between CPU spikes and user input delay—as the CPU gets more usage, the user input delay increases. Also, as more users get added to the system, CPU usage gets closer to 100%, leading to more frequent user input delay spikes. While this counter is very useful in cases where the server runs out of resources, you can also use it to track user input delay related to a specific application.

Labs

Implement and manage Azure Virtual Desktop profiles (Azure AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)²⁷.

Direct link to the **Lab - Implement and manage Azure Virtual Desktop profiles (Azure AD DS)**.²⁸

Objectives

After completing this lab, you will be able to:

- Configure Azure Files to store profile containers for Azure Virtual Desktop in Azure AD DS environment
- Implement FSLogix based profiles for Azure Virtual Desktop in Azure AD DS environment

Lab prerequisites

- An Azure subscription
- A Microsoft account or an Azure AD account with the Global Administrator role in the Azure AD tenant associated with the Azure subscription and with the Owner or Contributor role in the Azure subscription
- An Azure Virtual Desktop environment provisioned in the lab **Introduction to Azure Virtual Desktop (Azure AD DS)**

Estimated time: 30 minutes

Lab files

- None

Exercise: Implement FSLogix based profiles for Azure Virtual Desktop

The main tasks for this exercise are as follows:

1. Configure local Administrators group on Azure Virtual Desktop session host VMs
2. Configure FSLogix-based profiles on Azure Virtual Desktop session host VMs
3. Test FSLogix-based profiles with Azure Virtual Desktop

Package Windows Azure Desktop applications (AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)²⁹.

²⁷ https://aka.ms/AZ-140_Labs

²⁸ https://aka.ms/AZ-140_04_Lab_01

²⁹ https://aka.ms/AZ-140_Labs

Direct link to the **Lab - Package Azure Virtual Desktop applications (AD DS)**.³⁰

Objectives

After completing this lab, you will be able to:

- Prepare for and create MSIX app packages
- Implement MSIX app attach container for Azure Virtual Desktop in AD DS environment
- Implement the MSIX app attach on Azure Virtual Desktop in AD DS environment

Lab prerequisites

- An Azure subscription
- A Microsoft account or an Azure AD account with the Global Administrator role in the Azure AD tenant associated with the Azure subscription and with the Owner or Contributor role in the Azure subscription
- The completed lab **Prepare for deployment of Azure Virtual Desktop (AD DS)** or **Prepare for deployment of Azure Virtual Desktop (Azure AD DS)**
- The completed lab **Azure Virtual Desktop profile management (AD DS)** or **Azure Virtual Desktop profile management (Azure AD DS)**

Estimated time: 90 minutes

Lab files

- \\AZ-140\\AllFiles\\Labs\\04\\az140-42_azuredeploycl42.json
- \\AZ-140\\AllFiles\\Labs\\04\\az140-42_azuredeploycl42.parameters.json

Exercise 1: Prepare for and create MSIX app packages

The main tasks for this exercise are as follows:

1. Prepare for configuration of Azure Virtual Desktop session hosts
2. Deploy an Azure VM running Windows 10 by using an Azure Resource Manager QuickStart template
3. Prepare the Azure VM running Windows 10 for MSIX packaging
4. Generate a signing certificate
5. Download software to package
6. Install the MSIX Packaging Tool
7. Create an MSIX package

³⁰ https://aka.ms/AZ-140_04_Lab_03

Exercise 2: Implement MSIX app attach container for Azure Virtual Desktop in Azure AD DS environment

The main tasks for this exercise are as follows:

1. Enable Hyper-V on the Azure VMs running Window 10 Enterprise Edition
2. Create an app attach container

Exercise 3: Implement MSIX app attach on Azure Virtual Desktop session hosts

The main tasks for this exercise are as follows:

1. Configure an Azure File share for MSIX app attach
2. Configure Active Directory groups containing Azure Virtual Desktop hosts
3. Set up the Azure Files share
4. Mount and register the MSIX App attach container on Azure Virtual Desktop session hosts
5. Publish MSIX apps to an application group
6. Validate the functionality of MSIX App attach

Exercise 4: Stop and deallocate Azure VMs provisioned and used in the lab

The main tasks for this exercise are as follows:

1. Stop and deallocate Azure VMs provisioned and used in the lab

Note: In this exercise, you will deallocate the Azure VMs provisioned and used in this lab to minimize the corresponding compute charges

Review Questions

Module review questions

Review Question 1

You manage a AVD instance.

You need to plan how to run containerized apps without running the apps on the session host.

What do you need to use?

- ☐ A. MSIX app packages
- ☐ B. Docker remote containers
- ☐ C. NuGet packages
- ☐ D. Azure Service Bus queues

Review Question 2

You have a AVD host pool named HostPoolNorth that has three session hosts.

The sessions hosts use FSLogix profile containers.

You want to configure Cloud Cache on each of the session hosts.

What do you do first?

- ☐ A. Create and MSIX package
- ☐ B. Configure the IsAVDEnvironment reg key on all Windows 2019 servers
- ☐ C. Remove VHDLocations entries from the Windows registry
- ☐ D. Copy the rule sets to the FSLogix profile containers for all users

Review Question 3

You have a AVD host pool named: HostPoolSouth.

You are investigating and issue for a Remote Desktop client that is no longer responding.

You want to the default Remote Desktop client settings restored to the defaults and unsubscribed for workspaces.

What should you do?

- ☐ A. Stop the RDAgentBootLoader (net stop RDAgentBootLoader)
- ☐ B. Create a second host pool
- ☐ C. Install the FSLogix agent on the session hosts in HostPoolSouth
- ☐ D. Run msrdcw.exe

Review Question 4

You have a AVD deployment with the following:

- A host pool named HostPool-1
- Two session hosts: AVDSessionHost1 and AVDSessionHost1

- App groups named AppRemoteGR1 and AppRemoteGR2 with a RemoteApp named UserRemoteApp-1

You don't want your users copying and pasting content from UserRemoteApp-1 to a local computer or device.

What should you do to prevent this?

- ☐ A. Modify the RDP Properties of HostPool-1
- ☐ B. Require multi-factor authentication (MFA)
- ☐ C. Re-register App groups AppRemoteGR1 and AppRemoteGR2
- ☐ D. Disable Storage Sense on both session hosts

Review Question 5

You have a AVD deployment that uses M365 services that include Microsoft Teams.

Users have the remote desktop client to connect to the deployment using Win 10 computers.

You want to support the A/V features in AVD and allow your users with access to Teams calling and meeting features.

Which of the following three things should you do first?

- ☐ A. Install the Teams WebSocket Service on the VMs
- ☐ B. Install the Teams WebSocket Service all Windows 2019 Servers
- ☐ C. Install the Teams desktop app on the Windows 10 computers
- ☐ D. Install the Teams desktop app on the VMs
- ☐ E. Install the Remote Desktop WebRTC Redirector Service

Review Question 6

You manage a AVD host pool containing two session hosts. You have the Microsoft Teams client installed on both host sessions.

You have the following situation:

- The meeting feature is disabled
- The call feature is disabled
- The Teams collaboration feature is working
- The Teams chat feature is working

You want to allow your users the ability to use call and meeting features.

What do you do to resolve this?

- ☐ A. Add VHDLocations entries for the Windows registry
- ☐ B. Create an AppLocker policy
- ☐ C. Install the Remote Desktop WebRTC Redirector Service
- ☐ D. Configure RDP Properties on the host pool

Review Question 7

You manage a AVD deployment with the following:

- A AVD host pool that contains 20 Windows 10 En. Multi-session hosts.
- Users connect to the AVD deployment from Windows 10 computers.
You plan on using FSLogix Application Masking.

You want to deploy Application Masking rule sets.

Where should you copy the rule sets?

- ☐ A. C:\Program Files\FSLogix\Apps\Rules on every session host
- ☐ B. FSLogix Office Container
- ☐ C. An Azure Storage account
- ☐ D. \store2\file.core.windows.net\profiles

Answers

Review Question 1

You manage a AVD instance.

You need to plan how to run containerized apps without running the apps on the session host.

What do you need to use?

- ☒ A. MSIX app packages
- ☐ B. Docker remote containers
- ☐ C. NuGet packages
- ☐ D. Azure Service Bus queues

Explanation

The answer is A, MSIX app packages.

As seen in the *MSIX App Attach* topic, the Azure Virtual Desktop service recommends FSLogix profile containers as a user profile solution.

FSLogix is designed to roam profiles in remote computing environments, such as Azure Virtual Desktop. It stores a complete user profile in a single container.

Review Question 2

You have a AVD host pool named HostPoolNorth that has three session hosts.

The sessions hosts use FSLogix profile containers.

You want to configure Cloud Cache on each of the session hosts.

What do you do first?

- ☐ A. Create and MSIX package
- ☐ B. Configure the IsAVDEnviroment reg key on all Windows 2019 servers
- ☒ C. Remove VHDLocations entries from the Windows registry
- ☐ D. Copy the rule sets to the FSLogix profile containers for all users

Explanation

The answer is C, Remove VHDLocations entries from the Windows registry.

As seen in the *Configure Profile Containers*, topic, the configuration of Profile Container is accomplished through registry settings and user groups.

VHDLocations (required setting), A list of file system locations to search for the user's profile VHD(X) file. If one isn't found, one will be created in the first listed location.

If the VHD path doesn't exist, it will be created before it checks if a VHD(X) exists in the path.

These values can contain variables that will be resolved. Supported variables are %username%, %userdomain%, %sid%, %osmajor%, %osminor%, %osbuild%, %osservicepack%, %profileversion%, and any environment variable available at time of use.

Review Question 3

You have a AVD host pool named: HostPoolSouth.

You are investigating an issue for a Remote Desktop client that is no longer responding.

You want to the default Remote Desktop client settings restored to the defaults and unsubscribed for workspaces.

What should you do?

- ☐ A. Stop the RDAgentBootLoader (net stop RDAgentBootLoader)
- ☐ B. Create a second host pool
- ☐ C. Install the FSLogix agent on the session hosts in HostPoolSouth
- ☒ D. Run msrdcw.exe

Explanation

The answer is D, Run msrdcw.exe.

As seen in the *Troubleshoot AVD clients* topic, if a remote Desktop client for Windows 10 stops responding or cannot be opened, you can reset the user data from the About page or using a command.

Use the following command to remove your user data, restore default settings and unsubscribe from all Workspaces.*msrdcw.exe /reset [/f]a8

Review Question 4

You have a AVD deployment with the following:

You don't want your users copying and pasting content from UserRemoteApp-1 to a local computer or device.

What should you do to prevent this?

- ☒ A. Modify the RDP Properties of HostPool-1
- ☐ B. Require multi-factor authentication (MFA)
- ☐ C. Re-register App groups AppRemoteGR1 and AppRemoteGR2
- ☐ D. Disable Storage Sense on both session hosts

Explanation

The answer is A, Modify the RDP Properties of HostPool-1.

As seen in the *Configure device redirections* topic, you set the following RDP property to configure clipboard redirection:

Review Question 5

You have a AVD deployment that uses M365 services that include Microsoft Teams. Users have the remote desktop client to connect to the deployment using Win 10 computers. You want to support the A/V features in AVD and allow your users with access to Teams calling and meeting features.

Which of the following three things should you do first?

- ☒ A. Install the Teams WebSocket Service on the VMs
- ☐ B. Install the Teams WebSocket Service all Windows 2019 Servers
- ☒ C. Install the Teams desktop app on the Windows 10 computers
- ☒ D. Install the Teams desktop app on the VMs
- ☐ E. Install the Remote Desktop WebRTC Redirector Service

Explanation

The answer is A, C, and D:

As seen in the *Using Microsoft Teams on Azure Virtual desktop* topic, you install the Teams desktop app, install the Teams WebSocket Service, and install Microsoft Teams.

Review Question 6

You manage a AVD host pool containing two session hosts. You have the Microsoft Teams client installed on both host sessions.

You have the following situation:

You want to allow your users the ability to use call and meeting features.

What do you do to resolve this?

- ☐ A. Add VHDLocations entries for the Windows registry
- ☐ B. Create an AppLocker policy
- ☒ C. Install the Remote Desktop WebRTC Redirector Service
- ☐ D. Configure RDP Properties on the host pool

Explanation

The answer is C, Install the Remote Desktop WebRTC Redirector Service.

As seen in the *Using Microsoft Teams on Azure Virtual desktop* topic, by installing the Teams WebSocket Service (version 1.0.2006.11001), you fix an issue with the Teams app during a call or meeting with incoming video to dropping.

Review Question 7

You manage a AVD deployment with the following:

You want to deploy Application Masking rule sets.
Where should you copy the rule sets?

- ☒ A. C:\Program Files\FSSLogix\Apps\Rules on every session host
- ☐ B. FSSLogix Office Container
- ☐ C. An Azure Storage account
- ☐ D. \store2\file.core.windows.net\profiles

Explanation

The answer is A, C:\Program Files\FSSLogix\Apps\Rules on every session host.

As seen in the Manage Rule Sets and application masking topic, by default the Rules and Rule Sets are accessed from C:\Program Files\FSSLogix\Apps\Rules.

The location where Rules and Rule Sets are accessed differ if the FSSLogix installation location is changed.

Module 5 Monitor and maintain a AVD infrastructure

Plan and implement business continuity and disaster recovery

VM replication

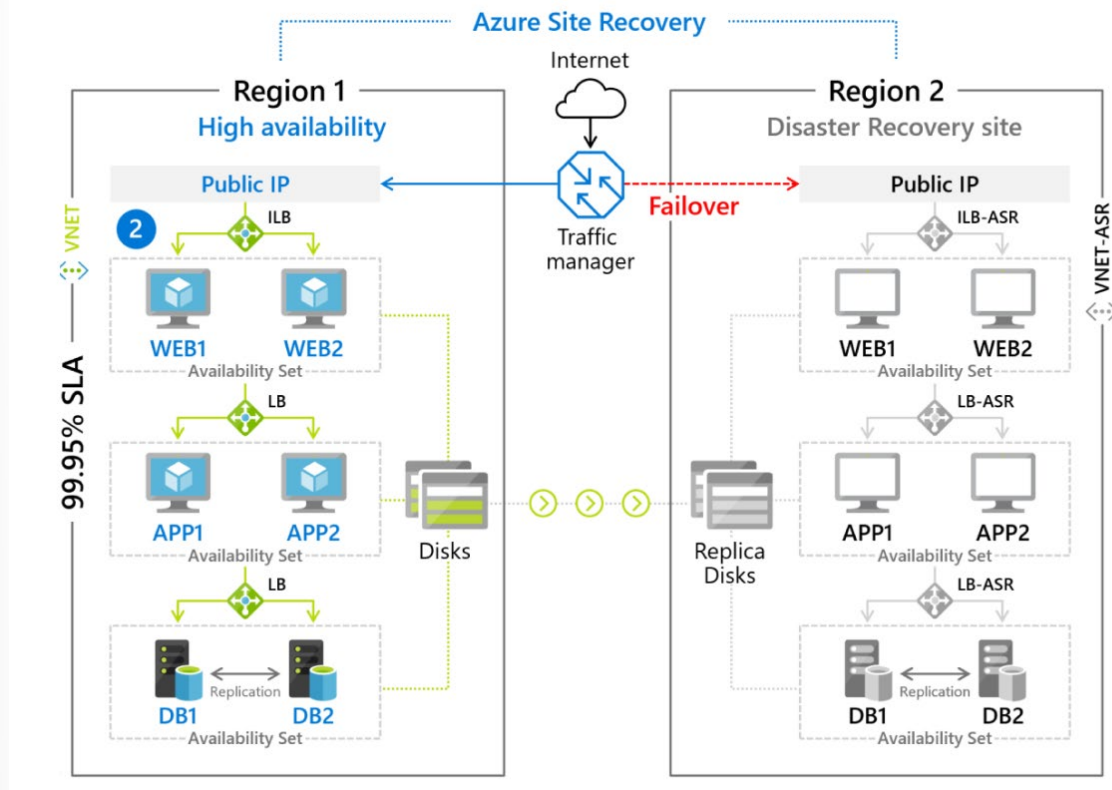
First, you'll need to replicate your VMs to the secondary location. Your options for doing so depend on how your VMs are configured:

- You can configure all your VMs for both pooled and personal host pools with Azure Site Recovery. With this method, you'll only need to set up one host pool and its related app groups and workspaces.
- You can create a new host pool in the failover region while keeping all resources in your failover location turned off.
 - You need to set up new app groups and workspaces in the failover region, then use an Azure Site Recovery plan to turn host pools on.
- You can create a host pool that's populated by VMs built in both the primary and failover regions while keeping the VMs in the failover region turned off.
 - You only need to set up one host pool and its related app groups and workspaces.
 - You can use an Azure Site Recovery plan to power on host pools with this method.

Use Azure Site Recovery to manage replicating VMs in other Azure locations, as described in **Azure-to-Azure disaster recovery architecture**¹.

Set up Azure Site Recovery by replicating an Azure VM to a different Azure region directly from the Azure portal. Site Recovery is automatically updated with new Azure features as they're released.

¹ <https://github.com/MicrosoftDocs/azure-docs/blob/master/articles/site-recovery/azure-to-azure-architecture.md>



You should use Azure Site Recovery for personal host pools, because Azure Site Recovery supports both server-based and client-based SKUs.

If you use Azure Site Recovery, you won't need to register VMs manually.

- The Azure Virtual Desktop agent in the secondary VM will automatically use the latest security token to connect to the service instance closest to it.
- The VM (session host) in the secondary location will automatically become part of the host pool.
- The end-user will have to reconnect during the process.

If there are existing user connections during the outage, before the admin can start failover to the secondary region, you need to end the user connections in the current region.

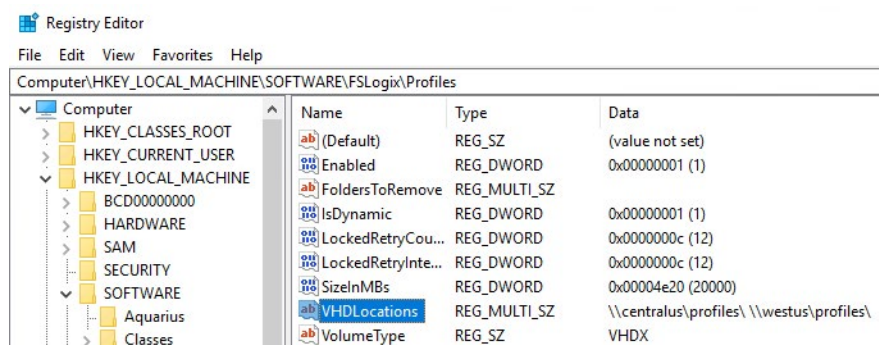
Once you've signed out all users in the primary region, you can fail over the VMs in the primary region and let users connect to the VMs in the secondary region.

FSLogix configuration

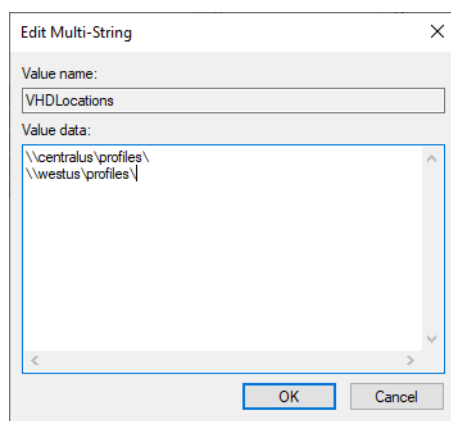
The FSLogix agent can support multiple profile locations if you configure the registry entries for FSLogix.

To configure the registry entries:

1. Open the **Registry Editor**.
2. Go to **Computer > HKEY_LOCAL_MACHINE > SOFTWARE > FSLogix > Profiles**.



3. Right-click on **VHDLocations** and select **Edit Multi-String**.



4. In the **Value Data** field, enter the locations you want to use.

5. When you're done, select **OK**.

If the first location is unavailable, the FSLogix agent will automatically fail over to the second, and so on.

It's recommended you configure the FSLogix agent with a path to the secondary location in the main region. Once the primary location shuts down, the FLogix agent will replicate as part of the VM Azure Site Recovery replication. Once the replicated VMs are ready, the agent will automatically attempt to path to the secondary region.

Example scenario

Let's say your primary session host VMs are in the Central US region, but your profile container is in the Central US region for performance reasons.

In this case, you would configure the FSLogix agent with a path to the storage in Central US. You would configure the session host VMs to replicate in West US. Once the path to Central US fails, the agent will try to create a new path for storage in West US instead.

S2D

Since S2D handles replication across regions internally, you don't need to manually set up the secondary path.

Network drives (VM with extra drives)

If you replicate the network storage VMs using Azure Site Recovery like the session host VMs, then the recovery keeps the same path, which means you don't need to reconfigure FSLogix.

Azure Files

Azure Files supports cross-region asynchronous replication that you can specify when you create the storage account. If the asynchronous nature of Azure Files already covers your disaster recovery goals, then you don't need to do additional configuration.

If you need synchronous replication to minimize data loss, then we recommend you use FSLogix Cloud Cache instead.

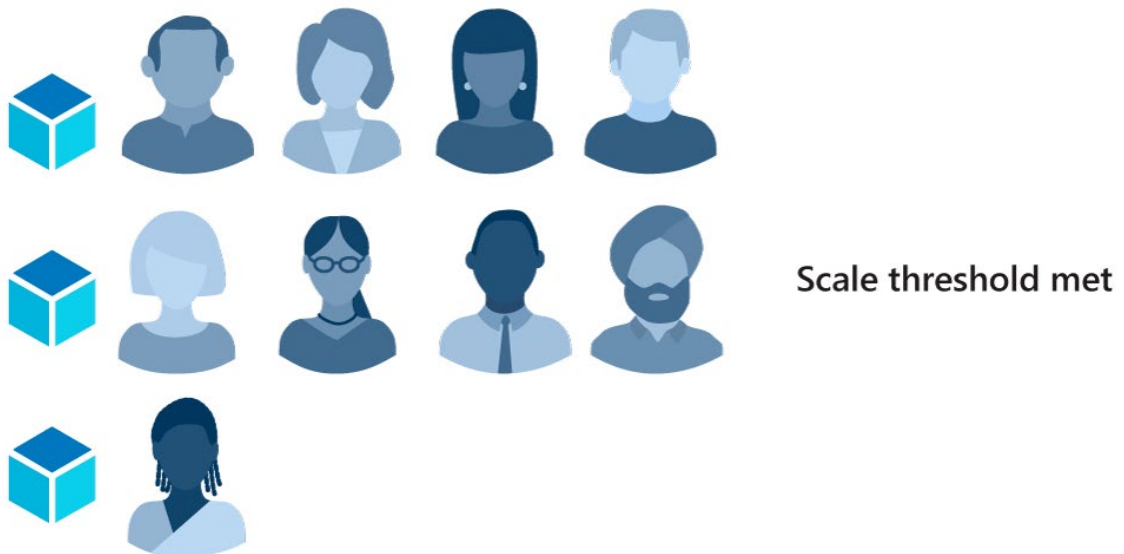
Automate AVD management tasks

Scale session hosts using Azure Automation

You can reduce your total Azure Virtual Desktop deployment cost by scaling your virtual machines (VMs). This means shutting down and deallocating session host VMs during off-peak usage hours, then turning them back on and reallocating them during peak hours.



Early morning, employees arriving at work

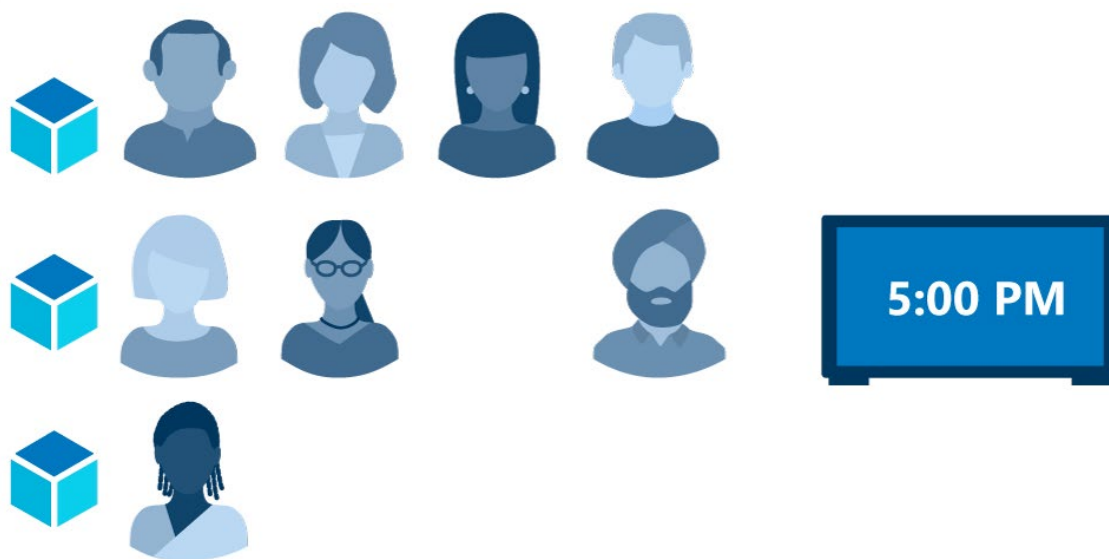


The threshold is met, increased VM allocation to meet demand

Ideal optimization for VM allocation



The work day winds down



Remaining workers begin leaving for home



In this topic, you'll learn about the scaling tool built with the Azure Automation account and Azure Logic App that automatically scales session host VMs in your Azure Virtual Desktop environment.

How the scaling tool works

The scaling tool provides a low-cost automation option for customers who want to optimize their session host VM costs.

You can use the scaling tool to:

- Schedule VMs to start and stop based on Peak and Off-Peak business hours.
- Scale out VMs based on number of sessions per CPU core.
- Scale in VMs during Off-Peak hours, leaving the minimum number of session host VMs running.

The scaling tool uses a combination of an Azure Automation account, a PowerShell runbook, a webhook, and the Azure Logic App to function. When the tool runs, Azure Logic App calls a webhook to start the Azure Automation runbook. The runbook then creates a job.

During peak usage time, the job checks the current number of sessions and the VM capacity of the current running session host for each host pool. It uses this information to calculate if the running session host VMs can support existing sessions based on the `SessionThresholdPerCPU` parameter defined for the `CreateOrUpdateAzLogicApp.ps1` file.

- If the session host VMs can't support existing sessions, the job starts additional session host VMs in the host pool.

During the off-peak usage time, the job determines how many session host VMs should be shut down based on the `MinimumNumberOfRDSH` parameter. If you set the `LimitSecondsToForceLogOffUser` parameter to a non-zero positive value, the job will set the session host VMs to drain mode to prevent new sessions from connecting to the hosts.

- The job will notify any currently signed in users to save their work, wait the configured amount of time, and then force the users to sign out.
- Once all user sessions on the session host VM have been signed out, the job will shut down the VM.
- After the VM shuts down, the job will reset its session host drain mode.

Create or update an Azure Automation account

You'll need an Azure Automation account to run the PowerShell runbook. The process this section describes is useful even if you have an existing Azure Automation account that you want to use to set up the PowerShell runbook.

Note: To setup a standalone automation account and Run As account using the Azure portal, see **Create a standalone Azure Automation account**².

Here's how to set it up:

1. Open Windows PowerShell.
2. Run the following cmdlet to sign in to your Azure account.

```
Login-AzAccount
```

Note: Your account must have contributor rights on the Azure subscription where you want to deploy the scaling tool.

3. Run the following cmdlet to download the script for creating the Azure Automation account:

```
New-Item -ItemType Directory -Path "C:\Temp" -Force
Set-Location -Path "C:\Temp"
$Uri = "https://raw.githubusercontent.com/Azure/RDS-Templates/master/
wvd-templates/wvd-scaling-script/CreateOrUpdateAzAutoAccount.ps1"
# Download the script
Invoke-WebRequest -Uri $Uri -OutFile ".\CreateOrUpdateAzAutoAccount.ps1"
```

4. Run the following cmdlet to execute the script and create the Azure Automation account. You can either fill in values for the parameters or comment them to use their defaults.

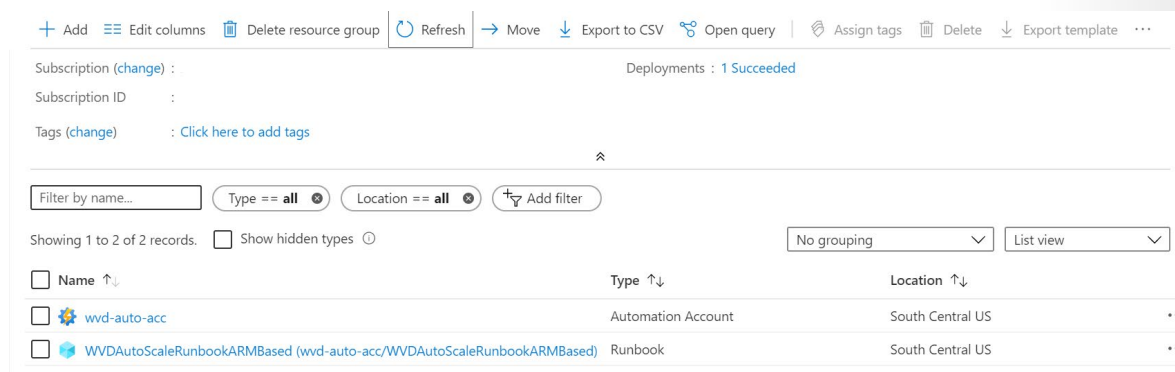
```
$Params = @{
    "AADTenantId"          = "<Azure_Active_Directory_tenant_ID>" #
    Optional. If not specified, it will use the current Azure context
    "SubscriptionId"       = "<Azure_subscription_ID>"           #
    Optional. If not specified, it will use the current Azure context
    "UseARMAPI"            = $true
    "ResourceGroupName"    = "<Resource_group_name>"             #
    Optional. Default: "AVDAutoScaleResourceGroup"
    "AutomationAccountName" = "<Automation_account_name>"       #
    Optional. Default: "AVDAutoScaleAutomationAccount"
    "Location"             = "<Azure_region_for_deployment>"
    "WorkspaceName"        = "<Log_analytics_workspace_name>"    #
    Optional. If specified, Log Analytics will be used to configure the custom
    log table that the runbook PowerShell script can send logs to
}

.\CreateOrUpdateAzAutoAccount.ps1 @Params
```

5. The cmdlet's output will include a webhook URI. Make sure to keep a record of the URI because you'll use it as a parameter when you set up the execution schedule for the Azure Logic App.

² <https://docs.microsoft.com/en-us/azure/automation/automation-create-standalone-account>

- If you specified the parameter **WorkspaceName** for Log Analytics, the cmdlet's output will also include the Log Analytics Workspace ID and its Primary Key. Make sure to remember URI because you'll need to use it again later as a parameter when you set up the execution schedule for the Azure Logic App.
- After you've set up your Azure Automation account, sign in to your Azure subscription and check to make sure your Azure Automation account and the relevant runbook have appeared in your specified resource group, as shown in the following image:



To check if your webhook is where it should be, select the name of your runbook. Next, go to your runbook's Resources section and select Webhooks.

Create an Azure Automation Run As account

Now that you have an Azure Automation account, you'll also need to create an Azure Automation Run As account.

The Azure Automation Run As account will allow you access to your Azure resources.

An Azure Automation Run As account provides authentication for managing resources in Azure with Azure cmdlets. When you create a Run As account, it creates a new service principal user in Azure Active Directory and assigns the Contributor role to the service principal user at the subscription level. An Azure Run As account is a great way to authenticate securely with certificates and a service principal name without needing to store a username and password in a credential object.

Any user who's a member of the Subscription Admins role and coadministrator of the subscription can create a Run As account.

To create a Run As account in your Azure Automation account:

- In the Azure portal, select **All services**. In the list of resources, enter and select **Automation accounts**.
- On the **Automation accounts** page, select the name of your Azure Automation account.

Microsoft Azure

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Automation Accounts

Default Directory

+ Add

⚙️ Manage view ▾

🔄 Refresh

⬇️ Export to CSV

🔗 Open query

💙 Feedback

|

🏷️ Assign tags

Filter for any field...

Subscription == all

Resource group == all ✕

Location == all ✕

+ Add filter

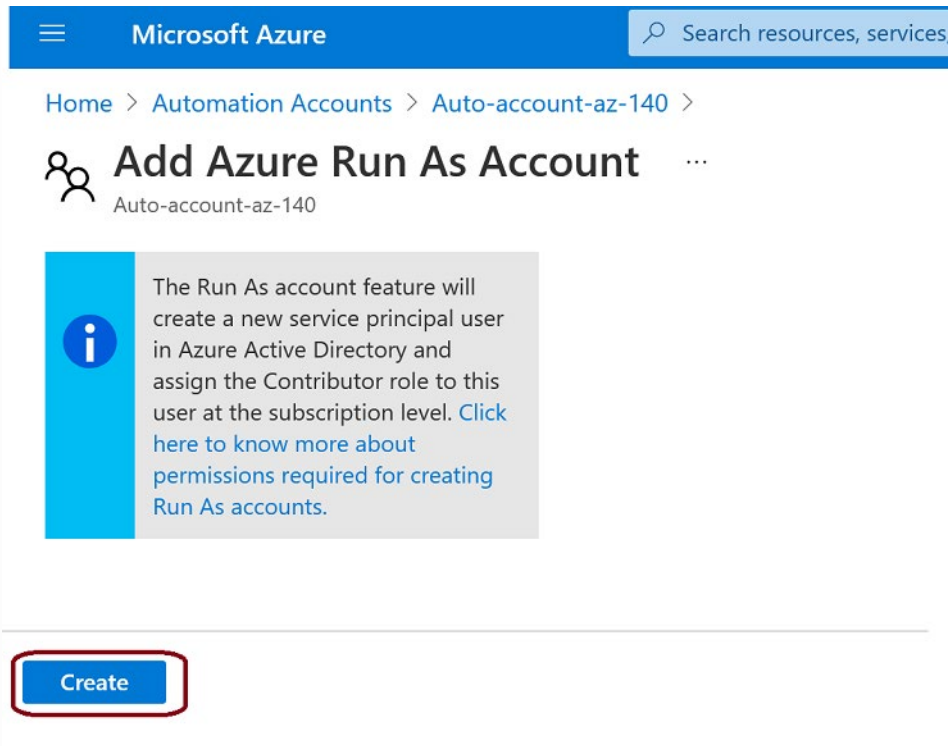
Showing 1 to 1 of 1 records.

☐ Name ↑↓	Type ↑↓	Resource group ↑↓	Location ↑↓
☐ ⚙️ Auto-account-az-140	Automation Account	avd-dm-rg1	East US

3. In the pane on the left side of the window, select **Run As accounts** under the **Account Settings** section.

The screenshot displays the Microsoft Azure portal interface. At the top, the navigation bar includes the Microsoft Azure logo and a search bar. The breadcrumb trail indicates the current location: Home > Automation Accounts > Auto-account-az-140. The left-hand navigation pane shows the 'Automation Accounts' section with a search bar and a list of accounts. The 'Auto-account-az-140' account is selected and highlighted. The main content area on the right provides details for the selected account, including a search bar, a list of related resources (Linked workspace, Event grid, Start/Stop VM), and a list of account settings (Properties, Network Isolation, Keys, Pricing, Source control, Run as accounts, Identity). The 'Run as accounts' option is currently selected and highlighted.

4. Select **Azure Run As account**. When the **Add Azure Run As account** pane appears, review the overview information, and then select **Create** to start the account creation process.



Microsoft Azure

Search resources, services

Home > Automation Accounts > Auto-account-az-140 >

Add Azure Run As Account ...

Auto-account-az-140

i The Run As account feature will create a new service principal user in Azure Active Directory and assign the Contributor role to this user at the subscription level. [Click here to know more about permissions required for creating Run As accounts.](#)

Create

5. Wait a few minutes for Azure to create the Run As account. You can track the creation progress in the menu under Notifications.
6. When the process finishes, it will create an account in the specified Azure Automation account. Select **Azure Run As account**.

 **Azure Run As Account**  ...

Properties

 Renew certificate  Delete

Azure Active Directory Application ⓘ

Display Name

Auto-account-az-140_6d0+Armi6Gqidn ... 

Certificate ⓘ

Name

AzureRunAsCertificate

Expiration

4/25/2022, 5:00 PM

Thumbprint

6E036553F70D306726974A8AB10DCD8 ... 

Connection ⓘ

Name

AzureRunAsConnection

Application ID

9bc0ca8e-daaa-466a-94f4-38166fafd70a 

Thumbprint

6E036553F70D306726974A8AB10DCD8 ... 

Subscription ID

d615bd6f-0b17-455c-b637-fb0b1a15b9 ... 

Tenant ID

6168e3b4-025e-4996-8615-1312f5f2293a 

Service Principal ⓘ

Service Principal Object ID

44530d10-c1f4-4e15-944c-693a189d6ec0

Roles ⓘ

Role	Assigned at
Contributor	Azure Pass - Sponsorship

Tutorial Runbook ⓘ

PowerShell script

AzureAutomationTutorialScript

Graphical runbook

AzureAutomationTutorial

Create the Azure Logic App and execution schedule

Finally, you'll need to create the Azure Logic App and set up an execution schedule for your new scaling tool.

1. Open Windows PowerShell.

2. Run the following cmdlet to sign in to your Azure account.

```
Login-AzAccount
```

3. Run the following cmdlet to download the script for creating the Azure Logic App.

```
New-Item -ItemType Directory -Path "C:\Temp" -Force
Set-Location -Path "C:\Temp"
$Uri = "https://raw.githubusercontent.com/Azure/RDS-Templates/master/
wvd-templates/wvd-scaling-script/CreateOrUpdateAzLogicApp.ps1"
# Download the script
Invoke-WebRequest -Uri $Uri -OutFile ".\CreateOrUpdateAzLogicApp.ps1"
```

4. Run the following PowerShell script to create the Azure Logic App and execution schedule for your host pool:

Note: You'll need to run this script for each host pool you want to autoscale, but you need only one Azure Automation account.

```
$AADTenantId = (Get-AzContext).Tenant.Id

$AzSubscription = Get-AzSubscription | Out-GridView -OutputMode:Single
-Title "Select your Azure Subscription"
Select-AzSubscription -Subscription $AzSubscription.Id

$ResourceGroup = Get-AzResourceGroup | Out-GridView -OutputMode:Single
-Title "Select the resource group for the new Azure Logic App"

$AVDHostPool = Get-AzResource -ResourceType "Microsoft.DesktopVirtualiza-
tion/hostpools" | Out-GridView -OutputMode:Single -Title "Select the host
pool you'd like to scale"

$LogAnalyticsWorkspaceId = Read-Host -Prompt "If you want to use Log Ana-
lytics, enter the Log Analytics Workspace ID returned by when you created
the Azure Automation account, otherwise leave it blank"
$LogAnalyticsPrimaryKey = Read-Host -Prompt "If you want to use Log Analyt-
ics, enter the Log Analytics Primary Key returned by when you created the
Azure Automation account, otherwise leave it blank"
$RecurrenceInterval = Read-Host -Prompt "Enter how often you'd like the job
to run in minutes, e.g. '15'"
$BeginPeakTime = Read-Host -Prompt "Enter the start time for peak hours in
local time, e.g. 9:00"
$EndPeakTime = Read-Host -Prompt "Enter the end time for peak hours in
local time, e.g. 18:00"
$TimeDifference = Read-Host -Prompt "Enter the time difference between local
time and UTC in hours, e.g. +5:30"
$SessionThresholdPerCPU = Read-Host -Prompt "Enter the maximum number of
sessions per CPU that will be used as a threshold to determine when new
session host VMs need to be started during peak hours"
$MinimumNumberOfRDSH = Read-Host -Prompt "Enter the minimum number of
session host VMs to keep running during off-peak hours"
$MaintenanceTagName = Read-Host -Prompt "Enter the name of the Tag associ-
```

```

ated with VMs you don't want to be managed by this scaling tool"
$LimitSecondsToForceLogOffUser = Read-Host -Prompt "Enter the number of
seconds to wait before automatically signing out users. If set to 0, any
session host VM that has user sessions, will be left untouched"
$LogOffMessageTitle = Read-Host -Prompt "Enter the title of the message sent
to the user before they are forced to sign out"
$LogOffMessageBody = Read-Host -Prompt "Enter the body of the message sent
to the user before they are forced to sign out"

$AutoAccount = Get-AzAutomationAccount | Out-GridView -OutputMode:Single
-Title "Select the Azure Automation account"
$AutoAccountConnection = Get-AzAutomationConnection -ResourceGroupName
$AutoAccount.ResourceGroupName -AutomationAccountName $AutoAccount.Automa-
tionAccountName | Out-GridView -OutputMode:Single -Title "Select the Azure
RunAs connection asset"

$WebhookURIAutoVar = Get-AzAutomationVariable -Name 'WebhookURIARMBased'
-ResourceGroupName $AutoAccount.ResourceGroupName -AutomationAccountName
$AutoAccount.AutomationAccountName

$Params = @{
    "AADTenantId"                = $AADTenantId
    # Optional. If not specified, it will use the current Azure context
    "SubscriptionID"              = $AzSubscription.Id
    # Optional. If not specified, it will use the current Azure context
    "ResourceGroupName"          = $ResourceGroup.ResourceGroupName
    # Optional. Default: "AVDAutoScaleResourceGroup"
    "Location"                    = $ResourceGroup.Location
    # Optional. Default: "West US2"
    "UseARMAPI"                   = $true
    "HostPoolName"                = $AVDHostPool.Name
    "HostPoolResourceGroupName"   = $AVDHostPool.ResourceGroupName
    # Optional. Default: same as ResourceGroupName param value
    "LogAnalyticsWorkspaceId"     = $LogAnalyticsWorkspaceId
    # Optional. If not specified, script will not log to the Log Analytics
    "LogAnalyticsPrimaryKey"      = $LogAnalyticsPrimaryKey
    # Optional. If not specified, script will not log to the Log Analytics
    "ConnectionAssetName"        = $AutoAccountConnection.Name
    # Optional. Default: "AzureRunAsConnection"
    "RecurrenceInterval"          = $RecurrenceInterval
    # Optional. Default: 15
    "BeginPeakTime"               = $BeginPeakTime
    # Optional. Default: "09:00"
    "EndPeakTime"                 = $EndPeakTime
    # Optional. Default: "17:00"
    "TimeDifference"              = $TimeDifference
    # Optional. Default: "-7:00"
    "SessionThresholdPerCPU"      = $SessionThresholdPerCPU
    # Optional. Default: 1
    "MinimumNumberOfRDSH"         = $MinimumNumberOfRDSH
    # Optional. Default: 1

```

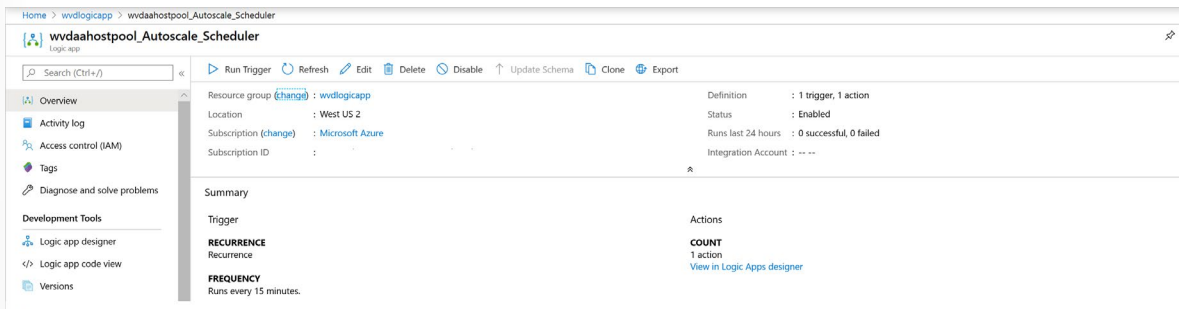
```

    "MaintenanceTagName"                = $MaintenanceTagName
# Optional.
    "LimitSecondsToForceLogOffUser" = $LimitSecondsToForceLogOffUser
# Optional. Default: 1
    "LogOffMessageTitle"                = $LogOffMessageTitle
# Optional. Default: "Machine is about to shutdown."
    "LogOffMessageBody"                 = $LogOffMessageBody
# Optional. Default: "Your session will be logged off. Please save and close
everything."
    "WebhookURI"                        = $WebhookURIAutoVar.Value
}

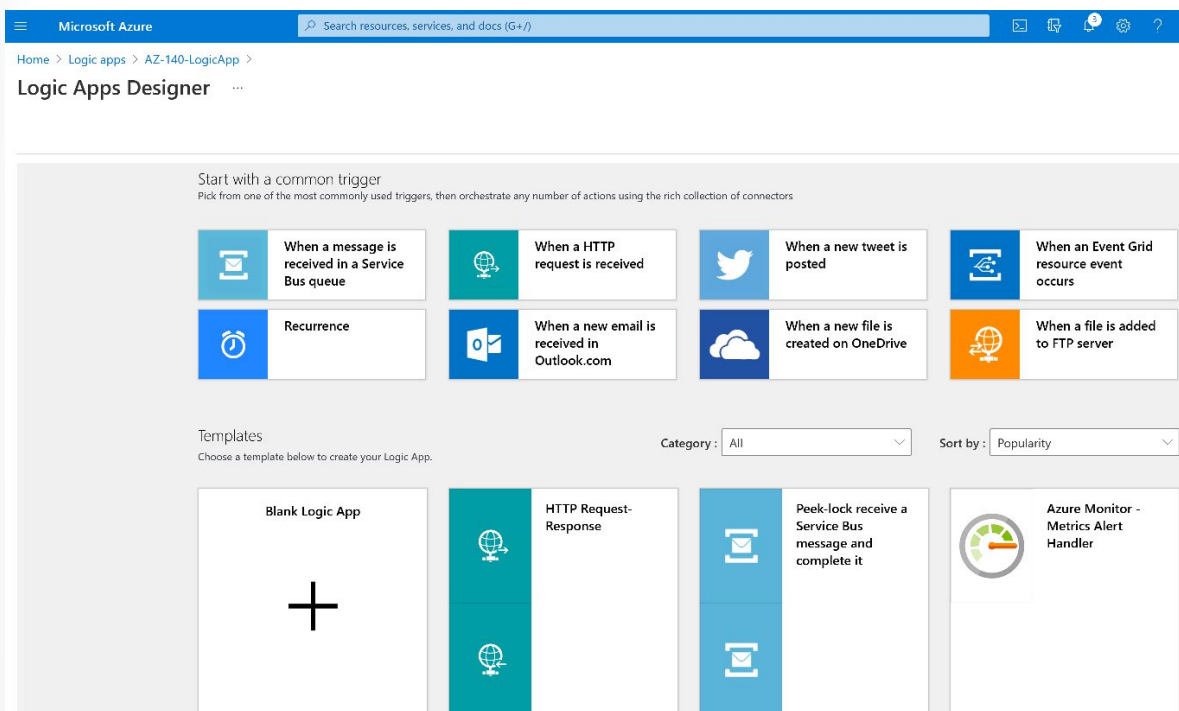
.\CreateOrUpdateAzLogicApp.ps1 @Params

```

After you run the script, the Azure Logic App should appear in a resource group, as shown in the following image.



To make changes to the execution schedule, such as changing the recurrence interval or time zone, go to the Azure Logic Apps autoscale scheduler and select **Edit** to go to the Azure Logic Apps Designer.



Monitor and manage performance and health

Monitor Azure Virtual Desktop by using Azure Monitor

This topic will walk you through how to set up Azure Monitor for Azure Virtual Desktop to monitor your Azure Virtual Desktop environments.

Before you start using Azure Monitor for Azure Virtual Desktop, you'll need to set up the following things:

- At least one configured Log Analytics Workspace. Use a designated Log Analytics workspace for your Azure Virtual Desktop session hosts to ensure that performance counters and events are only collected from session hosts in your Azure Virtual Desktop deployment.
- Enable data collection for the following things in your Log Analytics workspace:
 - Diagnostics from your Azure Virtual Desktop environment
 - Recommended performance counters from your Azure Virtual Desktop session hosts
 - Recommended Windows Event Logs from your Azure Virtual Desktop session hosts

The data setup process described in this topic is the only one you'll need to monitor Azure Virtual Desktop. You can disable all other items sending data to your Log Analytics workspace to save costs.

Anyone monitoring Azure Monitor for Azure Virtual Desktop for your environment will also need the following read-access permissions:

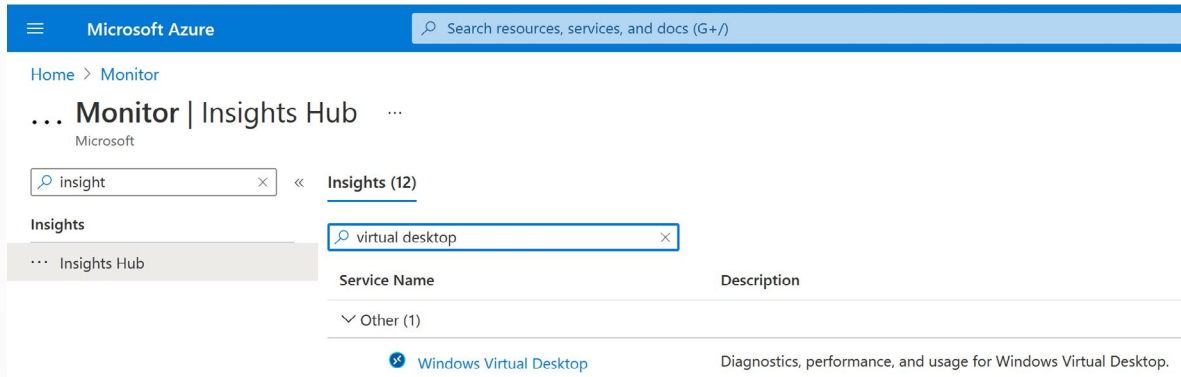
- **Read-access** to the Azure subscriptions that hold your Azure Virtual Desktop resources
- **Read-access** to the subscription's resource groups that hold your Azure Virtual Desktop session hosts
- **Read-access** to the Log Analytics workspace or workspaces

Read access only lets admins view data. They'll need different permissions to manage resources in the Azure Virtual Desktop portal.

Open Azure Monitor for Azure Virtual Desktop

You can open Azure Monitor for Azure Virtual Desktop by doing the following:

- Go to the Azure portal.
- Search for and select **Azure Monitor** from the Azure portal. Select **Insights Hub** under **Insights**, then select **Azure Virtual Desktop**. Once you have the page open, enter the **Subscription**, **Resource group**, **Host pool**, and **Time range** of the environment you want to monitor.



Log Analytics settings

To start using Azure Monitor for Azure Virtual Desktop, you'll need at least one Log Analytics workspace. Use a designated Log Analytics workspace for your Azure Virtual Desktop session hosts to ensure that performance counters and events are only collected from session hosts in your Azure Virtual Desktop deployment.

Log Analytics workspace for Azure Monitor

To start using Azure Monitor for Azure Virtual Desktop, you'll need at least one Log Analytics workspace.

Use a designated Log Analytics workspace for your Azure Virtual Desktop session hosts to ensure that performance counters and events are only collected from session hosts in your Azure Virtual Desktop deployment.

Resource diagnostic settings

To collect information on your Azure Virtual Desktop infrastructure, you'll need to enable several diagnostic settings on your Azure Virtual Desktop host pools and workspaces (this is your Azure Virtual Desktop workspace, not your Log Analytics workspace).

To set your resource diagnostic settings:

1. Select the **Diagnostic settings** host pool.
2. Under **Monitoring**, select **Diagnostic settings**.

Home > Host pools > az140-21a-hp1

Host pools

Default Directory

+ Create Manage view ...

Filter for any field...

Name ↑

az140-21a-hp1 ...

az140-21a-hp1 | Diagnostic settings

Host pool

Search (Ctrl+/) Refresh Feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

- RDP Properties
- Properties
- Locks

Manage

- Application groups
- MSIX packages
- Session hosts

Monitoring

- Diagnostic settings**
- Logs

Diagnostic settings

Name	Storage account	Event hub	Log Analytics works...
log3	-	-	serverlogs
serverlog1	-	-	demo-test-1

+ Add diagnostic setting

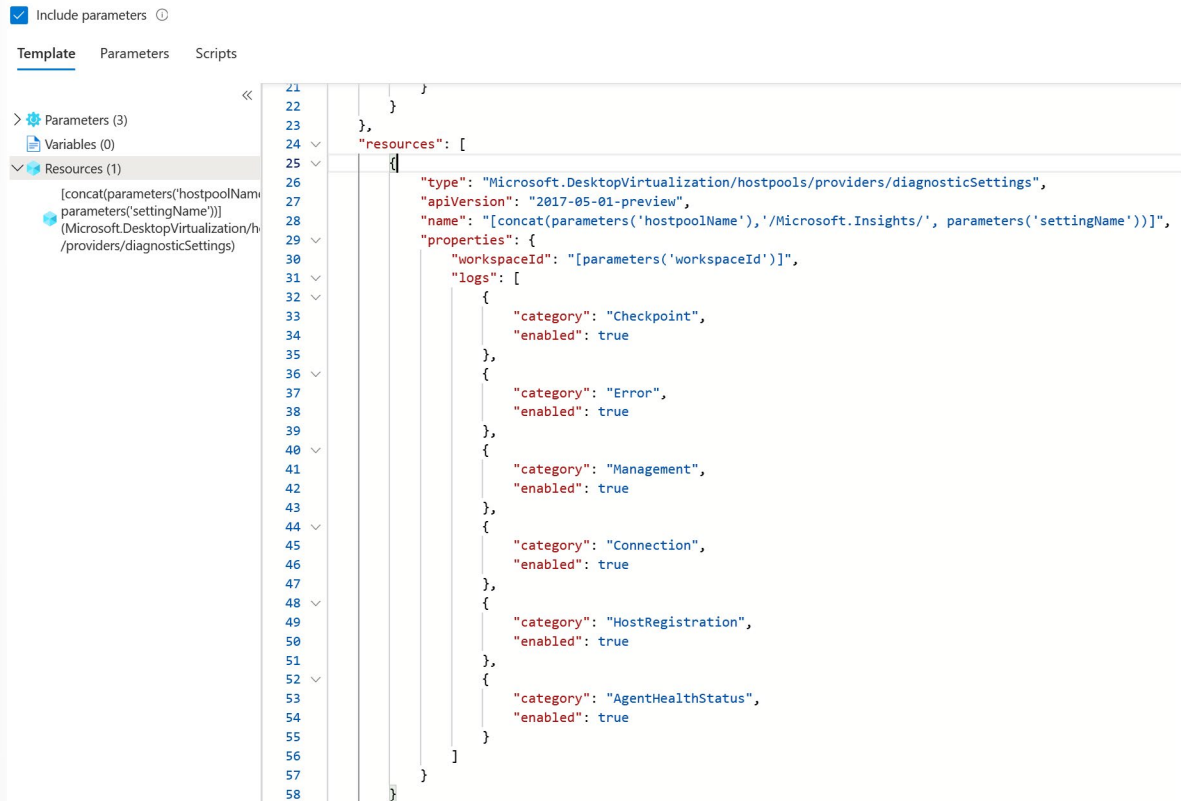
Click 'Add Diagnostic setting' above to configure the collection of the following data:

- Checkpoint
- Error
- Management
- Connection
- HostRegistration
- AgentHealthStatus

Host pool diagnostic settings

To set up host pool diagnostics using the resource diagnostic settings section in the configuration workbook:

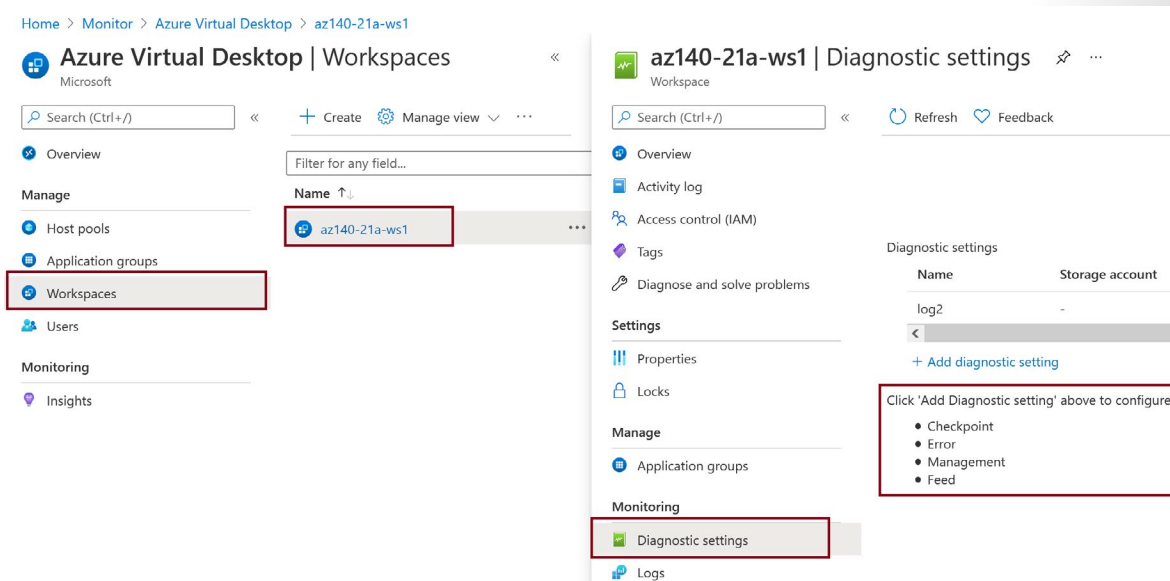
- Under **Host pool**, check to see whether Azure Virtual Desktop diagnostics are enabled. If they aren't, an error message will appear that says "No existing diagnostic configuration was found for the selected host pool." You'll need to enable the following supported diagnostic tables:
 - Checkpoint
 - Error
 - Management
 - Connection
 - HostRegistration
 - AgentHealthStatus



2. Select **Configure host pool**.
3. Select **Deploy**.
4. Refresh the configuration workbook.

Workspace diagnostic settings

To set up workspace diagnostics using the resource diagnostic settings section in the configuration workbook.



1. Under **Workspace**, check to see whether Azure Virtual Desktop diagnostics are enabled for the Azure Virtual Desktop workspace. If they aren't, an error message will appear that says "No existing diagnostic configuration was found for the selected workspace." You'll need to enable the following supported diagnostics tables:
 - Checkpoint
 - Error
 - Management
 - Feed
2. Select **Configure workspace**.
3. Select **Deploy**.
4. Refresh the configuration workbook.

Session host data settings

To collect information on your Azure Virtual Desktop session hosts, you'll need to install the Log Analytics agent on all session hosts in the host pool, make sure the session hosts are sending to a Log Analytics workspace, and configure your Log Analytics agent settings to collect performance data and Windows Event Logs.

The Log Analytics workspace you send session host data to doesn't have to be the same one you send diagnostic data to. If you have Azure session hosts outside of your Azure Virtual Desktop environment, we recommend having a designated Log Analytics workspace for the Azure Virtual Desktop session hosts.

To set the Log Analytics workspace where you want to collect session host data:

1. Select the **Session host data settings** tab in the configuration workbook.
2. Select the **Log Analytics workspace** you want to send session host data to.

Session hosts

You'll need to install the Log Analytics agent on all session hosts in the host pool and send data from those hosts to your selected Log Analytics workspace. If Log Analytics isn't configured for all the session

hosts in the host pool, you'll see a **Session hosts** section at the top of **Session host data settings** with the message "Some hosts in the host pool are not sending data to the selected Log Analytics workspace."

To set up your remaining session hosts using the configuration workbook:

1. Select **Add hosts to workspace**.
2. Refresh the configuration workbook.

Workspace performance counters

You'll need to enable specific performance counters to collect performance information from your session hosts and send it to the Log Analytics workspace.

To set up performance counters using the configuration workbook:

1. Under **Workspace performance counters** in the configuration workbook, check **Configured counters** to see the counters you've already enabled to send to the Log Analytics workspace. Check **Missing counters** to make sure you've enabled all required counters.
2. If you have missing counters, select **Configure performance counters**.
3. Select **Apply Config**.
4. Refresh the configuration workbook.
5. Make sure all the required counters are enabled by checking the **Missing counters** list.

Configure Windows Event Logs

You'll also need to enable specific Windows Event Logs to collect errors, warnings, and information from the session hosts and send them to the Log Analytics workspace.

To set up Windows Event Logs using the configuration workbook:

1. Under **Windows Event Logs configuration**, check **Configured Event Logs** to see the Event Logs you've already enabled to send to the Log Analytics workspace. Check **Missing Event Logs** to make sure you've enabled all Windows Event Logs.
2. If you have missing Windows Event Logs, select **Configure Events**.
3. Select **Deploy**.
4. Refresh the configuration workbook.
5. Make sure all the required Windows Event Logs are enabled by checking the **Missing Event Logs** list.

Monitor Azure Virtual Desktop by using Azure Advisor

Whenever you come across an issue in Azure Virtual Desktop, always check Azure Advisor first. Azure Advisor will give you directions for how to solve the problem, or at least point you towards a resource that can help.

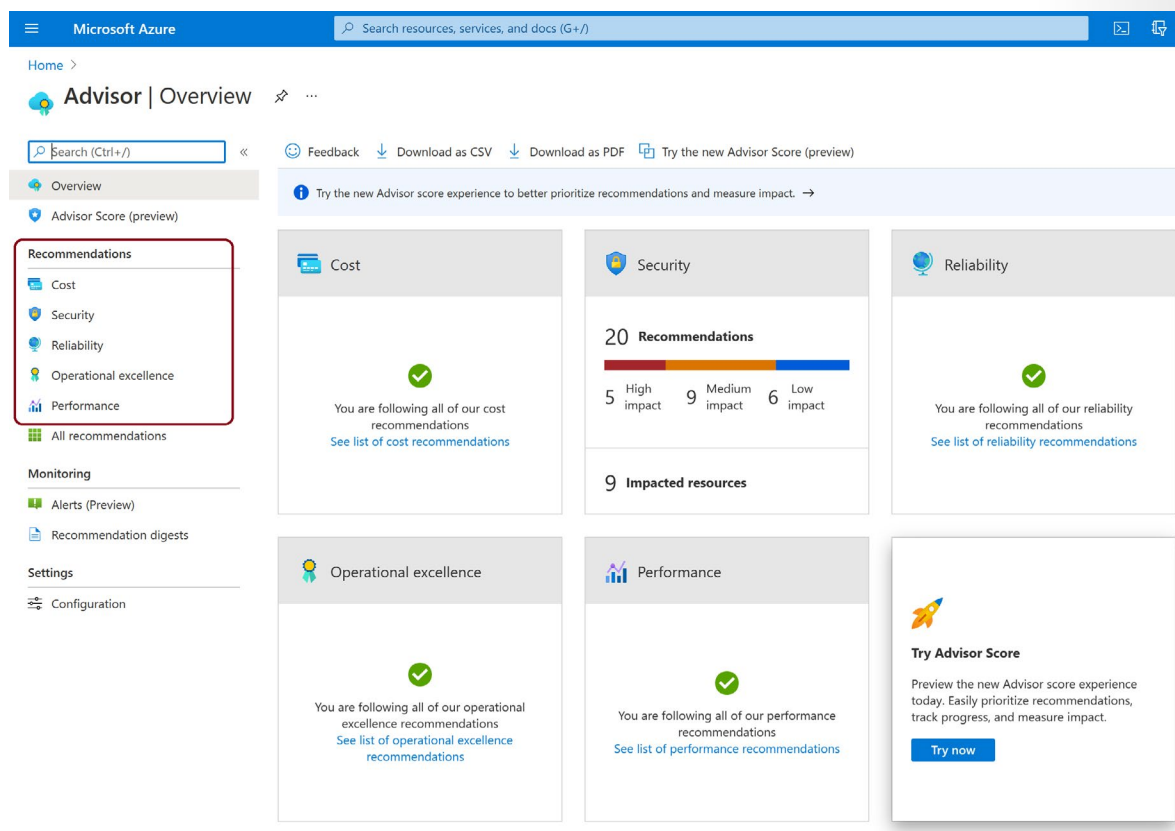
This topic will tell you how to set up Azure Advisor in your Azure Virtual Desktop deployment to help your users.

What is Azure Advisor?

Azure Advisor analyzes your configurations and telemetry to offer personalized recommendations to solve common problems. With these recommendations, you can optimize your Azure resources for reliability, security, operational excellence, performance, and cost.

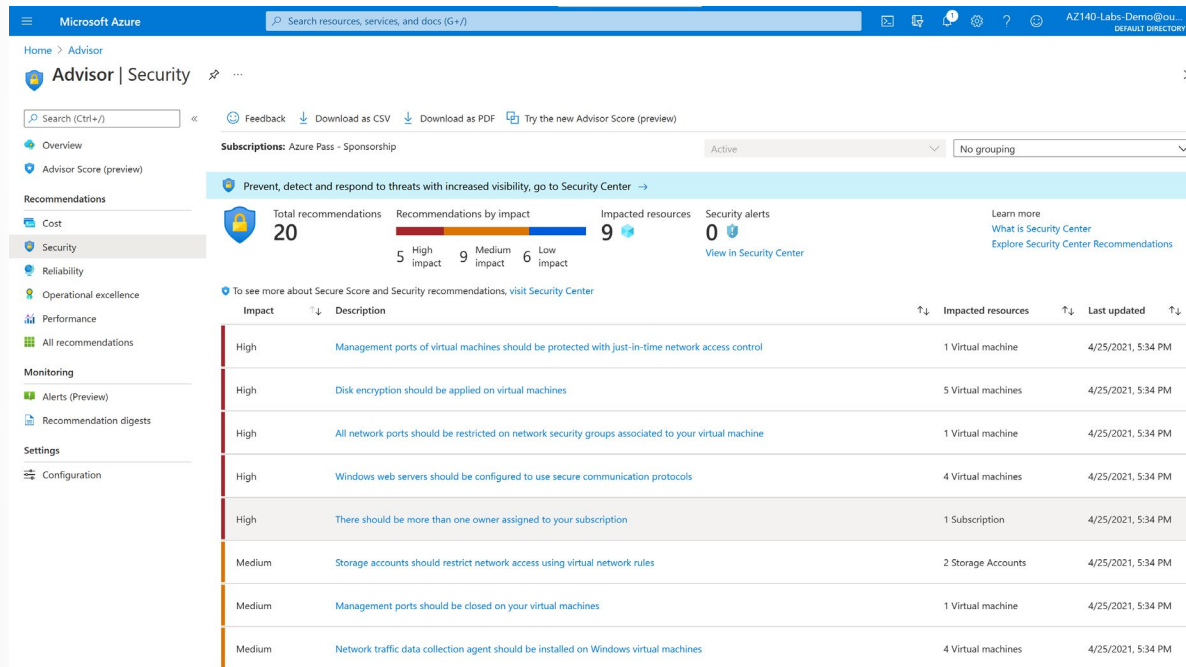
How to start using Azure Advisor

All you need to get started is an Azure account on the Azure portal. First, open the Azure portal then select **Advisor** under **Azure Services**, as shown in the following image. You can also enter "Azure Advisor" into the search bar in the Azure portal.



When you open Azure Advisor, you'll see five categories:

- Cost
- Security
- Reliability
- Operational Excellence
- Performance



Additional tips for Azure Advisor

- Make sure to check your recommendations frequently, at least more than once a week. Azure Advisor updates its active recommendations multiple times per day. Checking for new recommendations can prevent larger issues by helping you spot and solve smaller ones.
- Always try to solve the issues with the highest priority level in Azure Advisor. High priority issues are marked with red. Leaving high-priority recommendations unresolved can lead to problems down the line.
- If a recommendation seems less important, you can dismiss it or postpone it.
- Don't dismiss recommendations until you know why they're appearing and are sure it won't have a negative impact on you or your users.

How to resolve Azure Advisor recommendations

This topic describes how you can resolve recommendations that appear in Azure Advisor for Azure Virtual Desktop.

Recommendations to be resolved can include:

- No validation environment enabled
- Not enough production (non-validation) environments enabled
- Not enough links are unblocked to successfully implement your VM

No validation environment enabled

The screenshot shows the Microsoft Azure Advisor | Operational Excellence interface. The left sidebar contains navigation options: Overview, Recommendations, Monitoring, and Settings. The Recommendations section is active, showing a list of recommendations. A red box highlights a recommendation with the title "No validation environment enabled" and a medium impact level. The recommendation description states: "Ensure business continuity through WVD service deployments". The impacted resource is "1 Host Pool" and it was updated at "8/12/2020, 11:03:25 AM".

IMPACT	DESCRIPTION	POTENTIAL BENEFITS	IMPACTED RESOURCE	UPDATED AT
High	Enforce 'Audit VMs that do not use managed disks' using Azure Policy	This specific policy audits VMs that do not use managed disks.	1 Subscription	8/12/2020, 5:02:34 AM
Medium	No validation environment enabled	Ensure business continuity through WVD service deployments	1 Host Pool	8/12/2020, 11:03:25 AM
Medium	Enforce 'Inherit a tag from the resource group' using Azure Policy	This specific policy allows for resources to inherit tags from the parent resource group.	1 Subscription	8/12/2020, 5:01:46 AM
Medium	Enforce 'Allowed virtual machine SKUs' using Azure Policy	This specific policy restricts the virtual machine SKUs that your organization can deploy.	1 Subscription	8/12/2020, 5:02:26 AM
Medium	Enforce 'Add or replace a tag on resources' using Azure Policy	This specific policy allows for resources to be tagged or for tags to be replaced.	1 Subscription	8/12/2020, 5:01:34 AM
Medium	Enforce 'Allowed locations' using Azure Policy	This specific policy restricts the available locations where your organization can deploy new resources.	1 Subscription	8/12/2020, 5:01:38 AM
Low	Create an Azure service health alert	Get notified when Azure service issues affect you	1 Subscription	8/12/2020, 9:50:43 AM

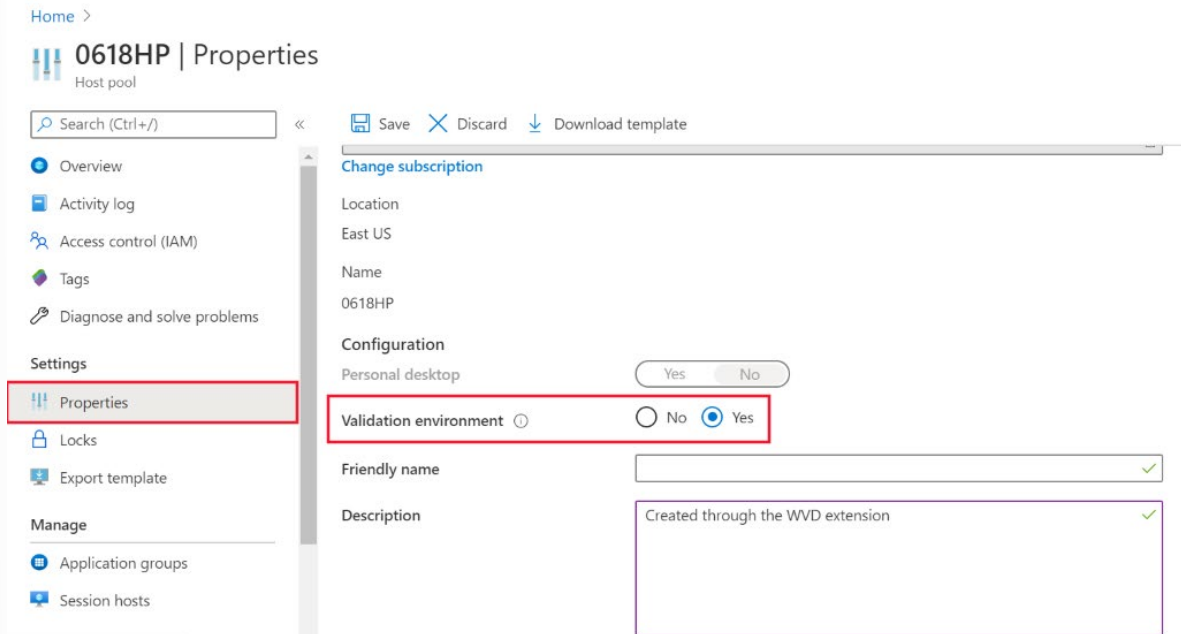
This recommendation appears under Operational Excellence. The recommendation should also show you a warning message like this:

*You don't have a validation environment enabled in this subscription. When you made your host pools, you selected **No** for "Validation environment" in the Properties tab. To ensure business continuity through Azure Virtual Desktop service deployments, make sure you have at least one host pool with a validation environment where you can test for potential issues.*

You can make this warning message go away by enabling a validation environment in one of your host pools.

To enable a validation environment:

1. Go to your Azure portal home page and select the host pool you want to change.
2. Next, select the host pool you want to change from a production environment to a validation environment.
3. In your host pool, select **Properties** on the left column. Next, scroll down until you see "Validation environment." Select **Yes**, then select **Apply**.



What happens next:

- These changes won't make the warning go away immediately.
- Azure Advisor updates twice a day.
- Allow enough time for the recommendations to go away on their own.

Not enough production (non-validation) environments enabled

This recommendation appears under Operational Excellence.

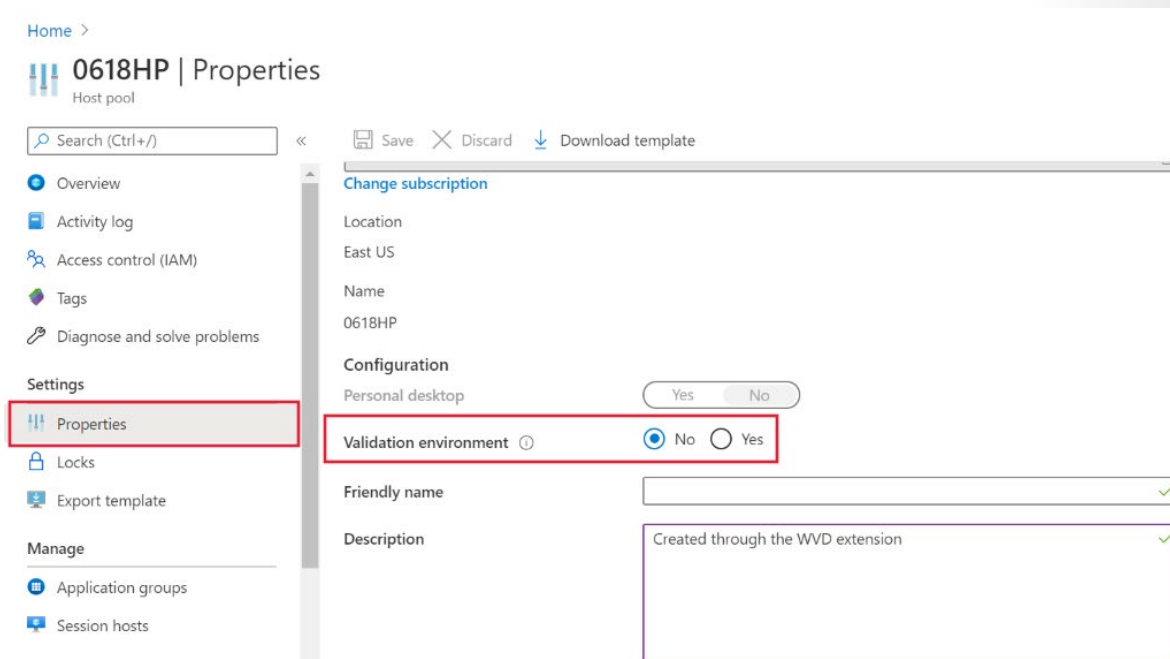
For this recommendation, the warning message appears for one of these reasons:

- You have too many host pools in your validation environment.
- You don't have any production host pools.

We recommend users have fewer than half of their host pools in a validation environment.

To resolve this warning:

1. Go to your Azure portal home page.
2. Select the host pools you want either want to change from validation to production.
3. In your host pool, select the **Properties** tab in the column on the right side of the screen. Next, scroll down until you see *Validation environment*. Select **No**, then select **Apply**.



Not enough links are unblocked to successfully implement your VM

This recommendation appears under Operational Excellence.

You need to unblock specific URLs to make sure that your virtual machine (VM) functions properly. You can see the list at Safe URL list. If the URLs aren't unblocked, then your VM won't work properly.

To solve this recommendation, make sure you unblock all the URLs on the Safe URL list. You can use Service Tag or FQDN tags to unblock URLs, too.

Diagnose graphics performance issues

To diagnose experience quality issues with your remote sessions, counters have been provided under the RemoteFX Graphics section of Performance Monitor. This topic helps you pinpoint and fix graphics-related performance bottlenecks during Remote Desktop Protocol (RDP) sessions using these counters.

Find your remote session name

You'll need your remote session name to identify the graphics performance counters. Follow the instructions in this section to identify your instance of each counter.

1. Open the Windows command prompt from your remote session.
2. Run the **qwinsta** command and find your session name.
 - If your session is hosted in a multi-session virtual machine (VM): Your instance of each counter is suffixed by the same number that suffixes your session name, such as "rdp-tcp 37."

- If your session is hosted in a VM that supports virtual Graphics Processing Units (vGPU): Your instance of each counter is stored on the server instead of in your VM. Your counter instances include the VM name instead of the number in the session name, such as "Win8 Enterprise VM."

Access performance counters

After you've determined your remote session name, follow these instructions to collect the RemoteFX Graphics performance counters for your remote session.

1. Select **Start > Administrative Tools > Performance Monitor**.
2. In the **Performance Monitor** dialog box, expand **Monitoring Tools**, select **Performance Monitor**, and then select **Add**.
3. In the **Add Counters** dialog box, from the **Available Counters** list, expand the section for RemoteFX Graphics.
4. Select the counters to be monitored.
5. In the **Instances of selected object** list, select the specific instances to be monitored for the selected counters and then select **Add**. To select all available counter instances, select **All instances**.
6. After adding the counters, select **OK**.

The selected performance counters will appear on the Performance Monitor screen.

Diagnose issues

Graphics-related performance issues generally fall into four categories:

- Low frame rate
- Random stalls
- High input latency
- Poor frame quality

Addressing low frame rate, random stalls, and high input latency

First check the Output Frames/Second counter. It measures the number of frames made available to the client. If this value is less than the Input Frames/Second counter, frames are being skipped. To identify the bottleneck, use the Frames Skipped/Second counters.

There are three types of Frames Skipped/Second counters:

- Frames Skipped/Second (Insufficient **Server** Resources)
- Frames Skipped/Second (Insufficient **Network** Resources)
- Frames Skipped/Second (Insufficient **Client** Resources)

A high value for any of the Frames Skipped/Second counters implies that the problem is related to the resource the counter tracks.

If the Output Frames/Second counter matches the Input Frames/Second counter, yet you still notice unusual lag or stalling, Average Encoding Time may be the culprit. Encoding is a synchronous process that occurs on the server in the single-session (vGPU) scenario and on the VM in the multi-session scenario. Average Encoding Time should be under 33 ms.

Because RDP supports an Average Encoding Time of 33 ms, it supports an input frame rate up to 30 frames/second. Note that 33 ms is the maximum supported frame rate. In many cases, the frame rate experienced by the user will be lower, depending on how often a frame is provided to RDP by the source.

Addressing poor frame quality

Use the Frame Quality counter to diagnose frame quality issues. This counter expresses the quality of the output frame as a percentage of the quality of the source frame. The quality loss may be due to RemoteFX, or it may be inherent to the graphics source. If RemoteFX caused the quality loss, the issue may be a lack of network or server resources to send higher-fidelity content.

Lab

Implement autoscaling in host pools (AD DS)

Important: To download the most recent version of this lab, please visit the AZ-140 [GitHub repository](#)³.

Direct link to the **Lab - Implement autoscaling in host pools (AD DS)**.⁴

Objectives

After completing this lab, you will be able to:

- Configure autoscaling of Azure Virtual Desktop session hosts
- Verify autoscaling of Azure Virtual Desktop session hosts

Lab prerequisites

- An Azure subscription you will be using in this lab.
- A Microsoft account or an Azure AD account with the Owner or Contributor role in the Azure subscription you will be using in this lab and with the Global Administrator role in the Azure AD tenant associated with that Azure subscription.
- The completed lab **Prepare for deployment of Azure Virtual Desktop (AD DS)**
- The completed lab **Deploy host pools and session hosts by using the Azure portal (AD DS)**

Estimated time: 60 minutes

Lab files

- None

Exercise 1: Configure autoscaling of Azure Virtual Desktop session hosts

The main tasks for this exercise are as follows:

1. Prepare for autoscaling of Azure Virtual Desktop session hosts
2. Create and configure an Azure Automation account
3. Create an Azure Logic app

Exercise 2: Verify and review autoscaling of Azure Virtual Desktop session hosts

The main tasks for this exercise are as follows:

1. Verify autoscaling of Azure Virtual Desktop session hosts
2. Use Azure Log Analytics to track Azure Virtual Desktop events

³ https://aka.ms/AZ-140_Labs

⁴ https://aka.ms/AZ-140_05_Lab_01

Exercise 3: Stop and deallocate Azure VMs provisioned in the lab

The main tasks for this exercise are as follows:

1. Stop and deallocate Azure VMs provisioned in the lab

Note: In this exercise, you will deallocate the Azure VMs provisioned in this lab to minimize the corresponding compute charges

Review questions

Module review questions

Review Question 1

You have an AVD deployment.

The session hosts are joined to an on-premises AD domain named westwind.com.

You need to limit your user sessions to three hours.

What needs to be configured?

- ☐ A. A GPO in westwind.com
- ☐ B. Run Register-AzWvdApplicationGroup
- ☐ C. Create an Azure Storage account that uses GRS
- ☐ D. Configure a Run As account in Azure Automation

Review Question 2

You manage an AVD deployment.

You deploy and configure WBD in a secondary location.

You plan to perform a test failover to the secondary location, but discover existing user sessions to the primary location.

*You need to sign out the users from the session hosts in the primary location.
which PowerShell cmdlet should you use?*

- ☐ A. Register-AzWvdApplicationGroup
- ☐ B. Update-AzWvdApplicationGroup
- ☐ C. Remove-AzWvdUserSession
- ☐ D. Register-AzWvdApplicationGroup

Review Question 3

You have an AVD deployment with the following host pools:

WestPool5

- Windows 10 Ent (personal)
- West US Azure region

WestPool7

- Windows Server 2019 (pooled)

- West US Azure region

*You need to implement a disaster recovery plan in the Central US region.
What do you need to include in the plan?*

- ☐ A. An Azure Site Recovery plan
- ☐ B. Create a new host pool in the Central US Azure region
- ☐ C. Enable Azure Backup in the Central US Azure region
- ☐ D. Add another host pool in the West US Azure region

Review Question 4

*You have an AVD host pool in the Central US Azure region.
You want to make sure that the host pool can failover to the US East Azure region.
What do you do first?*

- ☐ A. Configure the RDP properties the Central US host pool
- ☐ B. Run Update-AZWvdApplication group
- ☐ C. Create a new host pool in the East US
- ☐ D. Create a Recovery Services vault

Review Question 5

*You have an Azure Virtual Desktop host pool named host_poolTmp and an Azure automation account named autoaccount2. host_poolTmp is integrated with an Azure AD DS (westwind.com).
You plan to configure scaling for host_poolTmp using Azure Automation runbooks.
You need to authorize the runbooks to manage the scaling of host_poolTmp.
You should configure?*

- ☐ A. An additional host pool
- ☐ B. A Run As account in Azure Automation
- ☐ C. An Azure Site recovery plan
- ☐ D. A token to re-register the VMs in the host pool

Review Question 6

*You have an AVD deployment.
You use the Start/Stop VMs during off- hours solution in Azure.
You need to configure which virtual machines must never be stopped by the solution.
What should you configure?*

- ☐ A. An Azure Automation account variable
- ☐ B. A connection shared resource in Azure Automation
- ☐ C. A managed identity in AAD
- ☐ D. An Azure NetApp account

Review Question 7

You have an AVD host pool running a Win10 enterprise multi session.

You want to configure automatic scaling of the host pool to fulfill the following:

- Distribute new user sessions across all running session hosts
- Automatically start a new session host when concurrent user sessions exceed 30 users per host

What should you include in the solution?

- ☐ A. Azure Front Door with depth-first load balancing
- ☐ B. Azure traffic manager with weighted and performance traffic routing
- ☐ C. An Azure Automation account and the breadth-first load balancing algorithm
- ☐ D. Azure load balancer with cross-region load balancing

Review Question 8

You have an AVD instance.

You want to monitor the AVD instance using Azure Virtual Desktop Insights that is included in Azure Monitor.

What should you use as a diagnostic settings destination for the host pool?

- ☐ A. Azure Data Lake Storage Gen1
- ☐ B. Azure Files
- ☐ C. Azure Queues
- ☐ D. Log Analytics Workspace

Review Question 9

You have an AVD deployment.

Using Azure Advisor, you are given the following recommendation related to AVD.

- Impact: Medium
- Description: No validation environment enabled
- Benefits: Ensure business continuity through AVD service deployments
- Impacted resources: HOST-West-Pool2

What are the benefit of following the recommendation?

- ☐ A. You can validate allowed locations
- ☐ B. You can validate preview features for Azure Virtual Desktop
- ☐ C. You can validate audited VMs that do not use managed disks
- ☐ D. You can validate log analytics file storage access

Review Question 10

You have an AVD host pool named NewPool1. The pool contains Session Hosts that run Windows 10 Enterprise multi-session.

You need to use Performance Monitor to troubleshoot a low frame quality issue that is affecting a user in NewPool1.

What should you run to retrieve the user ID?

- ☐ A. Remove-AzWvdApplication
- ☐ B. Disconnect-AzWvdUserSession
- ☐ C. Get-AzWvdWorkspace
- ☐ D. qwinsta

Review Question 11

You manage an AVD host pool. The pool contains Session Hosts that run Windows 10 Enterprise multi-session.

You connect to a Remote Desktop session on hostpool5 and find an issue with the frequency of screen updates.

You need to identify whether the issue relates to insufficient server, network, or client resources.

The solution should minimize the time it takes to identify the resource type.

What should you do?

- ☐ A. In the current session, use Performance Monitor to display the values of all the `RemoteFX Graphics(*)\frames skipped/second` counters
- ☐ B. From within the RegEdit, navigate to `Computer\HKEY_LOCAL_MACHINE\software\FsLogix` and create a key named Profiles for path to VHDLocations
- ☐ C. From Azure Cloud Shell, run the `Update-AzWvdDesktopcmdlet` and specify the `InputObject` parameter
- ☐ D. From Azure Cloud Shell, run the `Remove-AzWvdApplicationGroup` cmdlet and specify the `-DefaultProfile` parameter

Review Question 12

You have an AVD deployment with the following host pools:

WestPool5

- Windows 10 Enterprise (personal)
- West US Azure region

WestPool7

- Windows Server 2019 (pooled)

- West US Azure region

You need to implement a disaster recovery plan in the Central US region. What do you need to include in the plan?

- ☐ A. An Azure Site Recovery plan
- ☐ B. Create a new host pool in the Central US Azure region
- ☐ C. Enable Azure Backup in the Central US Azure region
- ☐ D. Add another host pool in the West US Azure region

Answers

Review Question 1

You have an AVD deployment.

The session hosts are joined to an on-premises AD domain named westwind.com.

You need to limit your user sessions to three hours.

What needs to be configured?

- ☒ A. A GPO in westwind.com
- ☐ B. Run Register-AzWvdApplicationGroup
- ☐ C. Create an Azure Storage account that uses GRS
- ☐ D. Configure a Run As account in Azure Automation

Explanation

The answer is A, you need to configure a GPO in westwind.com.

As seen in the *Scale session hosts using Azure Automation* topic, if you set the `LimitSecondsToForceLogOffUser` parameter to zero, the job allows the session configuration setting in specified group policies to handle signing off user sessions.

To see these group policies, go to *Computer Configuration > Policies > Administrative Templates > Windows Components > Remote Desktop Services > Remote Desktop Session Host > Session Time Limits*. If there are any active sessions on a session host VM, the job will leave the session host VM running. If there aren't any active sessions, the job will shut down the session host VM.

During any time, the job also takes host pool's `MaxSessionLimit` into account to determine if the current number of sessions is more than 90% of the maximum capacity. If it is, the job will start additional session host VMs.

During any time, the job also takes host pool's `MaxSessionLimit` into account to determine if the current number of sessions is more than 90% of the maximum capacity.

Review Question 2

You manage an AVD deployment.

You deploy and configure WBD in a secondary location.

You plan to perform a test failover to the secondary location, but discover existing user sessions to the primary location.

You need to sign out the users from the session hosts in the primary location. Which PowerShell cmdlet should you use?

- ☐ A. Register-AzWvdApplicationGroup
- ☐ B. Update-AzWvdApplicationGroup
- ☒ C. Remove-AzWvdUserSession
- ☐ D. Register-AzWvdApplicationGroup

Explanation

The answer is C, you should use `Remove-AzWvdUserSession`.

As seen in the *VM replication* topic, to disconnect users in the Azure-integrated version of Azure Virtual Desktop, run this cmdlet:

Once you've signed out all users in the primary region, you can fail over the VMs in the primary region and let users connect to the VMs in the secondary region.

Review Question 3

You have an AVD deployment with the following host pools:
WestPool5

WestPool7

You need to implement a disaster recovery plan in the Central US region.
What do you need to include in the plan?

- ☒ A. An Azure Site Recovery plan
- ☐ B. Create a new host pool in the Central US Azure region
- ☐ C. Enable Azure Backup in the Central US Azure region
- ☐ D. Add another host pool in the West US Azure region

Explanation

The answer is A, you should include an Azure Site Recovery plan.

As seen in the *VM replication and Virtual networks, user identities, and data topics*, it is recommended you use Azure Site Recovery to manage replicating VMs in other Azure locations (as described in *Azure-to-Azure disaster recovery architecture*).

It is also recommended that you use Azure Site Recovery for personal host pools, because Azure Site Recovery supports both server-based and client-based SKUs.

Review Question 4

You have an AVD host pool in the Central US Azure region.
You want to make sure that the host pool can failover to the US East Azure region.
What do you do first?

- ☐ A. Configure the RDP properties of the Central US host pool
- ☐ B. Run Update-AZWvdApplication group
- ☐ C. Create a new host pool in the East US
- ☒ D. Create a Recovery Services vault

Explanation

The answer is D, you should create a Recovery Services vault.

As seen in the *Configure backup and for FSLogix user profiles, personal VDI, and images and Configure backup from the Recovery Services vault topics*, Recovery Services vault is a management entity that stores recovery points created over time and provides an interface to perform backup-related operations. These include taking on-demand backups, performing restores, and creating backup policies.

Review Question 5

You have an Azure Virtual Desktop host pool named *host_poolTmp* and an Azure automation account named *autoaccount2*. *host_poolTmp* is integrated with an Azure AD DS (westwind.com). You plan to configure scaling for *host_poolTmp* using Azure Automation runbooks. You need to authorize the runbooks to manage the scaling of *host_poolTmp*. You should configure?

- ☐ A. An additional host pool
- ☒ B. A Run As account in Azure Automation
- ☐ C. An Azure Site recovery plan
- ☐ D. A token to re-register the VMs in the host pool

Explanation

The answer is B, you should configure a Run As account in Azure Automation.

As seen in the *Create an Azure Automation Run As account* topic, an Azure Automation Run As account provides authentication for managing resources in Azure with Azure cmdlets.

When you create a Run As account, it creates a new service principal user in Azure Active Directory and assigns the Contributor role to the service principal user at the subscription level.

An Azure Run As account is a great way to authenticate securely with certificates and a service principal name without needing to store a username and password in a credential object.

Review Question 6

You have an AVD deployment.
You use the Start/Stop VMs during off- hours solution in Azure.
You need to configure which virtual machines must never be stopped by the solution.
What should you configure?

- ☒ A. An Azure Automation account variable
- ☐ B. A connection shared resource in Azure Automation
- ☐ C. A managed identity in AAD
- ☐ D. An Azure NetApp account

Explanation

The answer is A, you should configure an Azure Automation account variable.

As seen in the *Scale session hosts using Azure Automation* topic, the scaling tool in Azure Automation account provides a low-cost automation option for customers who want to optimize their session host VM costs.

You can use the scaling tool to:

Review Question 7

You have an AVD host pool running a Win10 enterprise multi session.
You want to configure automatic scaling of the host pool to fulfill the following:

What should you include in the solution?

- ☐ A. Azure Front Door with depth-first load balancing
- ☐ B. Azure traffic manager with weighted and performance traffic routing
- ☒ C. An Azure Automation account and the breadth-first load balancing algorithm
- ☐ D. Azure load balancer with cross-region load balancing

Explanation

The answer is C, you should include an Azure Automation account and the breadth-first load balancing algorithm.

As seen in the *Scale session hosts using Azure Automation* topic, you can use the scaling tool to:

Review Question 8

You have an AVD instance.

You want to monitor the AVD instance using Azure Virtual Desktop Insights that is included in Azure Monitor.

What should you use as a diagnostic settings destination for the host pool?

- ☐ A. Azure Data Lake Storage Gen1
- ☐ B. Azure Files
- ☐ C. Azure Queues
- ☒ D. Log Analytics Workspace

Explanation

The answer is D, you should use Log Analytics Workspace as a diagnostic settings destination.

As seen in the *Monitor AVD by using Azure Monitor* topic, you use Log Analytics Workspace.

To start using Azure Monitor for Azure Virtual Desktop, you'll need at least one Log Analytics workspace. Use a designated Log Analytics workspace for your Azure Virtual Desktop session hosts to ensure that performance counters and events are only collected from session hosts in your Azure Virtual Desktop deployment.

Review Question 9

You have an AVD deployment.

Using Azure Advisor, you are given the following recommendation related to AVD.

What are the benefits of following the recommendation?

- ☐ A. You can validate allowed locations
- ☒ B. You can validate preview features for Azure Virtual Desktop
- ☐ C. You can validate audited VMs that do not use managed disks
- ☐ D. You can validate log analytics file storage access

Explanation

The answer is B, you can validate preview features for Azure Virtual Desktop.

As seen in the *How to resolve Azure Advisor recommendations* topic, the message says: You don't have a validation environment enabled in this subscription. When you made your host pools, you selected No for "Validation environment" in the Properties tab. To ensure business continuity through Azure Virtual Desk-

top service deployments, make sure you have at least one host pool with a validation environment where you can test for potential issues.

You can make this warning message go away by enabling a validation environment in one of your host pools.

Review Question 10

You have an AVD host pool named *NewPool1*. The pool contains Session Hosts that run Win 10 Ent multi-session.

You need to use Performance Monitor to troubleshoot a low frame quality issue that is affecting a user in *NewPool1*.

What should you run to retrieve the user ID?

- ☐ A. Remove-AzWvdApplication
- ☐ B. Disconnect-AzWvdUserSession
- ☐ C. Get-AzWvdWorkspace
- ☒ D. qwinsta

Explanation

The answer is D, you should run qwinsta.

As seen in the topic [Diagnose graphics performance issues](#), you can run the qwinsta command and find your session name if your session is hosted in a multi-session virtual machine (VM), or your session is hosted in a VM that supports virtual Graphics Processing Units (vGPU).

See: <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/qwinsta>

Review Question 11

You manage an AVD host pool. The pool contains Session Hosts that run Win 10 Ent multi-session.

You connect to a RD session on *hostpool5* and find an issue with the frequency of screen updates.

You need to identify whether the issue relates to insufficient server, network, or client resources.

The solution should minimize the time it takes to identify the resource type.

What should you do?

- ☒ A. In the current session, use Performance Monitor to display the values of all the RemoteFX Graphics(*)\frames skipped/second counters
- ☐ B. From within the RegEdit, navigate to Computer\HKEY_LOCAL_MACHINE\software\FsLogix and create a key named Profiles for path to VHDLocations
- ☐ C. From Azure Cloud Shell, run the Update-AzWvdDesktopcmdlet and specify the InputObject parameter
- ☐ D. From Azure Cloud Shell, run the Remove-AzWvdApplicationGroup cmdlet and specify the -Default-Profile parameter

Explanation

The answer is A, while in the current session, use Performance Monitor to display the values of all the RemoteFX Graphics()\frames skipped/second counters.*

As seen in the [Diagnose graphics performance issues](#) topic, the Output Frames/Second counter measures the number of frames made available to the client. If this value is less than the Input Frames/Second counter, frames are being skipped.

There are three types of Frames Skipped/Second counters:

Review Question 12

You have an AVD deployment with the following host pools:
WestPool5

WestPool7

You need to implement a disaster recovery plan in the Central US region.
What do you need to include in the plan?

- ☒ A. An Azure Site Recovery plan
- ☐ B. Create a new host pool in the Central US Azure region
- ☐ C. Enable Azure Backup in the Central US Azure region
- ☐ D. Add another host pool in the West US Azure region

Explanation

The answer is A, an Azure Site Recovery plan

As seen in the VM replication and Virtual networks, user identities, and data topics, it is recommended you use Azure Site Recovery to manage replicating VMs in other Azure locations (as described in Azure-to-Azure disaster recovery architecture).

It is also recommended that you use Azure Site Recovery for personal host pools, because Azure Site Recovery supports both server-based and client-based SKUs.